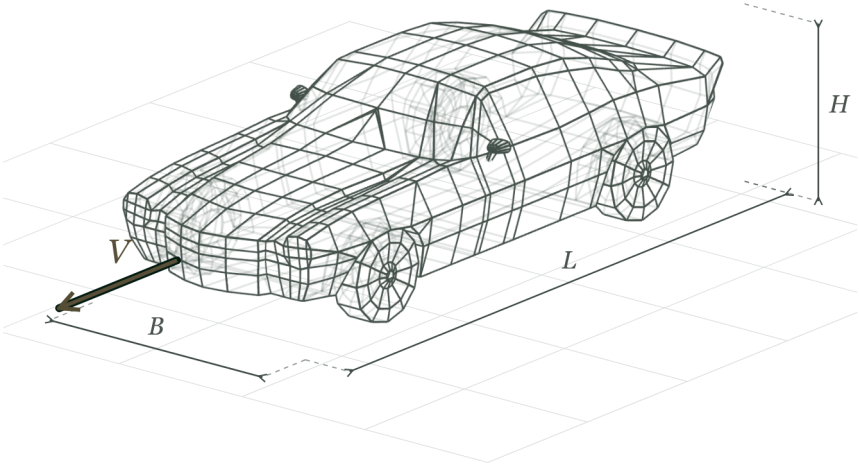
Ministry of Education and Science of Ukraine
Zhytomyr Agrotechnical Professional College
Department of Automotive Transport

Vehicle Design Calculation

Guidelines for the course project in Automobiles

Complete vehicle calculation: formulas, alternatives, tables and examples



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Zhytomyr Agrotechnical Professional College

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Interactive guidelines and calculation examples for the course project.

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About this document

Who prepared these guidelines, how they were compiled and where they apply.

These guidelines describe the course project workflow for Automobiles: which quantities are specified, which are calculated, in what order and using which formulas.

This document and the teaching application are built from a single source. Formulas, relationships, units and numerical examples come from the same calculation engine used by the application; they are not copied here by hand. This keeps the book consistent with the software: changing a formula also changes its printed version.

Reference tables in this guide use the chapter number and the prefix М, for example Table М.4.1, to distinguish them from the current project's tables. Input-value tables use the prefix Д and the associated formula number, for example Table Д.4.20; an additional final digit distinguishes calculations or branches. The prefix П identifies supplementary tables of symbols and design assumptions. A numerical substitution retains the number of its general formula.

Authors of these guidelines and development contributors are listed on the credits page. **Authors of the original sources** are the authors of the classical traction-calculation methods used by individual chapters; each formula cites its source.

Scope. The guidelines cover vehicle design calculations at course-project level. They do not replace design documentation or testing. The resulting numbers are design estimates suitable for selecting components and checking them against allowable limits.

Examples. The worked example follows one conventional vehicle with an internal combustion engine. Where the design branches, short examples of other powertrains appear alongside it, each clearly identified.

Quantity symbols

Names of the quantities used in the formulas in these guidelines.

Symbols are listed together with their units. Where the same letter is used in different chapters, the explanation beside the formula defines its meaning.

Table П.1 — Quantity symbols in the guidelines

Symbol	Quantity	Unit
$\omega_{дж,max}$	maximum angular speed of the traction source	s^{-1}
$M_{дж,max}$	maximum torque of the traction source	N·m
$u_{кmin}$	lowest gearbox ratio	—
$u_{к1}$	highest gearbox ratio	—
$k_{ω}$	rotational-speed safety factor	—
$K_{зап}$	critical-speed margin above operating speed	—
$L_{в}$	propeller-shaft length	m
d	shaft inner diameter	mm
β	clutch torque reserve factor	—
C_{w}	constant coefficient in the critical-speed formula	mm/s
$A_{z}ero$	zero area used to clip a negative square-root argument	mm^2
G_{s}	shear modulus of the shaft material	Pa
$k_{d}eg$	conversion factor from radians to degrees	deg
$\omega_{m}ax$	maximum shaft speed	s^{-1}
$\omega_{k}r$	critical shaft speed	s^{-1}
$D_{потр}$	required shaft outer diameter	mm
D	adopted shaft outer diameter	mm
$M_{р}$	design shaft torque	N·m
φ	shaft twist angle	deg
$n_{осіб}$	number of occupants, including the driver	number
$m_{особи}$	assumed mass per person	kg
$M_{багаж}$	luggage mass	kg
$M_{в}$	cargo mass	kg
q	curb-mass coefficient	—
$M_{кор}$	payload	kg
M_{0}	vehicle curb mass	kg
M_{a}	gross vehicle mass	kg

Symbol	Quantity	Unit
G_{a}	total vehicle weight force	kN
G_{i}	axle weight forces	kN
k_1	front-axle weight fraction	—
f_0	rolling-resistance coefficient at low speed	—
f	rolling-resistance coefficient at maximum speed	—
s	road gradient	m/m
B	vehicle wheelbase	m
B_1	rear-axle spacing	m
E_{u}	usable battery energy within the BMS 0–100% SOC window	kWh
ρ_{u}	usable specific energy of the complete battery pack	Wh/kg
M_{b}	vehicle mass excluding the battery and payload	kg
$M_{\text{кор,ref}}$	adopted reference occupant, luggage and cargo load of the BEV	kg
$E_{\text{u,ref}}$	usable capacity of the reference BEV pack	kWh
$\rho_{\text{u,ref}}$	usable specific energy of the complete reference BEV pack	Wh/kg
s_0	initial state of charge within the usable BMS window	—
$s_{\text{m}}in$	reserve state of charge within the usable BMS window	—
$M_{\text{l}}im$	allowable gross vehicle mass	kg
c_{b}	average chemical energy consumption while repeating the profile to the reserve	kWh/km
E_{trip}	trip energy available above the SOC reserve	kWh
φ	tyre–road adhesion coefficient	—
$u_{\text{тр}}$	adopted overall transmission ratio in top gear	—
C_{d}	adopted aerodynamic drag coefficient	—
A	adopted frontal projected area	m²
η_{i}	adopted indicated engine efficiency	—
$\psi_{\max}$	maximum road-resistance coefficient for the intended use	—
$\eta_{\text{ем}}^{\max}$	adopted maximum electric-machine efficiency	—
$i_{\text{g,front}}$	overall reduction ratio from the electric machine to the front wheels	—
$i_{\text{g,rear}}$	overall reduction ratio from the electric machine to the rear wheels	—
k	adopted gearshift strategy	—
$u_{\text{к,min}}$	adopted top-gear ratio	—
$u_{\text{рк,min}}$	top-range transfer-case ratio	—

Symbol	Quantity	Unit
$\eta_{м}$	adopted mechanical driveline efficiency	—
$\lambda_{ст}$	tyre deflection coefficient under load	—
V_{f}	target final acceleration speed	m/s
t_{a}	target acceleration time	s
ρ	air density for the adopted atmospheric conditions	kg/m³
$P_{доп}$	electrical auxiliary power	kW
l	number of bevel or hypoid gear pairs in the power path	number
K_{e}	braking-system efficiency class	—
$V_{г}$	initial braking speed	m/s
m	number of universal joints in the power path	number
h_{g}	adopted centre-of-mass height above the road	m
k	number of cylindrical gear pairs in the power path	number
ω_{x}	engine idle angular speed	rad/s
P_{max}	adopted maximum engine power	kW
T_{max}	manufacturer's maximum engine torque	N·m
ω_{1}	angular speed at the start of the torque plateau	s^{-1}
ω_{2}	angular speed at the end of the torque plateau	s^{-1}
$\omega_{Ркін}$	angular speed at the end of the specified constant-power range	s^{-1}
ω_{Pmax}	angular speed at maximum engine power	s^{-1}
u_{front}	adopted front-axle torque fraction for a single traction source	—
$\rho_{п}$	adopted fuel density	g/cm³
$\Delta W_{бат}$	battery energy balance over the cycle	kWh
x	fraction of cycle traction energy supplied by the electric machine	—
x_{e}	electric fraction of the preliminary installed hybrid power	—
V_{max}	target maximum vehicle speed	m/s
ω_{b}	base angular speed of the electric machine	s^{-1}
ω_{max}	maximum angular speed of the electric machine	s^{-1}
$P_{м}$	adopted electric-machine power	kW
P_{cont}	continuous electric-machine power in S1 duty	kW
t_{peak}	allowable electric-machine peak-duty duration	s
$\eta_{рек}$	reverse-path regenerative efficiency	—
δ	rotating-mass factor	—

Symbol	Quantity	Unit
g_{N}	specific fuel consumption at rated power	g/kWh
p	number of splined joints in the power path	number
$K_{\text{ср}}$	mean front and rear track width	m
$u_{\text{рк,max}}$	maximum transfer-case ratio	—
R	cornering radius used for the stability check	m
F_0	constant coastdown road-load term	N
F_1	linear coastdown road-load coefficient	N·s/m
F_2	quadratic coastdown road-load coefficient	N·s²/m²
$\eta_{\text{ем}}^{\text{зап}}$	explicitly assumed fallback electric-machine efficiency outside the map	—
j	vehicle acceleration versus speed in each gear	m/s²
F_{φ}	adhesion-limited tractive force	kN
$u_{\text{тр},\varphi}$	adhesion-limited transmission ratio	—
$G_{\text{зч}}$	weight force on the driven axles	kN
$j_{\text{сп}}$	design deceleration	m/s²
$F_{\text{т},\Sigma}$	total tractive force from the adopted sources versus speed	—
W	aerodynamic resistance factor	N·s²/m²
$e_{\text{ц}}$	driveline input energy over the cycle, per 100 km	kWh/100 km
D	dynamic factor versus speed in each gear	—
$r_{\text{д}}$	dynamic wheel radius	m
η_{e}	engine efficiency versus speed and load	—
u_0	required final-drive ratio	—
$u_{\text{к1}}$	first-gear ratio	—
n	adopted number of gearbox ratios	number
$u_{\text{кj}}$	adopted gear-ratio series	—
$G_{\text{ш}}$	maximum load per tyre	kN
$E_{\text{дв}}$	mechanical energy that the engine must supply over the cycle	kWh
$u_{\text{тр,max}}$	adopted maximum overall transmission ratio	—
$\eta_{\text{ем}}$	electric-machine efficiency versus torque and speed	—
N_{v}	required traction-source power for the target maximum speed	kW
$u_{\text{тр,дор}}$	required transmission ratio for road resistance	—
ψ	total road-resistance coefficient	—

Symbol	Quantity	Unit
$V_{\text{перем}}$	shift speeds and allowable gear-speed bands	m/s
$r_{\text{ст}}$	static loaded-tyre radius	m
$e_s(V)$	energy consumption versus steady-driving speed	kWh/100 km
$Q_s(V)$	fuel consumption versus steady-driving speed	L/100 km
$P_{\text{д}}(\omega)$	full-load internal-combustion engine characteristic	—
$P_{\text{м}}(\omega)$	traction electric-machine characteristic	—
$\eta_{\text{м}}$	mechanical transmission efficiency based on its component count	—
$F_{\text{т}}$	engine tractive force in top gear	—
$F_{\text{т}}$	front-wheel electric-machine tractive force	—
$F_{\text{т}}$	rear-wheel electric-machine tractive force	—
$u_{\text{к,min}}$	Lowest motor-reduction ratio upstream of the wheel reduction gear	number
$u_{\text{к,max}}$	Highest motor-reduction ratio upstream of the wheel reduction gear	number
$u_{\text{п}}$	adopted constant-mesh gear pair	—
Z_{ij}	gearbox tooth counts and actual ratios	—
β'	adopted helix angle	—
is	adopted gear face width	mm
m	adopted gear module	mm
m	common gearbox gear module	mm
Z_{ij}	adopted integer tooth counts	—
k_{α}	preliminary centre-distance coefficient	—
m,n,c	counts of gears, bearings and couplings on the shafts	—
is,B,H	relative widths of gearbox components	mm
a_{w}	geometric centre distance	mm
$M_{\text{к,max}}$	maximum internal-combustion engine torque	N·m
β	clutch torque reserve factor	—
$q_{\text{l}}ining$	adopted initial lining pressure	MPa
μ	friction-lining coefficient of friction	—
k_{w}	lining width to mean-radius ratio	—
z	number of active friction surfaces	—
M_{p}	design clutch torque	N·m
$R_{\text{p}}relim$	initial mean lining radius	m

Symbol	Quantity	Unit
$b_{l}ining$	calculated lining width	m
D_{a}	adopted lining outer diameter	m
d_{a}	adopted lining inner diameter	m
R_{c}	mean working radius of the adopted lining	m
$A_{l}ining$	area of one annular lining	m^2
P_{n}	required lining clamping force	N
$z_{s}pring$	number of cylindrical pressure springs	—
$c_{r}elease$	auxiliary-spring opposition factor	—
$k_{u}neven$	spring-load unevenness factor	—
$P_{s}pring$	required force per pressure spring	N
$D_{s}pring$	mean coil-spring diameter	m
$\tau_{a}llow$	allowable wire torsional stress	MPa
$P_{d}esign$	design force per spring	N
$d_{w}ire$	adopted spring-wire diameter	m
$travel$	additional deflection during disengagement	m
$shear_{m}odulus$	shear modulus of the wire material	MPa
$dP_{s}pring$	calculated spring-force increase	N
$dead_{c}oils$	number of inactive coils	—
$n_{w}orking$	calculated number of active coils	—
$n_{a}dopted$	adopted number of active coils	—
$f_{i}nstalled$	initial spring deflection in the assembly	m
$C_{s}pring$	stiffness of the adopted spring	N/m
$P_{d}eveloped$	actual spring force when engaged	N
$P_{n}et$	actual lining clamping force from the spring set	N
$L_{i}nstalled$	spring length in the installed, engaged state	m
$L_{s}olid$	preliminary solid length at coil contact	m
Z_1	adopted final-drive pinion tooth count	—
σ	allowable final-drive tooth bending stress	MPa
$M_{\text{розр}}$	design torque at the final-drive input	N·m
Z	tooth counts of the final-drive stages	—
β_1	adopted pinion spiral angle	deg

Symbol	Quantity	Unit
L	adopted cone distance	mm
m_{n}	final-drive gear module	mm
m_{n}	adopted normal module of the cylindrical stage	mm
$m_{\text{c}}andidate$	normal module of a candidate cylindrical pair	mm
M_3	design torque at the cylindrical-stage input	N·m
Z_3	cylindrical-stage pinion tooth count	—
Z_4	cylindrical-stage wheel tooth count	—
β	helix angle of the cylindrical pair	deg
k_{b}	face-width to normal-module ratio	—
y_3	tooth-form factor of the driving cylindrical pinion	—
y_4	tooth-form factor of the driven cylindrical wheel	—
$\sigma_{\text{Fa}}llow$	allowable cylindrical-pair bending stress	MPa
$\sigma_{\text{Ha}}llow$	allowable cylindrical-pair contact stress	MPa
α	pressure angle of the cylindrical pair	deg
E	elastic modulus of the gear material	MPa
k_{H}	adopted Hertz contact-formula coefficient	—
$K_{\text{від}}$	adopted lateral stiffness of one tyre	N/deg
R_{e}	cornering radius including tyre slip	m
$K_{\text{б}}$	differential locking factor	—
σ	allowable half-shaft bending stress	MPa
b	distance from the wheel plane to the critical section	m
$\gamma_{\text{д}}$	half-shaft dynamic-load factor	—
$\varphi_{\max}$	adhesion coefficient for the limiting half-shaft load	—
K_{M}	maximum fraction of axle torque on one wheel	—
ℓ	design half-shaft length	m
$M_{\text{розр}}$	design torque in one half-shaft	N·m
d	diameter of a load-bearing half-shaft	mm
$[\tau]$	allowable torsional stress of a fully floating half-shaft	MPa
d	adopted fully floating half-shaft diameter	mm
$D_{\text{o}}uter$	diaphragm-spring outer diameter	m
$D_{\text{c}}ontact$	pressure-plate contact diameter	m
$D_{\text{i}}nner$	inner diameter of the spring fingers	m

Symbol	Quantity	Unit
$D_{d}isc$	inner diameter of the continuous spring portion	m
$t_{s}pring$	diaphragm-spring thickness	m
$h_{s}pring$	initial diaphragm-spring height	m
$E_{m}aterial$	elastic modulus of the diaphragm-spring material	MPa
$\mu_{m}aterial$	Poisson's ratio of the diaphragm-spring material	—
$c_{a}lpha$	cone-angle multiplier for the selected method	—
$c_{b}ending$	bending-term multiplier for the selected method	—
$c_{t}hickness$	signed thickness-term multiplier for the selected method	—
$c_{m}odulus$	elastic-modulus multiplier for the selected method	—
$P_{r}elease$	release-bearing force	N
$D_{l}og$	logarithmic mean diameter of the continuous portion	m
$Z_{к}$	Adopted wheel-reduction ring-gear tooth count	—
m	Adopted wheel-reduction gear module	mm
$f_{ст}$	adopted static suspension deflection	m
g	gravitational acceleration	m/s²
$\nu_{п}$	sprung-mass natural frequency	Hz
$\nu_{в}$	unsprung-mass natural frequency	Hz
$\nu_{в,min}$	lower source limit for unsprung-mass natural frequency	Hz
$\nu_{в,max}$	upper source limit for unsprung-mass natural frequency	Hz
$p_{ш}$	adopted tyre pressure for the preliminary ground-pressure estimate	kPa
Z_{0}	bump height for the teaching comparison	m
$\Sigma C_{ш}$	total radial tyre stiffness on the axle	N/m
$m_{м}$	unsprung axle mass	kg
$F_{к}$	contact area of one wheel on the supporting surface	m²
$n_{к}$	number of wheels on the design axle	—
a	distance from the front axle to the centre of mass	m
$h \times b$	adopted standard leaf-spring section	mm
$L_{р}$	adopted leaf-spring length	m
$Z_{дин}$	design dynamic suspension load	N
f	total static and dynamic suspension deflection	m
$R_{р.к}$	adopted steering-wheel radius	m
C	adopted steered-wheel scrub radius	m

Symbol	Quantity	Unit
$\eta_{р.м}$	adopted steering-gear efficiency	—
$\eta_{р.п}$	adopted steering-linkage efficiency	—
$\Sigma M_{к.к}$	total steering resistance moment at the steered wheels	N·m
$U_{р}$	steering-system force ratio	—
$K_{т}$	adopted braking-force distribution coefficient	—
$\varphi_{розр}$	design braking adhesion coefficient	—
μ	adopted brake friction coefficient	—
$M_{т}$	braking torque at one front wheel	N·m
$S_{пед.роб}$	adopted service brake-pedal travel	mm
$U_{р}$	adopted brake-actuation force ratio	—
$\eta_{пр}$	adopted brake-actuation efficiency	—
x	steering-trapezoid position relative to the front axle	—
$m_{тр}/n$	adopted steering-arm to tie-rod length ratio	—
n	transverse tie-rod length	m
Z_1	adopted rack-and-pinion pinion tooth count	—
H	adopted rack height	mm
$m_{р}$	adopted rack tooth module	mm
$D_{ср}$	mean coil-spring diameter	m
$[\tau]_{пр}$	adopted allowable spring shear stress	MPa
d	calculated spring-wire diameter	m
$n_{п}$-$n_{р}$	adopted number of inactive spring coils	—
h_{θ}	torsion-bar lever arm	m
$[\tau]_{т}$	adopted allowable shear stress of the round torsion bar	MPa
d	calculated round torsion-bar diameter	m
l	adopted torsion-bar working length	m
h	torsion-strip thickness	m
b	torsion-strip width	m
$n_{п}$	number of torsion strips	—
$U_{р}$	adopted wheel-reduction ratio	—
$n_{с}$	adopted number of wheel-reduction planet gears	—
$[\sigma]$	adopted allowable tooth contact stress	MPa
a_{w}	adopted wheel-reduction centre distance	mm

Symbol	Quantity	Unit
b_c/b_2	adopted progressive face-width multiplier	—
$Z_{по}$	adopted sun-gear tooth count	—

Chapter 1. Project assignment and workflow

From vehicle purpose to consistent calculations, drawings and conclusions.

The course project starts with an assignment: who will use the vehicle, under what conditions, what payload it will carry and what traction performance it must achieve. Specify these requirements before selecting an engine or body outline. A prototype helps justify initial parameters, but its specification does not replace the assignment for the new vehicle.

Present the inputs, layout, traction and energy performance, calculations for the selected components and final conclusions in order. First establish the purpose, category, load and road conditions; then determine masses, resistances and tyres; finally coordinate the powertrain, ratios and operating modes. Running-gear and control-system calculations use the same masses and geometry.

What qualifies as an input parameter

For each number, state whether it comes from the assignment, a specific source, a designer's assumption or an earlier calculation. Retain prototype reference values together with the exact variant and record or page citation. If a reference gives only total payload capacity, do not attribute an occupant, luggage and cargo breakdown to it without separate evidence.

Record a selection from a range as a design assumption: state the range, selected value and reason. Present a formula with symbol definitions, units, numerical substitution and a brief conclusion. If the substitution already shows the result, a separate repeated result label is unnecessary.

Powertrain alternatives

Table M.1.1 — Powertrain alternatives and required initial decisions

Variant	Decisions required before the traction calculation
Internal combustion engine	Fuel type, engine characteristic and mechanical path to the driven wheels
Battery electric vehicle	Traction-machine characteristic, reduction ratio, battery energy reserve and allowable energy exchange
Parallel hybrid	Which machines drive the wheels and how their combined torque is limited
Series hybrid	Electric traction path and a separate engine-to-generator path
Plug-in hybrid	The same structural connections, plus the battery charge-depleting or charge-sustaining operating mode

A hybrid's name does not define its energy operating mode. When comparing fuel consumption, also state the change in stored battery energy; otherwise the same kilometre can be supplied by different energy sources. Chapters 4–8 give detailed conditions and examples.

Construct figures from the adopted dimensions and results. A schematic body or tooth shape may explain the layout if it is not described as manufacturer geometry. The final report should contain static vector figures and tables sufficient for checking without running the application.

Chapter 2. Design input data

Category, load, speed and road conditions with explicit units and sources.

Complete the input-data table before calculation. Specify masses in kilograms and lengths in metres; speeds may be entered in km/h or m/s with explicit conversion. Mass and weight force are different quantities: convert to force using gravitational acceleration. Do not merge data for different prototype variants merely because their model names are similar.

Table M.2.1 — Project input-data groups

Input-data group	Information to record
Purpose and category	Passenger car, bus or truck; transport purpose and adopted category
Payload	Occupants including the driver, luggage and cargo; sourced and assumed components separately
Prototype	Exact variant, masses, wheelbase and known overall dimensions, with provenance
Powertrain	Type, energy carrier, and mechanical and electrical connections to the driven axles
Maximum speed	Value, unit and road conditions under which it must be reached
Road conditions	Road network, surface, low-speed rolling-resistance coefficient and longitudinal gradient

Select the road network and surface material independently. A surface name does not determine allowable weight loads, the bump profile or roughness class. Positive gradient denotes uphill; negative gradient denotes downhill. Convert percentages, height ratios and angles in degrees to the ratio used in the calculation rather than treating them as identical numbers.

2.1. Rolling-resistance coefficient at low speed

The following table reproduces the ranges in Table 4, p. 44 of A. K. Birulia's Road Design, Part 1. They apply to the stated surface conditions. For example, an unbound crushed-stone highway must not be equated with loose crushed stone or compacted gravel.

Table M.2.2 — Rolling-resistance coefficient ranges by surface condition

Surface in the source	Rolling-resistance coefficient range
Asphalt concrete and cement concrete pavement	0,01–0,02
Smooth bitumen-bound crushed-stone highway	0,020–0,025
Unbound crushed-stone highway	0,03–0,05
Stone paving	0,04–0,05
Dry, even earth road	0,04–0,05
Uneven earth road with wet soil	0,07–0,15
Road over loose sand	0,15–0,30

The project catalogue also includes individual reference values from Table 2, Suggested values for rolling resistance, published by the US Federal Highway Administration. These point values come from the document in the US transportation repository, Table 2. A single listed value does not establish a corresponding Birulia range.

Table M.2.3 — Individual reference rolling-resistance coefficients

Catalogue surface	Individual reference value
Dry, even asphalt concrete	0,012
Compacted gravel	0,015
Loose sandy earth	0,037
Loose crushed stone	0,050
Loose gravel	0,100
Deep sand	0,150

For worn or uneven asphalt, the application provides the teaching assumption 0,020. This is not a separate Birulia table entry for that condition. If a custom surface or measured value is selected, state the source, tyre and loading conditions, and measurement conditions. Do not use the range for a similarly named surface as unsupported evidence.

2.2. Speed correction of the coefficient

Birulia gives the speed relationship on p. 45. The coefficient remains constant up to and including 50 km/h. An empirical correction applies above 50 and up to 150 km/h. Above 150 km/h, the application extends the same relationship and explicitly labels it as extrapolation. Enter speed in km/h in this formula even when other chapters use m/s.

Rolling-resistance coefficient at maximum speed

Store the surface input coefficient separately from the coefficient at maximum speed. The corrected coefficient feeds the total road resistance in Chapter 3 and the required-power calculation. Changing maximum speed must recalculate this chain, not merely a chart label.

Applicability: Maximum speed is no greater than 50 km/h; the rolling-resistance coefficient remains at its reference value.

This separate example assumes 40 km/h. No linear correction is used, so the coefficient does not fall below its initial value.

Table П.2.1 — Inputs to the adopted decision: Rolling-resistance coefficient at maximum speed

Symbol	Quantity	Unit	Provenance
f_0	rolling-resistance coefficient at low speed	—	input data
V_{max}	target maximum vehicle speed	m/s	input data

Example — vehicle with a maximum speed of 40 km/h

$$f = f_0$$

$$f = f_0 = 0.0150$$

Source: [1]

(2.1) **Applicability:** Maximum speed is above 50 and no greater than 150 km/h; the interval specified by the source is used.

The worked example assumes 120 km/h and an initial coefficient of 0,015 for compacted gravel. The initial coefficient and the speed correction have separate source bases.

Table Д.2.1.1 — Inputs for (2.1): Rolling-resistance coefficient at maximum speed

Symbol	Quantity	Unit	Provenance
f_0	rolling-resistance coefficient at low speed	—	input data
V_{max}	target maximum vehicle speed	m/s	input data

Example — conventional internal-combustion vehicle

$$f=f_0(1+0,01(V-50)) \quad (2.1)$$

$$f=0.0150(1+0,01(120.0-50))=0.0255$$

Source: [1]

(2.1) **Applicability:** Maximum speed exceeds 150 km/h; the relationship has been extended beyond the source interval.

The electric example specifies 160 km/h. Speed, rather than the electric drive type, causes the extrapolation. Check this result using a suitable model or tyre data for the operating condition.

Table Д.2.1.2 — Inputs for (2.1): Rolling-resistance coefficient at maximum speed

Symbol	Quantity	Unit	Provenance
f_0	rolling-resistance coefficient at low speed	—	input data
V_{max}	target maximum vehicle speed	m/s	input data

Example — electric vehicle

$$f=f_0(1+0,01(V-50)) \quad (2.1)$$

$$f=0.0150(1+0,01(160.0-50))=0.0315$$

- Speed 160.0 km/h exceeds 150 km/h, the upper bound of the interval specified for this relationship. The value of f is an extrapolation outside the source's applicability range.

Source: [1]

Chapter 3. Input analysis and vehicle layout

Vehicle masses, axle-weight distribution, centre-of-mass position, wheel configuration and road resistance.

Vehicle calculation starts with **mass**. Mass affects everything that follows: required power, gear ratios, component dimensions and brakes. At this stage it is still unknown and must be estimated.

The estimate follows this reasoning. The assignment specifies **what** the vehicle must carry: occupants, luggage and cargo. This is its payload. Select an existing prototype with a similar purpose and determine the ratio of its curb mass to its payload. Apply that ratio to the vehicle being designed.

This estimate is intentionally approximate. Refinement becomes meaningful once component masses are known, which they are not at this stage. Every number in this chapter is therefore a **design estimate**to be checked later.

Masses

Determining payload

Payload consists of what the vehicle actually carries. Its composition depends on the vehicle's purpose, so the method branches here.

(3.1) **Applicability:** The vehicle carries people: a passenger car or bus.

Calculate occupant mass as the number of seats, including the driver's, multiplied by the assumed mass per person. Add luggage separately.

Table Д.3.1.3 — Inputs for (3.1): Determining payload

Symbol	Quantity	Unit	Provenance
$n_{\text{осіб}}$	number of occupants, including the driver	number	input data
$m_{\text{особи}}$	assumed mass per person	kg	input data
$M_{\text{багаж}}$	luggage mass	kg	input data

Example — conventional internal-combustion vehicle

$$M_{\text{кор}}=M_{\text{п}}+M_{\text{багаж}} \tag{3.1}$$

$$M_{\text{кор}}=375+249=624 \text{ kg}$$

Source: [2]

(3.1) **Applicability:** The vehicle carries goods: a truck.

This branch has neither passengers nor luggage: payload is the specified cargo capacity. Adding the driver would count the driver twice, since the curb mass already includes the driver.

Table Д.3.1.1 — Inputs for (3.1): Determining payload

Symbol	Quantity	Unit	Provenance
$M_{\text{в}}$	cargo mass	kg	input data

Example — three-axle truck (KamAZ-53215, 6×4)

$$M_{\text{кор}}=M_{\text{в}} \tag{3.1}$$

$$M_{\text{кор}}=11155\ \text{kg}$$

- The payload of 11 155 kg comes from the prototype specification.

Source: [3]; [4]

(3.1) **Applicability:** The vehicle carries both people and goods: a dual-purpose vehicle.

Add all three terms. This explains the separate branches: each uses exactly the quantities that the vehicle actually carries.

Table Д.3.1.2 — Inputs for (3.1): Determining payload

Symbol	Quantity	Unit	Provenance
$n_{\text{осіб}}$	number of occupants, including the driver	number	input data
$m_{\text{особи}}$	assumed mass per person	kg	input data
$M_{\text{багаж}}$	luggage mass	kg	input data
$M_{\text{в}}$	cargo mass	kg	input data

Example — dual-purpose vehicle

$$M_{\text{кор}}=M_{\text{п}}+M_{\text{багаж}}+M_{\text{в}} \tag{3.1}$$

$$M_{\text{кор}}=675+60+200=935\ \text{kg}$$

- The prototype gives a total of 935 kg and 9 occupants including the driver. The author assumes 60 kg of luggage and 200 kg of cargo because the source does not provide that breakdown.

Source: [3]; [5]; [2]

Compare the three input lists above. The passenger branch has no cargo; the cargo branch has neither occupants nor luggage; the combined branch has all three. This is part of the calculation method: a quantity the vehicle does not carry should not enter the calculation, even as zero.

Determining curb mass

Estimate curb mass using a prototype. The curb-mass coefficient is a ratio obtained from an existing vehicle with a similar purpose: a passenger car weighs roughly twice its payload, while a truck weighs roughly as much as its payload.

This is the project's largest source of uncertainty. An error in this coefficient propagates into power, brakes and every driveline component.

(3.3) **Applicability:** Preliminary curb mass from the tare coefficient q.

The coefficient q describes the entire vehicle in running order. For an electric vehicle, battery mass is already included; do not add the pack again.

Table Д.3.3.1 — Inputs for (3.3): Determining curb mass

Symbol	Quantity	Unit	Provenance
q	curb-mass coefficient The ratio of vehicle curb mass to payload. Adopt it from a prototype.	—	adopted by the designer
$M_{\text{кор}}$	payload	kg	calculated earlier (3.1)

Example — conventional internal-combustion vehicle

$$M_0 = qM_{\text{кор}} \tag{3.3}$$

$$M_0 = 1.800 \cdot 624 \approx 1123 \text{ kg}$$

Source: [6]

(3.3) **Applicability:** Specified battery-free BEV mass and usable specific energy of the selected pack.

The vehicle in running order consists of the battery pack and the remaining structure. Battery-free mass excludes both the pack and payload. Add the new pack once; the next gross-mass equation adds passengers and cargo. These examples are teaching assumptions, not specifications of a particular prototype.

Determine the battery-free mass M_{b} from the reference configuration of the same BEV. The tare coefficient q and reference occupant, luggage and cargo load $M_{\text{кор,ref}}$ give the curb mass including the reference pack $M_{0,\text{ref}} = qM_{\text{кор,ref}}$. This is M_0 before battery replacement; the battery is not payload. The reference pack's usable capacity $E_{\text{u,ref}}$, kWh, and usable specific energy $\rho_{\text{u,ref}}$, Wh/kg, give its estimated mass. Before adopting a new pack, subtract this reference battery using (3.11).

Adopt the reference load $M_{\text{кор,ref}}$ at I-39 immediately after the result of (3.1). By default it equals that result and follows it until manually edited. To change trip cargo without redefining the vehicle, first fix I-39 or M_{b} manually. New-pack capacity and specific energy are separate inputs; (3.3) accounts for changing the pack without subtracting the reference battery again. A manually adopted base mass remains fixed until a new decision is made.

Initial NMC and LFP values are editable teaching assumptions for the usable energy of the complete pack, including casing and cooling. Gross manufacturer capacity and cell-level specific energy have a different basis. Specify the 400 or 800 V architecture separately; it does not determine chemistry or specific energy. Electrical-check mode (I-53) controls voltage and current accounting; the curb-mass model is identical in both modes.

Table Д.3.3.2 — Inputs for (3.3): Determining curb mass

Symbol	Quantity	Unit	Provenance
E_{u}	usable battery energy within the BMS 0–100% SOC window	kWh	input data
ρ_{u}	usable specific energy of the complete battery pack	Wh/kg	input data
M_{b}	vehicle mass excluding the battery and payload	kg	input data

Example — electric-vehicle battery, mass and range

$$M_0 = M_b + \frac{1000 E_u}{\rho_u} \tag{3.3}$$

$$M_0 = 923{,}2 + \frac{1000 \cdot 30}{150} = 1123{,}2 \text{ kg}$$

Source: [7]

Reference BEV mass and battery-free base mass

Reference curb mass includes the reference pack. First adopt the occupant, luggage and cargo load from (3.1), then subtract the pack. Reference capacity and specific energy follow the installed values until the reader explicitly changes the pack. A smaller or larger battery changes curb and gross mass through (3.3) and (3.4); the reference battery is not subtracted again. Manually adopting a different base mass is a separate project decision.

Table Д.3.11 — Inputs for (3.11): Reference BEV mass and battery-free base mass

Symbol	Quantity	Unit	Provenance
q	curb-mass coefficient The ratio of vehicle curb mass to payload. Adopt it from a prototype.	—	adopted by the designer
$M_{кор,ref}$	adopted reference occupant, luggage and cargo load of the BEV	kg	input data
$E_{u,ref}$	usable capacity of the reference BEV pack	kWh	input data
$\rho_{u,ref}$	usable specific energy of the complete reference BEV pack	Wh/kg	input data

Example — electric-vehicle battery, mass and range

$$M_b = M_{0,ref} - M_{bat,ref} = q M_{кор,ref} - \frac{1000 E_{u,ref}}{\rho_{u,ref}} \tag{3.11}$$

$$M_{0,ref} = 1{,}8 \cdot 624 = 1123{,}2 \text{ kg}$$

$$M_{bat,ref} = \frac{1000 \cdot 30}{150} = 200 \text{ kg}$$

$$M_b = 1123{,}2 - 200 = 923{,}2 \text{ kg}$$

Source: [2]

Determining gross mass

Gross mass is curb mass plus payload. This sum represents the complete vehicle in all subsequent calculations.

Table Д.3.4 — Inputs for (3.4): Determining gross mass

Symbol	Quantity	Unit	Provenance
M_0	vehicle curb mass	kg	calculated earlier (3.3)
$M_{кор}$	payload	kg	calculated earlier (3.1)

Example — conventional internal-combustion vehicle

$$M_{\text{a}}=M_0+M_{\text{кор}} \tag{3.4}$$

$$M_{\text{a}}\approx 1123+624=1747 \text{ kg}$$

Source: [6]

BEV battery and allowable payload

Battery-pack mass from usable specific energy.

Capacity divided by specific energy gives the mass of the complete pack. Both inputs use usable energy: specific energy includes the casing and cooling, and capacity covers the BMS 0–100% window. Gross manufacturer capacity is not a substitute for usable capacity. The unit calculator reconciles kWh and Wh.

Table Д.3.9 — Inputs for (3.9): Battery-pack mass from usable specific energy.

Symbol	Quantity	Unit	Provenance
E_{u}	usable battery energy within the BMS 0–100% SOC window	kWh	input data
ρ_{u}	usable specific energy of the complete battery pack	Wh/kg	input data

Example — electric-vehicle battery, mass and range

$$M_{\text{bat}}=\frac{1000E_{\text{u}}}{\rho_{\text{u}}} \tag{3.9}$$

$$M_{\text{bat}}=\frac{1000\cdot 30}{150}=200 \text{ kg}$$

Source: [7]

Allowable payload after installing the battery.

Subtract battery-free mass and the selected pack from allowable gross mass. The remainder is the maximum payload, not actual trip mass. Actual passengers, luggage and cargo are specified separately; if they exceed the remainder, the design fails the gross-mass check.

Table Д.3.10 — Inputs for (3.10): Allowable payload after installing the battery.

Symbol	Quantity	Unit	Provenance
E_{u}	usable battery energy within the BMS 0–100% SOC window	kWh	input data
ρ_{u}	usable specific energy of the complete battery pack	Wh/kg	input data
M_{b}	vehicle mass excluding the battery and payload	kg	input data
$M_{\text{l}}im$	allowable gross vehicle mass	kg	input data

Example — electric-vehicle battery, mass and range

$$Q_{\text{lim}}=M_{\text{lim}}-M_{\text{b}}-\frac{1000E_{\text{u}}}{\rho_{\text{u}}} \tag{3.10}$$

$$Q_{\text{lim}}=2047{,}2-923{,}2-\frac{1000\cdot 30}{150}=924 \text{ kg}$$

Source: [7]

Maximum battery mass for the specified load

Subtract fixed battery-free vehicle mass and actual occupants, luggage and cargo from allowable gross mass. The remainder limits the new pack mass; a negative remainder leaves no room even for a battery. A smaller pack leaves more capacity for cargo, but the check in §8.1 must also confirm the required range and successful completion of the selected driving profile.

Table Д.3.12 — Inputs for (3.12): Maximum battery mass for the specified load

Symbol	Quantity	Unit	Provenance
$M_{l}im$	allowable gross vehicle mass	kg	input data
M_{b}	vehicle mass excluding the battery and payload	kg	input data
$M_{кор}$	payload	kg	calculated earlier (3.1)

Example — electric-vehicle battery, mass and range

$$M_{bat,max}=\max(0,M_{lim}-M_{b}-M_{кор}) \tag{3.12}$$

$$M_{bat,max}=\max(0,2047{,}2-923{,}2-624)=500\ \text{kg}$$

Source: [2]

Weight and its distribution

Total vehicle weight force

Convert mass to weight force because the equations of motion use force. Multiply mass by gravitational acceleration.

Table Д.3.5.2 — Inputs for (3.5): Total vehicle weight force

Symbol	Quantity	Unit	Provenance
M_{a}	gross vehicle mass	kg	calculated earlier (3.4)

Example — conventional internal-combustion vehicle

$$G_{a}=M_{a}g \tag{3.5}$$

$$G_{a}=1747\cdot 9{,}81/1000=17.14\ \text{kN}$$

Source: [6]

Preliminary axle-load distribution

Weight is distributed unevenly between the axles. This distribution determines the tractive force the driven wheels can transmit and the vehicle's braking behaviour.

This step gives a **list**of axle forces rather than a single number: one force per axle. There is therefore no separate single-value result row; the substitution shows each axle as its own term.

Table Д.3.5.1 — Inputs for (3.5): Preliminary axle-load distribution

Symbol	Quantity	Unit	Provenance
G_a	total vehicle weight force	kN	calculated earlier (3.5)
—	adopted wheel configuration	—	adopted by the designer
k_1	front-axle weight fraction	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$k_i = \frac{1-k_1}{n-1}, \qquad G_i = k_i G_a \tag{3.5}$$

$$G_1 = 0.450 \cdot 17.14 = 7.71;\ G_2 = 0.550 \cdot 17.14 = 9.43$$

Source: [6]

Longitudinal centre-of-mass position

The centre-of-mass position follows from weight distribution: a larger rear-axle weight fraction moves it rearwards. Calculate its distance from the front axle and subtract this from the wheelbase to obtain the remaining distance.

The number of axles changes the equation itself, not just its numerical inputs.

(3.6) **Applicability:** Two-axle layout.

One rear axle produces a moment about the front axle, giving an equation with two terms.

Table Д.3.6 — Inputs for (3.6): Longitudinal centre-of-mass position

Symbol	Quantity	Unit	Provenance
G_i	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
G_a	total vehicle weight force	kN	calculated earlier (3.5)
B	vehicle wheelbase	m	input data

Example — conventional internal-combustion vehicle

$$a = \frac{G_2 B}{G_a}, \qquad b = B - a \tag{3.6}$$

$$a = \frac{9.43 \cdot 2.70}{17.14} = 1.485 \text{ m};\ b = 2.70 - 1.485 = 1.215 \text{ m}$$

- The result moves with the adopted fraction k_1 and remains preliminary until the layout drawing is prepared.

Source: [6]

(3.7) **Applicability:** Three-axle layout with a two-axle rear bogie.

The third axle lies beyond the wheelbase by the rear-axle spacing, so its moment is calculated separately. This is why rear-axle spacing enters only this calculation; a two-axle layout has no such distance.

Table Д.3.7 — Inputs for (3.7): Longitudinal centre-of-mass position

Symbol	Quantity	Unit	Provenance
G_i	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
G_a	total vehicle weight force	kN	calculated earlier (3.5)
B	vehicle wheelbase	m	input data
B_1	rear-axle spacing Applicable only to a three-axle layout; this distance does not exist in a two-axle layout.	m	input data

Example — three-axle truck (KamAZ-53215, 6×4)

$$a = \frac{G_2 B + G_3 (B + B_1)}{G_a}, \qquad b = B - a \tag{3.7}$$

$$a = \frac{69.30 \cdot 3.69 + 69.30 \cdot 5.01}{189.87} = 3.175 \text{ m}; \; b = 3.69 - 3.175 = 0.514 \text{ m}$$

- The result moves with the adopted fraction k_1 and remains preliminary until the layout drawing is prepared.

Source: [6]

Wheel configuration and road resistance

Wheel configuration and tyres

Select the number of axles using the **gross-mass limits**in the road traffic rules. An axle can transmit only a limited load to the road; add axles when the mass exceeds that limit.

Passenger cars and small buses bypass this table: their purpose calls for two axles.

This step has **no printed equation number**because it selects a value from a table rather than calculating one.

Selection rule. Use the smallest number of axles for which gross mass does not exceed the limit for the selected road network; then determine the tyre count.

Truck gross-mass limits under the road traffic rules — 2 axles: 18 t; 3 axles: 25 t; 4 axles: 32 t (roads of national importance)

Applicability: Passenger car or small passenger vehicle with two axles.

Two axles are adopted for this design group. Wheel configuration and tyre count describe different things: 4×2 gives wheel positions and driven positions. Explain the choice of driven axle separately.

Table П.3.2 — Inputs to the adopted decision: Wheel configuration and tyres

Symbol	Quantity	Unit	Provenance
M_a	gross vehicle mass	kg	calculated earlier (3.4)
—	vehicle category	—	input data
—	road network used by the vehicle	—	input data
—	road-surface type	—	input data
f_0	rolling-resistance coefficient at low speed	—	input data
—	adopted wheel configuration	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$n_{\text{осей}}=2$$

$$n_{\text{осей}}=2;\ 4\times2$$

$$= \mathbf{4}\ \text{number}$$

Source: [8]; [6]

Applicability: Truck whose preliminary axle count is selected from mass and the chosen road network.

The truck example uses a three-axle 6×4 layout with ten tyres: rear wheel positions have twin tyres. Wheel configuration therefore differs from the physical tyre count. This selection is preliminary; checking individual and grouped axle loads and the applicability of specific road limits remains a separate task.

Table П.3.1 — Inputs to the adopted decision: Wheel configuration and tyres

Symbol	Quantity	Unit	Provenance
M_a	gross vehicle mass	kg	calculated earlier (3.4)
—	vehicle category	—	input data
—	road network used by the vehicle	—	input data
—	road-surface type	—	input data
f_0	rolling-resistance coefficient at low speed	—	input data
—	adopted wheel configuration	—	adopted by the designer

Example — three-axle truck (KamAZ-53215, 6×4)

$$n_{\text{осей}}=\min\{n{:}M_a\le M_{\max}(n)\}$$

$$n_{\text{осей}}=\min\{n{:}19.355\le M_{\max}(n)\}=3;\ 6\times4$$

$$= \mathbf{10}\ \text{number}$$

Source: [8]; [6]

Total road resistance

Total road resistance combines two different effects: wheel rolling resistance and climbing resistance. The former depends on the surface; the latter on gradient.

Result **dimensionless** and represents the fraction of vehicle weight that must be overcome. It can therefore be compared directly with the adhesion coefficient: both are fractions of the same weight.

Table Д.3.8 — Inputs for (3.8): Total road resistance

Symbol	Quantity	Unit	Provenance
f	rolling-resistance coefficient at maximum speed	—	calculated earlier (2.1)
s	road gradient	m/m	input data

Example — conventional internal-combustion vehicle

$$\psi = f\cos\alpha + \sin\alpha \tag{3.8}$$

$$\psi = 0.0255\cos(0.00^\circ) + \sin(0.00^\circ) = 0.0255$$

Source: [9]; [1]

Vehicle layout

The numbers in this chapter describe geometry as a list of distances. The layout shows that geometry as a designer sees it: axle positions, the centre of mass and the dimensions used subsequently.

Both drawings below are generated **from the numbers calculated above** by the same software that draws them in the application. These are not independently prepared illustrations: changing the wheelbase or weight distribution changes the drawing accordingly.

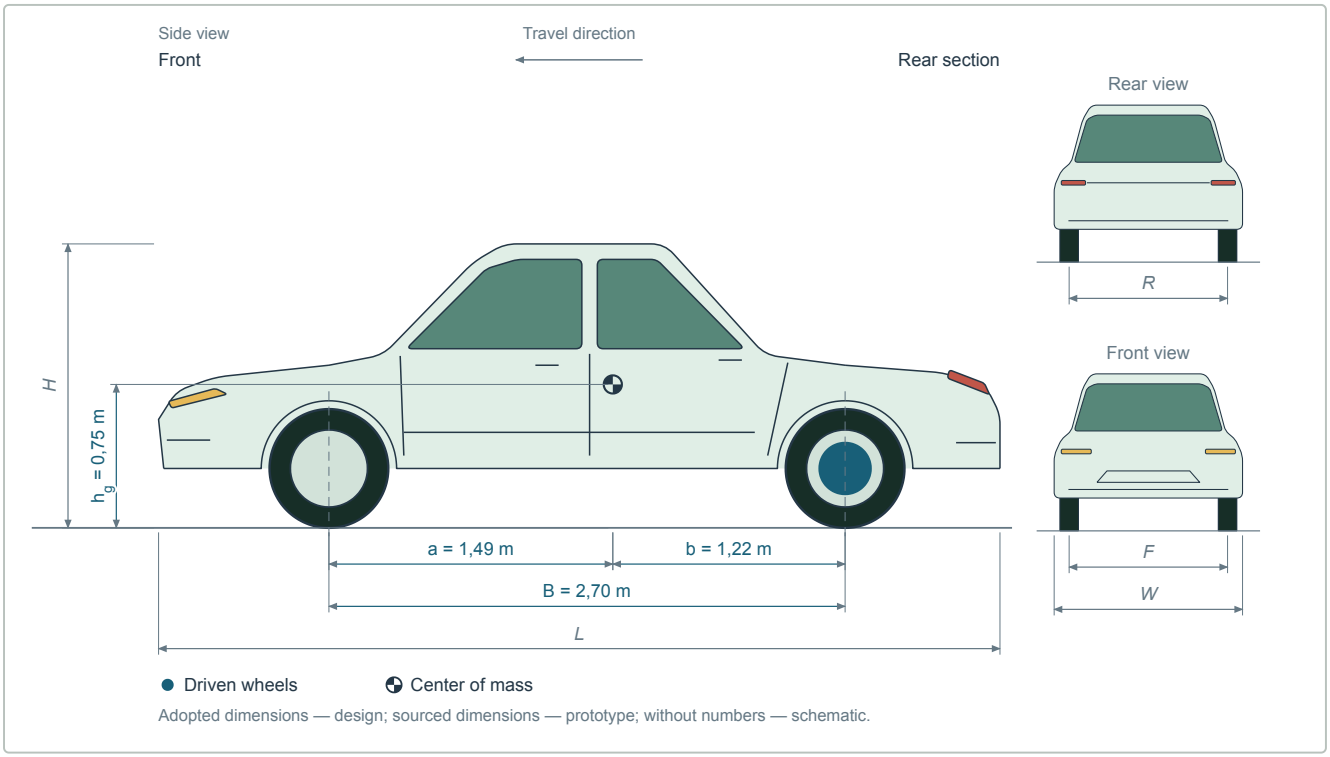


Figure 3.2 — Two-axle vehicle layout: wheelbase, longitudinal centre-of-mass position and height above the road.
Author's parametric body outline: overhangs, height, width and wheels are schematic. The outline does not establish manufacturer-specified prototype dimensions.
Source: [6].

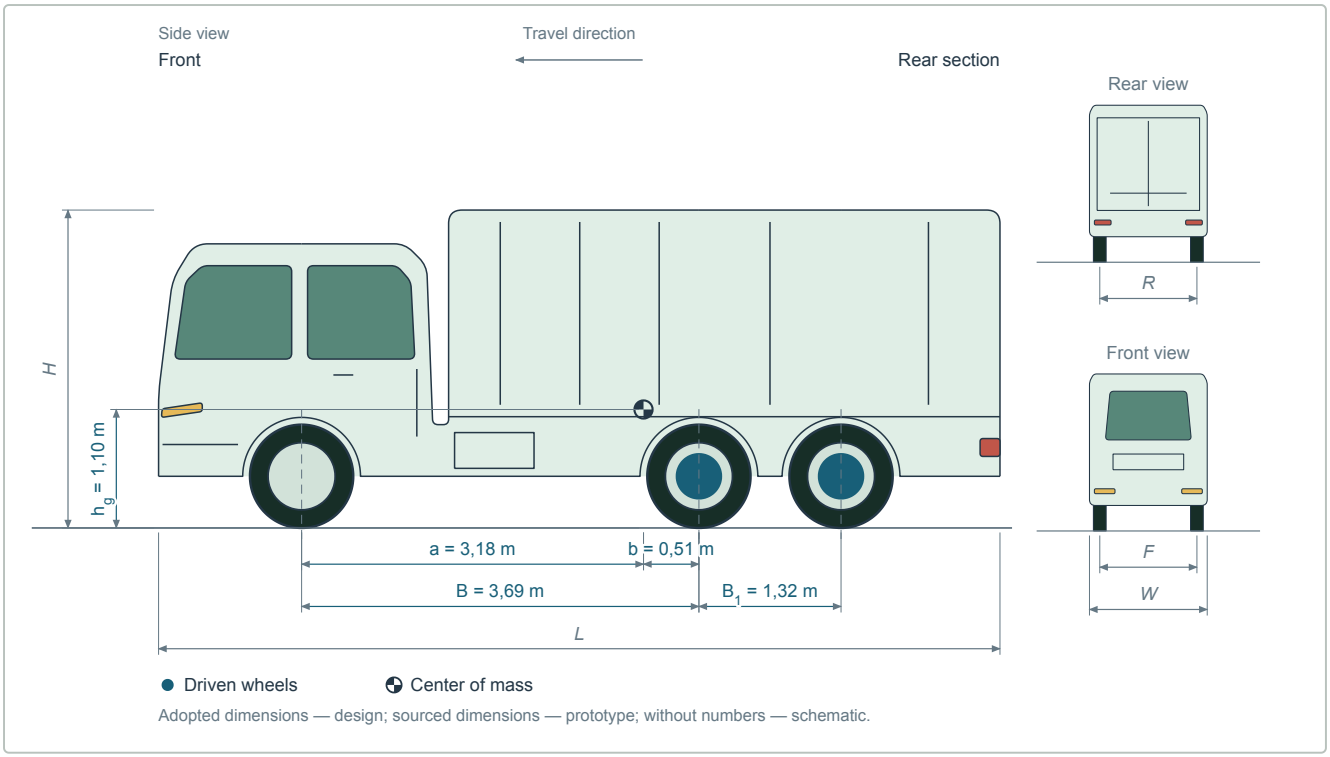


Figure 3.3 — Corresponding three-axle vehicle with a rear bogie: rear-axle spacing B_1 is added, moving the centre of mass rearwards.
Author's parametric body outline: overhangs, height, width and wheels are schematic. The outline does not establish manufacturer-specified prototype dimensions.
Source: [6].

Compare the centre-of-mass positions in the two drawings. The third axle carries part of the weight beyond the wheelbase and moves the centre of mass rearwards. Equation (3.7) accounts for this; (3.6) does not.

Chapter 4. Vehicle traction calculation

From tyre load and motion resistance to traction-source characteristics, gear ratios and speed verification.

Start with the masses and axle loads from Chapter 3. Determine wheel size, driveline losses and aerodynamic resistance before comparing required power with the selected machine characteristic.

Describe a powertrain using two independent features: which energy sources the vehicle carries, and which mechanically drive the wheels. In a series hybrid, the engine drives a generator; do not add its torque to traction-machine torque. Plug-in hybrids can have different mechanical layouts: a charging socket does not define shaft connections.

The examples below are teaching designs. An assumed value does not become a manufacturer's specification merely because it is substituted into a formula. The curves describe the selected peak or continuous duty; check peak duration, battery and inverter limits separately.

4.1. Loading, tyre selection and wheel radius

Load per tyre

Calculate the load per tyre on every axle and select against the largest result. Use two tyres on a single-tyred axle and four on a twin-tyred axle. Do not replace tyre count with driven-wheel count: undriven wheels also carry load.

Table Д.4.1 — Inputs for (4.1): Load per tyre

Symbol	Quantity	Unit	Provenance
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
—	adopted wheel configuration	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$G_{\text{ш}} = \frac{G_{\text{i}}}{n} \tag{4.1}$$

$$G_{\text{ш1}} = 7.71/2 = 3.86;\ G_{\text{ш2}} = 9.43/2 = 4.71;\ G_{\text{ш}} = 4.71 \text{ kN}$$

Source: [10]

Tyre load and speed requirements

This step establishes load capacity and speed requirements rather than selecting a finished tyre. For twin mounting, check the second index in the dual rating; the first applies to single mounting. If the requirement exceeds the table, its last row is not an allowable substitute.

Selection rule. From Tables M.4.1 and M.4.2, select the lowest load index and speed category whose limits meet or exceed the calculated requirements. Verify the actual tyre using its manufacturer's specification.

Table П.4.16 — Inputs to the adopted decision: Tyre load and speed requirements

Symbol	Quantity	Unit	Provenance
$G_{ш}$	maximum load per tyre	kN	calculated earlier (4.1)
V_{max}	target maximum vehicle speed	m/s	input data

Example — conventional internal-combustion vehicle

$$ИНС \geq G_{ш}, \qquad V_{кат} \geq V_{max}$$

$$G_{ш} = 480 \text{ kgf}, \; V_{max} = 120 \text{ km/h} \Rightarrow \text{minimum required LI 83, speed category M}$$

- The project does not include GOST 4754-80 and 5513-86, cited by the source, so this subsection gives a tyre requirement rather than a specific tyre designation from those standards.

Source: [11]; [10]

Tables M.4.1 and M.4.2 provide a working set of load indices and speed categories. UN Regulation No. 54, Annex 4 gives the official load-index-to-allowable-mass correspondence. Table M.4.1 expresses load in kgf for the adopted calculation notation; the UN Regulation gives mass in kg. Check the speed category and allowable load of the actual tyre against the manufacturer's specification, accounting for single or twin mounting.

Table M.4.1 — Load index (LI) and load per tyre

LI	kgf	LI	kgf	LI	kgf	LI	kgf	LI	kgf
0	45	40	140	80	450	120	1400	160	4500
1	46,2	41	145	81	462	121	1450	161	4625
2	47,5	42	150	82	475	122	1500	162	4750
3	48,7	43	155	83	487	123	1550	163	4875
4	50	44	160	84	500	124	1600	164	5000
5	51,5	45	165	85	515	125	1650	165	5150
6	53	46	170	86	530	126	1700	166	5300
7	54,5	47	175	87	545	127	1750	167	5450
8	56	48	180	88	560	128	1800	168	5600
9	58	49	185	89	580	129	1850	169	5800
10	60	50	190	90	600	130	1900	170	6000
11	61,5	51	195	91	615	131	1950	171	6150
12	63	52	200	92	630	132	2000	172	6300
13	65	53	206	93	650	133	2060	173	6500
14	67	54	212	94	670	134	2120	174	6700
15	69	55	218	95	690	135	2180	175	6900
16	71	56	224	96	710	136	2240	176	7100
17	73	57	230	97	730	137	2300	177	7300
18	75	58	236	98	750	138	2360	178	7500

LI	kgf	LI	kgf	LI	kgf	LI	kgf	LI	kgf
19	77,5	59	243	99	775	139	2430	179	7750
20	80	60	250	100	800	140	2500	180	8000
21	82,5	61	257	101	825	141	2575	181	8250
22	85	62	265	102	850	142	2650	182	8500
23	87,5	63	272	103	875	143	2725	183	8750
24	90	64	280	104	900	144	2800	184	9000
25	92,5	65	290	105	925	145	2900	185	9250
26	95	66	300	106	950	146	3000	186	9500
27	97	67	307	107	975	147	3075	187	9750
28	100	68	315	108	1000	148	3150	188	10000
29	103	69	325	109	1030	149	3250	189	10300
30	106	70	335	110	1060	150	3350	190	10600
31	109	71	345	111	1090	151	3450	191	10900
32	112	72	355	112	1120	152	3550	192	11200
33	115	73	365	113	1150	153	3650	193	11500
34	118	74	375	114	1180	154	3750	194	11800
35	121	75	387	115	1215	155	3875	195	12150
36	125	76	400	116	1250	156	4000	196	12500
37	128	77	412	117	1285	157	4125	197	12850
38	132	78	425	118	1320	158	4250	198	13200
39	136	79	437	119	1360	159	4375	199	13600

Table M.4.2 — Tyre speed category

Category	km/h	Category	km/h
A1	5	J	100
A2	10	K	110
A3	15	L	120
A4	20	M	130
A5	25	N	140
A6	30	P	150
A7	35	Q	160
A8	40	R	170
B	50	S	180
C	60	T	190
D	65	U	200
E	70	H	210

Category	km/h	Category	km/h
F	80	V	240
G	90	W	270

Static tyre radius from its designation

Decode the tyre designation first: convert rim diameter and section width to the same units. Where an aspect ratio is specified, use it; a fallback table value is then unnecessary. Select the deflection coefficient for the tyre construction and record the basis for the assumption.

Table Д.4.2 — Inputs for (4.2): Static tyre radius from its designation

Symbol	Quantity	Unit	Provenance
—	tyre size and construction designation	—	input data
$\lambda_{ст}$	tyre deflection coefficient under load	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$r_{ст}=0{,}5d+\Delta\lambda_{ст}B \tag{4.2}$$

$$r_{ст}=0{,}5\cdot 406.4+0.550\cdot 0.825\cdot 205.0=296.2 \text{ mm}=0.296 \text{ m}$$

- The source gives a range for $\lambda_{ст}$ rather than a point value; the midpoint is adopted as the default.
- Determine static radius using Volkov and Vilskyi (2010), p. 25, equation (1.8). Equating dynamic radius to static radius and applying additional rolling coefficients remain project assumptions.

Source: [12]

Table M.4.3 — Adopted values for approximate tyre radius. Static radius follows Volkov and Vilskyi, p. 25, equation (1.8). Use a fallback aspect ratio only when the designation omits it.

Condition	Quantity	Value
Truck or bus, excluding wide-section tyres	H/B	1,0
Passenger car, inch-based designation	H/B	0,95
Passenger car, mixed designation without an explicit aspect ratio	H/B	0,80–0,85
Truck and bias-ply passenger-car tyres	Deflection coefficient	0,85–0,90
Radial passenger-car tyres	Deflection coefficient	0,80–0,85
Free rolling	Rolling-radius to static-radius ratio	1,03–1,06

Volkov and Vilskyi, pp. 24–25, give H/B = 1 for truck and bus tyres, deflection coefficients of 0,85–0,90 for these tyres and 0,80–0,85 for radial passenger-car tyres. Other fallback H/B values, extension of 0,85–0,90 to bias-ply passenger-car tyres, and the free-rolling multiplier 1,03–1,06 are project teaching assumptions to be refined using actual tyre data. Selecting the midpoint is the calculation author's assumption.

Dynamic wheel radius

For preliminary hard-surface calculations, adopt dynamic radius equal to static radius as in Opanasiuk, subsection 2.1.1. This approximation does not describe pressure, speed or torque effects; use a measured rolling radius where available.

Table П.4.1 — Inputs to the adopted decision: Dynamic wheel radius

Symbol	Quantity	Unit	Provenance
$r_{ст}$	static loaded-tyre radius	m	calculated earlier (4.2)

Example — conventional internal-combustion vehicle

$$r_{д} \approx r_{ст}$$

$$r_{д} \approx r_{ст} = 0.296 \ \text{m}$$

Source: [12]; [10]

4.2. Transmission losses, aerodynamics and required power

Mechanical transmission efficiency

Multiply the efficiencies of the components that transmit power in the gear being considered. Do not count pairs that rotate without transmitting power. Following Opanasiuk, this estimate neglects bearing losses. For electric drives, adopted mechanical efficiency is a separate input decision: this formula does not provide electric-machine, inverter or battery efficiency.

Table Д.4.3 — Inputs for (4.3): Mechanical transmission efficiency

Symbol	Quantity	Unit	Provenance
k	number of cylindrical gear pairs in the power path	number	input data
l	number of bevel or hypoid gear pairs in the power path	number	input data
m	number of universal joints in the power path	number	input data
p	number of splined joints in the power path	number	input data

Example — conventional internal-combustion vehicle

$$\eta_{м} = 0{,}98^{k} \cdot 0{,}97^{l} \cdot 0{,}995^{m} \cdot 0{,}999^{p} \tag{4.3}$$

$$\eta_{м} = 0{,}98^{2} \cdot 0{,}97^{1} \cdot 0{,}995^{2} \cdot 0{,}999^{1} = 0.921$$

- The source table gives no \eta_{м} range for this vehicle class; no cross-check was performed.

Source: [10]

Table М.4.4 — Efficiency of one component following Opanasiuk, subsection 2.1.2

Component	Efficiency
Cylindrical gear pair	0,98
Bevel or hypoid gear pair	0,97
Universal joint	0,995
Splined joint	0,999

Aerodynamic resistance factor

Multiply frontal projected area by drag coefficient and half the air density. A density of 1,225 kg/m³ corresponds to the standard sea-level atmosphere at 15 °C; other conditions need a different input. Do not describe the overall bounding-rectangle area as a measured frontal area. The resistance factor has units N·s²/m²: multiplying by squared speed in m/s gives force in newtons.

Table Д.4.4 — Inputs for (4.4): Aerodynamic resistance factor

Symbol	Quantity	Unit	Provenance
ρ	air density for the adopted atmospheric conditions	kg/m³	input data
C_{d}	adopted aerodynamic drag coefficient	—	adopted by the designer
A	adopted frontal projected area	m²	adopted by the designer

Example — conventional internal-combustion vehicle

$$W = \frac{1}{2}\rho\ C_{\text{d}}A \tag{4.4}$$

$$W = 0{,}5 \cdot 1.225 \cdot 0.320 \cdot 2.20 = 0.431\ \text{N}\cdot\text{s}^2/\text{m}^2$$

- The source table gives no W range for this vehicle class; no cross-check was performed.

Source: [10]; [13]

Power required at maximum speed

Wheel power overcomes road and aerodynamic resistance; dividing by mechanical efficiency transfers the requirement to the driveline input. Reconcile forces in kilonewtons and power in kilowatts before substitution. Use the road coefficient for this particular speed.

(4.5) **Applicability:** The engine drives the wheels through a mechanical transmission.

Efficiency was calculated from the component count in the preceding step. The result is a steady-driving requirement; the engine specification may provide greater power.

Table Д.4.5.1 — Inputs for (4.5): Power required at maximum speed

Symbol	Quantity	Unit	Provenance
ψ	total road-resistance coefficient	—	calculated earlier (3.8)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)
$V_{\max}$	target maximum vehicle speed	m/s	input data
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_{\text{м}}$	mechanical transmission efficiency based on its component count	—	calculated earlier (4.3)

Example — conventional internal-combustion vehicle

$$N_{\text{v}} = \frac{\psi_{\text{V}} G_{\text{a}} V_{\max} + W V_{\max}^{3}}{1000 \eta_{\text{м}}} \tag{4.5}$$

$$N_{\text{v}} = \frac{0.0255 \cdot 17.14 \cdot 10^{3} \cdot 33.33 + 0.4312 \cdot 33.33^{3}}{1000 \cdot 0.9214} = 33.15 \text{ kW}$$

Source: [10]

(4.5) **Applicability:** An electric machine drives the wheels with the adopted mechanical driveline efficiency.

This is a continuous cruise-duty requirement. Also compare it with the acceleration requirement in Chapter 5, without treating it as the peak rating.

Table Д.4.5.2 — Inputs for (4.5): Power required at maximum speed

Symbol	Quantity	Unit	Provenance
ψ	total road-resistance coefficient	—	calculated earlier (3.8)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)
$V_{\max}$	target maximum vehicle speed	m/s	input data
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_{\text{м}}$	adopted mechanical driveline efficiency	—	adopted by the designer

Example — electric vehicle

$$N_{\text{v}} = \frac{\psi_{\text{V}} G_{\text{a}} V_{\max} + W V_{\max}^{3}}{1000 \eta_{\text{м}}} \tag{4.5}$$

$$N_{\text{v}} = \frac{0.0315 \cdot 17.14 \cdot 10^{3} \cdot 44.44 + 0.4312 \cdot 44.44^{3}}{1000 \cdot 0.9500} = 65.11 \text{ kW}$$

- For an electrified drive, this is the minimum continuous cruise power. Determine machine ratings and power sharing separately.

Source: [10]; [14]

(4.5) **Applicability:** Both the engine and electric machine contribute mechanically to traction.

The total requirement does not determine each source's share. Specify the shares and operating layout separately.

Table Д.4.5.3 — Inputs for (4.5): Power required at maximum speed

Symbol	Quantity	Unit	Provenance
ψ	total road-resistance coefficient	—	calculated earlier (3.8)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)
$V_{\max}$	target maximum vehicle speed	m/s	input data
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_{\text{м}}$	adopted mechanical driveline efficiency	—	adopted by the designer

Example — hybrid

$$N_{\text{v}} = \frac{\psi_{\text{V}} G_{\text{a}} V_{\max} + W V_{\max}^{3}}{1000 \eta_{\text{м}}} \tag{4.5}$$

$$N_{\text{v}} = \frac{0.0315 \cdot 17.14 \cdot 10^{3} \cdot 44.44 + 0.4312 \cdot 44.44^{3}}{1000 \cdot 0.9500} = 65.11 \text{ kW}$$

- For an electrified drive, this is the minimum continuous cruise power. Determine machine ratings and power sharing separately.

Source: [10]; [14]

(4.5) **Applicability:** The engine drives only a generator; the electric machine drives the wheels.

The traction-drive input requirement does not establish the generator-set rating: its losses and energy balance are also required.

Table Д.4.5.4 — Inputs for (4.5): Power required at maximum speed

Symbol	Quantity	Unit	Provenance
ψ	total road-resistance coefficient	—	calculated earlier (3.8)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)
$V_{\max}$	target maximum vehicle speed	m/s	input data
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_{\text{м}}$	adopted mechanical driveline efficiency	—	adopted by the designer

Example — series hybrid

$$N_{\text{v}} = \frac{\psi_{\text{V}} G_{\text{a}} V_{\max} + W V_{\max}^{3}}{1000 \eta_{\text{м}}} \tag{4.5}$$

$$N_{\text{v}} = \frac{0.0315 \cdot 17.14 \cdot 10^{3} \cdot 44.44 + 0.4312 \cdot 44.44^{3}}{1000 \cdot 0.9500} = 65.11 \text{ kW}$$

- For an electrified drive, this is the minimum continuous cruise power. Determine machine ratings and power sharing separately.

Source: [10]; [14]

Preliminary maximum engine power based on N_v

This rule applies only before selecting a specific engine. The teaching example retains the manufacturer power and prints the preliminary value alongside it. The reproducible examples below use historical course-method multipliers. A new project in the application instead initially adopts $k_{\text{N}}=1{,}15$ for an internal-combustion powertrain: an explicit design margin of 15 %, not a standard or manufacturer specification. Power and rated speed are editable; automatic selection does not overwrite a measured or explicitly specified plateau characteristic.

(4.22) **Applicability:** Hybrid power is distributed using the specified electric fraction.

The engine receives the fraction $1\text{-}x_{\text{e}}$ of preliminary installed power $k_{\text{N}} N_{\text{v}}$. This initial distribution requires an energy-balance check.

Table Д.4.22.2 — Inputs for (4.22): Preliminary maximum engine power based on N_v

Symbol	Quantity	Unit	Provenance
N_v	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)
—	engine fuel system and charging arrangement	—	adopted by the designer
—	vehicle category	—	input data
x_e	electric fraction of the preliminary installed hybrid power	—	input data

Example — hybrid

$$P_{ДВЗ,розр}=(1-x_e)k_N N_v \quad (4.22)$$

$$P_{ДВЗ,розр}=(1-0.50)\cdot 1.10\cdot 65.11=35.81 \text{ kW}$$

- Preliminary rating: this step replaces the manufacturer's rating only while that rating is unavailable.

Source: [6]

(4.22) **Applicability:** The engine supplies all traction power.

Select the multiplier k_N from the method's table according to engine type and vehicle purpose.

Table Д.4.22.1 — Inputs for (4.22): Preliminary maximum engine power based on N_v

Symbol	Quantity	Unit	Provenance
N_v	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)
—	engine fuel system and charging arrangement	—	adopted by the designer
—	vehicle category	—	input data

Example — conventional internal-combustion vehicle

$$P_{ДВЗ,розр}=k_N N_v \quad (4.22)$$

$$P_{ДВЗ,розр}=1.10\cdot 33.15=36.46 \text{ kW}$$

- Preliminary rating: this step replaces the manufacturer's rating only while that rating is unavailable.

Source: [6]

Preliminary calculated electric-machine power

For an electric drive, preliminary power equals $N_v$without the spark-ignition engine multiplier. Check the rating against heating, acceleration and allowable speed.

(4.22) **Applicability:** The hybrid powertrain's electric power fraction is specified.

The electric machine receives the fraction x_e from $k_N N_v$; the engine supplies the remainder.

Table Д.4.22.4 — Inputs for (4.22): Preliminary calculated electric-machine power

Symbol	Quantity	Unit	Provenance
N_{v}	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)
x_{e}	electric fraction of the preliminary installed hybrid power	—	input data
—	engine fuel system and charging arrangement	—	adopted by the designer

Example — hybrid

$$P_{\text{e,розр}}=x_{\text{e}}k_{\text{N}}N_{\text{v}} \tag{4.22}$$

$$P_{\text{e,розр}}=0.50\cdot1.10\cdot65.11=35.81\ \text{kW}$$

- After selecting a real machine, replace the calculated value with an adopted manufacturer rating.

Source: [6]; [14]

(4.22) **Applicability:** An electric machine drives the wheels.

The preliminary rating equals the driveline input requirement. Check acceleration margin and thermal duty separately.

Table Д.4.22.3 — Inputs for (4.22): Preliminary calculated electric-machine power

Symbol	Quantity	Unit	Provenance
N_{v}	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)

Example — electric vehicle

$$P_{\text{e,розр}}=N_{\text{v}} \tag{4.22}$$

$$P_{\text{e,розр}}=65.11=65.11\ \text{kW}$$

- After selecting a real machine, replace the calculated value with an adopted manufacturer rating.

Source: [6]; [14]

Preliminary speed at maximum power

Historical teaching examples use the midpoint of the source range and explicitly identify it as a preliminary assumption. The guide gives no passenger-diesel range; the examples use the 3250—4000 rpm envelope of three production engines. This is not a regulatory requirement. New application projects use separate starting profiles M1/M2/M3/N1/N2/N3: diesel ratings 3800/3500/2200/3500/2400/2100 rpm and spark-ignition ratings 5500/5000/4000/5000/4200/4000 rpm. Lower operating speeds are respectively 800/780/650/780/700/600 rpm. These are project assumptions; an adopted specific characteristic defines its own band.

Table П.4.7 — Inputs to the adopted decision: Preliminary speed at maximum power

Symbol	Quantity	Unit	Provenance
—	engine fuel system and charging arrangement	—	adopted by the designer
—	vehicle category	—	input data

Example — conventional internal-combustion vehicle

$$\omega_N \in [\omega_{N,min};\ \omega_{N,max}], \qquad \omega_v = k_\omega\ \omega_N$$

$$\omega_N = \frac{440+550}{2} = 495\ \text{s}^{-1}\ (4727\ \text{rev/min})$$

- Preliminary rating: this step replaces the manufacturer's rating only while that rating is unavailable.

Source: [6]

Table M.4.5 — Historical preliminary ratings for teaching examples from the course method, printed p. 13

Engine and purpose	Power multiplier	Speed at maximum power, s^{-1}	Maximum-to-rated speed ratio
Spark ignition, passenger car or bus	1,1	440–550	1,15–1,20
Truck spark-ignition engine with governor	1,0	310–330	1,0
Truck diesel	1,0	200–260	1,0
Passenger-car diesel	1,0	340—419 (3250—4000 rpm), envelope of production examples	1,0

Historical diesel-bus examples use the truck row according to the grouping in Table 5 of the method. Record this scope extension in the adopted decision; it is not the application's current class profile. In a series hybrid, this rough estimate does not replace generator sizing.

4.3. Engine and electric-machine characteristics

Full-load engine speed characteristic

Power and torque are related by $P=M\omega$: when power is specified in kilowatts, the torque numerator includes a factor of 1000. The approximation introduces $\xi=\omega/\omega_N$. Choose the curve form according to the available data; a smooth plot does not create measured points.

(4.7) **Applicability:** Maximum power, its speed and the fuel system are known, but a full dynamometer characteristic is unavailable.

Use an approximation based on the coefficients in Table M.4.6. The cubic-term coefficient ensures that power equals the rating at unit relative speed.

Table Д.4.7.1 — Inputs for (4.7): Full-load engine speed characteristic

Symbol	Quantity	Unit	Provenance
—	engine fuel system and charging arrangement	—	adopted by the designer
P_{max}	adopted maximum engine power	kW	input data
ω_{Pmax}	angular speed at maximum engine power	s^{-1}	input data
—	basis of the engine rating: peak or continuous duty	—	adopted by the designer
—	internal-combustion engine connection method	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$P=P_{max}[A\xi+B\xi^2-C\xi^3],\ \xi=\omega/\omega_N,\ T=1000P/\omega \tag{4.7}$$

$$\xi=\frac{80.000000}{575.958653}=0.138899$$

$$P=110.000000[0.511162\cdot0.138899+1.617855\cdot0.138899^2-1.129017\cdot0.138899^3]=10.910617\ \text{kW}$$

$$T=\frac{1000\cdot10.910617}{80.000000}=136.382712\ \text{N·m}$$

- Numerical substitutions are shown for representative operating points. The remaining tabulated and plotted points are calculated using the same formula; power P is in kW, torque T in N·m and angular speed ω in rad/s.
- This machine transmits torque to the wheels.
- The characteristic is approximated from the fuel system; coefficients were determined on a chassis dynamometer.
- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.

Source: [15]

Applicability: Measured power or torque values with speeds are available.

Retain reference points with provenance and units. Specified points alone do not establish behaviour outside the measured range; identify interpolation in later calculations as the project's chosen method.

Table П.4.9 — Inputs to the adopted decision: Full-load engine speed characteristic

Symbol	Quantity	Unit	Provenance
—	measured points of the engine's full-load characteristic	—	input data
—	basis of the engine rating: peak or continuous duty	—	adopted by the designer
—	internal-combustion engine connection method	—	adopted by the designer

Example — conventional vehicle with a measured characteristic

$$P(\omega) \text{ from specified points}$$

$$P_{max}=110.000 \text{ kW } (\omega=575.959 \text{ rad/s})$$

$$T_{max}=216.451 \text{ N·m from specified points}$$

- The specified-point table is input data. The displayed maxima are found among those points, not by substituting into an approximation.
- This machine transmits torque to the wheels.
- The characteristic uses measured points; source: engine specification, dynamometer points.
- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.
- The characteristic consists of specified points: values between them are not interpolated, and it does not extend beyond the end points.
- The user records the point source: dynamometer, engine specification or another source.

- The characteristic consists of specified points: values between them are not interpolated, and it does not extend beyond the end points.
- The user records the point source: dynamometer, engine specification or another source.

Source: [2]

Applicability: The specification provides a torque plateau and a separate maximum constant-power range.

Reference points are the start and end of the torque plateau, the maximum-power point and the end of constant-power operation. The project interpolates between these points; the low-speed region has the greatest uncertainty.

Table П.4.10 — Inputs to the adopted decision: Full-load engine speed characteristic

Symbol	Quantity	Unit	Provenance
P_{max}	adopted maximum engine power	kW	input data
ω_{Pmax}	angular speed at maximum engine power	s^{-1}	input data
T_{max}	manufacturer's maximum engine torque	N·m	input data
ω_1	angular speed at the start of the torque plateau	s^{-1}	input data
ω_2	angular speed at the end of the torque plateau	s^{-1}	input data
$\omega_{Ркін}$	angular speed at the end of the specified constant-power range	s^{-1}	input data
—	basis of the engine rating: peak or continuous duty	—	adopted by the designer
—	internal-combustion engine connection method	—	adopted by the designer

Example — conventional vehicle with a plateau characteristic

$$P(\omega) \text{ from specified points}$$

$$P_{max}=110.000 \text{ kW } (\omega=418.879 \text{ rad/s})$$

$$T_{max}=320.000 \text{ N·m from specified points}$$

- The specified-point table is input data. The displayed maxima are found among those points, not by substituting into an approximation.
- This machine transmits torque to the wheels.
- Straight segments connect the manufacturer's points. This is project interpolation, not a method from the source.
- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.
- Straight segments connect the manufacturer's engine points rather than approximating them. This is the project's explicit interpolation, not a method stated by the source.
- The region below the plateau is the least certain because specifications usually omit low-speed torque.
- The constant-power region follows the manufacturer's maximum-power range.

- Straight segments connect the manufacturer's engine points rather than approximating them. This is the project's explicit interpolation, not a method stated by the source.
- The region below the plateau is the least certain because specifications usually omit low-speed torque.
- The constant-power region follows the manufacturer's maximum-power range.

Source: [2]

Applicability: The truck-engine specification gives torque and power, but no maximum constant-power range.

End the characteristic at the specified maximum-power speed. Inventing a horizontal extension would increase achievable speed and change gear selection.

Table П.4.8 — Inputs to the adopted decision: Full-load engine speed characteristic

Symbol	Quantity	Unit	Provenance
P_{max}	adopted maximum engine power	kW	input data
ω_{Pmax}	angular speed at maximum engine power	s^{-1}	input data
T_{max}	manufacturer's maximum engine torque	N·m	input data
ω_1	angular speed at the start of the torque plateau	s^{-1}	input data
ω_2	angular speed at the end of the torque plateau	s^{-1}	input data
—	basis of the engine rating: peak or continuous duty	—	adopted by the designer
—	internal-combustion engine connection method	—	adopted by the designer

Example — three-axle truck (KamAZ-53215, 6×4)

$$P(\omega) \text{ from specified points}$$

$$P_{max} = 169.000 \text{ kW } (\omega = 230.383 \text{ rad/s})$$

$$T_{max} = 814.000 \text{ N·m from specified points}$$

- The specified-point table is input data. The displayed maxima are found among those points, not by substituting into an approximation.
- This machine transmits torque to the wheels.
- Straight segments connect the manufacturer's points. This is project interpolation, not a method from the source.

- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.
- Straight segments connect the manufacturer's engine points rather than approximating them. This is the project's explicit interpolation, not a method stated by the source.
- The region below the plateau is the least certain because specifications usually omit low-speed torque.
- Above n_{Pmax} the characteristic is not extended: the specification gives no maximum-power range.

- Straight segments connect the manufacturer's engine points rather than approximating them. This is the project's explicit interpolation, not a method stated by the source.
- The region below the plateau is the least certain because specifications usually omit low-speed torque.
- Above n_{Pmax} the characteristic is not extended: the specification gives no maximum-power range.

Source: [2]

(4.7) **Applicability:** The engine drives a generator and has no mechanical path to the wheels.

Its full-load characteristic is still needed to establish engine operation, but is not added to wheel tractive force.

Table Д.4.7.2 — Inputs for (4.7): Full-load engine speed characteristic

Symbol	Quantity	Unit	Provenance
—	engine fuel system and charging arrangement	—	adopted by the designer
P_{max}	adopted maximum engine power	kW	input data
ω_{Pmax}	angular speed at maximum engine power	s^{-1}	input data
—	basis of the engine rating: peak or continuous duty	—	adopted by the designer
—	internal-combustion engine connection method	—	adopted by the designer

Example — series hybrid

$$P=P_{\text{max}}[A\xi+B\xi^2-C\xi^3],\ \xi=\omega/\omega_{\text{N}},\ T=1000P/\omega \tag{4.7}$$

$$\xi=\frac{80.000000}{575.958653}=0.138899$$

$$P=110.000000[0.511162\cdot 0.138899+1.617855\cdot 0.138899^2-1.129017\cdot 0.138899^3]=10.910617\ \text{kW}$$

$$T=\frac{1000\cdot 10.910617}{80.000000}=136.382712\ \text{N·m}$$

- Numerical substitutions are shown for representative operating points. The remaining tabulated and plotted points are calculated using the same formula; power P is in kW, torque T in N·m and angular speed ω in rad/s.
- This machine does not transmit torque to the wheels; its characteristic is shown for completeness.
- The characteristic is approximated from the fuel system; coefficients were determined on a chassis dynamometer.
- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.

Source: [15]

The application's current initial mode is an explicit project approximation. A smooth envelope describes increasing torque, the torque plateau and approach to rated power. Shape parameters do not come from Euro 5/6 standards: Euro classifies emissions. Blank

manufacturer-torque and plateau-limit fields remain blank; calculating power does not create manufacturer data.

In the current initial mode, relationship (4.7) is expressed using $P=T\omega/1000$ and three torque regions:

- at $\omega_x \le \omega < \omega_1$: $T=T_m[c+(1-c)(3x^2-2x^3)]$;
- at $\omega_1 \le \omega \le \omega_2$: $T=T_m$;
- at $\omega_2 < \omega \le \omega_N$: $T=1000[P_N-(P_N-P_2)(1-y)^p]/\omega$.

Here $T_m=k_T 1000 P_N/\omega_N$, $\omega_1=a\omega_N$, $\omega_2=b\omega_N$, $P_2=T_m\omega_2/1000$, $x=(\omega-\omega_x)/(\omega_1-\omega_x)$, $y=(\omega-\omega_2)/(\omega_N-\omega_2)$, $p=k_T(1-b)/(1-k_T b)$. For diesel, the adopted values are $(k_T,a,b,c)=(1,40;0,43;0,62;0,45)$; for turbocharged petrol, $(1,50;0,32;0,58;0,40)$; for naturally aspirated spark ignition and the adopted LPG model, $(1,15;0,65;0,72;0,40)$. All these numbers are explicit project assumptions. Other fuels require an explicitly specified or measured characteristic. The remainder of this subsection gives separate historical polynomial and manufacturer-based teaching examples; their formulas are not the default envelope used by a new project.

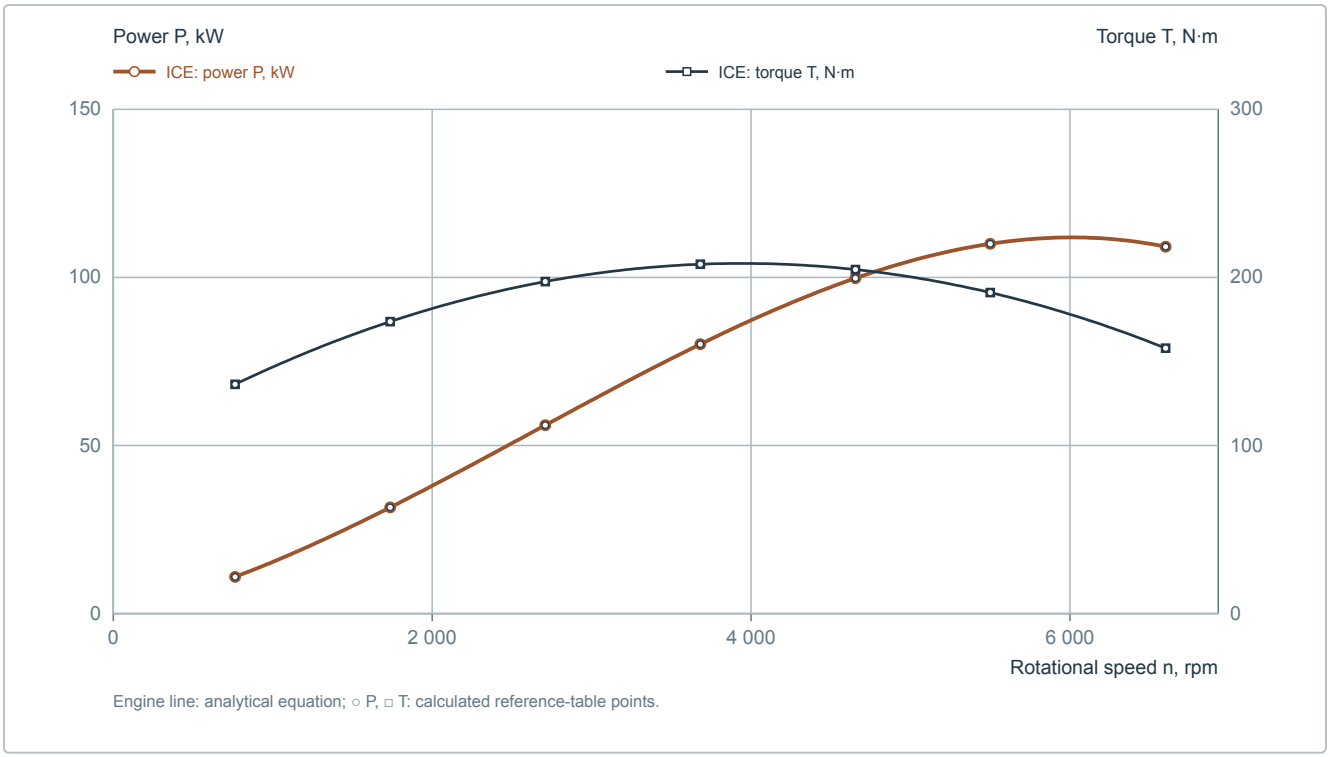


Figure 4.1 — Power and torque of the teaching engine approximation; reference values belong to the same worked example.
Source: [15].

Table M.4.6 — Shpitsa approximation coefficients (2019), relationships (8)–(9); C = A + B − 1

Power supply system	A	B	C	Sample count
Naturally aspirated petrol, fuel injection	0,511162	1,617855	1,129017	237
Turbocharged petrol	0,295132	2,120719	1,415851	9
First- and second-generation gas equipment	0,570199	1,574281	1,144480	64
Fourth-generation gas equipment	0,542335	1,544406	1,086741	23
Naturally aspirated diesel	0,738505	1,194464	0,932969	11
Turbocharged diesel	0,261225	2,568185	1,829410	175

These coefficients describe engine samples rather than the selected model's specification. The source notes insufficient samples for turbocharged petrol and naturally aspirated diesel groups. It recommends a different function for turbocharged petrol engines; turbo-diesel low-speed deviations can reach 50 %. Use the appropriate branch when manufacturer plateau data or dynamometer points are available.

Maximum engine torque

Find maximum torque on the constructed full-load characteristic, rather than dividing maximum power by an arbitrary speed. The speed at maximum torque is also needed to distribute gearbox ratios. For a measured characteristic, conclusions are limited to its points and selected interpolation method.

Table П.4.5 — Inputs to the adopted decision: Maximum engine torque

Symbol	Quantity	Unit	Provenance
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)

Example — conventional internal-combustion vehicle

$$M_{к,max} = \max_{\omega} T(\omega)$$

$$M_{к,max} = 207.8 \text{ N·m at } 3682 \text{ min}^{-1}$$

- Maximum torque occurs at 3 682 min⁻¹; this also defines the common ratio of the geometric gear series.
- This is full-load torque from the external characteristic.

Source: [15]

Maximum engine speed

Maximum speed is the upper limit of the adopted characteristic. It differs from both the maximum-power speed and the maximum-torque speed. The manufacturer's limit does not permit extending the curve beyond it.

Table П.4.4 — Inputs to the adopted decision: Maximum engine speed

Symbol	Quantity	Unit	Provenance
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)

Example — conventional internal-combustion vehicle

$$\omega_{д,\max}=\max_{\omega\in\text{characteristic}}\omega$$

$$\omega_{д,\max}=691.2\ \text{s}^{-1}$$

- Maximum speed is 6 600 min⁻¹.
- This is the upper limit of the full-load characteristic, not the speed of an operating point.

Source: [15]

Electric-motor external characteristic

For an idealised electric machine, torque is constant up to base speed, so power rises in proportion to speed; thereafter power is constant and torque decreases inversely with speed. Inputs are power, base and maximum speeds, and peak or continuous duty. A dynamometer characteristic may instead be entered as individual points with provenance; do not replace it with the two idealised regions.

(4.23) **Applicability:** Idealised electric machine with constant-torque and constant-power regions.

The two regions define a teaching envelope, not a complete map of thermal and electrical limits.

Table Д.4.23.1 — Inputs for (4.23): Electric-motor external characteristic

Symbol	Quantity	Unit	Provenance
$P_{м}$	adopted electric-machine power	kW	input data
ω_{b}	base angular speed of the electric machine	s⁻¹	input data
ω_{max}	maximum angular speed of the electric machine	s⁻¹	input data
—	basis of the electric-machine rating: peak or continuous duty	—	adopted by the designer

Example — electric vehicle

$$\begin{gathered}T_{b}=1000P_{max}/\omega_{b}\\ T=T_{b},\ P=T_{b}\omega/1000\ \ (\omega\le\omega_{b})\\ P=P_{max},\ T=1000P/\omega\ \ (\omega>\omega_{b})\end{gathered}\tag{4.23}$$

$$T_{b}=\frac{1000\cdot125.000000}{418.879020}=298.415518\ \text{N·m}$$

$$P(9.163)=\frac{298.415518\cdot9.162979}{1000}=2.734375\ \text{kW}$$

$$T(1466.077)=\frac{1000\cdot125.000000}{1466.076572}=85.261577\ \text{N·m}$$

- Numerical substitutions are shown for representative operating points. The remaining tabulated and plotted points are calculated using the same formula; power P is in kW, torque T in N·m and angular speed ω in rad/s.
- This machine transmits torque to the wheels.
- The curve comprises two ideal regions: constant torque and constant power. It does not show thermal limits.

- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.

Source: [14]

(4.23) **Applicability:** The electric-machine characteristic is entered as individual points.

In this teaching example, the author assumed the points to demonstrate dynamometer-table input. No measurements of a specific machine are claimed. Calculate each point's power from its torque and speed; the two-region model does not replace the points.

Table Д.4.23.2 — Inputs for (4.23): Electric-motor external characteristic

Symbol	Quantity	Unit	Provenance
—	specified electric-machine characteristic points	—	input data
—	basis of the electric-machine rating: peak or continuous duty	—	adopted by the designer

Example — tabulated electric-machine characteristic

$$P(\omega) \text{ from specified points} \tag{4.23}$$

$$P_{\max}=119.700 \text{ kW } (\omega=420.000 \text{ rad/s})$$

$$T_{\max}=290.000 \text{ N·m from specified points}$$

- The specified-point table is input data. The displayed maxima are found among those points, not by substituting into an approximation.
- This machine transmits torque to the wheels.
- Point-based characteristic; source: a teaching table assumed by the author to demonstrate point entry, not a manufacturer specification or measurement report.
- Manufacturer values are treated as short-term peak ratings; the continuous characteristic lies below them.
- The characteristic consists of specified points: values between them are not interpolated, and it does not extend beyond the end points.
- The user records the point source: dynamometer, engine specification or another source.
- The characteristic consists of specified points: values between them are not interpolated, and it does not extend beyond the end points.
- The user records the point source: dynamometer, engine specification or another source.

Source: [2]

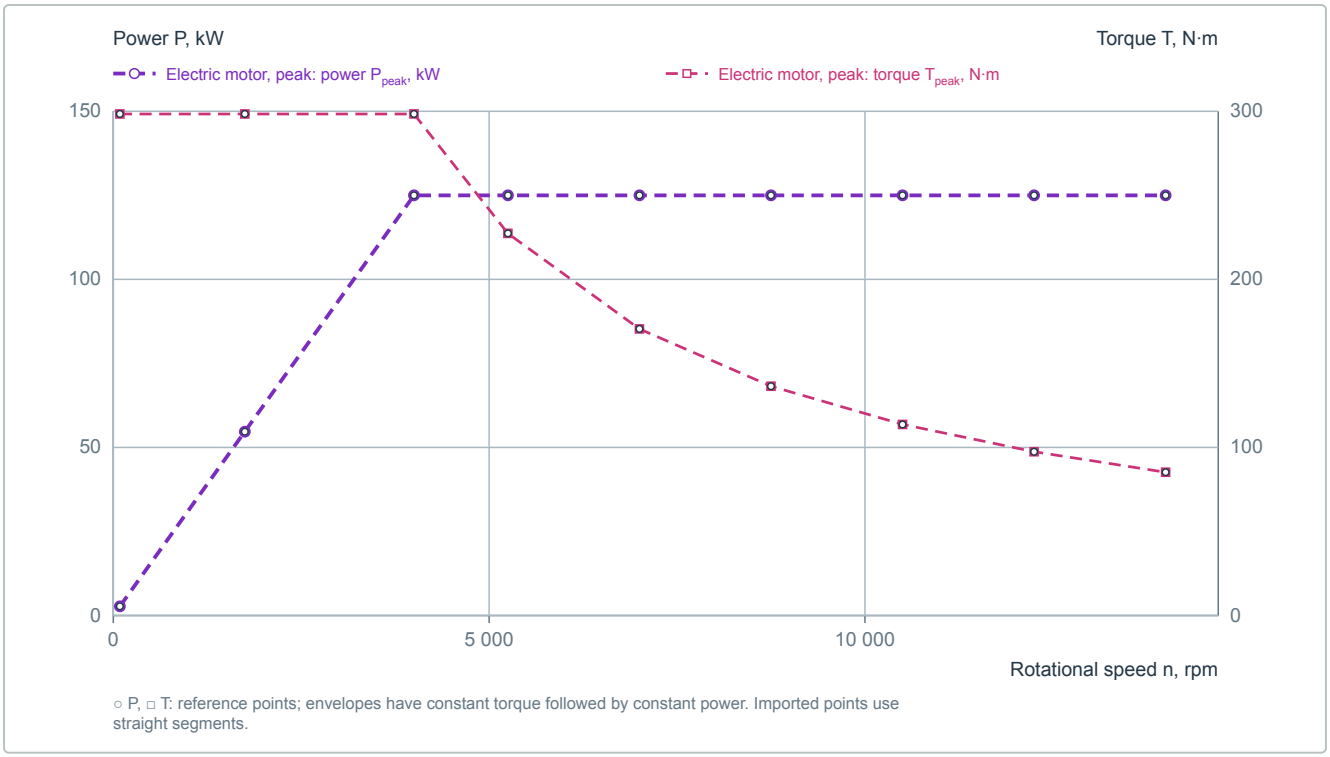


Figure 4.2 — Two regions of the traction-machine characteristic: constant torque and constant power.
Source: [14].

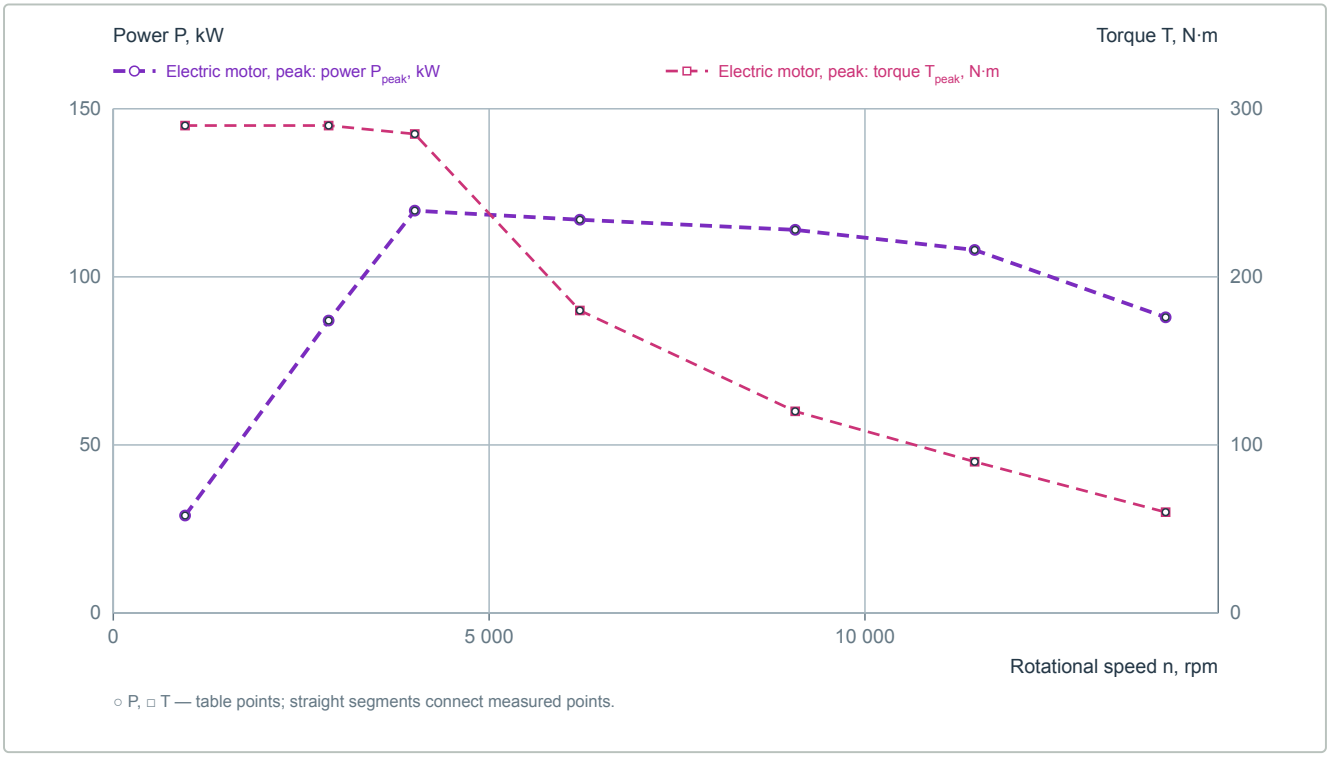


Figure 4.3 — Specified electric-machine points: the author's teaching table, without extension beyond its end points.
Source: [2].

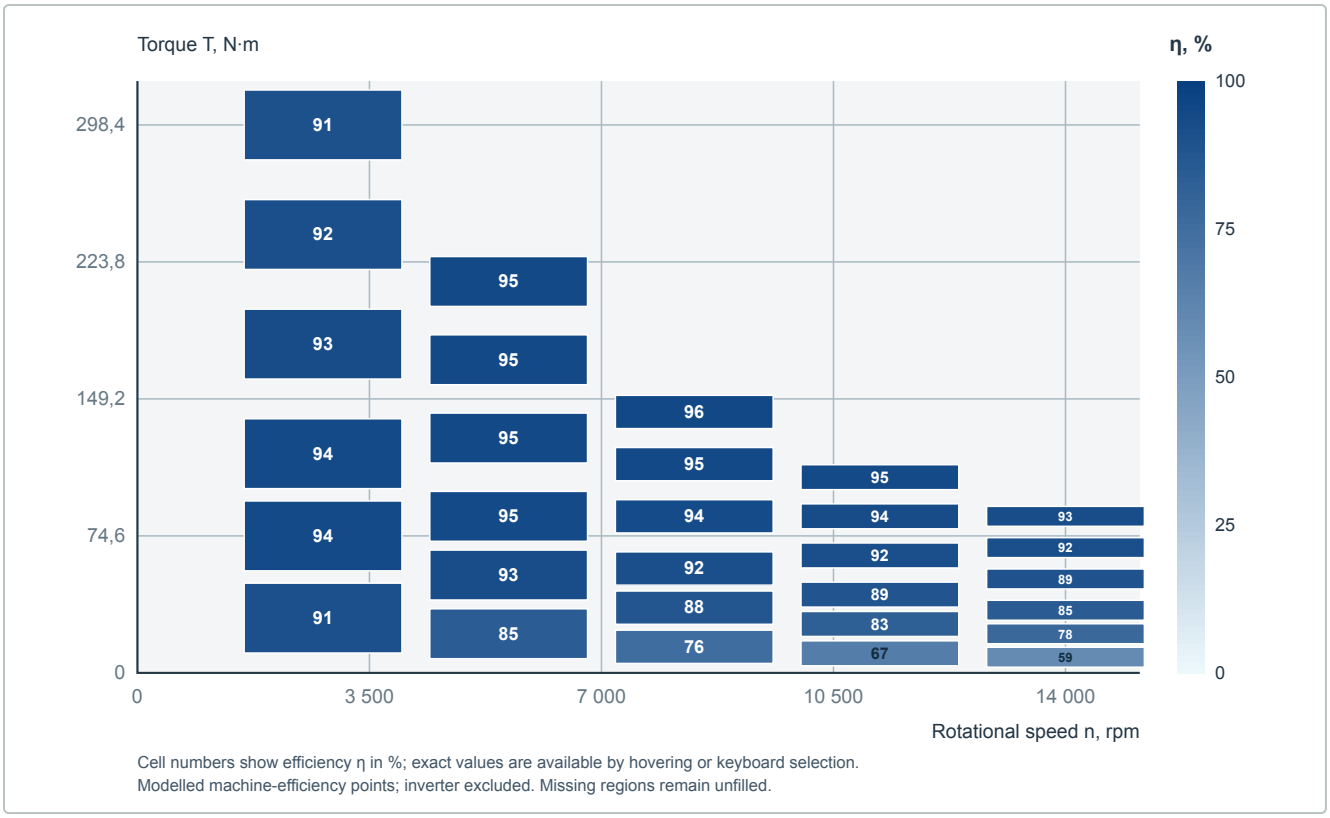


Figure 4.4 — Modelled electric-machine efficiency-map points from the same teaching example; the inverter is outside this boundary. Missing regions remain unfilled.
Source: [16].

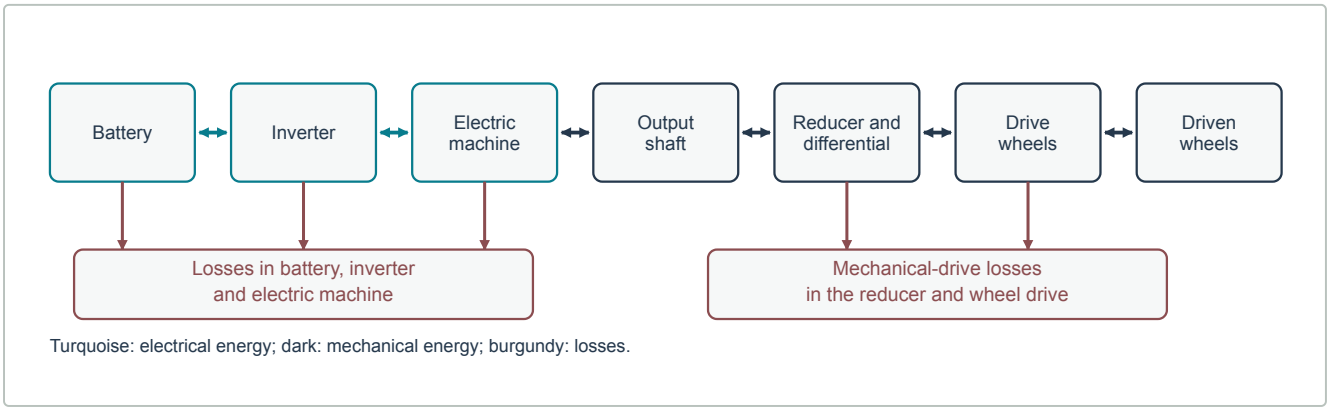


Figure 4.5 — BEV electrical and mechanical paths from battery to driven wheels, including reduction gear, differential and wheel drive. Components may be integrated into one assembly; path efficiency accounts for mechanical losses. Reverse arrows indicate permitted regeneration; arrow width does not represent power.
Source: [2].

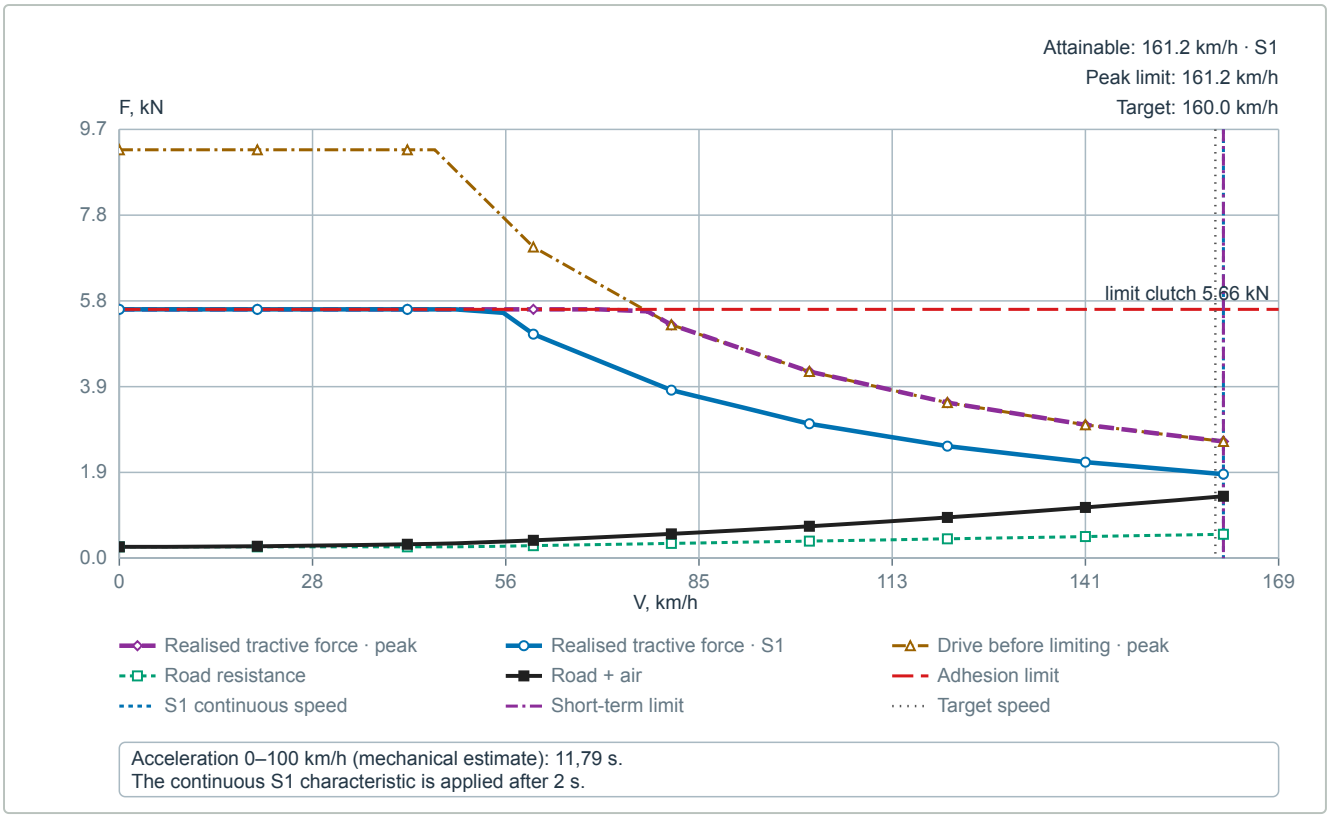


Figure 4.6 — BEV peak and continuous traction in the adopted gear, motion resistances and the independent adhesion limit. The author assumes 90 kW continuous power and 2 s peak duration; §5.1 explains the mechanical acceleration estimate with transition to S1. This figure does not check battery, inverter or thermal limits.
Source: [2].

Table M.4.10 — Teaching electric-machine points: all torques are author assumptions; power is calculated from them

Angular speed, s^{-1}	Torque, N·m	Power, kW
100	290	29
300	290	87
420	285	119,7
650	180	117
950	120	114
1200	90	108
1466,08	60	87,965

Maximum electric-motor torque

This calculation uses the two-region characteristic of the electric-vehicle example in Figure 4.2. Table M.4.10 and Figure 4.3 belong to a separate tabulated-machine example whose inputs are individual points. In the two-region model, power at base speed determines maximum torque, which remains constant below that speed. The speed printed beside the maximum identifies one calculated point on the plateau. Peak torque is available for brief acceleration only for its permitted duration.

Table Д.4.24 — Inputs for (4.24): Maximum electric-motor torque

Symbol	Quantity	Unit	Provenance
$P_{м}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)

Example — electric vehicle

$$M_{м,max}=\max_{\omega} T(\omega) \tag{4.24}$$

$$M_{м,max}=298.4\ \text{N·m at } 88\ \text{min}^{-1}$$

- Maximum torque occurs at 88 min⁻¹.
- This is full-load torque from the external characteristic.

Source: [14]

Maximum electric-motor speed

The upper machine-speed limit bounds vehicle speed through the adopted reduction ratio. It does not verify allowable torque, winding temperature or inverter voltage.

Table Д.4.25 — Inputs for (4.25): Maximum electric-motor speed

Symbol	Quantity	Unit	Provenance
$P_{м}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)

Example — electric vehicle

$$\omega_{м,max}=\max_{\omega \in \text{characteristic}} \omega \tag{4.25}$$

$$\omega_{м,max}=1466.1\ \text{s}^{-1}$$

- Maximum speed is 14 000 min⁻¹.
- This is the upper limit of the full-load characteristic, not the speed of an operating point.

Source: [14]

Required maximum front-axle electric-motor speed

For the front path, convert wheel speed to machine speed using the overall reduction and dynamic radius. Check the front axle specifically; do not substitute the rear-path ratio. This example has one machine splitting torque between two axles.

Table Д.4.6.1 — Inputs for (4.6): Required maximum front-axle electric-motor speed

Symbol	Quantity	Unit	Provenance
V_{max}	target maximum vehicle speed	m/s	input data
$r_{д}$	dynamic wheel radius	m	calculated earlier
$i_{g,front}$	overall reduction ratio from the electric machine to the front wheels	—	adopted by the designer
$P_{м}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)

Example — electric vehicle with two driven axles

$$n_{\text{m,max}} = \frac{30 V_{\max} i_{\text{g}}}{\pi r_{\text{д}}} \tag{4.6}$$

$$n_{\text{m,max,front}} = \frac{30 \cdot 44.44 \cdot 9.700}{\pi \cdot 0.296} = 13898 \ \text{min}^{-1}$$

- The motor passes: required 13 898 min⁻¹, available 14 000 min⁻¹ (margin 1,01).
- This checks rotational speed only. Required torque and overheating are separate questions.

Source: [14]

Required maximum rear-axle electric-motor speed

Perform the same kinematic check for the rear path using its own ratio. If required speed exceeds the manufacturer's limit, the target vehicle speed is unattainable regardless of power reserve. For a two-speed drive, check the active gear rather than multiplying two alternative ratios.

Table Д.4.6.2 — Inputs for (4.6): Required maximum rear-axle electric-motor speed

Symbol	Quantity	Unit	Provenance
$V_{\max}$	target maximum vehicle speed	m/s	input data
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier
$i_{\text{g,rear}}$	overall reduction ratio from the electric machine to the rear wheels	—	adopted by the designer
$P_{\text{м}}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)

Example — electric vehicle

$$n_{\text{m,max}} = \frac{30 V_{\max} i_{\text{g}}}{\pi r_{\text{д}}} \tag{4.6}$$

$$n_{\text{m,max,rear}} = \frac{30 \cdot 44.44 \cdot 9.700}{\pi \cdot 0.296} = 13898 \ \text{min}^{-1}$$

- The motor passes: required 13 898 min⁻¹, available 14 000 min⁻¹ (margin 1,01).
- This checks rotational speed only. Required torque and overheating are separate questions.

Source: [14]

4.4. Efficiency maps: dependence on operating point

Engine efficiency map

The Willans-line map follows from the balance of shaft power, indicated power and friction. The author specifies indicated efficiency; friction torque is matched to rated specific fuel consumption. Relative losses increase at light load. Comparison with the empirical corrections in Chapter 7 assesses model sensitivity rather than accuracy against an independent experiment. Constant friction torque and the absence of a mixture-enrichment model limit applicability.

Table Д.4.20 — Inputs for (4.20): Engine efficiency map

Symbol	Quantity	Unit	Provenance
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
g_N	specific fuel consumption at rated power	g/kWh	input data
η_i	adopted indicated engine efficiency	—	adopted by the designer
—	Energy carrier Fuel or electricity adopted for the powertrain; for an engine, this choice determines the fuel's lower heating value.	—	input data

Example — conventional internal-combustion vehicle

$$\eta_N = \frac{3600}{g_N H_u}$$
$$M_{тр} = \frac{P_N}{\omega_N}\left(\frac{\eta_i}{\eta_N} - 1\right) \qquad (4.20)$$
$$\eta_e = \frac{\eta_i P_e}{P_e + M_{тр}\omega}$$

$$\eta_N = \frac{3600}{250.00000 \cdot 44.00000} = 0.327273$$
$$M_{тр} = \frac{10^3 \cdot 110.00000}{575.958653}\left(\frac{0.400000}{0.327273} - 1\right) = 42.44132 \ \text{N·m}$$
$$\eta_e(\omega_*, P_*) = \frac{0.400000 \cdot 83.529546 \cdot 10^3}{83.529546 \cdot 10^3 + 42.441318 \cdot 403.171057} = 0.331991 = 33.2\%$$

- Power-balance formulas use W: multiply characteristic values in kW by 10^3. For η_N specific fuel consumption g_N is specified in g/(kWh), and lower heating value H_u in MJ/kg.
- After determining friction, a substitution is shown for the calculated grid's best point: angular speed 403,171 rad/s and power 83,53 kW. Other points use the same equation; the result is the entire map, not just its maximum.
- Shaft power equals indicated power minus friction power (Ehsani, Eq. 3.6). Dividing by fuel power gives $\eta_e = \eta_i \cdot P/(P + M_{тр} \cdot \omega)$: the numerator is power delivered to the shaft; the denominator adds friction, which increases with speed but not load.
- Friction torque was not fitted arbitrarily: the engine's rated point gives $M_{тр}$ = 42,4 N·m, equivalent to 24,4 kW at rated speed.
- Mechanical efficiency at the rated point is 81,8 %. Ehsani gives 75…90 % at full throttle, so the result lies within this reference range. This does not validate the particular engine map.
- The best map point is 33,2 % at 100 % load and 70 % rated speed: high load and relatively low speed, as expected.
- Check: the same map was calculated at 35 points using the course method ($g_e = g_N \cdot K_\omega \cdot K_B$). The models were constructed independently, one from power balance and the other from polynomial fits to experiments. Their maximum difference is 23 %. This is model disagreement, not accuracy validation. Without a specified tolerance and independent measurements, agreement cannot be concluded.
- The map omits mixture enrichment at full load: the Willans line is monotonic in power and therefore places the best point at full load. A real spark-ignition engine's minimum consumption is near 80 %, where the course method's K_B falls below unity.

Source: [14]; [7]

Electric-machine efficiency map

Winding losses approximately follow torque squared, while iron losses depend on speed. Determine the two model constants from the assumed best point and equality of the loss components there. Record the source of best efficiency separately. Bound the map by the same machine's characteristic: positive formula results do not make points beyond allowable torque or speed valid operating points.

Table Д.4.21 — Inputs for (4.21): Electric-machine efficiency map

Symbol	Quantity	Unit	Provenance
$P_{м}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)
$\eta_{ем}^{\max}$	adopted maximum electric-machine efficiency	—	adopted by the designer

Example — electric vehicle

$$k_{м}M_{*}^{2}=k_{ст}\omega_{*}^{1,5}=\frac{M_{*}\omega_{*}(1/\eta_{*}-1)}{2}$$
$$\eta_{ем}=\frac{M\omega}{M\omega+k_{м}M^{2}+k_{ст}\omega^{1,5}} \quad (4.21)$$

$$k_{м}=\frac{119.366207\cdot 418.879020(1/0.950000-1)}{2\cdot 119.366207^{2}}=0.09234718$$
$$k_{ст}=\frac{119.366207\cdot 418.879020(1/0.950000-1)}{2\cdot 418.879020^{1,5}}=0.15348060$$
$$P=142.102628\cdot 879.645943=125000.0000\ \text{W}$$
$$P_{м}=0.09234718\cdot 142.102628^{2}=1864.7810\ \text{W}$$
$$P_{ст}=0.15348060\cdot 879.645943^{1,5}=4004.1962\ \text{W}$$
$$\eta_{ем}=\frac{125000.0000}{125000.0000+1864.7810+4004.1962}=0.955154=95.5\%$$

- First determine both loss coefficients from the calibration point, then calculate efficiency at the computed grid's best point. Torque is in N·m, speed in rad/s and Mω in W; calculate the rest of the map analogously.
- Zheng divides machine losses into two main components: winding losses increase with current squared, with current proportional to torque; iron losses increase with magnetisation frequency and therefore rotational speed. Hence $P_{втр} = k_{м}\cdot M^{2} + k_{ст}\cdot\omega^{1,5}$; efficiency is useful power divided by useful power plus losses.
- One point cannot determine two coefficients, but the maximum-efficiency condition provides a second equation: setting the efficiency derivative with respect to torque to zero gives $k_{м}\cdot M^{2} = k_{ст}\cdot\omega^{1,5}$. At maximum efficiency, winding losses equal iron losses. This follows from the assumed loss form rather than being an independent assertion.
- Calibration point: 119 N·m at 4 000 min⁻¹, with 95 % efficiency. Total loss is 2,63 kW, half winding and half iron loss.
- Map efficiency ranges from 59,2 to 95,5 %. The worst point is 10 % torque at 100 % speed: iron losses remain at high speed while useful power is very small.
- This illustrates why constant driveline efficiency is a crude replacement: the best-to-worst map difference is several times larger than the error often attributed to such a calculation.
- The speed exponent 1,5 is a compromise: Zheng treats hysteresis losses (∝ω) and eddy-current losses (∝ω²) separately, but one calibration point cannot distinguish them.
- Field weakening above the corner speed reduces flux and iron losses. The model omits this and therefore underestimates efficiency at the highest speeds.

- The map describes motoring. Regeneration has losses of similar form, but efficiency uses the reverse power ratio; this document does not perform that step.

Source: [16]; [7]

4.5. Adhesion, reduction and traction paths

Adhesive weight

Sum the weight forces of driven axles only. Wheel configuration gives their count, but a 4×2 layout must still specify front or rear drive. For two-axle all-wheel drive, include both axles without doubling vehicle weight. Static distribution omits acceleration load transfer.

Table Д.4.11 — Inputs for (4.11): Adhesive weight

Symbol	Quantity	Unit	Provenance
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
—	list of axles transmitting traction to the road	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$G_{\text{зч}}=\Sigma_{\text{i}\in\text{вед}}G_{\text{i}} \tag{4.11}$$

$$G_{\text{зч}}=G_2=9.43\ \text{kN}$$

- Driven axles: 2. Adhesive-weight coefficient m = G_зч/G = 0,55.
- Axle fractions are static. Acceleration and uphill load transfer are omitted, so the moving vehicle's $G_{\text{зч}}$ is greater for rear driven axles and smaller for front driven axles than this estimate.

Source: [17]

Driven-wheel adhesion force

Multiplying adhesive weight by the adopted adhesion coefficient gives the force the road can accept without excessive slip. This limit is independent of engine torque. A different surface requires a different coefficient even for the same vehicle.

Table Д.4.14 — Inputs for (4.14): Driven-wheel adhesion force

Symbol	Quantity	Unit	Provenance
$G_{\text{зч}}$	weight force on the driven axles	kN	calculated earlier (4.11)
φ	tyre–road adhesion coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$F_{\varphi}=\varphi\ G_{\text{зч}} \tag{4.14}$$

$$F_{\varphi}=0.60\cdot 9.43=5.66\ \text{kN}$$

- This is a static limit: acceleration transfers weight rearwards, so the moving vehicle's F_{φ} is greater for rear driven wheels and smaller for front driven wheels than this estimate.
- The adhesion coefficient is constant; this model does not describe its dependence on slip.

Source: [17]; [18]

Table M.4.7 — Adhesion coefficients following Birulia, §15, Table 6, printed p. 60

Surface and conditions	Coefficient
Dry, clean; particularly favourable	0,7
Dry, clean; normal	0,5
Wet, dirty; unfavourable	0,2–0,3
Icy; particularly unfavourable	0,1

The course method also uses a separate calculation value of 0,6. It is not the midpoint of every row and does not describe an arbitrary modern surface. For the range 0,2–0,3, the value 0,25 is an adopted midpoint, not a published individual measurement.

Kinematic ratio limit in top gear

Compare target vehicle speed with the traction source's maximum speed through the wheel radius. The lowest-ratio notation refers to top gear in the series. At fixed vehicle speed, a larger overall ratio raises engine speed; equality corresponds to its maximum speed. Power verification is a separate final step.

Table Д.4.10 — Inputs for (4.10): Kinematic ratio limit in top gear

Symbol	Quantity	Unit	Provenance
$r_д$	dynamic wheel radius	m	calculated earlier
V_{max}	target maximum vehicle speed	m/s	input data
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)

Example — conventional internal-combustion vehicle

$$u_{тр,min} = \frac{\omega_{max}\ r_к}{V_{max}} \qquad (4.10)$$

$$u_{тр,min} = \frac{691.2 \cdot 0.296}{33.33} = 6.14$$

- At the specified speed this is the greatest ratio that does not exceed maximum engine speed: ω = V·u_тр/r_к ≤ ω_{max}. The notation $u_{тр}$,min denotes top gear in the series. The power balance (4.16) checks whether power is sufficient at this speed.
- Using the free-rolling radius, $u_{тр}$,min would be up to 1,06 times greater; the difference provides a maximum-speed margin.

Source: [6]

Final-drive ratio

Remove the gearbox, active transfer-range and separate wheel-reduction factors from the required overall ratio. Show the required final-drive ratio separately from the actual product of integer tooth-count ratios. Use the adopted actual reduction in speed, traction, acceleration, consumption and component-load calculations.

Table Д.4.12 — Inputs for (4.12): Final-drive ratio

Symbol	Quantity	Unit	Provenance
$u_{тр}$	adopted overall transmission ratio in top gear	—	adopted by the designer
$u_{к,min}$	adopted top-gear ratio	—	adopted by the designer
$u_{рк,min}$	top-range transfer-case ratio	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$u_{0,потр}=\frac{u_{тр,потр}}{u_{к,min}\ u_{рк,min}},\qquad u_0=\prod_j \frac{z_{2,j}}{z_{1,j}} \tag{4.12}$$

$$u_{0,потр}=\frac{6.142}{1.00\cdot 1.000}=6.1420;\ u_0=\frac{55}{9}=6.1111$$

- Required reduction u_0=6,142; adopted stages: 55/9, giving u_0=6,1111. This actual product is used in every traction path; the final drive is not multiplied again.
- Tooth counts establish kinematics. Chapter 12 checks geometry, bending and contact; this does not yet confirm durability or manufacturability.

Source: [6]; [7]

Use one chain for each gear: $u_{\Sigma,k}=u_{к,k}u_{рк}u_0u_{кол}$. An absent stage has a factor of unity. The overall ratio already includes the final drive; multiplying by it again would be an error.

Table M.4.8 — Top gear from the course method, printed p. 17

Layout and purpose	Transmission ratio
Rear-wheel-drive passenger car	1,0
Front-wheel-drive passenger car	0,75–0,95
Diesel truck	0,66–1,00
Diesel truck with splitter or range unit	0,71–0,82
Truck with direct top gear	1,0
Urban or suburban bus	1,0
Intercity bus	0,72–0,78

For the high range of an all-wheel-drive truck transfer case, the same source gives 0,917–1,00. Without a transfer case, its factor in the product is unity: this denotes an absent reduction stage, not an imaginary component.

Ratio required for maximum road resistance

Low gear must provide sufficient force for the maximum adopted road resistance. This preliminary estimate uses maximum engine torque. Also check the ratio against adhesion: reduction alone cannot compensate for insufficient adhesive weight.

(4.13) **Applicability:** Mechanical efficiency was calculated from transmission components.

Vehicle purpose determines the maximum road-resistance row.

Table Д.4.13.1 — Inputs for (4.13): Ratio required for maximum road resistance

Symbol	Quantity	Unit	Provenance
ψ_{max}	maximum road-resistance coefficient for the intended use	—	adopted by the designer
G_a	total vehicle weight force	kN	calculated earlier (3.5)
$r_д$	dynamic wheel radius	m	calculated earlier
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
$\eta_м$	mechanical transmission efficiency based on its component count	—	calculated earlier (4.3)

Example — conventional internal-combustion vehicle

$$u_{тр,дор} = \frac{\psi_{max}\ G_a\ r_д}{M_{max}\ \eta_м} \tag{4.13}$$

$$u_{тр,дор} = \frac{0.3000 \cdot 17.14003 \cdot 10^3 \cdot 0.296219}{207.8406 \cdot 0.92137} = 7.954$$

- G_a from Chapter 3 is expressed in kN; the factor 10^3 converts it to N before substitution with torque in N·m and radius in m.
- This is the ratio required by road resistance, not an automatically accepted first gear; adhesion also limits it.

Source: [6]

(4.13) **Applicability:** The parallel hybrid's mechanical engine path is designed using the adopted efficiency.

Additional electric traction does not replace this path's requirement; account for it separately during combined operation.

Table Д.4.13.2 — Inputs for (4.13): Ratio required for maximum road resistance

Symbol	Quantity	Unit	Provenance
ψ_{max}	maximum road-resistance coefficient for the intended use	—	adopted by the designer
G_a	total vehicle weight force	kN	calculated earlier (3.5)
$r_д$	dynamic wheel radius	m	calculated earlier
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer

Example — hybrid

$$u_{тр,дор} = \frac{\psi_{max}\ G_a\ r_д}{M_{max}\ \eta_м} \tag{4.13}$$

$$u_{тр,дор} = \frac{0.3000 \cdot 17.14003 \cdot 10^3 \cdot 0.296219}{207.8406 \cdot 0.95000} = 7.714$$

- G_a from Chapter 3 is expressed in kN; the factor 10^3 converts it to N before substitution with torque in N·m and radius in m.
- This is the ratio required by road resistance, not an automatically accepted first gear; adhesion also limits it.

Source: [6]

Table M.4.9 — Maximum road-resistance coefficient from the course method, printed p. 18

Purpose	Range
Passenger cars, trucks and intercity buses	0,25–0,35
Other trucks and buses	0,35–0,45
Enhanced off-road capability	0,45–0,55

Adhesion-limited transmission ratio

Determine the ratio at which source torque produces the limiting adhesion force. Further reduction under the same conditions causes wheelspin rather than a proportional increase in useful traction.

(4.15) **Applicability:** Mechanical transmission with calculated efficiency.

Use the static weight of the driven axles specifically.

Table Д.4.15.1 — Inputs for (4.15): Adhesion-limited transmission ratio

Symbol	Quantity	Unit	Provenance
$F_φ$	adhesion-limited tractive force	kN	calculated earlier (4.14)
$r_д$	dynamic wheel radius	m	calculated earlier
$P_д(ω)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
$η_м$	mechanical transmission efficiency based on its component count	—	calculated earlier (4.3)

Example — conventional internal-combustion vehicle

$$u_{тр,φ} = \frac{F_φ \; r_д}{M_{max} \; η_м} \tag{4.15}$$

$$u_{тр,φ} = \frac{5.65621 \cdot 10^3 \cdot 0.296219}{207.8406 \cdot 0.92137} = 8.749$$

- Take the force $F_φ$ from (4.14); the factor 10^3 converts kN to N. This is the adhesion force limit, not another multiplication by the final drive.
- This limit uses static adhesive weight; acceleration load transfer changes front- and rear-drive limits in opposite directions.

Source: [6]; [18]

(4.15) **Applicability:** The parallel hybrid's mechanical engine path with adopted efficiency.

This branch excludes additional electric-machine torque; limit combined traction separately.

Table Д.4.15.2 — Inputs for (4.15): Adhesion-limited transmission ratio

Symbol	Quantity	Unit	Provenance
F_{φ}	adhesion-limited tractive force	kN	calculated earlier (4.14)
$r_{д}$	dynamic wheel radius	m	calculated earlier
$P_{д}(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
$\eta_{м}$	adopted mechanical driveline efficiency	—	adopted by the designer

Example — hybrid

$$u_{тр,\varphi} = \frac{F_{\varphi}\ r_{д}}{M_{max}\ \eta_{м}} \tag{4.15}$$

$$u_{тр,\varphi} = \frac{5.65621 \cdot 10^{3} \cdot 0.296219}{207.8406 \cdot 0.95000} = 8.486$$

- Take the force F_{φ} from (4.14); the factor 10^{3} converts kN to N. This is the adhesion force limit, not another multiplication by the final drive.
- This limit uses static adhesive weight; acceleration load transfer changes front- and rear-drive limits in opposite directions.

Source: [6]; [18]

Adopted maximum transmission ratio

The historical teaching example compares road demand with the adhesion limit and takes the smaller ratio. Current automatic mode separately checks normal level-road launch from low engine speed and launch on the adopted gradient with available torque. The characteristic gives required gradient-launch speed after tooth counts are adopted. Closed-clutch crawl kinematics are included only when explicitly required. A shorter first gear may need torque limiting to avoid wheelspin. Additional reduction cannot remedy insufficient adhesive weight.

Selection rule. The current application determines first gear from road, launch and low-speed requirements; adhesion limits realised wheel force.

Table П.4.6 — Inputs to the adopted decision: Adopted maximum transmission ratio

Symbol	Quantity	Unit	Provenance
$u_{тр,дор}$	required transmission ratio for road resistance	—	calculated earlier (4.13)
$u_{тр,\varphi}$	adhesion-limited transmission ratio	—	calculated earlier (4.15)

Example — conventional internal-combustion vehicle

$$u_{тр,max} = \min\left(u_{тр,дор}, u_{тр,\varphi}\right)$$

$$u_{тр,max} = \min(7.95, 8.75) = 7.95$$

- The adhesion condition is satisfied: 7,95 ≤ 8,75.
- M_{max} comes from the full-load characteristic: maximum engine torque, not torque at an arbitrary calculation speed.

Source: [6]

Wheel tractive force: engine-top-gear

Convert engine torque to wheel force using adopted reduction, efficiency and radius. Also calculate vehicle speed at each point. This characteristic belongs to one path in top gear, not the complete multi-gear envelope.

Applicability: The engine drives the wheels through a mechanical transmission.

The source's full force goes through its single path; the adhesion limit is not yet imposed.

Table П.4.11 — Inputs to the adopted decision: Wheel tractive force: engine-top-gear

Symbol	Quantity	Unit	Provenance
$P_{д}(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
$r_{д}$	dynamic wheel radius	m	calculated earlier
—	list of axles transmitting traction to the road	—	adopted by the designer
$u_{тр}$	adopted overall transmission ratio in top gear	—	adopted by the designer
$\eta_{м}$	mechanical transmission efficiency based on its component count	—	calculated earlier (4.3)
V_{max}	target maximum vehicle speed	m/s	input data

Example — conventional internal-combustion vehicle

$$F_{т}=\frac{T\ i\ \eta\ u}{r_{д}},\qquad V=\frac{\omega r_{д}}{i}$$

$$F_{т,max}=\frac{207.84063\cdot 6.111111\cdot 0.92137\cdot 1.00000}{0.296219}=3950.696\ \text{N}=3.9507\ \text{kN}$$

$$V_{*}=\frac{385.57519\cdot 0.296219}{6.111111}=18.68966\ \text{m/s}$$

$$V_{*}=3{,}6\cdot 18.68966=67.283\ \text{km/h}$$

- Substitution of path values into (4.8) shows the maximum-force point; the remaining curve points use the same rule.
- Path engine-top-gear: source engine drives axle 2 through ratio 6,11 with efficiency 0,921.
- This path receives the source's entire force.
- Axle dynamic radius: 0,296 m (the document's common tyre).
- Adhesion is not checked here: the curve shows the force the driveline can deliver; allowable tyre–road force is determined separately.

Source: [6]

Applicability: The engine has its own mechanical path in the parallel hybrid.

Combine this engine's flow with the electric machine only in a compatible layout at the same vehicle speed.

Table П.4.12 — Inputs to the adopted decision: Wheel tractive force: engine-top-gear

Symbol	Quantity	Unit	Provenance
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
$r_д$	dynamic wheel radius	m	calculated earlier
—	list of axles transmitting traction to the road	—	adopted by the designer
$u_{тр}$	adopted overall transmission ratio in top gear	—	adopted by the designer
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
V_{max}	target maximum vehicle speed	m/s	input data

Example — hybrid

$$F_т = \frac{T\ i\ \eta\ u}{r_д}, \qquad V = \frac{\omega r_д}{i}$$

$$F_{т,max} = \frac{207.84063 \cdot 4.555556 \cdot 0.95000 \cdot 1.00000}{0.296219} = 3036.567\ \text{N} = 3.0366\ \text{kN}$$

$$V_* = \frac{385.57519 \cdot 0.296219}{4.555556} = 25.07150\ \text{m/s}$$

$$V_* = 3{,}6 \cdot 25.07150 = 90.257\ \text{km/h}$$

- Substitution of path values into (4.8) shows the maximum-force point; the remaining curve points use the same rule.
- Path engine-top-gear: source engine drives axle 2 through ratio 4,56 with efficiency 0,95.
- This path receives the source's entire force.
- Axle dynamic radius: 0,296 m (the document's common tyre).
- Adhesion is not checked here: the curve shows the force the driveline can deliver; allowable tyre–road force is determined separately.

Source: [6]

Wheel tractive force: motor-rear

Calculate rear-wheel electric force from the machine's own characteristic and reduction. A single machine's torque may be split between axles, but cannot be used in full on each axle.

Applicability: The single electric path drives the rear axle.

The rear axle receives the machine's full adopted force.

Table П.4.14 — Inputs to the adopted decision: Wheel tractive force: motor-rear

Symbol	Quantity	Unit	Provenance
$P_{м}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)
$r_{д}$	dynamic wheel radius	m	calculated earlier
—	list of axles transmitting traction to the road	—	adopted by the designer
$i_{g,rear}$	overall reduction ratio from the electric machine to the rear wheels	—	adopted by the designer
$\eta_{м}$	adopted mechanical driveline efficiency	—	adopted by the designer

Example — electric vehicle

$$F_{т}=\frac{T\ i\ \eta\ u}{r_{д}},\qquad V=\frac{\omega r_{д}}{i}$$

$$F_{т,max}=\frac{298.41552\cdot 9.700000\cdot 0.95000\cdot 1.00000}{0.296219}=9283.339\ \text{N}=9.2833\ \text{kN}$$

$$V_{*}=\frac{9.16298\cdot 0.296219}{9.700000}=0.27982\ \text{m/s}$$

$$V_{*}=3{,}6\cdot 0.27982=1.007\ \text{km/h}$$

- Substitution of path values into (4.8) shows the maximum-force point; the remaining curve points use the same rule.
- Path motor-rear: source motor drives axle 2 through ratio 9,7 with efficiency 0,95.
- This path receives the source's entire force.
- Axle dynamic radius: 0,296 m (the document's common tyre).
- Adhesion is not checked here: the curve shows the force the driveline can deliver; allowable tyre–road force is determined separately.

Source: [6]

Applicability: One machine drives both front and rear axles.

The rear path receives the remainder after the adopted front share. Here this is 60 %, not a second full machine output.

Table П.4.15 — Inputs to the adopted decision: Wheel tractive force: motor-rear

Symbol	Quantity	Unit	Provenance
$P_{м}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)
$r_{д}$	dynamic wheel radius	m	calculated earlier
—	list of axles transmitting traction to the road	—	adopted by the designer
$i_{g,rear}$	overall reduction ratio from the electric machine to the rear wheels	—	adopted by the designer
$\eta_{м}$	adopted mechanical driveline efficiency	—	adopted by the designer
u_{front}	adopted front-axle torque fraction for a single traction source	—	input data

Example — electric vehicle with two driven axles

$$F_т = \frac{T\ i\ \eta\ u}{r_д}, \qquad V = \frac{\omega r_д}{i}$$

$$F_{т,max} = \frac{298.41552 \cdot 9.700000 \cdot 0.95000 \cdot 0.60000}{0.296219} = 5570.003\ \text{N} = 5.5700\ \text{kN}$$

$$V_* = \frac{9.16298 \cdot 0.296219}{9.700000} = 0.27982\ \text{m/s}$$

$$V_* = 3{,}6 \cdot 0.27982 = 1.007\ \text{km/h}$$

- Substitution of path values into (4.8) shows the maximum-force point; the remaining curve points use the same rule.
- Path motor-rear: source motor drives axle 2 through ratio 9,7 with efficiency 0,95.
- This path receives 60 % of the source's force.
- Axle dynamic radius: 0,296 m (the document's common tyre).
- Adhesion is not checked here: the curve shows the force the driveline can deliver; allowable tyre–road force is determined separately.

Source: [6]

Wheel tractive force: motor-front

The front path of a single machine driving two axles receives an explicit torque share. The example uses 40 % front and 60 % rear. Two independent motors with different characteristics require two separate sources.

Table П.4.13 — Inputs to the adopted decision: Wheel tractive force: motor-front

Symbol	Quantity	Unit	Provenance
$P_м(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)
$r_д$	dynamic wheel radius	m	calculated earlier
—	list of axles transmitting traction to the road	—	adopted by the designer
$i_{g,front}$	overall reduction ratio from the electric machine to the front wheels	—	adopted by the designer
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
u_{front}	adopted front-axle torque fraction for a single traction source	—	input data

Example — electric vehicle with two driven axles

$$F_т = \frac{T\ i\ \eta\ u}{r_д}, \qquad V = \frac{\omega r_д}{i}$$

$$F_{т,max} = \frac{298.41552 \cdot 9.700000 \cdot 0.95000 \cdot 0.40000}{0.296219} = 3713.336\ \text{N} = 3.7133\ \text{kN}$$

$$V_* = \frac{9.16298 \cdot 0.296219}{9.700000} = 0.27982\ \text{m/s}$$

$$V_* = 3{,}6 \cdot 0.27982 = 1.007\ \text{km/h}$$

- Substitution of path values into (4.8) shows the maximum-force point; the remaining curve points use the same rule.
- Path motor-front: source motor drives axle 1 through ratio 9,7 with efficiency 0,95.

- This path receives 40 % of the source's force.
- Axle dynamic radius: 0,296 m (the document's common tyre).
- Adhesion is not checked here: the curve shows the force the driveline can deliver; allowable tyre–road force is determined separately.

Source: [6]

Total tractive force

Sum forces at the same vehicle speed after referring each source to the wheels. The simultaneous full-output envelope limits source capability; it is not a control law and does not automatically include heating, SOC, battery limits or adhesion.

(4.9) **Applicability:** One mechanical engine path acts on the wheels.

The sum has one term; summation does not create extra traction.

Table Д.4.9.1 — Inputs for (4.9): Total tractive force

Symbol	Quantity	Unit	Provenance
$F_{т}$	engine tractive force in top gear	—	calculated earlier
—	connection layout of traction sources and axles	—	input data
—	joint traction-source operating strategy	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$F_{\text{adopted}}(V)=\Sigma_{s} u_{s}(V)\ F_{s,\max}(V) \tag{4.9}$$

$$V_{*}=3{,}6\cdot 18.68966=67.283\ \text{km/h}$$

$$F_{т,\Sigma}(V_{*})=1.0000\cdot 3950.6960=3950.696\ \text{N}=3.9507\ \text{kN}$$

- The substitution shows the combined curve's maximum. Every term is evaluated at the same vehicle speed V_*; do not add individual source maxima occurring at different speeds.
- One traction source transmits force through the adopted path to the driven wheels. No independent engines are added here.
- Adopted strategy: simultaneous full-output upper envelope.
- Thermal power derating is excluded.
- Shared gearbox limits are excluded.
- Tyre–road adhesion is excluded: this is a source limit rather than a vehicle-motion limit.
- Curve points use linear interpolation, a project method rather than a source method.
- All sources operate together only within 14.0…120.6 km/h; outside this interval some sources are marked inactive.
- The source is evaluated at full load: its capability limit rather than controller operation.

Source: [2]

(4.9) **Applicability:** A single rear-axle traction machine acts on the wheels.

The electric example has no mechanical engine-force term.

Table Д.4.9.2 — Inputs for (4.9): Total tractive force

Symbol	Quantity	Unit	Provenance
$F_т$	rear-wheel electric-machine tractive force	—	calculated earlier
—	connection layout of traction sources and axles	—	input data
—	joint traction-source operating strategy	—	adopted by the designer

Example — electric vehicle

$$F_{adopted}(V)=\Sigma_s u_s(V)\ F_{s,max}(V) \tag{4.9}$$

$$V_*=3{,}6\cdot 0.27982=1.007\ \text{km/h}$$

$$F_{т,\Sigma}(V_*)=1.0000\cdot 9283.3388=9283.339\ \text{N}=9.2833\ \text{kN}$$

- The substitution shows the combined curve's maximum. Every term is evaluated at the same vehicle speed V_*; do not add individual source maxima occurring at different speeds.
- One traction source transmits force through the adopted path to the driven wheels. No independent engines are added here.
- Adopted strategy: simultaneous full-output upper envelope.

- Thermal power derating is excluded.
- Shared gearbox limits are excluded.
- Tyre–road adhesion is excluded: this is a source limit rather than a vehicle-motion limit.
- Battery state of charge is excluded.
- Battery and inverter limits are excluded.
- Curve points use linear interpolation, a project method rather than a source method.
- All sources operate together only within 1.0…161.2 km/h; outside this interval some sources are marked inactive.
- The source is evaluated at full load: its capability limit rather than controller operation.

Source: [2]

(4.9) **Applicability:** Engine and electric-machine torques join in a common driveline.

Check speed and connecting-gear compatibility before adding forces. Simultaneous full output is a separately adopted strategy.

Table Д.4.9.4 — Inputs for (4.9): Total tractive force

Symbol	Quantity	Unit	Provenance
$F_т$	rear-wheel electric-machine tractive force	—	calculated earlier
$F_т$	engine tractive force in top gear	—	calculated earlier
—	connection layout of traction sources and axles	—	input data
—	joint traction-source operating strategy	—	adopted by the designer

Example — hybrid

$$F_{adopted}(V)=\Sigma_s u_s(V)\ F_{s,max}(V) \tag{4.9}$$

$$V_*=3{,}6\cdot 12.79173=46.050\ \text{km/h}$$

$$F_{т,\Sigma}(V_*)=1.0000\cdot 9283.3388+1.0000\cdot 2588.4119=11871.751\ \text{N}=11.8718\ \text{kN}$$

- The substitution shows the combined curve's maximum. Every term is evaluated at the same vehicle speed V_*; do not add individual source maxima occurring at different speeds.
- Connection layout: torque coupling, with the sources joining one driveline. Torques are added at the coupling (Ehsani, Eqs. (6.2)—(6.3)).
- Adopted strategy: simultaneous full-output upper envelope.

- Thermal power derating is excluded.
- Shared gearbox limits are excluded.
- Tyre–road adhesion is excluded: this is a source limit rather than a vehicle-motion limit.
- Battery state of charge is excluded.
- Battery and inverter limits are excluded.
- Curve points use linear interpolation, a project method rather than a source method.
- All sources operate together only within 18.7…161.2 km/h; outside this interval some sources are marked inactive.
- All sources are evaluated at full load simultaneously: a capability limit rather than controller operation.

Source: [2]; [14]

(4.9) **Applicability:** One electric machine splits torque between two axles.

Sum shares of one machine. The two paths recover its full force without doubling power.

Table Д.4.9.3 — Inputs for (4.9): Total tractive force

Symbol	Quantity	Unit	Provenance
$F_{т}$	front-wheel electric-machine tractive force	—	calculated earlier
$F_{т}$	rear-wheel electric-machine tractive force	—	calculated earlier
—	connection layout of traction sources and axles	—	input data
—	joint traction-source operating strategy	—	adopted by the designer

Example — electric vehicle with two driven axles

$$F_{\text{adopted}}(V)=\Sigma_{s} u_{s}(V)\ F_{\text{s,max}}(V) \tag{4.9}$$

$$V_{*}=3{,}6\cdot 0.27982=1.007\ \text{km/h}$$

$$F_{т,\Sigma}(V_{*})=1.0000\cdot 3713.3355+1.0000\cdot 5570.0033=9283.339\ \text{N}=9.2833\ \text{kN}$$

- The substitution shows the combined curve's maximum. Every term is evaluated at the same vehicle speed V_*; do not add individual source maxima occurring at different speeds.
- Connection layout: separate axles, each driven by its own source. Forces add at wheel level because the axles are mechanically independent.
- Adopted strategy: simultaneous full-output upper envelope.

- Thermal power derating is excluded.
- Shared gearbox limits are excluded.
- Tyre–road adhesion is excluded: this is a source limit rather than a vehicle-motion limit.
- Battery state of charge is excluded.
- Battery and inverter limits are excluded.
- Source motor drives 2 axles as one machine: its force is split using the adopted fraction, not doubled. This model does not describe two independent machines with different characteristics; those require separate sources and curves.
- Curve points use linear interpolation, a project method rather than a source method.

- All sources operate together only within 1.0…161.2 km/h; outside this interval some sources are marked inactive.
- All sources are evaluated at full load simultaneously: a capability limit rather than controller operation.

Source: [2]; [14]

(4.9) **Applicability:** The engine drives a generator; only the electric machine provides the traction path.

The project includes the engine characteristic, but no mechanical engine-force term acts on the wheels.

Table Д.4.9.5 — Inputs for (4.9): Total tractive force

Symbol	Quantity	Unit	Provenance
$F_{т}$	rear-wheel electric-machine tractive force	—	calculated earlier
—	connection layout of traction sources and axles	—	input data
—	joint traction-source operating strategy	—	adopted by the designer

Example — series hybrid

$$F_{\text{adopted}}(V)=\Sigma_{s} u_{s}(V)\ F_{s,\max}(V) \tag{4.9}$$

$$V_{*}=3{,}6\cdot 0.27982=1.007\ \text{km/h}$$

$$F_{т,\Sigma}(V_{*})=1.0000\cdot 9283.3388=9283.339\ \text{N}=9.2833\ \text{kN}$$

- The substitution shows the combined curve's maximum. Every term is evaluated at the same vehicle speed V_*; do not add individual source maxima occurring at different speeds.
- One traction source transmits force through the adopted path to the driven wheels. No independent engines are added here.
- Adopted strategy: simultaneous full-output upper envelope.
- Thermal power derating is excluded.
- Shared gearbox limits are excluded.
- Tyre–road adhesion is excluded: this is a source limit rather than a vehicle-motion limit.
- Battery state of charge is excluded.
- Battery and inverter limits are excluded.
- Curve points use linear interpolation, a project method rather than a source method.
- All sources operate together only within 1.0…161.2 km/h; outside this interval some sources are marked inactive.
- The source is evaluated at full load: its capability limit rather than controller operation.

Source: [2]

If the author specifies partial use of each source, construct the sum using those fractions; retain the full envelope as a separate upper limit. Speed coupling through a planetary mechanism cannot be represented by simply adding two independent forces; its kinematics are required. Outside its working characteristic, mark a source inactive or undetermined instead of extending it arbitrarily.

4.6. Gear-ratio series and final verification

The automatic series starts with five working gears (eleven for N3), ends in direct drive and expands if overlap or tooth selection requires it. Add a separate cruising gear when it reduces fuel consumption and maintains at least 15 % power reserve. Its ratio is outside the working gears' geometric progression; maximum speed may occur in the preceding gear. Manual end-ratio selection retains a series between the specified end gears.

Specify reverse R separately: its initial magnitude equals actual first gear; enter a custom value at (I-147). Check kinematic speed, wheel torque and adhesion using the same complete reduction chain. For propeller-shaft, final-drive and half-shaft loading, use the larger of first and reverse ratios. R is excluded from the forward series, acceleration and fuel cycle. Checking this ratio does not replace layout and tooth calculations for the reverse idler gear.

First-gear ratio

Remove the final-drive, active transfer-case range and wheel-reduction factors from the adopted maximum overall ratio. An absent stage has a factor of unity. Specify transfer-case low range separately; it must not replace the active road range.

Table Д.4.17 — Inputs for (4.17): First-gear ratio

Symbol	Quantity	Unit	Provenance
$u_{\text{тр,max}}$	adopted maximum overall transmission ratio	—	calculated earlier
u_0	required final-drive ratio	—	calculated earlier (4.12)
$u_{\text{рк,min}}$	top-range transfer-case ratio	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$u_{\text{к1}} = \frac{u_{\text{тр,max}}}{u_{\text{рк}} u_0} \tag{4.17}$$

$$u_{\text{к1}} = \frac{7.95}{1.00 \cdot 6.11} = 1.302$$

Source: [6]

Number of gearbox ratios

In the teaching example, maximum-torque speed and the characteristic's upper limit determine the speed drop. The current application selects count using the established first and top gears, allowable shift step, minimum loaded speed and explicit class profile. The author may adopt a custom count, checked against the same speed band. Chapter 10 also checks the gear pairs.

Table Д.4.18 — Inputs for (4.18): Number of gearbox ratios

Symbol	Quantity	Unit	Provenance
$u_{\text{к1}}$	first-gear ratio	—	calculated earlier (4.17)
$u_{\text{к,min}}$	adopted top-gear ratio	—	adopted by the designer
$P_{\text{д}}(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)

Example — conventional internal-combustion vehicle

$$q=\frac{\omega_{\text{M}}}{\omega_{\max}}, \qquad n=\frac{\lg u_{\text{к,min}}-\lg u_{\text{к1}}}{\lg q}+1 \tag{4.18}$$

$$q=\frac{385.57519}{691.15038}=0.557875$$

$$n_{\text{розр}}=\frac{\lg(1.000000)-\lg(1.301545)}{\lg(0.557875)}+1=1.4516,\ n=2$$

- The formula gives 1,45; adopt 2 by rounding upwards.
- This estimates speed-band overlap, not a verified production-gearbox selection. Check launch, minimum closed-clutch speed, clutch heating, post-shift traction and continuous maximum-speed duty separately.

Source: [6]

Gearbox ratio series

For the current stepped-gearbox design, place intermediate ratios geometrically between adopted end ratios, retaining both ends for direct or overdrive top gear. The historical example below illustrates the direct-top special case. Electric reduction ratios have a different basis: they are adopted values, not results of an engine-specific formula.

(4.19) **Applicability:** Stepped engine gearbox with direct top gear.

The number of intervals between first and direct top gear is one less than the gear count. The current application uses the same interval count between adopted end ratios, preserving both.

Table Д.4.19 — Inputs for (4.19): Gearbox ratio series

Symbol	Quantity	Unit	Provenance
$u_{\text{к1}}$	first-gear ratio	—	calculated earlier (4.17)
n	adopted number of gearbox ratios	number	calculated earlier (4.18)
$u_{\text{к,min}}$	adopted top-gear ratio	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$u_{\text{кj}}=\sqrt[\text{n-1}]{u_{\text{к1}}^{\ \text{n-j}}} \tag{4.19}$$

$$u_{\text{к1}}=1.301545^{\frac{2-1}{2-1}}=1.301545$$

$$u_{\text{к2}}=1.301545^{\frac{2-2}{2-1}}=1.000000$$

- The series has 2 gears between $u_{\text{к1}}$ = 1,3 and $u_{\text{к}}$,min = 1.
- The series has n−1 geometric intervals between adopted end ratios.
- These numbers are gearbox requirements, not existing gearbox ratios; production ratios are discrete.
- The progression applies to a stepped engine gearbox, not to a single-stage electric-drive reduction gear.

Source: [6]

Applicability: Electric drive with one adopted ratio.

Transfer the adopted reduction to the series without a geometric progression. A second permanent reduction stage multiplies the first; a second selectable gear is an alternative ratio.

Example — electric vehicle

$$u_1 = 9.70$$

- These ratios cover the complete motor-to-wheel path. If wheel reduction is present, divide once by its adopted ratio to obtain the upstream gearbox ratio and shaft load.
- One selectable rear-axle gear: the single ratio was adopted in subsection 4.3.
- No progression is derived: it distributes gears across an engine's narrow operating band, which an electric machine does not have. The author adopts these ratios; the source does not cover this case.

Source: [2]

The report shows gearbox, transfer-case, final-drive and wheel-reduction ratios, overall reduction, minimum-speed kinematics and each gear's operating range. After tooth rounding, check overlap and post-shift speeds. Low-speed kinematics do not model clutch heating.

Power balance at the specified speed

Compare available power from all active paths with road demand at the target speed. Kinematically allowable source speed alone does not prove vehicle-speed attainability. Continuous driving requires continuous characteristics; excess peak power is only a temporary reserve.

Table Д.4.16 — Inputs for (4.16): Power balance at the specified speed

Symbol	Quantity	Unit	Provenance
N_{v}	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)
$F_{\text{т},\Sigma}$	total tractive force from the adopted sources versus speed	—	calculated earlier (4.9)
$V_{\max}$	target maximum vehicle speed	m/s	input data

Example — conventional internal-combustion vehicle

$$N_{\text{зап}} = \frac{\Sigma P_{\text{к}}(V_{\max})}{P_{\text{потр}}(V_{\max})} \tag{4.16}$$

$$N_{\text{зап}} = \frac{100.7}{30.5} = 3.30$$

- This checks power at one point only. Check launch, continuous range coverage, post-shift speeds and positive acceleration to this speed separately.
- The power balance passes at 120 km/h: wheel power is 100,7 kW versus 30,5 kW required.
- Steady motion in still air; headwind and climbing are excluded.
- Auxiliary power is not subtracted.

Source: [18]

Electric-drive gearbox ratio alone: minimum

The declared electric series gives complete motor-to-wheel reduction. The minimum motor-gearbox ratio upstream of wheel reduction equals the series minimum divided by actual wheel-reduction ratio; without wheel reduction the divisor is unity. This scalar determines maximum propeller-shaft speed. Do not add separate final-drive or transfer-case ratios to the declared

motor path. This is a kinematic decomposition of inputs, not adoption of integer motor-gearbox tooth counts.

Applicability: The declared motor series includes complete wheel reduction; its minimum upstream ratio is needed for shaft-speed verification.

This example has no separate wheel reduction, so its ratio is unity.

Table П.4.3 — Inputs to the adopted decision: Electric-drive gearbox ratio alone: minimum

Symbol	Quantity	Unit	Provenance
$u_{\text{кj}}$	adopted gear-ratio series	—	calculated earlier

Example — electric vehicle

$$u_{\text{к,min}} = \frac{\min_{\text{j}} u_{\text{мотор-колесо,j}}}{u_{\text{hub}}}$$

$$u_{\text{к,min}} = \frac{9.7000}{1.0000} = 9.7000$$

- The shaft is considered at the gearbox output, before separate wheel reduction. Minimum ratio determines maximum shaft speed; maximum ratio determines design torque. All declared selectable electric ratios are included.
- This required gearbox ratio comes from declared motor kinematics. Integer-tooth candidates have not yet been adopted into the traction path; their structural calculation remains preliminary.

Source: [7]

Electric-drive gearbox ratio alone: maximum

For maximum propeller-shaft torque, take the greatest declared selectable motor-to-wheel reduction and divide by actual wheel-reduction ratio. Input order does not determine the lowest gear. This scalar belongs to the same layout as the series minimum; wheel reduction must not multiply upstream shaft torque twice. The motor-gearbox integer-tooth candidates in Chapter 10 have not yet been adopted into this path.

Applicability: The declared motor series includes complete wheel reduction; its maximum upstream ratio is needed for shaft-torque verification.

This example has no separate wheel reduction, so its ratio is unity.

Table П.4.2 — Inputs to the adopted decision: Electric-drive gearbox ratio alone: maximum

Symbol	Quantity	Unit	Provenance
$u_{\text{кj}}$	adopted gear-ratio series	—	calculated earlier

Example — electric vehicle

$$u_{\text{к,max}} = \frac{\max_{\text{j}} u_{\text{мотор-колесо,j}}}{u_{\text{hub}}}$$

$$u_{\text{к,max}} = \frac{9.7000}{1.0000} = 9.7000$$

- The shaft is considered at the gearbox output, before separate wheel reduction. Minimum ratio determines maximum shaft speed; maximum ratio determines design torque. All declared selectable electric ratios are included.

- This required gearbox ratio comes from declared motor kinematics. Integer-tooth candidates have not yet been adopted into the traction path; their structural calculation remains preliminary.

Source: [7]

Chapter 5. Traction and speed performance

Per-gear acceleration, shift strategy, cornering stability and electric-machine requirements.

5.1. Dynamic factor, acceleration, time and distance

Vehicle dynamic factor

Subtract aerodynamic force from tractive force and divide by total vehicle weight. Compare this dynamic factor with road resistance: a positive difference is available for acceleration. Plot each gear separately. This series belongs to an engine's mechanical gearbox; do not invent absent electric-drive gears.

Table Д.5.1 — Inputs for (5.1): Vehicle dynamic factor

Symbol	Quantity	Unit	Provenance
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$F_т$	engine tractive force in top gear	—	calculated earlier

Example — conventional internal-combustion vehicle

$$D=\frac{P_т\text{-}P_w}{G_a}, \qquad P_w=WV^2 \tag{5.1}$$

$$V=2.979\ \text{m/s},\ D_1=\frac{3374.13\text{-}0.4312\cdot 2.979^2}{17140.03}=0.19663$$

- The substitution shows one point: gear 1, V = 10.73 km/h. Figure 5.1 shows all complete gear curves; these are not their maxima.
- The dynamic factor is dimensionless and can be compared directly with ψ: the vehicle accelerates where D exceeds road resistance.
- Aerodynamic resistance assumes still air; headwind is excluded.

Source: [6]

Vehicle acceleration

Convert the dynamic-factor margin to acceleration including rotating masses. Update rolling resistance at each speed rather than applying the maximum-speed value throughout acceleration. The empirical multiplier containing squared gear ratio describes the adopted mechanical-transmission model.

Table Д.5.2 — Inputs for (5.2): Vehicle acceleration

Symbol	Quantity	Unit	Provenance
D	dynamic factor versus speed in each gear	—	calculated earlier (5.1)
f_0	rolling-resistance coefficient at low speed	—	input data
s	road gradient	m/m	input data
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)

Example — conventional internal-combustion vehicle

$$j=\frac{D-\psi}{1{,}04+0{,}04u_{к}^{2}}\ g \tag{5.2}$$

$$V=2.979\ \text{m/s},\ j_1=\frac{0.196633-0.015000}{1{,}04+0{,}04\cdot 1.30155^2}\cdot 9{,}81=1.6085\ \text{m/s}^2$$

- The substitution shows one point in gear 1 at 10.73 km/h; Figure 5.2 gives the complete per-gear j(V) curves.
- Road resistance is recalculated at every speed: ψ = 0.015…0.026. The source's 0,02 is its own plot condition, not a universal constant.
- The denominator 1,04 + 0,04u_к² accounts for rotating masses: in lower gears the flywheel rotates faster relative to the wheels, so more than vehicle translation must be accelerated.
- Gear changes are instantaneous; power-interruption time is excluded.

Source: [6]

Acceleration time and distance

Integrate positive-acceleration intervals using mean acceleration between adjacent points while also calculating distance. If acceleration ceases to be positive before the target speed, no complete acceleration time exists. State the actual speed interval in the caption; a curve starting above zero is not a measured standing start.

Table Д.5.3 — Inputs for (5.3): Acceleration time and distance

Symbol	Quantity	Unit	Provenance
j	vehicle acceleration versus speed in each gear	m/s²	calculated earlier (5.2)

Example — conventional internal-combustion vehicle

$$\Delta\tau_i=\frac{2(V_{i+1}-V_i)}{j_{i+1}+j_i},\qquad \Delta S_i=\frac{V_i+V_{i+1}}{2}\Delta\tau_i \tag{5.3}$$

$$\Delta\tau_1=\frac{2(3.8778-2.9794)}{1.71948+1.60849}=0.5399\ \text{s}$$

$$\Delta S_1=\frac{2.9794+3.8778}{2}\cdot 0.5399=1.851\ \text{m}$$

$$\tau=0.5399+1.5250+\ldots+4.2261=16.162\ \text{s}$$

$$S=1.851+8.121+\ldots+129.781=317.25\ \text{m}$$

- The arithmetic of the first of 13 intervals and final totals are shown. Acceleration is integrated from 10.73 to 120.61 km/h; Figure 5.3 shows accumulated time and distance. Zero time is the first synchronous speed, not a stationary vehicle.
- Acceleration reaches 121 km/h.
- Gearshift time is excluded, so actual acceleration takes longer.

Source: [6]

Acceleration characteristic

Accumulated time and distance form the acceleration characteristic. Each point contains speed, time and distance, so the example retains a list. The ideal per-gear envelope excludes shift delays; calculate the strategy-based result separately.

Table П.5.1 — Inputs to the adopted decision: Acceleration characteristic

Symbol	Quantity	Unit	Provenance
j	vehicle acceleration versus speed in each gear	m/s²	calculated earlier (5.2)

Example — conventional internal-combustion vehicle

$$\tau(V)=\Sigma\Delta\tau_i, \qquad S(V)=\Sigma\Delta S_i$$

$$\tau=0.5399+1.5250+\ldots+4.2261=16.162 \text{ s}$$
$$S=1.851+8.121+\ldots+129.781=317.25 \text{ m}$$

- The same equality (5.3) gives final time; here its sums are accumulated to show where acceleration slows.
- Graphical–analytical method: acceleration is replaced by a straight line on each interval, so denser points refine the result.

Source: [6]

BEV standing-start acceleration to target speed

Integrate mechanical BEV acceleration from rest in the adopted gear. Traction cannot exceed either wheel-referred motor torque or the independent adhesion limit. The teaching example assumes 90 kW continuous power and 2 s peak duty, followed by S1. Update road resistance with speed and shorten the 0,05 s step at the peak-time boundary and at 100 km/h. The analytical characteristic defines starting torque at zero speed; do not extrapolate a table beginning above zero to launch. This mechanical integral does not check battery, inverter, SOC change or thermal state and is not a manufacturer test.

Table П.5.2 — Inputs to the adopted decision: BEV standing-start acceleration to target speed

Symbol	Quantity	Unit	Provenance
$P_м(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)
—	basis of the electric-machine rating: peak or continuous duty	—	adopted by the designer
P_{cont}	continuous electric-machine power in S1 duty	kW	input data
$P_м$	adopted electric-machine power	kW	input data
ω_b	base angular speed of the electric machine	s⁻¹	input data
ω_{max}	maximum angular speed of the electric machine	s⁻¹	input data
t_{peak}	allowable electric-machine peak-duty duration	s	input data
$F_т$	rear-wheel electric-machine tractive force	—	calculated earlier
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
F_φ	adhesion-limited tractive force	kN	calculated earlier (4.14)

Symbol	Quantity	Unit	Provenance
f_0	rolling-resistance coefficient at low speed	—	input data
s	road gradient	m/m	input data
δ	rotating-mass factor	—	input data
V_f	target final acceleration speed	m/s	input data

Example — electric standing-start acceleration with peak-to-S1 transition

$$F_т=\min(F_м,F_φ),\ j=\frac{F_т-F_{дор}-WV^2}{M_a\delta},\ V_{k+1}=V_k+j_k\Delta t_k$$

$$V_0=0,\ V_f=100{,}00\ \text{km/h},\ \tau_{0\to V_f}=11{,}793\ \text{s}$$

- Acceleration uses the peak characteristic independently of the working envelope selected for continuous calculations.
- Acceleration starts from zero only if the analytical characteristic specifies constant torque to base speed or the entered table includes zero speed.
- The 0,05 s time step is shortened at peak-duration expiry and target-speed attainment.
- The continuous S1 characteristic is applied after 2 s.
- Mechanical estimate: battery and inverter limits, SOC changes and heating are not modelled.
- There is no torque-delivery or shift delay; this estimate is not a manufacturer test.

Source: [6]; [14]; [2]

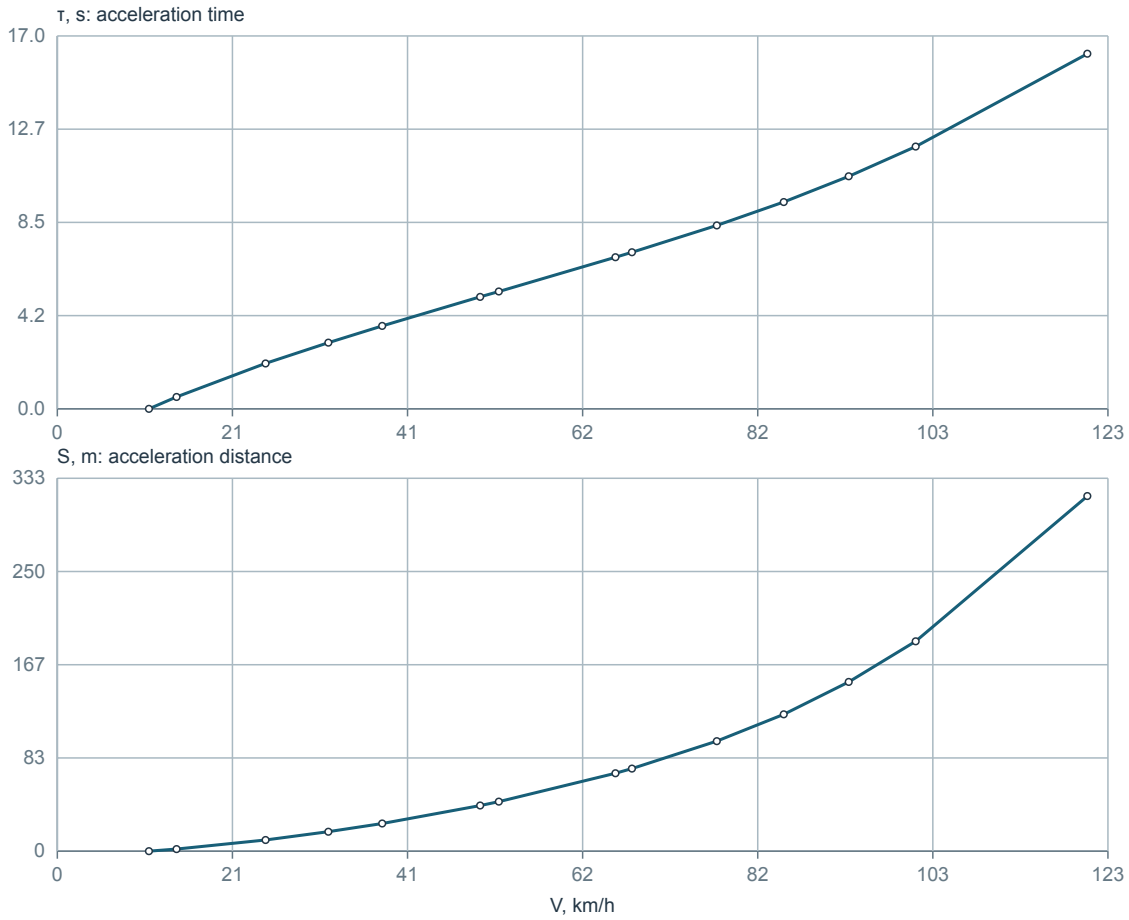


Figure 5.1 — Acceleration time and distance from calculated accelerations; the interval starts at the lowest physically defined speed.
Source: [6].

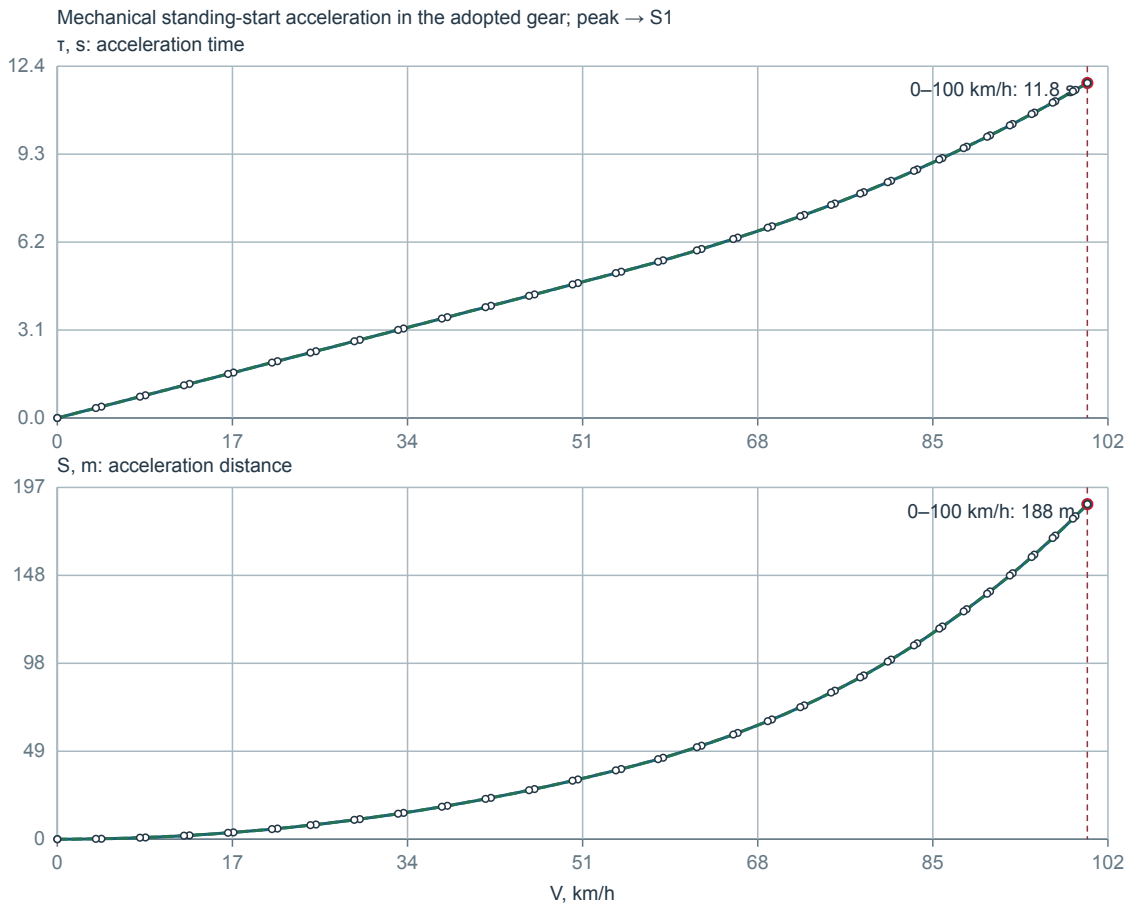


Figure 5.2 — Mechanical estimate of BEV standing-start acceleration time and distance to 100 km/h in the adopted gear. After 2 s of permitted peak duty, continuous S1 applies. Points, time and distance belong to the same teaching calculation. Assess battery and inverter limits separately.
Source: [2].

5.2. Gears over time and strategy-based acceleration

Gearshift map

Convert working engine-speed limits to vehicle speed in every gear. Normal, sport and economy strategies set shifts by a fraction of the working band. These fractions are teaching-project settings. The mode based on UN GTR No. 15, Annex 2 also checks available power; this shortened algorithm does not reproduce every certification requirement.

Table Д.5.4 — Inputs for (5.4): Gearshift map

Symbol	Quantity	Unit	Provenance
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)
k	adopted gearshift strategy	—	adopted by the designer
$r_д$	dynamic wheel radius	m	calculated earlier
$\omega_х$	engine idle angular speed	rad/s	input data
u_0	required final-drive ratio	—	calculated earlier (4.12)
$u_{рк,min}$	top-range transfer-case ratio	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$\omega_{\text{страт,i}}=\omega_{\min,\text{i}}+k(\omega_{95}-\omega_{\min,\text{i}}),\qquad V_{\text{перем,i}}=\max\left(\frac{\omega_{\text{страт,i}}r_{\text{д}}}{u_{\text{i}}u_{\text{рк}}u_0u_{\text{hub}}},\frac{\omega_{\min,\text{i+1}}r_{\text{д}}}{u_{\text{i+1}}u_{\text{рк}}u_0u_{\text{hub}}}\right) \tag{5.4}$$

$$\omega_{\text{страт,1}}=80.0000+0.650(691.1504-80.0000)=477.2477\ \text{rad/s}$$

$$V_{\text{страт,1}}=\frac{477.2477\cdot0.29622}{1.30155\cdot1.00000\cdot6.11111\cdot1.00000}=17.7737\ \text{m/s}$$

$$V_{\min,2}=\frac{72.0000\cdot0.29622}{1.00000\cdot1.00000\cdot6.11111\cdot1.00000}=3.4900\ \text{m/s}$$

$$V_{\text{перем,1}}=\max(17.7737,3.4900)=17.7737\ \text{m/s}\ (\times3{,}6=63.99\ \text{km/h})$$

- The substitution shows the first shift. All shift limits: 1 → 2: 63.99 km/h. Shift speed is the greater of strategy speed and the next gear's minimum usable speed.
- Normal strategy: shift after roughly two-thirds of the working band, representing unhurried urban driving.
- Vehicle speed uniquely sets engine speed in each gear: ω = V·u_{i}·$u_{\text{рк}}$·u₀·u_hub/r_д (without wheel reduction, u_{hub}=1). Shift timing is therefore a STRATEGY choice rather than a vehicle property: the same gearbox and vehicle can shift early or late.
- GTR 15, Annex 2, the prescribed WLTP gear-selection algorithm, gives working-speed limits. Lower limit: n_min_drive = n_{x} + 0,125·(n_{N} − n_{x}) above second gear (688 min⁻¹), n_{x} in first gear, 0,9·n_{x} in second gear (§2(k)). Upper limit: n_95_high is the highest speed where power remains at least 95 % of rated power (6600 min⁻¹, §2(g)).
- This strategy uses 65 % of the band before shifting. A fraction of the BAND is used instead of rated speed because rated-speed fraction alone ignores engine characteristics: 70 % of ω_{N} on an engine idling at 800 min⁻¹ and on one idling at 1400 min⁻¹ represent different driving styles; the same fraction of the working band represents the same style.
- Standard cycle files have no gear column, as checked for WLTP, NEDC, Artemis and EPA, so construct the map separately. Regulation R83 (NEDC) does not calculate gears: it provides a fixed time-based table (Annex 4a, §6.1.3), which cannot simply be transferred to a different vehicle.

Source: [19]; [7]

Table M.5.1 — Adopted project strategies; speed limits and power reserve follow UN GTR No. 15, Annex 2

Strategy	Working-band fraction before shifting	Condition
Economy	0,35	Early shifts with a traction-adequacy check
Normal	0,65	Intermediate mode
Sport	1,00	Top of the working band
Prescribed suitable-gear selection	Not a fixed fraction	Highest usable gear according to speed and power

The adopted prescription's lower limit above second gear is: $n_{\min}=n_{\text{x}}+0{,}125(n_{\text{N}}-n_{\text{x}})$. The upper limit relates to the highest speed retaining 95 % of rated power. The power check uses a 10 % reserve. These conditions have a different purpose from the project fractions 0,35; 0,65; 1,00. Cycle vehicle speed alone cannot determine gear without a specific engine characteristic.

Acceleration using the shift map

Integrate time using the actually selected gears and add shift durations. State initial and final speeds, strategy and individual shift time. Label a result 0–100 only when it actually starts at

zero and reaches 100 km/h; a shorter interval requires its own label.

Table Д.5.5 — Inputs for (5.5): Acceleration using the shift map

Symbol	Quantity	Unit	Provenance
j	vehicle acceleration versus speed in each gear	m/s²	calculated earlier (5.2)
k	adopted gearshift strategy	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$\tau_{\text{p}}=\int_0^{\text{V}} \frac{dV}{j(V,\ u_{\text{i}}(V))} + n_{\text{перем}} t_{\text{перем}} \tag{5.5}$$

$$\Delta\tau_1=\frac{0.10000}{1.60849}=0.06217\ \text{s}$$

$$\tau_{\text{p}}=13.72022+1\cdot 0.500=14.220\ \text{s}$$

- First interval: gear 1, from 0.000 to 0.100 m/s. Approximate the integral by summing ΔV/j over 278 intervals; 13.72022 s is motion time, with shift time added separately.
- Sport strategy: 1 shift to 100 km/h, with an assumed shift duration of 0.5 s.
- The document selects normal strategy; this acceleration is calculated separately using sport strategy.
- From 0 to the first synchronous point, an explicit approximation uses constant initial first-gear acceleration capability with clutch slip. This integral does not check clutch heating.
- Do not extend acceleration beyond other gears' operating-speed ranges. This is a preliminary calculation, not a manufacturer 0–100 test.

Source: [7]

5.3. Cornering stability

Critical cornering speed

At the adopted radius, compare lateral sliding and rollover limits; the lower governs in this model. Inputs are mean track, centre-of-mass height and adhesion coefficient. The approximation excludes suspension roll, road crossfall and dynamic load transfer. It applies to engine, electric and hybrid vehicles alike, but does not verify every handling condition.

Table Д.5.6 — Inputs for (5.6): Critical cornering speed

Symbol	Quantity	Unit	Provenance
R	cornering radius used for the stability check	m	input data
$K_{\text{ср}}$	mean front and rear track width	m	input data
h_{g}	adopted centre-of-mass height above the road	m	input data
φ	tyre–road adhesion coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$V_{\text{пер}}=\sqrt{\frac{gRK_{\text{ср}}}{2h_{\text{g}}}},\qquad V_{\text{ковз}}=\sqrt{gR\varphi},\qquad V_{\text{кр}}=\min\left(V_{\text{пер}},V_{\text{ковз}}\right) \tag{5.6}$$

$$V_{\text{пер}}=\sqrt{\frac{9{,}81\cdot 50.00\cdot 1.500}{2\cdot 0.550}}=25.862\ \text{m/s}$$

$$V_{\text{ковз}}=\sqrt{9{,}81\cdot 50.00\cdot 0.600}=17.155\ \text{m/s}$$

$$V_{\text{кр}}=\min\left(25.862,17.155\right)=17.155\ \text{m/s}\ \left(3{,}6\cdot 17.155=61.76\ \text{km/h}\right)$$

- The sliding limit is lower in this model: 61.8 km/h versus 93.1 km/h for rollover.
- These are quasi-static limits, not a safe driving speed. Sliding consequences, transients, suspension roll and interaction with obstacles are not assessed.
- The vehicle is a rigid body; suspension roll lowers the rollover speed.
- The adhesion coefficient is constant; this model excludes slip dependence.

Source: [6]

5.4. Power for specified electric-vehicle acceleration

Machine power selection from acceleration and cruising

For a single traction motor, compare continuous cruising and acceleration requirements. Ehsani's equation (5.12) uses target final speed, acceleration time, machine base speed and a constant rotating-mass factor. Fractions 2/3 and 1/5 arise from averaging resistances over the adopted profile $V=V_{\text{f}}\sqrt{t/t_{\text{a}}}$ rather than arbitrary margins. Rolling resistance is constant here, as in the derivation. Compare wheel power with wheel power; first reduce manufacturer machine power by its path losses.

(5.7) **Applicability:** The BEV's single traction machine.

Adopt the larger acceleration or cruising requirement. Check that base speed does not exceed final speed and that peak duty lasts long enough.

Table Д.5.7.1 — Inputs for (5.7): Machine power selection from acceleration and cruising

Symbol	Quantity	Unit	Provenance
N_{v}	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)
M_{a}	gross vehicle mass	kg	calculated earlier (3.4)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
f_0	rolling-resistance coefficient at low speed	—	input data
V_{f}	target final acceleration speed	m/s	input data
t_{a}	target acceleration time	s	input data
δ	rotating-mass factor	—	input data
$F_{\text{т},\Sigma}$	total tractive force from the adopted sources versus speed	—	calculated earlier (4.9)

Example — electric vehicle

$$P_{\text{t}}=\frac{\delta M}{2t_{\text{a}}}\left(V_{\text{f}}^{2}+V_{\text{b}}^{2}\right)+\frac{2}{3}Mgf_{\text{r}}V_{\text{f}}+\frac{1}{5}\rho C_{\text{D}}A_{\text{f}}V_{\text{f}}^{3} \tag{5.7}$$

$$P_{\text{t}}=70.8+4.8+3.7=79.3;\text{kW}$$

- ACCELERATION sizes the machine: 79,3 kW versus 61,9 kW for cruising.
- Adopted power is sufficient: 121,2 kW at the wheels, margin 1,53.
- Acceleration 0…100 km/h in 12 s; base speed 46 km/h.

- Equation (5.12) applies to one electric machine as the sole traction source: a BEV or series hybrid/REEV. Engine vehicles and multi-source parallel hybrids need another acceleration model.
- This is wheel power: account for driveline losses separately, using each path's efficiency.
- The rotating-mass factor is constant as in the source; in reality it depends on gear.

Source: [14]

(5.7) **Applicability:** The series hybrid's single traction machine; the engine is mechanically disconnected from the wheels.

The electric machine delivers acceleration power. Checking its battery and generator energy supply is a separate subsequent step.

Table Д.5.7.2 — Inputs for (5.7): Machine power selection from acceleration and cruising

Symbol	Quantity	Unit	Provenance
N_{v}	required traction-source power for the target maximum speed	kW	calculated earlier (4.5)
M_{a}	gross vehicle mass	kg	calculated earlier (3.4)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
f_0	rolling-resistance coefficient at low speed	—	input data
V_{f}	target final acceleration speed	m/s	input data
t_{a}	target acceleration time	s	input data
δ	rotating-mass factor	—	input data
$F_{\text{т},\Sigma}$	total tractive force from the adopted sources versus speed	—	calculated earlier (4.9)

Example — series hybrid

$$P_{\text{t}}=\frac{\delta M}{2t_{\text{a}}}\left(V_{\text{f}}^{2}+V_{\text{b}}^{2}\right)+\frac{2}{3}Mgf_{\text{r}}V_{\text{f}}+\frac{1}{5}\rho C_{\text{D}}A_{\text{f}}V_{\text{f}}^{3} \tag{5.7}$$

$$P_{\text{t}}=70.8+4.8+3.7=79.3;\text{kW}$$

- ACCELERATION sizes the machine: 79,3 kW versus 61,9 kW for cruising.
- Adopted power is sufficient: 121,2 kW at the wheels, margin 1,53.
- Acceleration 0…100 km/h in 12 s; base speed 46 km/h.

- Equation (5.12) applies to one electric machine as the sole traction source: a BEV or series hybrid/REEV. Engine vehicles and multi-source parallel hybrids need another acceleration model.
- This is wheel power: account for driveline losses separately, using each path's efficiency.
- The rotating-mass factor is constant as in the source; in reality it depends on gear.

Source: [14]

For a parallel hybrid, this single-machine formula does not determine acceleration with both sources operating. Explicit power-sharing strategy and time-dependent source characteristics are required. Two independent electric machines likewise need separately specified limits.

Chapter 6. Braking performance

Deceleration, braking distance and total stopping distance for the adopted road condition.

Assess braking from adhesion and brake actuation independently of the traction source. Regeneration does not remove an electric or hybrid vehicle's need for friction brakes: the electrical path may be limited by SOC, temperature and speed. This chapter assigns no guaranteed emergency-deceleration contribution to regeneration.

Vehicle deceleration

Positive gradient denotes uphill, which assists stopping; negative gradient denotes downhill, which opposes it. The sine already includes the gradient sign, so do not introduce an arbitrary second sign. Update rolling resistance for initial braking speed. If deceleration is nonpositive, this model gives no finite stopping distance.

Table Д.6.1 — Inputs for (6.1): Vehicle deceleration

Symbol	Quantity	Unit	Provenance
φ	tyre–road adhesion coefficient	—	adopted by the designer
f_0	rolling-resistance coefficient at low speed	—	input data
s	road gradient	m/m	input data
$V_г$	initial braking speed	m/s	input data

Example — conventional internal-combustion vehicle

$$j_{сп} = (\varphi \cos\alpha + f(V) \pm \sin\alpha) g \tag{6.1}$$

$$j_{сп} = (0.60 \cdot \cos(0.000000) + 0.019500 + \sin(0.000000)) \cdot 9{,}81 = 6.08 \ \text{m/s}^2$$

- Gradient angle α = 0.000000 rad retains its sign: a negative angle denotes downhill.
- Rolling resistance is recalculated at the initial braking speed of 80.0 km/h: f = 0.0195.
- The project adopts adhesion coefficient φ = 0.60.
- The adhesion coefficient remains constant throughout braking.

Source: [6]; [1]

Braking and stopping distances

Braking distance starts after braking force develops; total stopping distance also includes driver reaction and actuator delay. Half of brake-torque build-up time is included. Convert initial speed to m/s before substitution; apply the efficiency factor to braking distance rather than the whole reaction interval.

Table Д.6.2 — Inputs for (6.2): Braking and stopping distances

Symbol	Quantity	Unit	Provenance
$j_{сп}$	design deceleration	m/s²	calculated earlier (6.1)
$V_г$	initial braking speed	m/s	input data
—	brake-actuation type	—	input data
K_e	braking-system efficiency class	—	input data

Example — conventional internal-combustion vehicle

$$S_{г}=\frac{V^{2}}{2j_{сп}}, \qquad S_{зуп}=(t_{р}+t_{пр}+0{,}5t_{зр})V+K_{е}S_{г} \tag{6.2}$$

$$S_{г}=\frac{22.22222^{2}}{2\cdot 6.07729}=40.629\ \text{m}$$

$$S_{зуп}=(0.80+0.20+0{,}5\cdot 0.50)\cdot 22.22222+1.200\cdot 40.629=76.53\ \text{m}$$

- During the 1.25 s before braking, the vehicle travels 27.8 m due to driver reaction and actuator response, not braking itself.
- GOST 25478-82 limits are absent: the guide names the standard but prints no numerical limits. Enter a standard limit to perform a comparison.

- The document does not include GOST 25478-82 limits: the guide names the standard but gives no numbers, so the reader must supply the limit.
- Braking load transfer is excluded: front wheels become more loaded than their static share and rear wheels less loaded.

Source: [6]

Table M.6.1 — Course-method inputs, subsection 3.2, printed p. 24; scan «Курсовой проект+», sheet 13

Quantity	Condition	Value
Driver reaction time	Adopted by the method	0,8 s
Actuator response time	Hydraulic	0,2 s
Actuator response time	Pneumatic	0,6 s
Torque build-up time	Adopted by the method	0,5 s
Efficiency factor	Passenger car	1,2
Efficiency factor	Truck	1,4
Initial test speed	Passenger car	22,2 m/s, approximately 80 km/h
Initial test speed	Truck or bus	16,7 m/s, approximately 60 km/h

A compliance conclusion requires specific deceleration and stopping-distance limits and the standard's conditions. The course method cites GOST 25478-82, but the checked transcription omits its limits table. The application starts with editable teaching-design limits: 5,8 m/s² and 80 m for M1 at 80 km/h, or 65 m for other categories at 60 km/h. These are comparison assumptions, not standard values; refine stopping distance for other speeds. A blank limit means that check was not performed. These design calculations do not establish compliance with current type-approval requirements.

Chapter 7. Fuel and energy economy in steady driving

Consumption versus speed on a level road and its check points.

Determine motion resistance and required power at every speed. A level road has zero gradient, but rolling resistance is not constant: calculate speed correction at each point. An engine's top-gear fuel curve and a BEV's energy curve use different physical models and units.

7.1. Internal combustion engine

Steady-driving fuel-consumption characteristic

Use the full-load characteristic to find available power and utilisation fraction. Rated-point specific consumption cannot describe partial load without corrections. Include only speeds where the engine operates within its known range and can sustain the requested motion.

(7.1) **Applicability:** Steady engine-only driving; transmission efficiency is calculated.

This is a top-gear characteristic without cycle accelerations or shifts.

Table Д.7.1.1 — Inputs for (7.1): Steady-driving fuel-consumption characteristic

Symbol	Quantity	Unit	Provenance
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
f_0	rolling-resistance coefficient at low speed	—	input data
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_м$	mechanical transmission efficiency based on its component count	—	calculated earlier (4.3)
$u_{тр}$	adopted overall transmission ratio in top gear	—	adopted by the designer
$r_д$	dynamic wheel radius	m	calculated earlier
—	basis for adopted specific fuel consumption	—	input data
g_N	specific fuel consumption at rated power	g/kWh	input data
$\rho_п$	adopted fuel density	g/cm³	input data

Example — conventional internal-combustion vehicle

$$Q_s = \frac{g_N K_\omega {}^Q K_B}{3{,}6 \cdot 10^4 \eta_{тр} \rho_п} (P_\psi + P_w) \tag{7.1}$$

$$V = 5.000 \text{ m/s}: \; Q_s = \frac{270.00 \cdot 1.10275 \cdot 2.58112}{3{,}6 \cdot 10^4 \cdot 0.92137 \cdot 0.750} (257.100 + 10.780) = 8.275 \text{ L/100 km}$$

- The numerical example is the first usable point at V = 18.00 km/h. The figure shows the complete characteristic; determine its minimum separately among all calculated points.

- The curve covers only level-road speeds where the engine can sustain steady motion; f is updated at each speed.

Source: [6]; [1]

(7.1) **Applicability:** The parallel hybrid's engine path is assessed using adopted efficiency.

This curve is not total hybrid consumption with arbitrary electric-machine participation. Chapter 8's energy balance is needed for that comparison.

Table Д.7.1.2 — Inputs for (7.1): Steady-driving fuel-consumption characteristic

Symbol	Quantity	Unit	Provenance
$P_д(\omega)$	full-load internal-combustion engine characteristic	—	calculated earlier (4.7)
f_0	rolling-resistance coefficient at low speed	—	input data
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
$u_{тр}$	adopted overall transmission ratio in top gear	—	adopted by the designer
$r_д$	dynamic wheel radius	m	calculated earlier
—	basis for adopted specific fuel consumption	—	input data
g_N	specific fuel consumption at rated power	g/kWh	input data
$\rho_п$	adopted fuel density	g/cm³	input data

Example — hybrid

$$Q_s = \frac{g_N K_\omega {}^Q K_B}{3{,}6 \cdot 10^4 \eta_{тр} \rho_п} (P_\psi + P_w) \qquad (7.1)$$

$$V = 7.000 \text{ m/s}: \; Q_s = \frac{270.00 \cdot 1.09763 \cdot 2.38462}{3{,}6 \cdot 10^4 \cdot 0.95000 \cdot 0.750} (257.100 + 21.129) = 7.666 \text{ L/100 km}$$

- The numerical example is the first usable point at V = 25.20 km/h. The figure shows the complete characteristic; determine its minimum separately among all calculated points.
- The curve covers only level-road speeds where the engine can sustain steady motion; f is updated at each speed.

Source: [6]; [1]

Defining $z = \omega/\omega_N$ and $B = N_{потр}/N_i$, apply corrections from the Kyiv course method, subsection 2.3.7, printed pp. 50–53, equations (2.81)–(2.83):

$K_\omega = 1{,}25 - 0{,}99z + 0{,}98z^2 - 0{,}24z^3$.

For a petrol engine: $K_B = 3{,}27 - 8{,}22B + 9{,}13B^2 - 3{,}18B^3$.

For a diesel engine: $K_B = 1{,}20 + 0{,}14B - 1{,}80B^2 + 1{,}46B^3$.

These empirical corrections are separate from Chapter 4's loss map. Retain their historical applicability when transferring them to another engine; arbitrary extrapolation beyond

working loads is not permitted.

Table M.7.1 — Rated specific fuel consumption from the Kyiv course method, p. 50

Engine type	Specific consumption, g/(kWh)	Basis
Carburettor	330–360	Source range
Diesel	220–240	Source range
Modern fuel-injected petrol	Engine data required	The verified source gives no universal range

Table M.7.2 — Fuel properties adopted in the calculation model

Fuel	Density, g/cm³	Lower heating value, MJ/kg
Petrol	0,75	44,0
Diesel fuel	0,83	42,5

Densities appear in the Kyiv course method, p. 51. Heating values are separate tabulated inputs to the project model, not attributed to that guide. Other fuel compositions require corresponding properties. In some gas-fuel modes the application explicitly approximates using petrol properties; the litre result is then not the gas fuel's declared consumption.

Reference fuel consumption

Reference consumption is the lowest value among physically allowable points of the constructed curve. Report speed and evaluated range with it. A discrete-grid minimum is an estimate at the adopted resolution, not proof of a global optimum across all driving modes.

Table П.7.2 — Inputs to the adopted decision: Reference fuel consumption

Symbol	Quantity	Unit	Provenance
$Q_s(V)$	fuel consumption versus steady-driving speed	L/100 km	calculated earlier (7.1)

Example — conventional internal-combustion vehicle

$$Q_{s\min} = \min_V Q_s(V)$$

$$Q_{s\min} = \min\{8.275,\ 8.365,\ \ldots,\ 15.918,\ 16.034\} = 8.275 \text{ L/100 km},\ V = 18.0 \text{ km/h}$$

- The minimum is selected among all 15 usable characteristic points. The substitution shows end values and the minimum point; ellipses indicate omitted intermediate values.
- Minimum consumption occurs at the lowest calculated speed. A sample-boundary point does not locate the true minimum; lower speeds in this gear have not been checked.

Source: [7]

7.2. Electric vehicle

Steady-driving energy-consumption characteristic

On a level road, determine electric power from required wheel power and adopted efficiency, adding auxiliaries. Divide by speed to obtain energy per distance. Consumption per 100 km is not defined at zero speed even though auxiliaries may operate. This steady model uses constant efficiency; cycle integration considers the machine map separately.

Table П.7.3 — Inputs to the adopted decision: Steady-driving energy-consumption characteristic

Symbol	Quantity	Unit	Provenance
f_0	rolling-resistance coefficient at low speed	—	input data
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
$P_{доп}$	electrical auxiliary power	kW	input data

Example — electric vehicle

$$e_s = \frac{100}{3{,}6V}\left(\frac{P_к}{\eta_{тр}} + P_{доп}\right)$$

$$V = 3.000 \text{ m/s}: \; e_s = \frac{100}{3{,}6 \cdot 3.000}\left(\frac{0.78294}{0.95000} + 0.000\right) = 7.631 \text{ kWh/100 km}$$

- The numerical example is the first usable point at V = 10.80 km/h. The figure shows the complete characteristic; determine its minimum separately among all calculated points.
- This is driveline-input energy on a level road at constant efficiency; f is updated at each speed and the machine loss map is not applied.

Source: [1]; [2]

Minimum energy consumption

The electric characteristic's minimum also depends on auxiliary power: slow driving distributes it over less distance. State speed, efficiency and included auxiliaries with the result. Driveline-input energy does not automatically equal grid energy because charging losses are unspecified.

Table П.7.1 — Inputs to the adopted decision: Minimum energy consumption

Symbol	Quantity	Unit	Provenance
$e_s(V)$	energy consumption versus steady-driving speed	kWh/100 km	calculated earlier

Example — electric vehicle

$$e_{smin} = \min_V e_s(V)$$

$$e_{smin} = \min\{7.631,\ 7.833,\ \ldots,\ 44.330,\ 47.292\} = 7.631 \text{ kWh/100 km},\ V = 10.8 \text{ km/h}$$

- The minimum is selected among all 24 usable characteristic points. The substitution shows end values and the minimum point; ellipses indicate omitted intermediate values.
- Minimum consumption occurs at the lowest calculated speed. A sample-boundary point does not locate the true minimum; lower speeds in this gear have not been checked.

Source: [7]

Chapter 8. Energy and consumption over a driving cycle

Profile import, work integration, fuel-coverage limits and hybrid balance.

The cycle specifies speed over time. The calculation finds required power, checks operating-point availability and sums energy. The plotted trajectory is a demand, not a measured or automatically achieved vehicle speed.

8.1. Input file and motion resistance

The file contains time in seconds and speed explicitly labelled in km/h or m/s. An optional third gradient column must identify percent, ratio, degrees or radians. Time strictly increases and speed is nonnegative; actual steps may be uneven. Do not guess the meaning of an unlabelled gradient column.

Design resistance uses adopted mass, rolling coefficient and aerodynamic factor from earlier chapters. Coastdown independently specifies $F(v)=F_0+F_1v+F_2v^2$, where F_0 has unit N, F_1 N·s/m, F_2 and N·s²/m² respectively, with speed in m/s. Coefficients must match the particular test and its mass; zero is also an explicitly declared result or assumption.

Wheel-power components are: $P_к=[F(v)+\delta ma+mg\sin\alpha]v$. Acceleration comes from adjacent speed points and actual elapsed time. Positive power denotes traction and negative power braking.

Table M.8.1 — Locally stored cycles: file characteristics rather than complete test procedures

File	Duration	Application
NEDC.csv	1163 s	Specified trajectory of the older driving cycle
WLTC_Class_3_v53.csv	1800 s	Class 3 trajectory: approximately 23,26 km, maximum speed 131,3 km/h

Check durations and check values against Data/Vehicle/cycles files. Selecting a WLTC trajectory alone does not reproduce WLTP: test mass, coastdown resistance, initial state, conditioning and the applicable gear-selection procedure are needed. Align these conditions before comparison with a declared value.

Energy over the driving cycle

Include forward-path losses during traction, subtract recovered braking energy, and add auxiliary demand throughout its operation. When braking power is stored with a negative sign, add its integral as a negative contribution. Divide by actual distance and normalise to 100 km.

(8.5) **Applicability:** Conventional vehicle without a traction machine, using design motion resistance.

This vehicle's physical fuel balance has no traction regeneration. Electrical auxiliaries require a separate alternator model; zero auxiliary power explicitly excludes that load.

Table Д.8.5.1 — Inputs for (8.5): Driving-cycle energy

Symbol	Quantity	Unit	Provenance
—	cycle points: time, speed and gradient with declared units	—	input data
—	cycle road-load basis: design inputs or coastdown	—	input data
M_a	gross vehicle mass	kg	calculated earlier (3.4)
$\eta_м$	mechanical transmission efficiency based on its component count	—	calculated earlier (4.3)
δ	rotating-mass factor	—	input data
$P_{доп}$	electrical auxiliary power	kW	input data
f_0	rolling-resistance coefficient at low speed	—	input data
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)

Example — conventional internal-combustion vehicle

$$E_ц = \frac{1}{3{,}6\cdot 10^6}\left[\int_{P_к>0} \frac{P_к}{\eta}\,dt + \int P_{доп}\,dt\right], \qquad e_ц = \frac{100E_ц}{S_ц} \tag{8.5}$$

$$\Delta E_{тр} = \frac{9.946\cdot 1.0000}{0.921373\cdot 3{,}6\cdot 10^6} = 0.000003 \text{ kWh}$$

$$E_ц = 4.891864 - 0.000000 + 0.000000 = 4.891864 \text{ kWh}$$

$$e_ц = \frac{100\cdot 4.891864}{23.262389} = 21.029 \text{ kWh/100 km}$$

- The formula uses power in W and time in s; division by 3,6·10⁶ converts J to kWh. Sц is cycle distance in km. The numerical sum lists traction, recovered and auxiliary energy over the entire cycle.
- The integration example shows positive traction over 11.00…12.00 s at mean speed 0.028 m/s; the total includes all 1800 intervals.
- Cycle WLTC_class_3 is integrated using its actual time steps.
- For an engine vehicle, this is mechanical energy at transmission input, not fuel consumption; no specific-fuel-consumption map or shift algorithm is applied.
- This is not certified WLTP: test mass, conditioning, charge balance and prescribed shifts are not reproduced.

Source: [20]; [1]; [2]

(8.5) **Applicability:** Electric vehicle with machine-efficiency map and design resistance.

Multiply mechanical efficiency by machine efficiency at the operating point. Regenerative efficiency already covers the whole reverse path and must not be multiplied by the same map again.

Table Д.8.5.2 — Inputs for (8.5): Driving-cycle energy

Symbol	Quantity	Unit	Provenance
—	cycle points: time, speed and gradient with declared units	—	input data
—	cycle road-load basis: design inputs or coastdown	—	input data
M_a	gross vehicle mass	kg	calculated earlier (3.4)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
δ	rotating-mass factor	—	input data
$P_{доп}$	electrical auxiliary power	kW	input data
$\eta_{рек}$	reverse-path regenerative efficiency	—	input data
f_0	rolling-resistance coefficient at low speed	—	input data
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_{ем}$	electric-machine efficiency versus torque and speed	—	calculated earlier (4.21)

Example — electric vehicle

$$E_ц = \frac{1}{3{,}6\cdot 10^6}\left[\int_{P_к>0}\frac{P_к}{\eta(M,\omega)}\,dt + \int_{P_к<0} P_к \eta_{рек}\,dt + \int P_{доп}\,dt\right], \qquad e_ц = \frac{100E_ц}{S_ц}, \qquad \eta = \frac{\eta_м}{s/\eta_{ем}(M,\omega)+(1-s)} \quad (8.5)$$

$$\Delta E_{тр} = \frac{9.946\cdot 1.0000}{0.435537\cdot 3{,}6\cdot 10^6} = 0.000006 \text{ kWh}$$

$$E_ц = 5.476888 - 0.563061 + 0.000000 = 4.913827 \text{ kWh}$$

$$e_ц = \frac{100\cdot 4.913827}{23.262389} = 21.123 \text{ kWh/100 km}$$

- The formula uses power in W and time in s; division by 3,6·10⁶ converts J to kWh. Sц is cycle distance in km. The numerical sum lists traction, recovered and auxiliary energy over the entire cycle.
- The integration example shows positive traction over 11.00…12.00 s at mean speed 0.028 m/s; the total includes all 1800 intervals.
- Cycle WLTC_class_3 is integrated using its actual time steps.
- Traction efficiency is NOT constant: mechanical efficiency is fixed, while machine efficiency comes from Chapter 4's map at each torque and speed. Transmission efficiency here is strictly mechanical, covering gears, joints and splines; multiplying by the machine map does not count losses twice.
- Useful-work-weighted mean machine efficiency over the cycle: 87.3 %. Replacing the map with one constant overlooks operation at low torque, where efficiency is poorest.
- Regeneration does NOT pass through the machine map: recovered energy uses a constant $\eta_{рек}$ because this document defines it as whole reverse-path efficiency. Multiplication by $\eta_{ем}$ would count the machine twice.
- This is not certified WLTP: test mass, conditioning, charge balance and prescribed shifts are not reproduced.

Source: [20]; [1]; [2]

(8.5) **Applicability:** The parallel hybrid's electric machine supplies a fraction of traction energy.

The adopted electric traction share determines how much forward flow passes through the machine map. Do not apply electric-machine losses to engine flow that bypasses that machine.

Table Д.8.5.5 — Inputs for (8.5): Driving-cycle energy

Symbol	Quantity	Unit	Provenance
—	cycle points: time, speed and gradient with declared units	—	input data
—	cycle road-load basis: design inputs or coastdown	—	input data
M_a	gross vehicle mass	kg	calculated earlier (3.4)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
δ	rotating-mass factor	—	input data
$P_{доп}$	electrical auxiliary power	kW	input data
$\eta_{рек}$	reverse-path regenerative efficiency	—	input data
f_0	rolling-resistance coefficient at low speed	—	input data
G_a	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	N·s²/m²	calculated earlier (4.4)
$\eta_{ем}$	electric-machine efficiency versus torque and speed	—	calculated earlier (4.21)
x	fraction of cycle traction energy supplied by the electric machine	—	input data

Example — hybrid

$$E_ц = \frac{1}{3{,}6\cdot 10^6}\left[\int_{P_к>0} \frac{P_к}{\eta(M,\omega)}\,dt + \int_{P_к<0} P_к \eta_{рек}\,dt + \int P_{доп}\,dt\right], \qquad e_ц = \frac{100E_ц}{S_ц}, \qquad \eta = \frac{\eta_м}{s/\eta_{ем}(M,\omega)+(1-s)} \quad (8.5)$$

$$\Delta E_{тр} = \frac{9.946\cdot 1.0000}{0.662770\cdot 3{,}6\cdot 10^6} = 0.000004 \text{ kWh}$$

$$E_ц = 5.451979 - 0.563061 + 0.000000 = 4.888919 \text{ kWh}$$

$$e_ц = \frac{100\cdot 4.888919}{23.262389} = 21.016 \text{ kWh/100 km}$$

- The formula uses power in W and time in s; division by 3,6·10⁶ converts J to kWh. Sц is cycle distance in km. The numerical sum lists traction, recovered and auxiliary energy over the entire cycle.
- The integration example shows positive traction over 11.00…12.00 s at mean speed 0.028 m/s; the total includes all 1800 intervals.
- Cycle WLTC_class_3 is integrated using its actual time steps.
- Traction efficiency is NOT constant: mechanical efficiency is fixed, while machine efficiency comes from Chapter 4's map at each torque and speed. Transmission efficiency here is strictly mechanical, covering gears, joints and splines; multiplying by the machine map does not count losses twice.
- Useful-work-weighted mean machine efficiency over the cycle: 81.5 %. Replacing the map with one constant overlooks operation at low torque, where efficiency is poorest.

- The machine supplies 60 % of traction; the engine supplies the remainder without machine losses. Traction energy is therefore split: 3.55 kWh through the machine and 1.90 kWh through the engine shaft.
- Regeneration does NOT pass through the machine map: recovered energy uses a constant $\eta_{рек}$because this document defines it as whole reverse-path efficiency. Multiplication by $\eta_{ем}$ would count the machine twice.
- This is not certified WLTP: test mass, conditioning, charge balance and prescribed shifts are not reproduced.

Source: [20]; [1]; [2]

(8.5) **Applicability:** Electric vehicle with coastdown-coefficient road load.

To demonstrate the method, the author assumes F_0 = 160 N, F_1 = 0,6 N·s/m and F_2 = 0,38 N·s²/m². These are teaching inputs, not a real coastdown test. Replace them with test data under declared conditions in your project.

Table Д.8.5.3 — Inputs for (8.5): Driving-cycle energy

Symbol	Quantity	Unit	Provenance
—	cycle points: time, speed and gradient with declared units	—	input data
—	cycle road-load basis: design inputs or coastdown	—	input data
$M_а$	gross vehicle mass	kg	calculated earlier (3.4)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
δ	rotating-mass factor	—	input data
$P_{доп}$	electrical auxiliary power	kW	input data
$\eta_{рек}$	reverse-path regenerative efficiency	—	input data
F_0	constant coastdown road-load term	N	input data
F_1	linear coastdown road-load coefficient	N·s/m	input data
F_2	quadratic coastdown road-load coefficient	N·s²/m²	input data
$\eta_{ем}$	electric-machine efficiency versus torque and speed	—	calculated earlier (4.21)

Example — driving cycle with coastdown coefficients

$$E_ц = \frac{1}{3{,}6\cdot 10^6}\left[\int_{P_к>0}\frac{P_к}{\eta(M,\omega)}\,dt + \int_{P_к<0} P_к\eta_{рек}\,dt + \int P_{доп}\,dt\right], \qquad e_ц = \frac{100E_ц}{S_ц}, \qquad \eta = \frac{\eta_м}{s/\eta_{ем}(M,\omega)+(1-s)} \quad (8.5)$$

$$\Delta E_{тр} = \frac{7.249\cdot 1.0000}{0.508262\cdot 3{,}6\cdot 10^6} = 0.000004 \text{ kWh}$$

$$E_ц = 4.369414 - 0.669542 + 0.000000 = 3.699872 \text{ kWh}$$

$$e_ц = \frac{100\cdot 3.699872}{23.262389} = 15.905 \text{ kWh/100 km}$$

- The formula uses power in W and time in s; division by 3,6·10⁶ converts J to kWh. Sц is cycle distance in km. The numerical sum lists traction, recovered and auxiliary energy over the entire cycle.

- The integration example shows positive traction over 11.00…12.00 s at mean speed 0.028 m/s; the total includes all 1800 intervals.
- Cycle WLTC_class_3 is integrated using its actual time steps.
- Traction efficiency is NOT constant: mechanical efficiency is fixed, while machine efficiency comes from Chapter 4's map at each torque and speed. Transmission efficiency here is strictly mechanical, covering gears, joints and splines; multiplying by the machine map does not count losses twice.
- Useful-work-weighted mean machine efficiency over the cycle: 85.6 %. Replacing the map with one constant overlooks operation at low torque, where efficiency is poorest.
- Regeneration does NOT pass through the machine map: recovered energy uses a constant $\eta_{рек}$because this document defines it as whole reverse-path efficiency. Multiplication by $\eta_{ем}$ would count the machine twice.
- This is not certified WLTP: test mass, conditioning, charge balance and prescribed shifts are not reproduced.

Source: [20]; [2]

(8.5) **Applicability:** The motor power curve is entered as points; no efficiency map is supplied.

One external characteristic does not establish efficiency. The author separately assumes constant machine efficiency of 0,85 for this example; all positive traction work passes through it. This is a teaching assumption, not an efficiency derived from the entered points. Without a map, its domain limits do not prove cycle feasibility; separately check available torque and speed against the adopted characteristic.

Table Д.8.5.4 — Inputs for (8.5): Driving-cycle energy

Symbol	Quantity	Unit	Provenance
—	cycle points: time, speed and gradient with declared units	—	input data
—	cycle road-load basis: design inputs or coastdown	—	input data
$M_а$	gross vehicle mass	kg	calculated earlier (3.4)
$\eta_м$	adopted mechanical driveline efficiency	—	adopted by the designer
δ	rotating-mass factor	—	input data
$P_{доп}$	electrical auxiliary power	kW	input data
$\eta_{рек}$	reverse-path regenerative efficiency	—	input data
f_0	rolling-resistance coefficient at low speed	—	input data
$G_а$	total vehicle weight force	kN	calculated earlier (3.5)
W	aerodynamic resistance factor	$N·s^2/m^2$	calculated earlier (4.4)
$\eta_{ем}^{зап}$	explicitly assumed fallback electric-machine efficiency outside the map	—	input data

Example — tabulated electric-machine characteristic

$$E_{ц}=\frac{1}{3{,}6\cdot10^{6}}\left[\int_{P_к>0}\frac{P_к}{\eta}\,dt+\int_{P_к<0}P_к\eta_{рек}\,dt+\int P_{доп}\,dt\right],\qquad e_{ц}=\frac{100E_{ц}}{S_{ц}} \tag{8.5}$$

$$\Delta E_{тр}=\frac{9.946\cdot1.0000}{0.807500\cdot3{,}6\cdot10^{6}}=0.000003\ \text{kWh}$$

$$E_{ц}=5.581711-0.563061+0.000000=5.018651\ \text{kWh}$$

$$e_{ц}=\frac{100\cdot5.018651}{23.262389}=21.574\ \text{kWh/100 km}$$

- The formula uses power in W and time in s; division by 3,6·10⁶ converts J to kWh. Sц is cycle distance in km. The numerical sum lists traction, recovered and auxiliary energy over the entire cycle.
- The integration example shows positive traction over 11.00…12.00 s at mean speed 0.028 m/s; the total includes all 1800 intervals.
- Cycle WLTC_class_3 is integrated using its actual time steps.
- This is not certified WLTP: test mass, conditioning, charge balance and prescribed shifts are not reproduced.

Source: [20]; [1]; [2]

When a map misses operating points, fallback efficiency is permitted only as an explicit assumption with the affected energy share reported. Exceeding torque or speed limits makes the profile infeasible, rather than authorising fallback efficiency. Distinguish complete coverage, assumed fallback and rejection in the report.

8.2. Fuel: constant specific consumption and operating points

Fuel estimate using constant specific consumption

Constant-specific-consumption estimation converts engine mechanical work to fuel mass and volume. It excludes idling, warm-up, enrichment and operating-point efficiency changes. It is neither a guaranteed upper nor lower bound on real consumption: error direction depends on the adopted value and omitted processes.

(8.6) **Applicability:** Conventional-vehicle estimate using one adopted specific consumption.

Energy must belong to the engine, not combine fuel and battery contributions.

Table Д.8.6.1 — Inputs for (8.6): Fuel estimate using constant specific consumption

Symbol	Quantity	Unit	Provenance
$e_{ц}$	driveline input energy over the cycle, per 100 km	kWh/100 km	calculated earlier (8.5)
g_{N}	specific fuel consumption at rated power	g/kWh	input data
$\rho_{п}$	adopted fuel density	g/cm³	input data

Example — conventional internal-combustion vehicle

$$Q_{ц}=\frac{E_{ц}g_{e}}{\rho\cdot10^{3}}\cdot\frac{100}{s_{ц}} \tag{8.6}$$

$$Q_{ц}=\frac{4.89\cdot250}{0.750\cdot10^{3}}\cdot\frac{100}{23.26}=7.01\ \text{L/100 km}$$

- The engine supplies 4.89 kWh mechanical energy over 23.26 km. Multiplying by specific consumption and dividing by density gives 1.63 L.
- In (8.6), specific consumption is constant at its rated-point value. This simplified estimate excludes changes in engine operation and idle fuel.
- Section §8.4 separately integrates fuel by gear, speed and load; the launch model also includes warm idle and clutch slip. Result (8.7) may therefore differ from (8.6). These are calculated estimates, not certified WLTP consumption.
- Not certification consumption: coastdown road load, prescribed shifts and phase weighting are absent.

Source: [7]

(8.6) **Applicability:** Hybrid fuel contribution after the battery balance.

First isolate work the engine must supply. Do not multiply total vehicle energy by engine specific consumption without this separation.

Table Д.8.6.2 — Inputs for (8.6): Fuel estimate using constant specific consumption

Symbol	Quantity	Unit	Provenance
$e_{ц}$	driveline input energy over the cycle, per 100 km	kWh/100 km	calculated earlier (8.5)
g_{N}	specific fuel consumption at rated power	g/kWh	input data
$\rho_{п}$	adopted fuel density	g/cm³	input data

Example — hybrid

$$Q_{ц} = \frac{E_{ц} g_{e}}{\rho \cdot 10^{3}} \cdot \frac{100}{s_{ц}} \tag{8.6}$$

$$Q_{ц} = \frac{4.89 \cdot 250}{0.750 \cdot 10^{3}} \cdot \frac{100}{23.26} = 7.01 \text{ L/100 km}$$

- The engine supplies 4.89 kWh mechanical energy over 23.26 km. Multiplying by specific consumption and dividing by density gives 1.63 L.
- In (8.6), specific consumption is constant at its rated-point value. This simplified estimate excludes changes in engine operation and idle fuel.
- Section §8.4 separately integrates fuel by gear, speed and load; the launch model also includes warm idle and clutch slip. Result (8.7) may therefore differ from (8.6). These are calculated estimates, not certified WLTP consumption.
- Not certification consumption: coastdown road load, prescribed shifts and phase weighting are absent.

Source: [7]

Cycle fuel consumption with gears

For every interval use gear to find engine speed, required power and the engine-map point. Integrate fuel using actual time. Chapter 4's loss map is the current working basis; empirical speed/load corrections provide a separate comparison. Gearshift strategy changes operating points and must be an explicit example input.

Table Д.8.7 — Inputs for (8.7): Cycle fuel consumption with gears

Symbol	Quantity	Unit	Provenance
$e_{ц}$	driveline input energy over the cycle, per 100 km	kWh/100 km	calculated earlier (8.5)
$V_{перем}$	shift speeds and allowable gear-speed bands	m/s	calculated earlier (5.4)
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)
g_{N}	specific fuel consumption at rated power	g/kWh	input data
$\rho_{п}$	adopted fuel density	g/cm³	input data
η_{e}	engine efficiency versus speed and load	—	calculated earlier (4.20)

Example — conventional internal-combustion vehicle

$$\Delta m_{пал,i}=\frac{g_{e,i}N_{потр,i}\Delta t_{i}}{3600},\qquad Q_{ц}=\frac{100\Sigma_{i}\Delta m_{пал,i}}{10^{3}\rho S_{покр}} \tag{8.7}$$

$$\Delta m_{пал}=\frac{312.9116\cdot 6.87144\cdot 1.0000}{3600}=0.59726\ \text{g}$$

$$m_{пал}=\ldots+0.59726+0.86272+\ldots+1.23106+\ldots=1731.26896\ \text{g}$$

$$Q_{ц}=\frac{100\cdot 1731.26896}{10^{3}\cdot 0.7500\cdot 20.374083}=11.3299\ \text{L/100 km}\ \textit{(87.6\% шляху)}$$

- At every interval check selected-gear power including acceleration. Project strategies downshift where needed; do not replace a prescribed shift algorithm. Launch is quasi-steady: speed may exceed idle, clutch heat is included in engine energy, and engine spin-up inertia is not separately modelled.
- In the substitution, gₑ uses g/(kWh), Nпотр kW, Δt s, ρ g/cm³ and Sпокр km. Division by 3600 converts seconds to hours; the factor 1000 converts cm³ to litres.
- One traction interval, 15.00…16.00 s, is shown. Fuel summation includes all 1682 calculated intervals and normalises to 20.374083 km covered distance.
- Specific consumption is now NOT constant; it comes from Chapter 4's efficiency map rather than empirical corrections. The shift map gives speed at every cycle point, the Willans line gives efficiency at that speed and load, and g_{e} = 3600/(η·H_{u}) converts it to grams. The difference is physical rather than arithmetic: map speed and load are physical quantities, whereas K_{ω} and K_{B} are polynomial fits to engines of their period.
- The same cycle points give 10.96 L/100 km with course-method corrections versus 11.33 from the map, a 3 % difference. This compares two calculation models over the same intervals.
- Engine-work-weighted cycle specific consumption is 424 g/(kWh), versus 250 at the rated point. In §8.4 consumption depends on speed and load and inertia depends on gear; §8.3 uses constant specific consumption.
- The normal gearshift strategy determines gear, engine speed and load in every interval. Changing strategy changes operating points and calculated consumption.
- The engine supplies no traction for 655 s of the cycle: braking and stops. This chapter does not calculate idle fuel because the source gives no model. The vehicle also TRACTS for 118 s while the selected gear cannot maintain engine speed. These are clutch-assisted launches, which the chapter does not model, so their fuel is also omitted and the result is underestimated.
- Each step uses its gear-dependent δ, from 1.08 to 1.11; the δ input applies only to a single-speed path.
- The result covers 88 % of cycle distance (20.37 of 23.26 km), not the whole cycle. For 118 s the vehicle requires traction but the selected gear cannot maintain engine speed; these clutch-assisted launches

are not modelled. The denominator is the covered distance: dividing by total distance would include travel for which fuel was not calculated, with error direction depending on where the gap occurs.

- This is not WLTP consumption: that procedure uses coastdown resistance, prescribed gear selection and weighted phases.
- This consumption is of the TYPE-APPROVAL kind: it describes the vehicle on a prescribed cycle rather than in everyday use. Owner refuelling logs measure the difference: real petrol-engine consumption is about 14 % higher and increasing (8 % in 2018, 14 % in 2022). A standardised declared value typically underestimates fleet-average real consumption. This is a population property, not a bound for an individual vehicle: economical driving can produce a lower value than declared.

Source: [7]; [21]

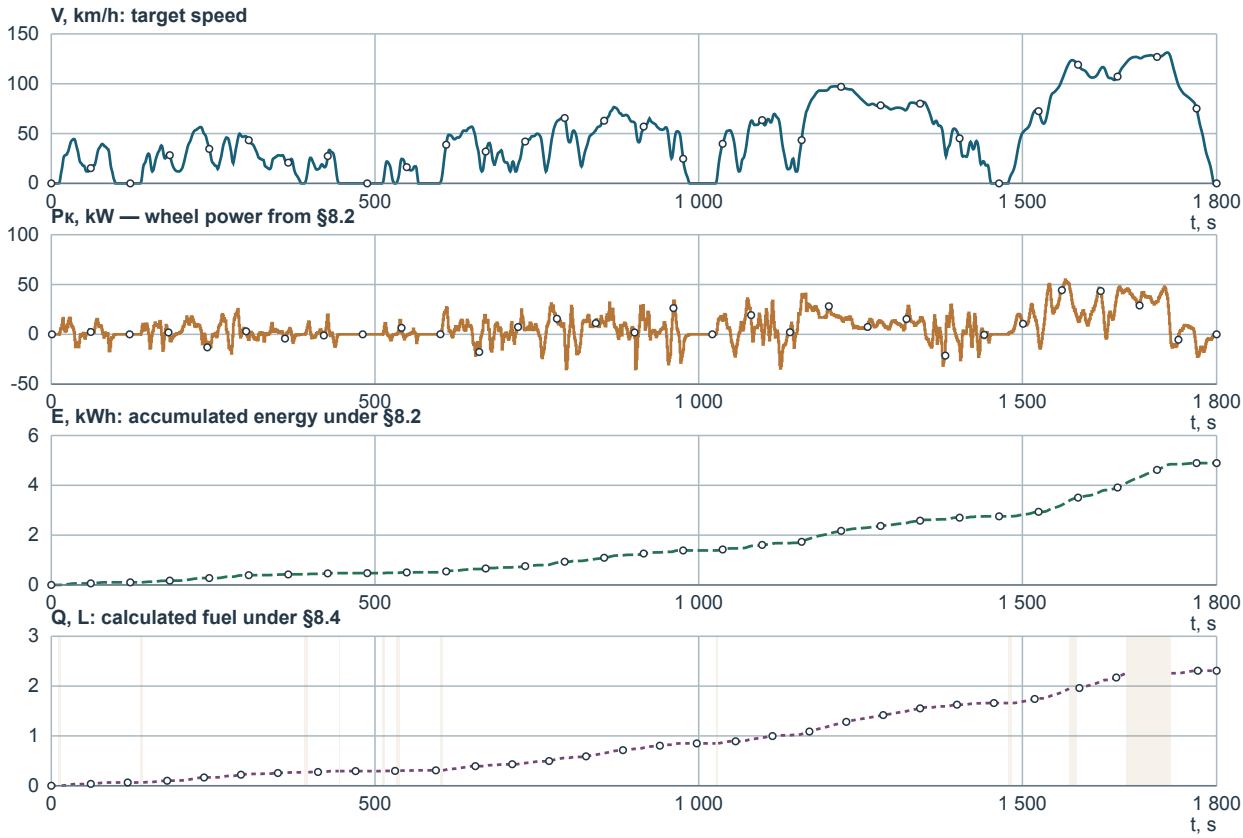


Figure 8.1 — Specified WLTC profile and accumulated fuel over covered intervals; uncovered intervals require separate assessment.
Source: [2].

Report consumption together with the time and distance covered. Stopping and braking do not automatically mean zero engine fuel: without idle and fuel-cutoff models, these intervals remain a separate limitation. If a gear cannot provide required power or an operating point is outside the characteristic, do not substitute zero for the omission. Do not describe a covered-part result as whole-cycle consumption.

8.3. Hybrid and plug-in hybrid

Cycle energy distribution between engine and electric machine

Electric traction share and battery energy balance are independent decisions. A large electric-machine work share does not necessarily mean high grid-energy use: in charge-sustaining mode, the engine ultimately supplies this work through the adopted conversion path. Specify the balance sign and battery accounting boundary explicitly.

(8.8) **Applicability:** The plug-in petrol hybrid preserves its initial battery energy.

The adopted electric traction share is 0,6 and the battery balance is 0 kWh. In this simplified balance the engine supplies the electric branch; the presence of an electric machine alone does not imply grid-energy savings.

Table Д.8.8.2 — Inputs for (8.8): Cycle energy distribution between engine and electric machine

Symbol	Quantity	Unit	Provenance
$e_{ц}$	driveline input energy over the cycle, per 100 km	kWh/100 km	calculated earlier (8.5)
x	fraction of cycle traction energy supplied by the electric machine	—	input data
$\Delta W_{бат}$	battery energy balance over the cycle	kWh	input data

Example — charge-sustaining hybrid

$$E_{ел}=E_{тр,ел}-E_{рек}+E_{доп},\ E_{мех}=E_{тр,дв},\qquad E_{бат}=\min(E_{ел},\max(0,-\Delta W_{бат})),\qquad E_{дв}=E_{мех}+\frac{E_{ел}-E_{бат}}{\eta_{бат}} \quad (8.8)$$

$$E_{ел}=3.554198-0.563061+0.000000=2.991137\ \text{kWh}$$

$$E_{мех}=E_{тр,дв}=1.897782\ \text{kWh}$$

$$E_{бат}=\min(2.991137,\max(0,-(0.000000)))=0.000000\ \text{kWh}$$

$$E_{дв}=1.897782+\frac{2.991137-0.000000}{1.0000}=4.888919\ \text{kWh}$$

- Mechanical and electrical traction energies are integrated separately in (8.5) because their efficiencies differ. Do not split the combined energy by the fraction x again.
- The document does NOT invent a control strategy: the project author specifies electric-machine share and charge balance as design decisions, not consequences of road resistance. The remainder is arithmetic.
- Zero charge balance means CHARGE-SUSTAINING mode: the battery finishes with the same energy it started with, so the engine ultimately supplied all energy. This is how fuel consumption is measured for a hybrid that is not charged from the grid.
- Of the 2.99 kWh supplied by the electric machine, 0.00 comes from the battery and 2.99 is generated by the engine.
- This is not a certification procedure: actual control determines the share rather than a single number, and the result is weighted across phases.
- A plug-in hybrid's declared fuel consumption is typically underestimated by multiples rather than small percentages, and the factor depends on the population: for NEDC values, 220…240 % for private cars and 360…410 % for company cars; for WLTP values, three to five times. The reason is not the model or measurement: the procedure assumes daily charging, while real electric-driving shares are about 45…49 % for private and 11…15 % for company cars. The same design and declared value can have different charging habits, so one figure cannot describe its consumption independently of its use.

Source: [2]; [21]

(8.8) **Applicability:** The same plug-in hybrid finishes with less battery energy.

Electric traction share remains 0,6; the author specifies a balance of −1 kWh. One kilowatt-hour comes from stored battery energy, reducing required engine work. State initial and final battery energy together with fuel litres.

Table Д.8.8.1 — Inputs for (8.8): Cycle energy distribution between engine and electric machine

Symbol	Quantity	Unit	Provenance
$e_{ц}$	driveline input energy over the cycle, per 100 km	kWh/100 km	calculated earlier (8.5)
x	fraction of cycle traction energy supplied by the electric machine	—	input data
$\Delta W_{бат}$	battery energy balance over the cycle	kWh	input data

Example — charge-depleting hybrid

$$E_{ел}=E_{тр,ел}-E_{рек}+E_{доп},\ E_{мех}=E_{тр,дв},\qquad E_{бат}=\min(E_{ел},\max(0,-\Delta W_{бат})),\qquad E_{дв}=E_{мех}+\frac{E_{ел}-E_{бат}}{\eta_{бат}} \quad (8.8)$$

$$E_{ел}=3.554198-0.563061+0.000000=2.991137\ \text{kWh}$$

$$E_{мех}=E_{тр,дв}=1.897782\ \text{kWh}$$

$$E_{бат}=\min(2.991137,\max(0,-(-1.000000)))=1.000000\ \text{kWh}$$

$$E_{дв}=1.897782+\frac{2.991137-1.000000}{1.0000}=3.888919\ \text{kWh}$$

- Mechanical and electrical traction energies are integrated separately in (8.5) because their efficiencies differ. Do not split the combined energy by the fraction x again.
- The document does NOT invent a control strategy: the project author specifies electric-machine share and charge balance as design decisions, not consequences of road resistance. The remainder is arithmetic.
- The battery supplies 1.00 kWh over the cycle: CHARGE-DEPLETING mode. That energy came from charging, so compare this fuel consumption with a conventional vehicle only alongside this number.
- Of the 2.99 kWh supplied by the electric machine, 1.00 comes from the battery and 1.99 is generated by the engine.

- This is not a certification procedure: actual control determines the share rather than a single number, and the result is weighted across phases.
- A plug-in hybrid's declared fuel consumption is typically underestimated by multiples rather than small percentages, and the factor depends on the population: for NEDC values, 220…240 % for private cars and 360…410 % for company cars; for WLTP values, three to five times. The reason is not the model or measurement: the procedure assumes daily charging, while real electric-driving shares are about 45…49 % for private and 11…15 % for company cars. The same design and declared value can have different charging habits, so one figure cannot describe its consumption independently of its use.

Source: [2]; [21]

In charge-sustaining mode, initial and final battery energy are equal. In a charge-depleting plug-in hybrid, the decrease in stored energy is a separate input; fuel litres alone do not describe vehicle consumption. Grid consumption additionally includes charging losses, which the battery-energy change alone excludes. The vehicle enters another mode after depletion; do not extrapolate a short trip starting fully charged to a long uncharged trip.

Combined cycle consumption

Convert engine mechanical work to chemical fuel energy using adopted efficiency. Determine fuel volume from density and lower heating value. Show battery energy used beside fuel litres and their sum at the declared accounting boundary. This sum is not primary supply-chain energy: extraction, production, delivery and charging losses are excluded.

(8.9) **Applicability:** Plug-in hybrid fuel and energy summary with unchanged battery energy.

The result describes charge-sustaining operation on this cycle and does not apply to charge-depleting operation.

Table Д.8.9.2 — Inputs for (8.9): Combined cycle consumption

Symbol	Quantity	Unit	Provenance
$E_{дв}$	mechanical energy that the engine must supply over the cycle	kWh	calculated earlier (8.8)
g_N	specific fuel consumption at rated power	g/kWh	input data
$\rho_п$	adopted fuel density	g/cm³	input data
$\Delta W_{бат}$	battery energy balance over the cycle	kWh	input data

Example — charge-sustaining hybrid

$$E_{пал}=\frac{E_{дв}}{\eta_e},\qquad Q_{100}=\frac{100\cdot E_{пал}\cdot 3{,}6}{H_u\rho S_ц},\qquad E_{бат}=\max(0,-\Delta W_{бат}),\qquad E_{разом,100}=\frac{100(E_{пал}+E_{бат})}{S_ц}\quad (8.9)$$

$$E_{пал}=\frac{4.888919}{0.327273}=14.938363\ \text{kWh}$$

$$Q_{100}=\frac{100\cdot 14.938363\cdot 3{,}6}{44.000\cdot 0.7500\cdot 23.262389}=7.0055\ \text{L/100 km}$$

$$E_{бат}=\max(0,-(0.000000))=0.000000\ \text{kWh}$$

$$E_{разом,100}=\frac{100(14.938363+0.000000)}{23.262389}=64.2168\ \text{kWh/100 km}$$

- A hybrid has two consumption quantities rather than one. While they remain two numbers in different units, neither a petrol vehicle nor a BEV can be compared directly with it. The common measure is PRIMARY ENERGY: what is purchased as fuel litres in the tank and kilowatt-hours from the grid.
- With CONSTANT engine efficiency, a charge-sustaining hybrid gives the same consumption as a conventional vehicle with the same engine. This is the only possible result of that model: the battery adds no energy, but shifts it in time.
- The real hybrid benefit depends on the EFFICIENCY MAP. At quarter load, engine consumption is about 415 g/kWh rather than 250, with efficiency falling from 33 to 20 percent. The electric machine helps avoid this region: the engine either stops or operates at a more favourable load. To quantify the benefit, specify the engine power-utilisation coefficient; efficiency in the expression above changes accordingly.
- A litre of fuel contains $\rho\cdot H_u$ of energy; engine efficiency of 32.7 % indicates how much reaches the wheels. Efficiency is essential to this sum, not a cosmetic detail.
- Zero charge balance means no energy has come from the grid, so all primary energy is fuel. This hybrid's consumption equals that of a conventional vehicle with the same engine: the battery shifts energy in time rather than adding it.
- This is not the certification procedure's equivalent fuel consumption: that converts electricity to litres using a prescribed coefficient and weights phases. Here the two quantities retain their own units.

Source: [7]

(8.9) **Applicability:** The same hybrid's fuel and energy summary with 1 kWh battery depletion.

Reduced fuel litres are accompanied by a separate battery-energy contribution. Show both; the electrical contribution shown excludes charging losses.

Table Д.8.9.1 — Inputs for (8.9): Combined cycle consumption

Symbol	Quantity	Unit	Provenance
$E_{дв}$	mechanical energy that the engine must supply over the cycle	kWh	calculated earlier (8.8)
g_N	specific fuel consumption at rated power	g/kWh	input data
$\rho_п$	adopted fuel density	g/cm³	input data
$\Delta W_{бат}$	battery energy balance over the cycle	kWh	input data

Example — charge-depleting hybrid

$$E_{пал}=\frac{E_{дв}}{\eta_e},\qquad Q_{100}=\frac{100\cdot E_{пал}\cdot 3{,}6}{H_u\rho S_ц},\qquad E_{бат}=\max(0,-\Delta W_{бат}),\qquad E_{разом,100}=\frac{100(E_{пал}+E_{бат})}{S_ц}\quad (8.9)$$

$$E_{пал}=\frac{3.888919}{0.327273}=11.882807\ \text{kWh}$$

$$Q_{100}=\frac{100\cdot 11.882807\cdot 3{,}6}{44.000\cdot 0.7500\cdot 23.262389}=5.5725\ \text{L/100 km}$$

$$E_{бат}=\max(0,-(-1.000000))=1.000000\ \text{kWh}$$

$$E_{разом,100}=\frac{100(11.882807+1.000000)}{23.262389}=55.3804\ \text{kWh/100 km}$$

- A hybrid has two consumption quantities rather than one. While they remain two numbers in different units, neither a petrol vehicle nor a BEV can be compared directly with it. The common measure is PRIMARY ENERGY: what is purchased as fuel litres in the tank and kilowatt-hours from the grid.
- With CONSTANT engine efficiency, a charge-sustaining hybrid gives the same consumption as a conventional vehicle with the same engine. This is the only possible result of that model: the battery adds no energy, but shifts it in time.
- The real hybrid benefit depends on the EFFICIENCY MAP. At quarter load, engine consumption is about 415 g/kWh rather than 250, with efficiency falling from 33 to 20 percent. The electric machine helps avoid this region: the engine either stops or operates at a more favourable load. To quantify the benefit, specify the engine power-utilisation coefficient; efficiency in the expression above changes accordingly.
- A litre of fuel contains $\rho\cdot H_u$ of energy; engine efficiency of 32.7 % indicates how much reaches the wheels. Efficiency is essential to this sum, not a cosmetic detail.
- The grid supplies 4.30 kWh/100 km, or 8 % of all primary energy. Compare this fuel consumption with a conventional vehicle only alongside this number.
- This is not the certification procedure's equivalent fuel consumption: that converts electricity to litres using a prescribed coefficient and weights phases. Here the two quantities retain their own units.

Source: [7]

A series hybrid requires generator-set, electrical-path and battery losses, together with the engine operating mode. The kinematic example in Chapter 4 does not supply a missing fuel balance. A fuel cell similarly requires its own converter map and hydrogen mass consumption; petrol specific consumption does not apply. These formulas provide teaching estimates from explicit inputs rather than certification results for every electrified layout.

8.4. BEV battery, SOC reserve and load-based sizing

Available energy from initial SOC to reserve.

Trip energy equals usable battery energy in the BMS 0–100% window multiplied by the difference between initial and reserve SOC fractions. Both energies use kWh. Convert percentages to fractions before calculation. Set the trip reserve within the usable window; do not subtract the factory gross-to-usable buffer again.

Table Д.8.1 — Inputs for (8.1): Available energy from initial SOC to reserve.

Symbol	Quantity	Unit	Provenance
E_{u}	usable battery energy within the BMS 0–100% SOC window	kWh	input data
s_0	initial state of charge within the usable BMS window	—	input data
$s_{\mathrm{m}}in$	reserve state of charge within the usable BMS window	—	input data

Example — electric-vehicle battery, mass and range

$$E_{\mathrm{trip}}=E_{\mathrm{u}}\cdot(\mathrm{SOC}_0\text{-}\mathrm{SOC}_{\mathrm{min}}) \tag{8.1}$$

$$E_{\mathrm{trip}}=30\cdot(0.9\text{-}0.1)=24\ \mathrm{kWh}$$

Source: [7]

Range from repeating the selected profile to the SOC reserve.

Calculated range is distance on the selected profile from initial to reserve SOC. Repeat the chosen WLTC 3a/3b or custom profile to that reserve. Determine average chemical-energy consumption from the distance and energy used. Shorten the final interval to the energy boundary, interpolating speed and distance. Auxiliaries, resistance, driveline and mass follow the preceding energy calculation.

Table Д.8.2 — Inputs for (8.2): Range from repeating the selected profile to the SOC reserve.

Symbol	Quantity	Unit	Provenance
c_{b}	average chemical energy consumption while repeating the profile to the reserve	kWh/km	input data
E_{trip}	trip energy available above the SOC reserve	kWh	calculated earlier (8.1)

Example — electric-vehicle battery, mass and range

$$L_{\mathrm{bat}}=\frac{E_{\mathrm{trip}}}{c_{\mathrm{b}}} \tag{8.2}$$

$$L_{\mathrm{bat}}=\frac{24}{0.211798}=113{,}3\ \mathrm{km}$$

Source: [7]

In energy mode, stored-energy change equals terminal energy. The optional constant-voltage, internal-resistance model solves P = (U − RI)I; chemical power UI equals terminal power plus heat I^2R. Current and voltage limits can make the profile infeasible. At upper SOC or the

charging-current limit, regeneration decreases and friction brakes absorb excess braking energy.

Payload comprises occupants, luggage and cargo, excluding the battery. Sizing keeps payload fixed and recalculates road resistance for each battery mass. Check 25 capacities within allowable gross mass, refine the first interval meeting the target, then recheck rounded capacity. The greatest range among sampled points does not prove a global optimum. Measured coastdown belongs to one mass and is not used for this sweep.

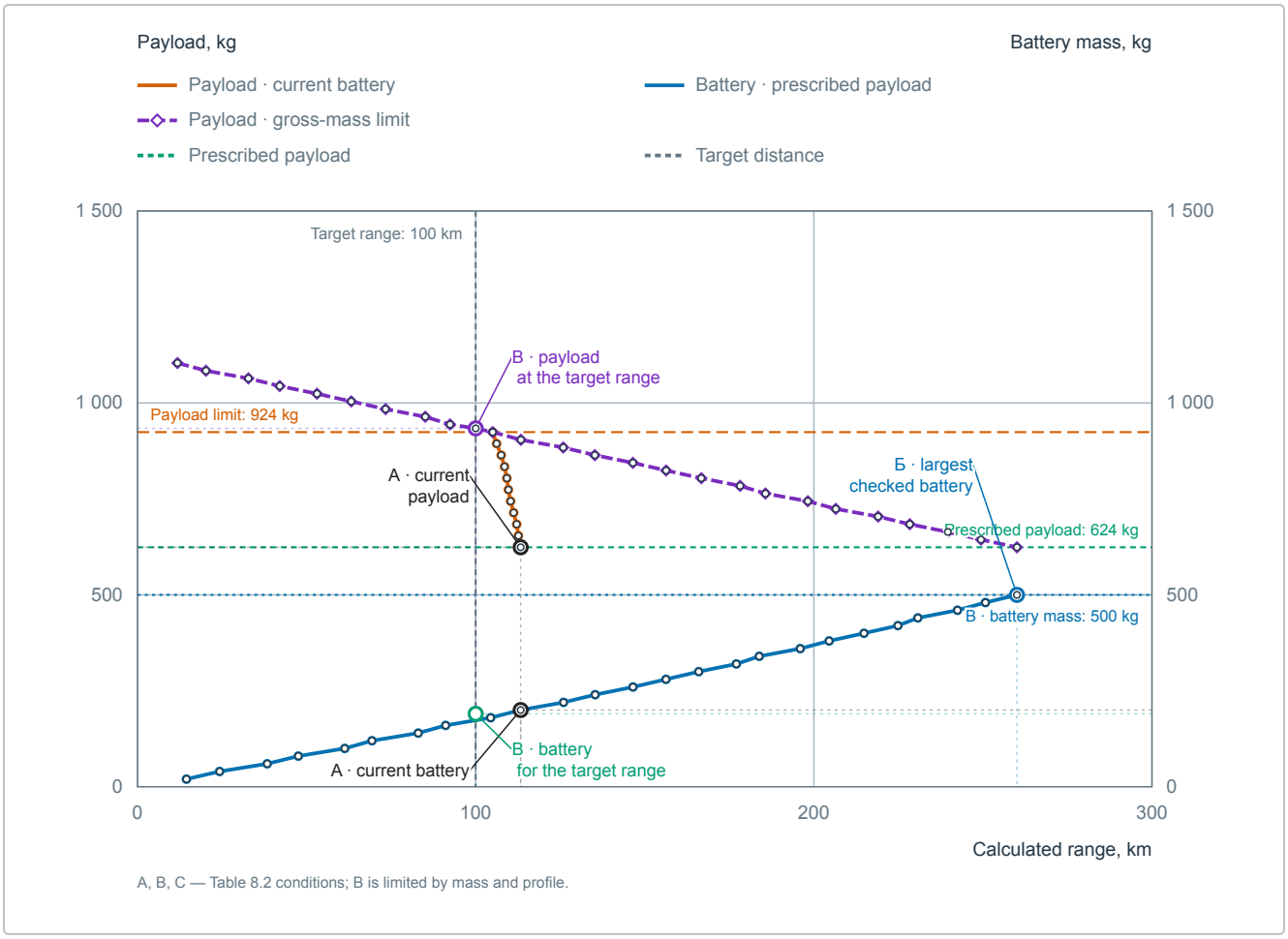


Figure 8.2 — Checked battery, payload and calculated-range alternatives: payload varies at fixed battery size, battery varies at fixed payload, and a third curve varies both masses at the gross-mass limit. Alternative B sizes the battery and largest checked payload for the target range. Both mass axes use the same scale; unconfirmed alternatives are not connected by a curve.
Source: [2].

This is a WLTC-profile estimate with declared assumptions. Air conditioning, temperature effects, ageing, measured charging energy and WLTP certification corrections are unspecified. Check loading and profile feasibility before adopting range.

8.5. Battery discharge and electrical limits

First specify usable pack capacity E_{u}, initial state of charge SOC_0, reserve SOC_{min}, upper limit SOC_{max} and the driving profile. Electrical verification also requires open-circuit voltage U_{b}, internal resistance R_{b}, terminal-current limits I and voltage limits U. Positive current denotes discharge; negative current denotes regenerative charging. Terminal power P, chemical power P_{b} and internal heat P_{R} are related by:

$$U=U_b-R_bI,\ P=\frac{UI}{1000},\ P_b=\frac{U_bI}{1000}=P+P_R,\ P_R=\frac{R_bI^2}{1000}.$$

Voltage is in V, current in A and resistance in Ω; dividing by 1000 gives power in kW.

The numerical example retains the previous example's 30 kWh and mass, but enables the electrical model using the application's common starting assumptions: 400 V, 0,05 Ω, discharge up to 800 A, charging up to 300 A and a 250…450 V window. These are assumed teaching-pack properties, not a specific battery specification. The same integrator checks every interval to the reserve: a traction-power deficit rejects the range result, while limited regenerative acceptance increases friction-brake work. Plots retain roughly one point per 30 s plus the first and last points; short peaks between them are not shown. Time is in hours and power in kW.

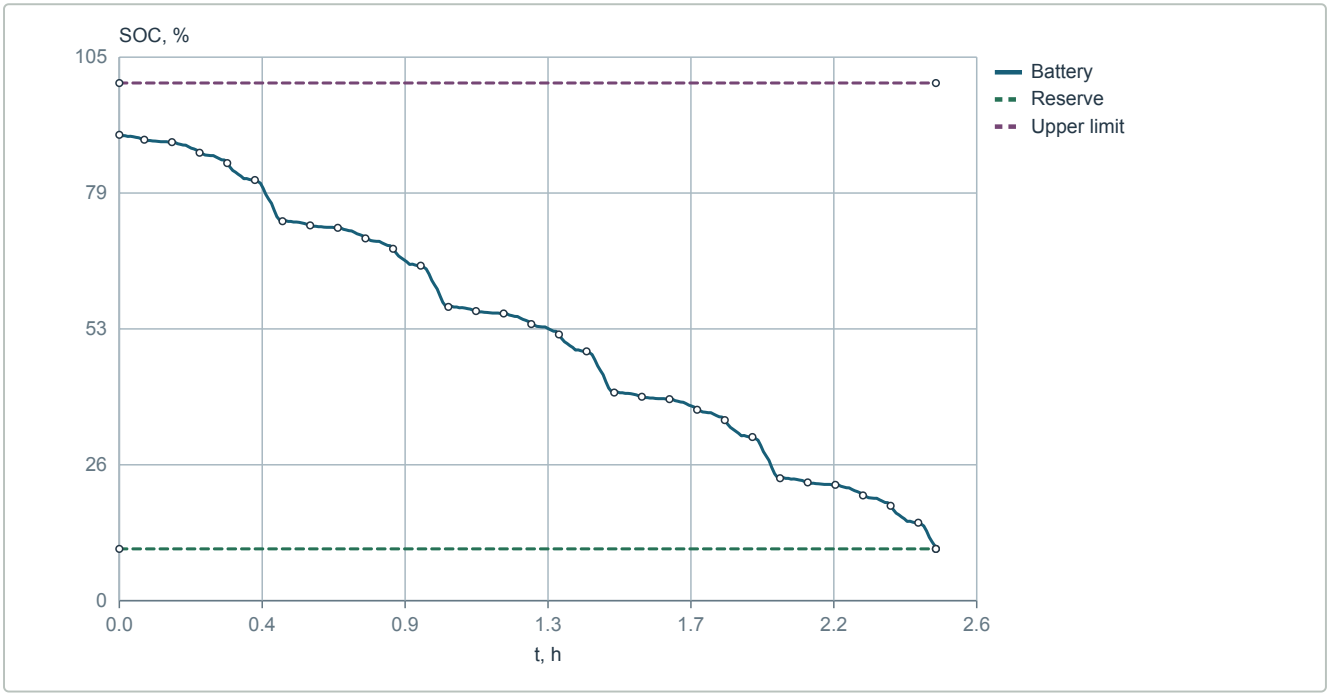


Figure 8.3 — Battery SOC while repeating the profile to reserve; dashed lines show specified SOC limits. Source: [2].

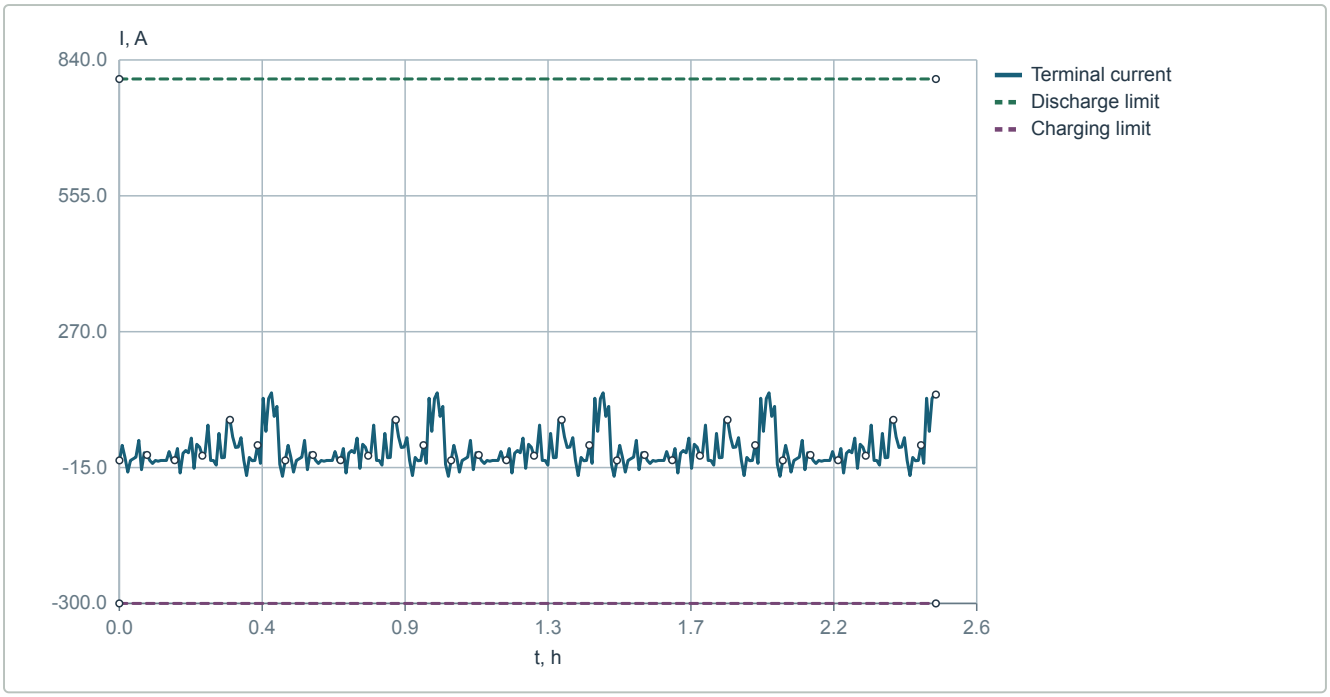


Figure 8.4 — Battery terminal current at retained discharge points: positive for discharge, negative for charging. Dashed lines show specified current limits.
Source: [2].

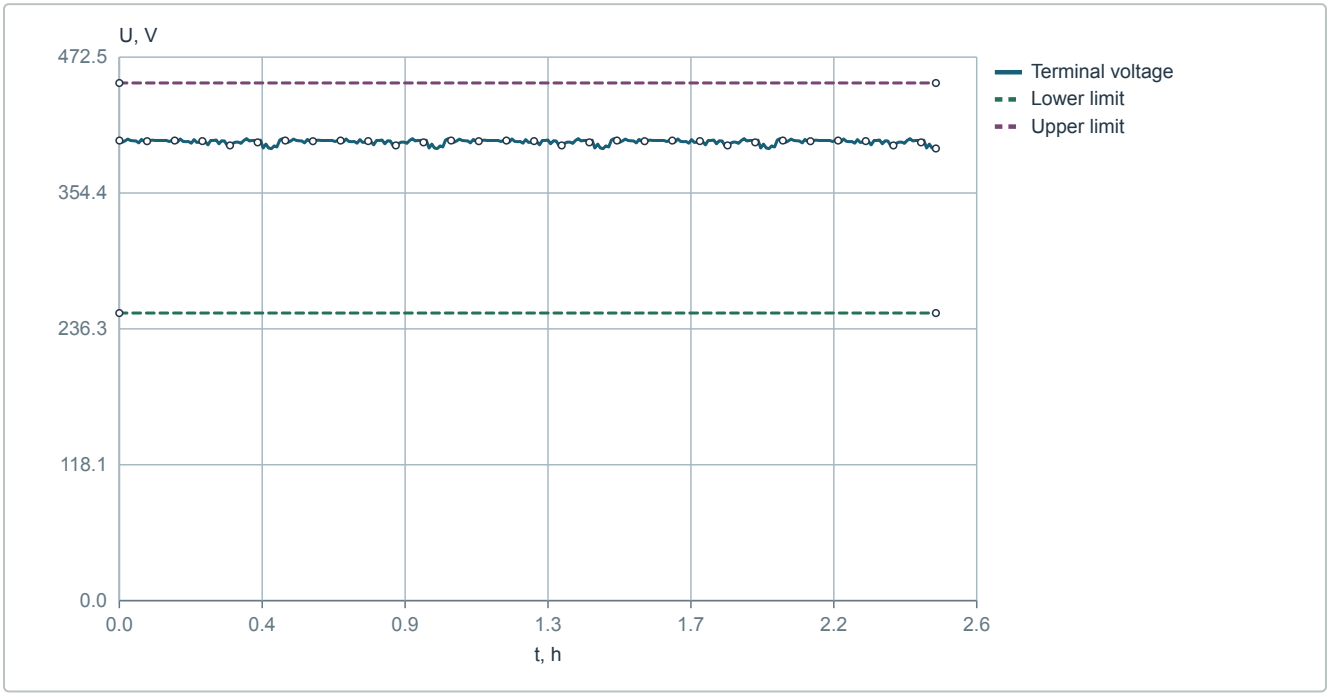


Figure 8.5 — Battery terminal voltage including internal resistance; dashed lines show the specified voltage window.
Source: [2].

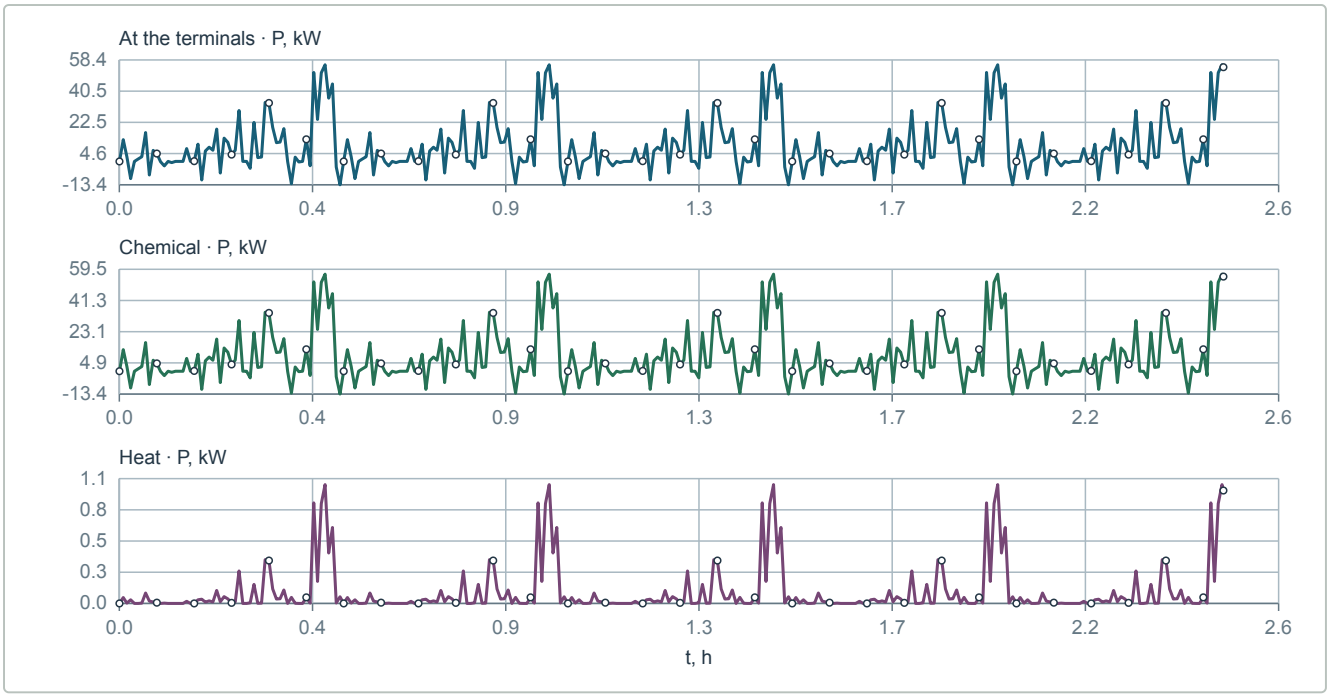


Figure 8.6 — Terminal, chemical and thermal power at retained points of the electrical battery model; chemical minus terminal power equals internal heat.
Source: [2].

A complete electric-machine package may declare its own DC-side power limit and maximum-speed dependence on terminal voltage. The shared component checks these properties together with the battery. A motor-plus-inverter map already includes their combined losses; a motor-only map requires a separate inverter efficiency without counting losses twice. Constant efficiency does not specify inverter power rating, and battery terminal current is not phase current. Without manufacturer phase, thermal and switching limits, independent inverter sizing remains unresolved. A constant-voltage, constant-resistance model also does not replace cell, BMS, protection and pack-cooling design.

8.6. Comparison with nearby prototypes

After calculating the main parameters, compare 0–100 km/h time and fuel or electricity consumption per 100 km with published values for nearby prototypes. Select up to 15 vehicles with the same category, powertrain type and energy carrier. Rank proximity using Chapter 2 payload, seat count and maximum speed. Missing values lower a record's priority.

Use the same prototype selection for every metric. Each figure retains the largest group sharing a declared procedure, phase, mode, market and energy boundary; prefer WLTP when coverage is equal. Count other contexts and missing values separately. If the source omits the procedure, state this explicitly; comparison then requires checking test conditions. Preserve both bounds of published ranges. Do not convert 0–60 mph to 0–100 km/h or replace missing consumption with zero.

Figures (8.7) and (8.8) use an M1 petrol-engine passenger-car example: five occupants, 624 kg payload and 120 km/h maximum speed. Displayed values belong to prototypes; the project's own calculation appears in preceding chapters.

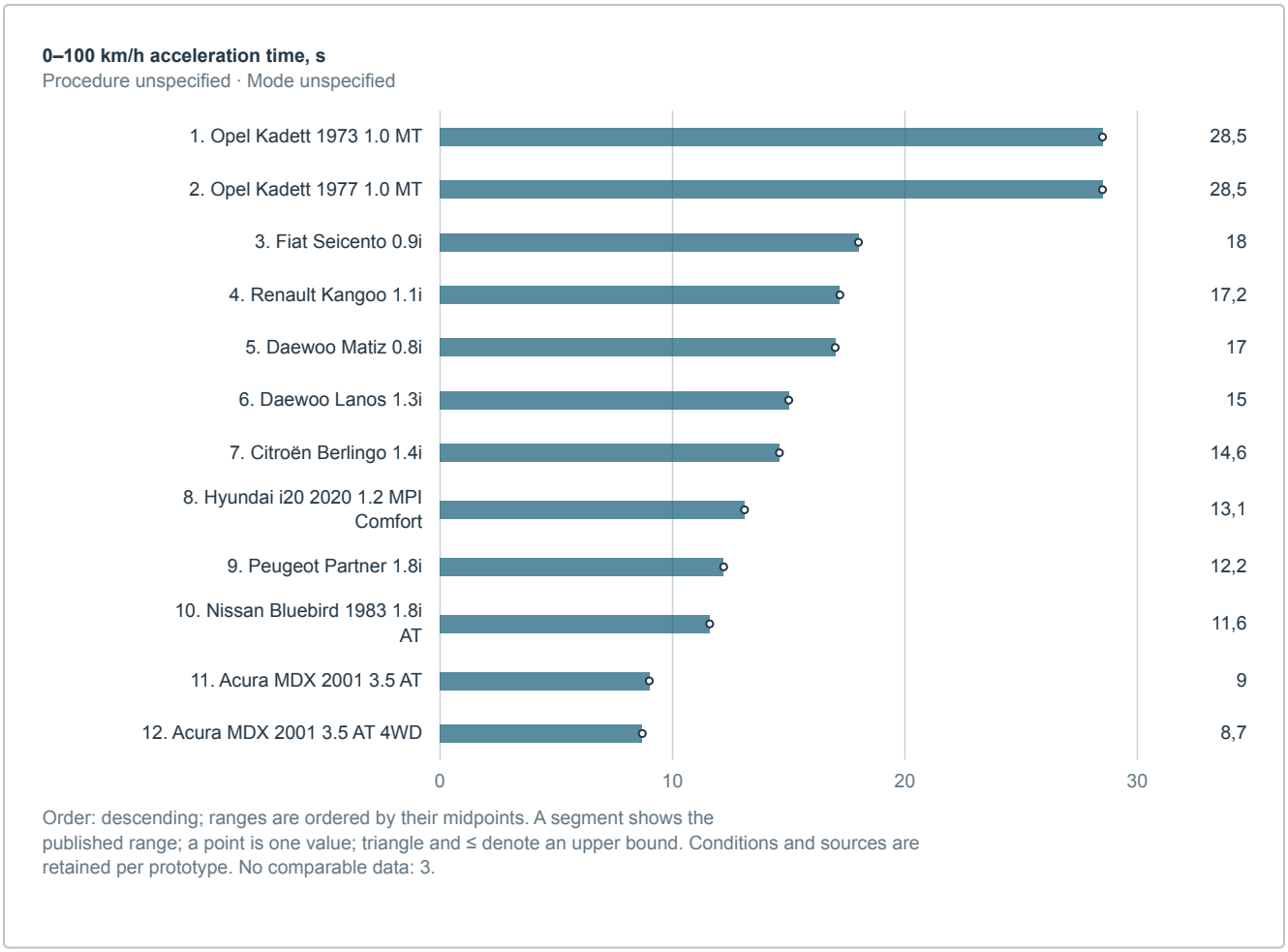


Figure 8.7 — Published 0–100 km/h times for nearby M1 petrol-engine prototypes. The adopted group's procedure and conditions are stated in the figure.
Source: [3].

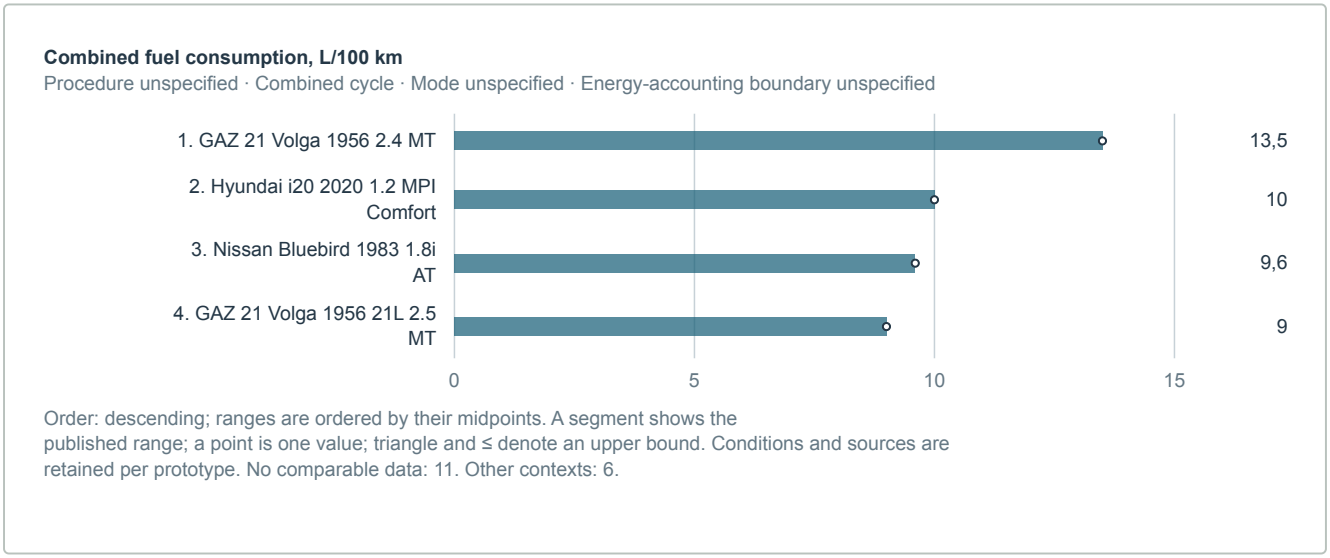


Figure 8.8 — Published combined fuel consumption per 100 km for the same nearby M1 petrol-engine prototypes. Litres retain the source's fuel basis.
Source: [3].

Figure (8.9) uses an M1 BEV with the same payload and 160 km/h maximum speed. From-grid electricity includes the source's charging path. Before comparison with calculated battery

consumption, align the energy boundary, driving profile, mass and initial conditions.

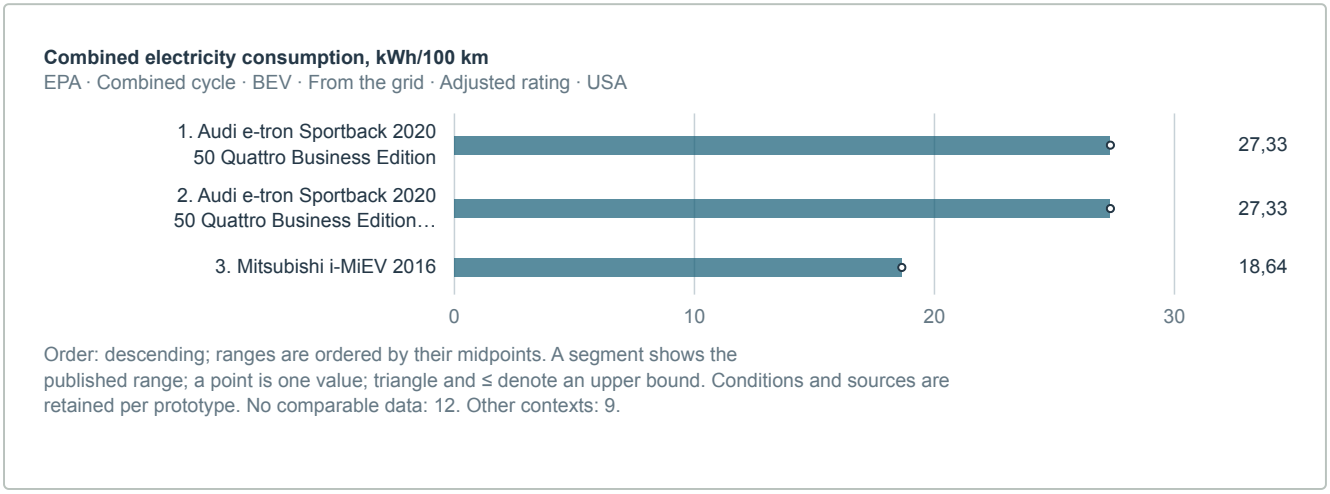


Figure 8.9 — Published combined electricity consumption per 100 km for nearby M1 BEV prototypes. The adopted group's energy boundary, procedure and mode are stated in the figure.
Source: [3].

Chapter 9. Clutch

From design torque to friction linings and the pressure mechanism; checking torque capacity and component strength.

Calculate the clutch after selecting the engine characteristic and transmission layout. First establish whether the engine transmits torque to the wheels and which device enables launch. This method covers dry friction clutches. Wet multi-plate clutches, torque converters and dual clutches require their own models.

Table M.9.1 — Applicability of dry-clutch calculation to different powertrains

Powertrain and traction path	Applicability of this chapter
Internal combustion engine with dry friction clutch	Calculation of the lining and selected pressure mechanism
Parallel hybrid, including plug-in variants	Only an explicitly selected dry clutch in the mechanical engine path
Series hybrid	The engine drives a generator; the wheel drive requires no dry launch clutch
Battery electric vehicle	The electric machine can launch from zero speed; do not add a clutch reserve factor to the reduction gear

A battery and external charging do not determine clutch construction. A hybrid requires a kinematic layout. Mark an unsuitable model as not applicable; zero design torque is not a substitute for that conclusion.

Selecting the method and friction linings

Adopt one methodological basis before calculation. Do not mix numbers from different sources: the profile determines coefficients, lining selection rules and spring checks. A range midpoint is the author's teaching assumption. Real materials require manufacturer data at the relevant temperatures and sliding speeds; historical tables do not replace these data.

Table M.9.2 — Comparison of initial inputs and clutch selection rules by method

Quantity or rule	Course guide, pp. 37–40	Opanasiuk, pp. 28–32
Clutch reserve factor	1,75…2,50	Passenger cars 1,20…1,75; trucks 1,50…2,20; off-road vehicles 1,80…3,00
Coefficient of friction	0,25…0,65	0,25…0,50
Initial pressure, MPa	0,15…0,30	0,14…0,30
Lining width to mean-radius ratio	0,375…0,620	0,38…0,63
Tabulated outer-diameter selection	Next larger value	Nearest value
Assistance recommendation threshold, N	9000	9000

Design clutch torque using the reserve factor.

Multiply maximum engine torque by the adopted reserve factor. This establishes the clutch torque requirement. Use it consistently for lining geometry and verification of the assembled clutch's torque capacity.

Table Д.9.1 — Inputs for (9.1): Design clutch torque using the reserve factor.

Symbol	Quantity	Unit	Provenance
M_{e} $M_{\text{к,max}}$	maximum internal-combustion engine torque	N·m	calculated earlier
β	clutch torque reserve factor	—	input data

Example — conventional internal-combustion vehicle

$$M_{\text{p}}=\beta\cdot M_{\text{e}} \tag{9.1}$$

$$M_{\text{p}}=1.475\cdot 207.841=306,6\ \text{N·m}$$

Source: [6]; [10]

Preliminary mean friction radius from allowable lining pressure.

Determine initial mean radius from design torque, pressure, friction, friction-surface count and relative lining width. A single-plate clutch has two working surfaces; friction-surface count is not pressure-spring count.

Table Д.9.2 — Inputs for (9.2): Preliminary mean friction radius from allowable lining pressure.

Symbol	Quantity	Unit	Provenance
$q_{\text{l}}ining$	adopted initial lining pressure	MPa	input data
μ	friction-lining coefficient of friction	—	input data
k_{w}	lining width to mean-radius ratio	—	input data
z	number of active friction surfaces	—	input data
M_{p}	design clutch torque	N·m	calculated earlier (9.1)

Example — conventional internal-combustion vehicle

$$R_{\text{prelim}}=\sqrt[3]{\frac{M_{\text{p}}}{(2\cdot\pi\cdot q_{\text{lining}}\cdot z\cdot k_{\text{w}}\cdot\mu)}} \tag{9.2}$$

$$R_{\text{prelim}}=\sqrt[3]{\frac{306.565}{(2\cdot\pi\cdot 0.22\cdot 2\cdot 0.505\cdot 0.375)}}=0,0837\ \text{m}$$

Source: [6]; [10]

Friction-lining width.

Calculate width from adopted relative width and initial mean radius. This is intermediate geometry for lining selection.

Table Д.9.3 — Inputs for (9.3): Friction-lining width.

Symbol	Quantity	Unit	Provenance
k_w	lining width to mean-radius ratio	—	input data
$R_p relim$	initial mean lining radius	m	calculated earlier (9.2)

Example — conventional internal-combustion vehicle

$$b_{lining} = k_w \cdot R_{prelim} \tag{9.3}$$

$$b_{lining} = 0.505 \cdot 0.0836609 = 0{,}0422 \text{ m}$$

Source: [6]; [10]

Preliminary lining outer diameter.

Preliminary outer diameter locates the relevant table region. After selecting a standard lining, its actual inner and outer diameters become new inputs; do not carry preliminary dimensions into later checks.

Table Д.9.4 — Inputs for (9.4): Preliminary lining outer diameter.

Symbol	Quantity	Unit	Provenance
$R_p relim$	initial mean lining radius	m	calculated earlier (9.2)
$b_l ining$	calculated lining width	m	calculated earlier (9.3)

Example — conventional internal-combustion vehicle

$$D_{prelim} = 2 \cdot R_{prelim} + b_{lining} \tag{9.4}$$

$$D_{prelim} = 2 \cdot 0.0836609 + 0.0422487 = 0{,}2096 \text{ m}$$

Source: [6]; [10]

The following reproduces the verified GOST 1786-88 extract in Opanasiuk's Table 11. These historical asbestos linings are included to reproduce the teaching method. Continuous allowable temperature is 200 °C for every row. A modern design requires the relevant asbestos-free lining specification and an explicit statement of the changed basis.

Table M.9.3 — Reference friction-lining dimensions and allowable operating conditions

Outer diameter, mm	Inner diameter, mm	Thickness, mm	Pressure, MPa	Short-term temperature, °C
160	110	3,3	0,14…0,25	300
180	125	3,5	0,14…0,25	300
184	125	3,5	0,14…0,25	300
190	130	3,5	0,14…0,25	350
200	130	3,3	0,14…0,25	300
200	140	3,5	0,14…0,25	300
204	146	3,3	0,14…0,25	300
254	150	3,5	0,14…0,25	350

Outer diameter, mm	Inner diameter, mm	Thickness, mm	Pressure, MPa	Short-term temperature, °C
280	164	3,5	0,14...0,25	300
300	164	4,0	0,14...0,25	350
340	186	4,0	0,14...0,30	350
342	186	4,0	0,14...0,30	350
350	200	4,7	0,14...0,30	350
400	220	4,2	0,14...0,30	250

For the 254 × 150 mm row, the course guide gives an upper pressure of 0,18 MPa and Opanasiuk gives 0,25 MPa. Check against the selected source. Choosing the nearest row alone does not guarantee capacity: if pressure is exceeded, adopt different geometry and repeat subsequent steps.

Actual mean friction radius from adopted diameters.

Refine mean radius using both adopted diameters. This is the point of application of resultant friction force in the selected model.

Table Д.9.5 — Inputs for (9.5): Actual mean friction radius from adopted diameters.

Symbol	Quantity	Unit	Provenance
D_a	adopted lining outer diameter	m	input data
d_a	adopted lining inner diameter	m	input data

Example — conventional internal-combustion vehicle

$$R_c = \frac{(D_a + d_a)}{4} \tag{9.5}$$

$$R_c = \frac{(0.254 + 0.15)}{4} = 0{,}101 \text{ m}$$

Source: [6]; [10]

Annular area of one friction lining.

Annular lining area determines mean contact pressure. This is the area of one surface; surface count is already included in the torque formula.

Table Д.9.6 — Inputs for (9.6): Annular area of one friction lining.

Symbol	Quantity	Unit	Provenance
D_a	adopted lining outer diameter	m	input data
d_a	adopted lining inner diameter	m	input data

Example — conventional internal-combustion vehicle

$$A_{lining} = \frac{\pi \cdot (D_a \cdot D_a - d_a \cdot d_a)}{4} \tag{9.6}$$

$$A_{lining} = \frac{\pi \cdot (0.254 \cdot 0.254 - 0.15 \cdot 0.15)}{4} = 0{,}033 \text{ m}^2$$

Source: [6]; [10]

Required normal force on the friction surfaces.

Find required axial force from the friction-torque equation. Increasing working radius or surface count reduces the required force. Compare it with the selected pressure mechanism and release actuator capability.

Table Д.9.7 — Inputs for (9.7): Required normal force on the friction surfaces.

Symbol	Quantity	Unit	Provenance
M_{p}	design clutch torque	N·m	calculated earlier (9.1)
μ	friction-lining coefficient of friction	—	input data
z	number of active friction surfaces	—	input data
R_{c}	mean working radius of the adopted lining	m	calculated earlier (9.5)

Example — conventional internal-combustion vehicle

$$P_{\text{n}} = \frac{M_{\text{p}}}{(\mu \cdot R_{\text{c}} \cdot z)} \tag{9.7}$$

$$P_{\text{n}} = \frac{306.565}{(0.375 \cdot 0.101 \cdot 2)} = 4047{,}1 \text{ N}$$

Source: [6]; [10]

Actual friction-lining pressure.

Compare actual pressure with the limits of the adopted row. Exceeding the upper limit requires a design change. A value below the recommended range indicates extra area and needs explanation, but does not mean lining overload.

Table Д.9.8 — Inputs for (9.8): Actual friction-lining pressure.

Symbol	Quantity	Unit	Provenance
$A_{\text{l}}ining$	area of one annular lining	m²	calculated earlier (9.6)
P_{n}	required lining clamping force	N	calculated earlier (9.7)

Example — conventional internal-combustion vehicle

$$q_{\text{actual}} = \frac{P_{\text{n}}}{A_{\text{lining}}} \tag{9.8}$$

$$q_{\text{actual}} = \frac{4047.06}{0.0329993} = 0{,}1226 \text{ MPa}$$

Source: [6]; [10]

Coil-spring set

Adopt spring count, mean diameter and positions from the layout. Required force determines minimum wire size; after adopting wire and coil count, check the actual assembly. Required force differs from the force developed by an installed spring.

Table M.9.4 — Coil pressure-spring parameters by method

Spring parameter	Course guide	Opanasiuk
Force multiplier for auxiliary-spring opposition	1	1,05...1,08
Load-unevenness factor	1,10...1,20	1,10...1,20
Allowable wire torsional stress, MPa	800	800
Wire shear modulus, GPa	85	85
Single-/twin-plate clutch release travel, mm	3 / 4	3 / 4
Inactive coils	1,5...2,0	1,5...2,0

Force per pressure spring.

Distribute clamping force among the springs. Opanasiuk's profile also includes return and separating spring resistance; the course profile omits this contribution. Spring count must be a positive integer.

Table Д.9.9 — Inputs for (9.9): Force per pressure spring.

Symbol	Quantity	Unit	Provenance
P_{n}	required lining clamping force	N	calculated earlier (9.7)
$z_{s}pring$	number of cylindrical pressure springs	—	input data
$c_{r}elease$	auxiliary-spring opposition factor	—	input data

Example — conventional internal-combustion vehicle

$$P_{\text{spring}} = \frac{c_{\text{release}} \cdot P_{\text{n}}}{z_{\text{spring}}} \tag{9.9}$$

$$P_{\text{spring}} = \frac{1.065 \cdot 4047.06}{12} = 359{,}2 \ \text{N}$$

Source: [6]; [10]

Design spring force during clutch release.

Increase design spring load using the method's release and unevenness factors. Use the result to determine minimum wire size.

Table Д.9.10 — Inputs for (9.10): Design spring force during clutch release.

Symbol	Quantity	Unit	Provenance
$k_{u}neven$	spring-load unevenness factor	—	input data
$P_{s}pring$	required force per pressure spring	N	calculated earlier (9.9)

Example — conventional internal-combustion vehicle

$$P_{\text{design}} = 1.2 \cdot k_{\text{uneVen}} \cdot P_{\text{spring}} \tag{9.10}$$

$$P_{\text{design}} = 1.2 \cdot 1.15 \cdot 359.177 = 495{,}7 \ \text{N}$$

Source: [6]; [10]

Required spring-wire diameter from allowable torsional stress.

Find wire diameter using allowable torsional stress. Then adopt an available diameter no smaller than required and record its source; a fractional calculated diameter is not yet a product selection.

Table Д.9.11 — Inputs for (9.11): Required spring-wire diameter from allowable torsional stress.

Symbol	Quantity	Unit	Provenance
D_{spring}	mean coil-spring diameter	m	input data
τ_{allow}	allowable wire torsional stress	MPa	input data
P_{design}	design force per spring	N	calculated earlier (9.10)

Example — conventional internal-combustion vehicle

$$d_{required} = \sqrt[3]{\frac{8 \cdot P_{design} \cdot D_{spring}}{(\pi \cdot \tau_{allow})}} \tag{9.11}$$

$$d_{required} = \sqrt[3]{\frac{8 \cdot 495.664 \cdot 0.03}{(\pi \cdot 800)}} = 0{,}0036 \text{ m}$$

Source: [6]; [10]

Limiting force at which adopted wire reaches allowable stress.

Rearranging the same strength condition gives allowable force for the adopted wire. Compare this project-derived limit with maximum actual force during release.

Table Д.9.12 — Inputs for (9.12): Limiting force at which adopted wire reaches allowable stress.

Symbol	Quantity	Unit	Provenance
D_{spring}	mean coil-spring diameter	m	input data
d_{wire}	adopted spring-wire diameter	m	input data
τ_{allow}	allowable wire torsional stress	MPa	input data

Example — conventional internal-combustion vehicle

$$P_{allow} = \frac{\pi \cdot d_{wire} \cdot d_{wire} \cdot d_{wire} \cdot \tau_{allow}}{(8 \cdot D_{spring})} \tag{9.12}$$

$$P_{allow} = \frac{\pi \cdot 0.004 \cdot 0.004 \cdot 0.004 \cdot 800}{(8 \cdot 0.03)} = 670{,}2 \text{ N}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Spring-force increase during clutch release.

The force increase between initial and design states, together with release travel, determines required spring compliance.

Table Д.9.13 — Inputs for (9.13): Spring-force increase during clutch release.

Symbol	Quantity	Unit	Provenance
$P_{s}pring$	required force per pressure spring	N	calculated earlier (9.9)
$P_{d}esign$	design force per spring	N	calculated earlier (9.10)

Example — conventional internal-combustion vehicle

$$dP_{\text{spring}}=P_{\text{design}}-P_{\text{spring}} \tag{9.13}$$

$$dP_{\text{spring}}=495.664-359.177=136,5\ \text{N}$$

Source: [6]; [10]

Active coil count of the pressure spring.

Determine active coil count from required force increase. After constructive rounding, recalculate stiffness using adopted count because rounding changes the force characteristic.

Table Д.9.14 — Inputs for (9.14): Active coil count of the pressure spring.

Symbol	Quantity	Unit	Provenance
$D_{s}pring$	mean coil-spring diameter	m	input data
$d_{w}ire$	adopted spring-wire diameter	m	input data
$travel$	additional deflection during disengagement	m	input data
$shear_{m}odulus$	shear modulus of the wire material	MPa	input data
$dP_{s}pring$	calculated spring-force increase	N	calculated earlier (9.13)

Example — conventional internal-combustion vehicle

$$n_{\text{working}}=\frac{traVel\cdot d_{\text{wire}}{}^{4}\cdot shear_{\text{modulus}}}{(8\cdot dP_{\text{spring}}\cdot D_{\text{spring}}{}^{3})} \tag{9.14}$$

$$n_{\text{working}}=\frac{0.003\cdot 0.004^{4}\cdot 85000}{(8\cdot 136.487\cdot 0.03^{3})}=2,214$$

Source: [6]; [10]

Total coil count of the pressure spring.

Add inactive end coils to active coils. Total count is needed for the drawing but must not replace active count in the stiffness calculation.

Table Д.9.15 — Inputs for (9.15): Total coil count of the pressure spring.

Symbol	Quantity	Unit	Provenance
$dead_{c}oils$	number of inactive coils	—	input data
$n_{w}orking$	calculated number of active coils	—	calculated earlier (9.14)

Example — conventional internal-combustion vehicle

$$n_{\text{total}}=n_{\text{working}}+dead_{\text{coils}} \tag{9.15}$$

$$n_{\text{total}}=2.21429+1.75=3,964$$

Source: [6]; [10]

Stiffness of the adopted pressure spring.

Calculate assembled spring stiffness from adopted diameters and active coil count. This project check is derived from the spring-deflection equation. Use consistent length units.

Table Д.9.16 — Inputs for (9.16): Stiffness of the adopted pressure spring.

Symbol	Quantity	Unit	Provenance
$D_{s}pring$	mean coil-spring diameter	m	input data
$d_{w}ire$	adopted spring-wire diameter	m	input data
$shear_{m}odulus$	shear modulus of the wire material	MPa	input data
$n_{a}dopted$	adopted number of active coils	—	input data

Example — conventional internal-combustion vehicle

$$C_{\text{spring}} = \frac{shear_{\text{modulus}} \cdot d_{\text{wire}}{}^{4}}{(8 \cdot D_{\text{spring}}{}^{3} \cdot n_{\text{adopted}})} \tag{9.16}$$

$$C_{\text{spring}} = \frac{85000 \cdot 0.004^{4}}{(8 \cdot 0.03^{3} \cdot 2)} = 50370{,}4 \ \text{N/m}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Force developed by one spring at working deflection.

Initial deflection comes from assembly geometry: free length minus installed length. Stiffness multiplied by this deflection gives actual engaged-state spring force.

Table Д.9.17 — Inputs for (9.17): Force developed by one spring at working deflection.

Symbol	Quantity	Unit	Provenance
$f_{i}nstalled$	initial spring deflection in the assembly	m	input data
$C_{s}pring$	stiffness of the adopted spring	N/m	calculated earlier (9.16)

Example — conventional internal-combustion vehicle

$$P_{\text{deVeloped}} = C_{\text{spring}} \cdot f_{\text{installed}} \tag{9.17}$$

$$P_{\text{deVeloped}} = 50370.4 \cdot 0.0075 = 377{,}8 \ \text{N}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Maximum spring force during clutch release.

Add release travel to initial deflection. Check maximum force against allowable wire force and coil-bind travel margin where the necessary geometry is specified.

Table Д.9.18 — Inputs for (9.18): Maximum spring force during clutch release.

Symbol	Quantity	Unit	Provenance
travel	additional deflection during disengagement	m	input data
$f_{i}nstalled$	initial spring deflection in the assembly	m	input data
$C_{s}pring$	stiffness of the adopted spring	N/m	calculated earlier (9.16)

Example — conventional internal-combustion vehicle

$$P_{\text{maximum}}=C_{\text{spring}}\cdot(f_{\text{installed}}+traVel) \tag{9.18}$$

$$P_{\text{maximum}}=50370.4\cdot(0.0075+0.003)=528{,}9\ \text{N}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Available clamping force on the friction surfaces.

Sum actual spring forces and account for auxiliary opposition using the same profile as at the start. This is the force available to the linings in the adopted design.

Table Д.9.19 — Inputs for (9.19): Available clamping force on the friction surfaces.

Symbol	Quantity	Unit	Provenance
$z_{s}pring$	number of cylindrical pressure springs	—	input data
$c_{r}elease$	auxiliary-spring opposition factor	—	input data
$P_{d}eveloped$	actual spring force when engaged	N	calculated earlier (9.17)

Example — conventional internal-combustion vehicle

$$P_{\text{net}}=\frac{z_{\text{spring}}\cdot P_{\text{deVeloped}}}{c_{\text{release}}} \tag{9.19}$$

$$P_{\text{net}}=\frac{12\cdot 377.778}{1.065}=4256{,}7\ \text{N}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Clutch torque capacity from available spring force.

The final check returns to torque: adopted geometry and actual force must provide at least the design torque. If not, change springs, assembly or lining and repeat force, pressure and strength checks.

Table Д.9.20 — Inputs for (9.20): Clutch torque capacity from available spring force.

Symbol	Quantity	Unit	Provenance
R_c	mean working radius of the adopted lining	m	calculated earlier (9.5)
μ	friction-lining coefficient of friction	—	input data
z	number of active friction surfaces	—	input data
P_net	actual lining clamping force from the spring set	N	calculated earlier (9.19)

Example — conventional internal-combustion vehicle

$$M_{capacity}=\mu \cdot P_{net} \cdot R_c \cdot z \tag{9.20}$$

$$M_{capacity}=0.375 \cdot 4256.65 \cdot 0.101 \cdot 2=322,4 \ \text{N·m}$$

Source: [6]; [10]

Actual lining pressure from the available force of the adopted spring set using (9.8).

Substitute the available spring-set force from (9.19) into (9.8) to obtain assembled pressure: $q_ф=P_{наяв}/A$. Use lining area from (9.6). Compare this pressure with the adopted limit separately from preliminary pressure based on required force. Adequate torque under (9.20) does not eliminate lining overload caused by excessive spring force.

Table П.9.1 — Inputs to the adopted decision: Actual lining pressure from the available force of the adopted spring set using (9.8).

Symbol	Quantity	Unit	Provenance
A_lining	area of one annular lining	m²	calculated earlier (9.6)
P_net	actual lining clamping force from the spring set	N	calculated earlier (9.19)

Example — conventional internal-combustion vehicle

$$q_{deVeloped}=\frac{P_{net}}{A_{lining}}$$

$$q_{deVeloped}=\frac{4256.65}{0.0329993}=0,129 \ \text{MPa}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Preliminary spring solid length with adopted active coils.

After adopting active coil count, return to the principle of (9.15) and add inactive coils. For preliminary geometry, the project rounds this total upwards to an integer and multiplies it by wire diameter: $L_{суц}=\lceil n_{пр}+n_{непр} \rceil d$. This is a conservative project convention for solid length, not a manufacturer spring specification or a manufacturing-process check.

Table П.9.3 — Inputs to the adopted decision: Preliminary spring solid length with adopted active coils.

Symbol	Quantity	Unit	Provenance
$d_{w}ire$	adopted spring-wire diameter	m	input data
$dead_{c}oils$	number of inactive coils	—	input data
$n_{a}dopted$	adopted number of active coils	—	input data

Example — conventional internal-combustion vehicle

$$L_{\text{solid}} = \lceil n_{\text{adopted}} + dead_{\text{coils}} \rceil \cdot d_{\text{wire}}$$

$$L_{\text{solid}} = \lceil 2 + 1.75 \rceil \cdot 0.004 = 0{,}016 \ \text{m}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Remaining coil-bind clearance during clutch release.

Subtract additional release travel and the solid length calculated above from installed spring length: $s_{\text{вим}} = L_{\text{уст}} - \Delta f - L_{\text{суц}}$. Use metres for all three lengths. A positive result is remaining travel within this geometric model; zero or a negative result fails. Without installed length, clearance remains unchecked rather than being replaced by zero.

Table П.9.2 — Inputs to the adopted decision: Remaining coil-bind clearance during clutch release.

Symbol	Quantity	Unit	Provenance
$travel$	additional deflection during disengagement	m	input data
$L_{i}nstalled$	spring length in the installed, engaged state	m	input data
$L_{s}olid$	preliminary solid length at coil contact	m	calculated earlier

Example — conventional internal-combustion vehicle

$$s_{\text{release}} = L_{\text{installed}} - traVel - L_{\text{solid}}$$

$$s_{\text{release}} = 0.034 - 0.003 - 0.016 = 0{,}015 \ \text{m}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Diaphragm spring

A diaphragm spring is an alternative pressure mechanism; do not add its force to the previous coil-spring set. Distinguish outer diameter, pressure-plate contact diameter, inner diameter of the continuous portion and release-bearing contact diameter on the drawing.

Table M.9.5 — Initial diaphragm-spring geometry by method

Diaphragm-spring assumptions	Course guide	Opanasiuk
Outer diameter to continuous-portion inner-diameter ratio	1,2…1,5	1,2…1,5
Outer diameter to finger inner-diameter ratio	2,5…3,5	2,5…3,5
Outer diameter to thickness ratio	75…100	75…100
Height to thickness ratio	1,5…2,0	1,6…2,8
Thickness for passenger cars / trucks, mm	2…2,5 / 3…5	No separate range provided
Finger count	8…20	No separate range provided
Allowable stress, MPa	1400	1400
Elastic modulus, GPa; Poisson's ratio	210; 0,26	210; 0,26

Force required to release a diaphragm-spring clutch.

Determine release-bearing force from lever equilibrium. The contact diameter must lie between inner and outer diameters. A zero or negative lever arm means invalid geometry rather than a finite release force.

Table Д.9.21 — Inputs for (9.21): Force required to release a diaphragm-spring clutch.

Symbol	Quantity	Unit	Provenance
P_{n}	required lining clamping force	N	calculated earlier (9.7)
$D_{o}uter$	diaphragm-spring outer diameter	m	input data
$D_{c}ontact$	pressure-plate contact diameter	m	input data
$D_{i}nner$	inner diameter of the spring fingers	m	input data

Example — diaphragm spring following Opanasiuk

$$P_{\text{release}} = \frac{P_{\text{n}} \cdot (D_{\text{outer}} - D_{\text{contact}})}{(D_{\text{contact}} - D_{\text{inner}})} \tag{9.21}$$

$$P_{\text{release}} = \frac{4047.06 \cdot (0.254 - 0.2)}{(0.2 - 0.079375)} = 1811{,}7 \text{ N}$$

Source: [6]; [10]

Logarithmic mean of continuous-portion outer and inner diameters.

The logarithmic mean of the continuous portion's two diameters enters the stress check. Both diameters must be positive and the outer larger than the inner.

Table Д.9.22 — Inputs for (9.22): Logarithmic mean of continuous-portion outer and inner diameters.

Symbol	Quantity	Unit	Provenance
$D_{o}uter$	diaphragm-spring outer diameter	m	input data
$D_{d}isc$	inner diameter of the continuous spring portion	m	input data

Example — diaphragm spring following Opanasiuk

$$D_{\log}=\frac{(D_{\text{outer}}-D_{\text{disc}})}{\ln\left(\frac{D_{\text{outer}}}{D_{\text{disc}}}\right)} \tag{9.22}$$

$$D_{\log}=\frac{(0.254-0.189552)}{\ln\left(\frac{0.254}{0.189552}\right)}=0{,}2202\ \text{m}$$

Source: [6]; [10]

Diaphragm-spring finger stress.

The books assess different section points and give different coefficients. The first term is shared; take the thickness-term sign and second-term factors entirely from the selected profile. Interpreting the sign difference as a section-point difference is a project interpretation, not the sources' explanation.

(9.23) **Applicability:** Diaphragm-spring check using the course guide, p. 40.

This separate comparison retains Opanasiuk's clamping force from the shared friction calculation and the same spring dimensions. The spring check uses the course guide's complete coefficient set, evaluating the finger-root midpoint when flattened.

Table Д.9.23.1 — Inputs for (9.23): Diaphragm-spring finger stress.

Symbol	Quantity	Unit	Provenance
$D_{o}uter$	diaphragm-spring outer diameter	m	input data
$D_{i}nner$	inner diameter of the spring fingers	m	input data
$D_{d}isc$	inner diameter of the continuous spring portion	m	input data
$t_{s}pring$	diaphragm-spring thickness	m	input data
$h_{s}pring$	initial diaphragm-spring height	m	input data
$E_{m}aterial$	elastic modulus of the diaphragm-spring material	MPa	input data
$\mu_{m}aterial$	Poisson's ratio of the diaphragm-spring material	—	input data
$c_{a}lpha$	cone-angle multiplier for the selected method	—	input data
$c_{b}ending$	bending-term multiplier for the selected method	—	input data
$c_{t}hickness$	signed thickness-term multiplier for the selected method	—	input data
$c_{m}odulus$	elastic-modulus multiplier for the selected method	—	input data
$P_{r}elease$	release-bearing force	N	calculated earlier (9.21)
$D_{l}og$	logarithmic mean diameter of the continuous portion	m	calculated earlier (9.22)

Example — diaphragm spring following the course guide

$$\sigma_{\text{spring}} = \frac{2 \cdot P_{\text{release}} \cdot D_{\text{disc}}}{(t_{\text{spring}}^{2} \cdot (D_{\text{inner}} + D_{\text{disc}}))} + \frac{(c_{\text{bending}} \cdot (D_{\text{log}} - D_{\text{disc}}) \cdot (\frac{c_{\alpha} \cdot h_{\text{spring}}}{(D_{\text{outer}} - D_{\text{disc}})})^{2} + c_{\text{thickness}} \cdot t_{\text{spring}} \cdot (\frac{c_{\alpha} \cdot h_{\text{spring}}}{(D_{\text{outer}} - D_{\text{disc}})}))}{D_{\text{disc}}} \cdot c_{\text{modulus}} \cdot E_{\text{material}} \quad (9.23)$$

$$\sigma_{\text{spring}} = \frac{2 \cdot 1811.74 \cdot 0.189552}{(0.0025^{2} \cdot (0.079375 + 0.189552))} + \frac{(0.5 \cdot (0.220207 - 0.189552) \cdot (\frac{1 \cdot 0.005}{(0.254 - 0.189552)})^{2} + -0.5 \cdot 0.0025 \cdot (\frac{1 \cdot 0.005}{(0.254 - 0.189552)}))}{0.189552} \cdot 1 \cdot 2.1 \cdot 10^{+5}}{(1 - 0.26^{2})} = 403 \text{ МПа}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

(9.23) **Applicability:** The same diaphragm spring checked using Opanasiuk, p. 32.

Use Opanasiuk's coefficients for the midpoint of fingers in the slotted portion at maximum deflection. Keeping geometry and force equal to the preceding example isolates the methods' difference.

Table Д.9.23.2 — Inputs for (9.23): Diaphragm-spring finger stress.

Symbol	Quantity	Unit	Provenance
$D_{o}uter$	diaphragm-spring outer diameter	m	input data
$D_{i}nner$	inner diameter of the spring fingers	m	input data
$D_{d}isc$	inner diameter of the continuous spring portion	m	input data
$t_{s}pring$	diaphragm-spring thickness	m	input data
$h_{s}pring$	initial diaphragm-spring height	m	input data
$E_{m}aterial$	elastic modulus of the diaphragm-spring material	MPa	input data
$\mu_{m}aterial$	Poisson's ratio of the diaphragm-spring material	—	input data
$c_{a}lpha$	cone-angle multiplier for the selected method	—	input data
$c_{b}ending$	bending-term multiplier for the selected method	—	input data
$c_{t}hickness$	signed thickness-term multiplier for the selected method	—	input data
$c_{m}odulus$	elastic-modulus multiplier for the selected method	—	input data
$P_{r}elease$	release-bearing force	N	calculated earlier (9.21)
$D_{l}og$	logarithmic mean diameter of the continuous portion	m	calculated earlier (9.22)

Example — diaphragm spring following Opanasiuk

$$\sigma_{\text{spring}} = \frac{2 \cdot P_{\text{release}} \cdot D_{\text{disc}}}{(t_{\text{spring}}^{2} \cdot (D_{\text{inner}} + D_{\text{disc}}))} + \frac{(c_{\text{bending}} \cdot (D_{\text{log}} - D_{\text{disc}}) \cdot (\frac{c_{\alpha} \cdot h_{\text{spring}}}{(D_{\text{outer}} - D_{\text{disc}})})^{2} + c_{\text{thickness}} \cdot t_{\text{spring}} \cdot (\frac{c_{\alpha} \cdot h_{\text{spring}}}{(D_{\text{outer}} - D_{\text{disc}})})) \cdot c_{\text{modulus}} \cdot E_{\text{material}}}{\frac{D_{\text{disc}}}{(1 - \mu_{\text{material}}^{2})}} \quad (9.23)$$

$$\sigma_{\text{spring}} = \frac{2 \cdot 1811.74 \cdot 0.189552}{(0.0025^{2} \cdot (0.079375 + 0.189552))} + \frac{(0.5 \cdot (0.220207 - 0.189552) \cdot (\frac{2 \cdot 0.005}{(0.254 - 0.189552)})^{2} + 1 \cdot 0.0025 \cdot (\frac{2 \cdot 0.005}{(0.254 - 0.189552)}))}{\frac{0.189552}{0.5 \cdot 2.1 \cdot 10^{+5}}} = 858{,}3 \text{ МПа}$$

$$\frac{}{(1 - 0.26^{2})}$$

- This relationship is derived within the project; the sources do not print it in this form.

Source: [7]

Table M.9.6 — Diaphragm-spring stress-check coefficients

Coefficient in the parameterised stress expression	Course guide	Opanasiuk
Cone-angle multiplier	1	2
Bending-term multiplier	0,5	0,5
Thickness-term multiplier	−0,5	+1
Elastic-modulus multiplier	1	0,5

Hub splines and release levers

Perform these checks during detailed clutch design. Take spline dimensions, lever arms and critical-section modulus from the design. Heat-treated steels 35 or 40X and hub length approximately equal to spline outer diameter remain teaching-example assumptions. Buriennikov, Kashkanov and Rebedailo, p. 35, (3.22)–(3.24), give spline checks using circumferential force and contact area.

Driven-disc hub splines

Check spline bearing pressure and shear. The respective ranges 15…30 and 5…15 MPa are retained as teaching-project parameters, not allowable stresses attributed to a new reference source. The shear expression uses mean diameter, so its denominator contains the sum of outer and inner diameters. This follows from torque equilibrium and shear area, consistent with (3.23)–(3.24), p. 35 of the cited text, at adopted load-sharing coefficient 1. Show the alternative expression using diameter difference separately.

Table Д.9.24 — Inputs for (9.24): Driven-plate hub splines

Symbol	Quantity	Unit	Provenance
M_{p}	design clutch torque	N·m	calculated earlier (9.1)

Example — additional clutch checks

$$\sigma_{\text{см}} = \frac{8 M_{\max}}{(d_{\text{н}}^{2} - d_{\text{в}}^{2}) z l_{\text{ш}}}, \qquad \tau_{\text{ср}} = \frac{4 M_{\max}}{(d_{\text{н}} + d_{\text{в}}) z l_{\text{ш}} b_{\text{ш}}} \quad (9.24)$$

$$\sigma_{\text{см}} = 18.2 \text{ MPa}, \ \tau_{\text{ср}} = 13.7 \text{ MPa}$$

- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.
- Departure from the source: the shear denominator uses ($d_н + d_в$) rather than the printed ($d_н - d_в$).

Source: [22]

Clutch release levers

Calculate lever bending stress from force, lever-arm ratio, lever count and section modulus. The allowable range 320…400 MPa is a teaching-project assumption. Check section enlargement together with required release-bearing travel and clearances.

Table Д.9.25 — Inputs for (9.25): Clutch release levers

Symbol	Quantity	Unit	Provenance
P_n	required lining clamping force	N	calculated earlier (9.7)

Example — additional clutch checks

$$\sigma_{изг} = \frac{P_{пр.в} l}{u z W_{изг}}, \qquad u = \frac{l}{f} \tag{9.25}$$

$$u = 4.00: \ \sigma_{изг} = 81 \ \text{MPa}$$

- Lever bending stress must not exceed 320…400 MPa (the project's reference assumption). Levers use malleable cast iron or low- or medium-carbon steel.
- u = l/f is the lever ratio: the factor multiplying release-sleeve travel and dividing its force.
- Inner lever ends contacting the release bearing wear most. They are therefore case-hardened or nitrided and fitted with a thrust ring.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

Source: [22]

Thermal verification requires slip work, heat distribution and component heat capacities. Lining temperature limits remain example assumptions; no complete thermal model is constructed here. A temperature row alone does not prove thermal capability.

Chapter 10. Gearbox and electric-drive reduction gear

From specified ratios to integer teeth, common module, strength checks and housing dimensions.

Inputs are the selected traction-source characteristic and its path ratios. For an engine, the series comes from traction calculation. For an electric machine, use adopted reduction rather than deriving it from an engine's narrow speed band.

For parallel hybrids, including plug-in variants, explicitly select the engine gearbox or machine reduction gear. A series hybrid has an electric wheel path. Reduction stages work in series and multiply their ratios; selectable gears are alternative paths. These are different design features.

Kinematic layout and tooth counts

A two-shaft layout uses one working pair. A three-shaft layout uses the constant-mesh pair and selected-gear pair; direct drive couples shafts without transmitting through those pairs. A single-stage electric reducer corresponds to the two-shaft layout and a two-stage reducer to the three-shaft layout. Preliminary values and ratio-deviation checks follow the course guide, pp. 44–47, and Opanasiuk, pp. 33–37.

Table M.10.1 — Reference gearbox gear-pair selection parameters

Assumption	Value and condition
Constant-mesh ratio	1,8…2,5; applicable to a three-shaft layout
Driving pinion tooth count for this pair	17…23
Tooth-count sum in each pair	Equal with common module, helix angle and parallel shafts
Allowable actual-ratio deviation	Absolute relative deviation no greater than 5 %

These are starting recommendations in the current automatic design. If the common tooth sum is insufficient, the project algorithm increases it until every pinion and gear has at least 17 teeth, retaining first-gear, top-gear and shift-step requirements. Extension beyond 17…23 is not attributed to the books; check size and strength separately. A manually selected infeasible design is retained with its rejection reason.

Candidate gearbox tooth counts (not yet adopted)

Select integer counts using a common sum, calculate actual ratios and check deviation. Prefer pairs without a common tooth-count divisor. Meeting the ratio tolerance alone does not justify low tooth counts, particularly below the recommended range.

(10.2) **Applicability:** Three-shaft gearbox in the engine's mechanical path.

Every selectable gear has its own working pair and shares the constant-mesh pair. After rounding, check every gear separately.

Table Д.10.2.1 — Inputs for (10.2): Candidate gearbox tooth counts (not yet adopted)

Symbol	Quantity	Unit	Provenance
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)
—	adopted gearbox kinematic layout	—	adopted by the designer
$u_{п}$	adopted constant-mesh gear pair	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$u_{кi}=\frac{Z_2}{Z_1}\cdot\frac{Z_{iв}}{Z_{iп}} \qquad (10.2)$$

$$\Sigma Z=63$$

$$u_{к1}=\frac{43}{20}\cdot\frac{24}{39}=1.32308$$

- Every pair has the same tooth sum: 63. This maintains parallel shafts with equal module and helix angle.
- Torque passes through the constant-mesh pair and the selected-gear pair. Direct drive bypasses the gears entirely.
- The historical teaching example retains the specified traction ratios. Integer counts below are preliminary candidates; their ratios have not been adopted into the traction path and do not establish a finished design.
- Tooth counts cannot be fractional, so the exact calculated ratio is generally unattainable; the method allows up to 5 % deviation.
- Some pairs have a common divisor. Avoiding common divisors is secondary to integer counts, end ratios and shift-step requirements.

Source: [6]; [10]

(10.2) **Applicability:** Two-stage electric-machine or series-hybrid reduction gear.

Both pairs operate together. Divide a large overall ratio between stages to avoid a geometrically unsuitable small pinion. With separate wheel reduction, the gearbox target is overall ratio divided by wheel ratio. Electric-reducer integer pairs remain preliminary candidates: they do not automatically replace declared traction-path ratios or establish a completed design.

Table Д.10.2.2 — Inputs for (10.2): Candidate gearbox tooth counts (not yet adopted)

Symbol	Quantity	Unit	Provenance
$u_{кj}$	adopted gear-ratio series	—	calculated earlier
—	adopted gearbox kinematic layout	—	adopted by the designer
$u_{п}$	adopted constant-mesh gear pair	—	adopted by the designer

Example — electric vehicle

$$u_{кi}=\frac{Z_2}{Z_1}\cdot\frac{Z_{iв}}{Z_{iп}} \qquad (10.2)$$

$$\Sigma Z=70$$

$$u_{к1}=\frac{53}{17}\cdot\frac{53}{17}=9.71972$$

- Every pair has the same tooth sum: 70. This maintains parallel shafts with equal module and helix angle.
- Torque passes through the constant-mesh pair and the selected-gear pair. Direct drive bypasses the gears entirely.
- The historical teaching example retains the specified traction ratios. Integer counts below are preliminary candidates; their ratios have not been adopted into the traction path and do not establish a finished design.
- Tooth counts cannot be fractional, so the exact calculated ratio is generally unattainable; the method allows up to 5 % deviation.

Source: [6]; [10]

(10.3) **Applicability:** Two-shaft gearbox or single-stage reducer with one working pair.

The ratio follows directly from the two gears' tooth counts. This layout has no additional constant-mesh multiplier.

Table Д.10.3 — Inputs for (10.3): Candidate gearbox tooth counts (not yet adopted)

Symbol	Quantity	Unit	Provenance
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)
—	adopted gearbox kinematic layout	—	adopted by the designer
$u_{п}$	adopted constant-mesh gear pair	—	adopted by the designer

Example — two-shaft gearbox

$$u_{кi}=\frac{Z_{2i}}{Z_{1i}} \tag{10.3}$$

$$\Sigma Z=63$$

$$u_{к1}=\frac{36}{27}=1.33333$$

- Every pair has the same tooth sum: 63. This maintains parallel shafts with equal module and helix angle.
- Torque passes through one pair, so ratio is simply the tooth-count ratio.
- The historical teaching example retains the specified traction ratios. Integer counts below are preliminary candidates; their ratios have not been adopted into the traction path and do not establish a finished design.
- Tooth counts cannot be fractional, so the exact calculated ratio is generally unattainable; the method allows up to 5 % deviation.
- Some pairs have a common divisor. Avoiding common divisors is secondary to integer counts, end ratios and shift-step requirements.

Source: [6]; [10]

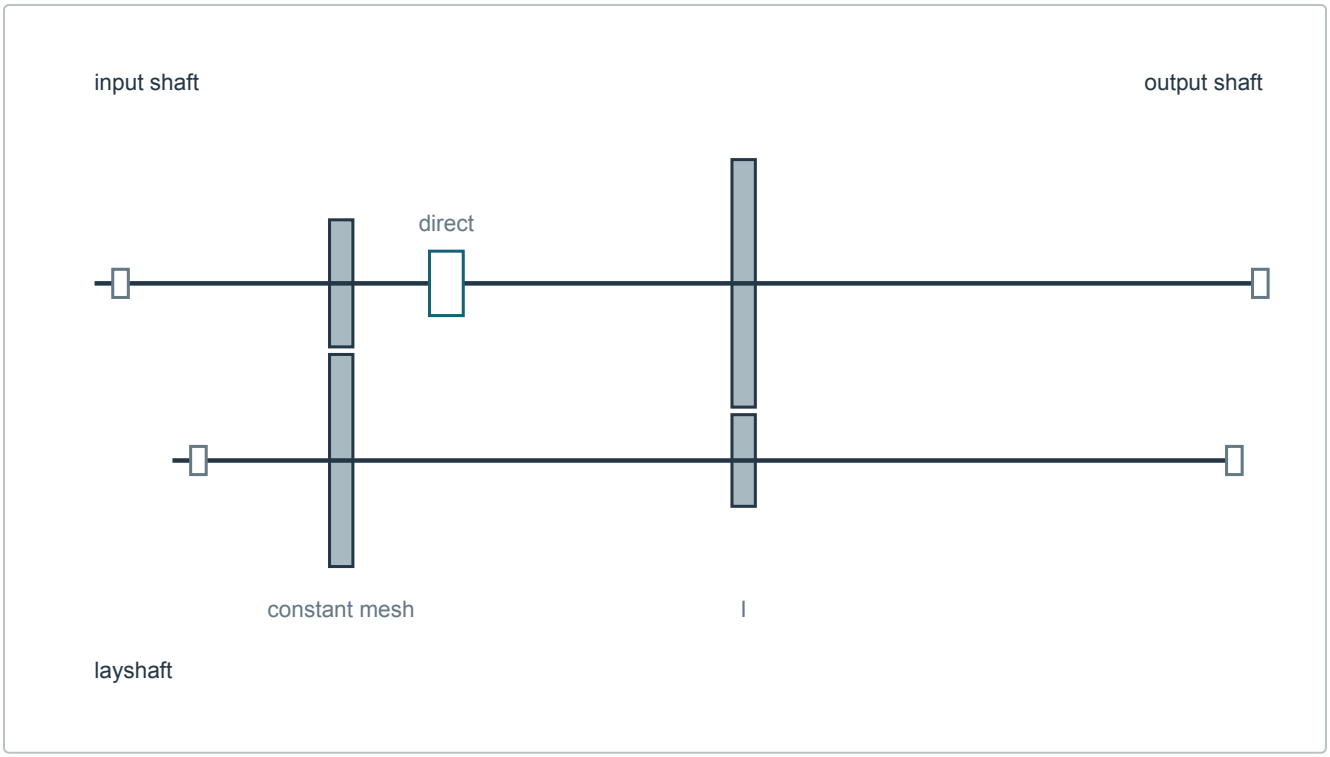


Figure 10.1 — Worked example's three-shaft gearbox kinematic layout. Torque paths are shown; the drawing does not establish tooth scale or structural dimensions.
Source: [6].

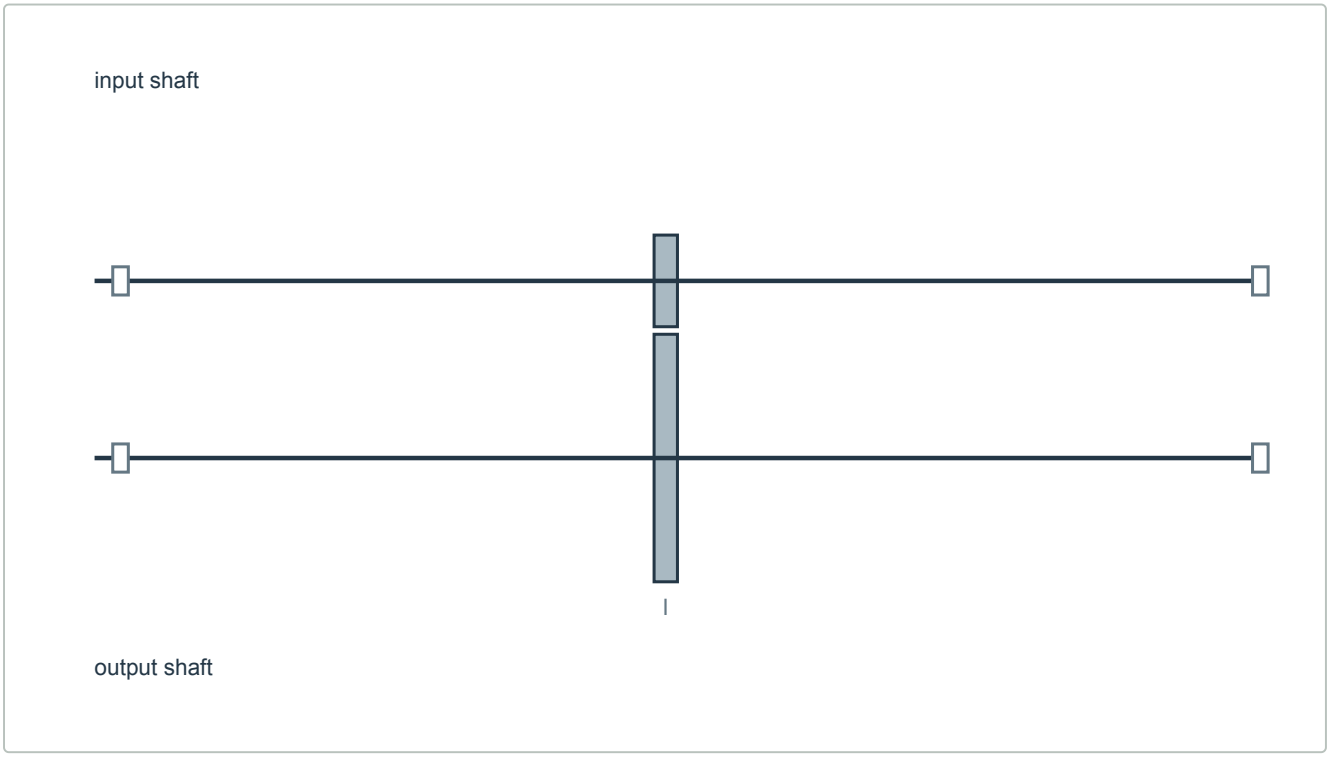


Figure 10.2 — Alternative two-shaft layout for the same example. One working pair transmits torque in the selected gear.
Source: [6].

Module and tooth strength

Determine module from bending in each loaded pair. Include the dry-clutch reserve factor only in a mechanical path containing that clutch. Electric reduction uses machine torque without

this factor. The selected method defines helix angle, width and contact coefficient.

Table M.10.2 — Helix angles and face widths by method

Method and design	Helix angle, deg	Face width, mm
Course guide: passenger car	25…35	15…25
Course guide: truck	20…30	20…25
Opanasiuk: passenger car, three shafts	22…34	15…25
Opanasiuk: passenger car, two shafts	20…25	15…25
Opanasiuk: truck or bus	18…26	20…40

Take tooth-form factor from Table 12 at the nearest virtual tooth count. For helical gears, divide actual tooth count by the cube of helix-angle cosine. Results outside the table remain preliminary and require separate justification.

Table M.10.3 — Tooth-form factor by virtual tooth count

Virtual tooth count	Form factor	Virtual tooth count	Form factor
16	0,101	28	0,117
17	0,102	30	0,120
18	0,104	32	0,123
19	0,105	35	0,128
20	0,106	37	0,131
21	0,108	40	0,136
22	0,110	45	0,142
23	0,112	50	0,145
24	0,114	60	0,150

Use allowable bending stresses from Table 13 of either method. The heavy group covers payload above 5000 kg; at the boundary retain the selected profile's rule and state the category.

Table M.10.4 — Allowable gear-pair bending stresses

Working pair	Light group, MPa	Heavy group, MPa
First gear and reverse	750…850	500…650
Constant mesh and remaining gears	350…450	175…275

Normal gear module

Adopt one common module no smaller than the greatest required value. Round upwards to an available standard series. Identify the governing pair and distinguish calculated minimum from adopted module.

Table Д.10.5 — Inputs for (10.5): Normal gear module

Symbol	Quantity	Unit	Provenance
Z_{ij}	gearbox tooth counts and actual ratios	—	calculated earlier (10.2)
β'	adopted helix angle	—	adopted by the designer
is	adopted gear face width	mm	adopted by the designer
m	adopted gear module	mm	adopted by the designer

Example — conventional internal-combustion vehicle

$$m = 10^3 \sqrt{\frac{2M_{розр}\cos\beta'}{\pi z y i s k_\sigma}} \tag{10.5}$$

$$m_1 = 10^3 \sqrt{\frac{2 \cdot 306.565 \cdot \cos(30.000^\circ)}{\pi \cdot 20 \cdot 0.1200 \cdot (20.000 \cdot 10^{-3}) \cdot (400.00 \cdot 10^6)}} = 2.967 \text{ mm}$$

$$m_2 = 10^3 \sqrt{\frac{2 \cdot 659.115 \cdot \cos(30.000^\circ)}{\pi \cdot 39 \cdot 0.1500 \cdot (20.000 \cdot 10^{-3}) \cdot (800.00 \cdot 10^6)}} = 1.970 \text{ mm}$$

$$m_{потр} = \max\{2.967,\ 1.970\} = 2.967 \text{ mm}$$

$$m_{прийн} = 3.00 \text{ mm (adopted)}$$

- m1 is the module required by the countershaft-drive pair. The substitution converts width from mm to m and stress from MPa to Pa; the outer factor 1000 returns module in mm.
- m2 is the module required by the first-gear pair. The substitution converts width from mm to m and stress from MPa to Pa; the outer factor 1000 returns module in mm.
- One module applies to the whole gearbox: equal tooth sums require equal modules, so adopt the largest required value.
- The countershaft-drive pair requires the largest module.
- Tooth-form factor comes from a discrete table at the nearest virtual tooth count.

Source: [6]; [10]

Table M.10.5 — Standard gear-module series

Module series	Value, mm
Preferred	2; 2,5; 3; 4; 5; 6; 8; 10; 12; 16
Supplementary	2,25; 2,75; 3,5; 4,5; 5,5; 7; 9; 11; 14; 18

The course guide cites GOST 9563-60; Opanasiuk cites OST 37.001.223-80. These historical source series are reproduced for teaching selection.

Tooth contact stresses

Check contact after adopting module. Steel elastic modulus is 210 GPa and pressure angle 20°. The contact coefficient is 0,418 in the course guide and 0,4118 in Opanasiuk. Limits are 1500…3000 MPa for spur and 1000…2500 MPa for helical gears. Exceeding the upper limit requires geometry changes and rechecking.

Both sources print a difference of reciprocal radii, whereas classical Hertz external-contact theory uses their sum. The calculation reproduces the printed expression and separately reports the sum-based result. Do not conceal this discrepancy when judging capability.

Table Д.10.6 — Inputs for (10.6): Tooth contact stresses

Symbol	Quantity	Unit	Provenance
m	common gearbox gear module	mm	calculated earlier (10.5)
β'	adopted helix angle	—	adopted by the designer
—	selected gearbox calculation method	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$\sigma_{ст}=k\cos\beta'\sqrt{\frac{PE}{is_0\sin\alpha\cos\alpha}\left(\frac{1}{r_{01}}-\frac{1}{r_{02}}\right)} \tag{10.6}$$

$$\sigma_{ст}=0.4180\cdot\cos(30.000^{\circ})\sqrt{\frac{8849.767\cdot210000}{37.3210\cdot\sin(20.00^{\circ})\cdot\cos(20.00^{\circ})}\left(\frac{1}{34.6410}-\frac{1}{74.4782}\right)}=559.91\ \text{MPa}$$

- The countershaft-drive pair is checked, as the most highly loaded of the calculated pairs.
- Coefficient 0.418 follows the selected method; the books print 0,418 and 0,4118.
- Both books print a MINUS in parentheses: (1/r₀₁ − 1/r₀₂). Classical Hertz external-contact theory uses plus. The printed form is calculated here; plus would give 927 MPa.

Source: [6]; [10]

Layout dimensions

Centre distance from gear geometry

Adopted gear geometry determines centre distance. It cannot be reduced arbitrarily while retaining module and tooth counts. The course guide prescribes the geometric approach; the explicit equality is derived within this project.

Table Д.10.7 — Inputs for (10.7): Centre distance from gear geometry

Symbol	Quantity	Unit	Provenance
Z_{ij}	adopted integer tooth counts	—	adopted by the designer
m	common gearbox gear module	mm	calculated earlier (10.5)
β'	adopted helix angle	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$a_w=\frac{m(Z_1+Z_2)}{2\cos\beta'} \tag{10.7}$$

$$a_w=\frac{3.0\cdot63}{2\cos30^{\circ}}=109.1\ \text{mm}$$

- The course guide prescribes deriving centre distance from gear dimensions but prints no separate equation.
- The distance is common to the entire gearbox: equal tooth sums give the same value for every pair.
- The equality is standard involute geometry, not an expression printed by the book. The book supplies the prescribed approach.

Source: [7]

Centre distance from torque

Use Opanasiuk's approximate output-torque estimate as an independent comparison. Its coefficient is 8,9…9,3 for passenger cars and 8,6…9,6 for trucks and buses. The source gives this formula for three-shaft gearboxes, not a two-shaft check. Explain any discrepancy with geometry before drawing the layout.

Table Д.10.8 — Inputs for (10.8): Centre distance from torque

Symbol	Quantity	Unit	Provenance
$u_{кj}$	adopted gear-ratio series	—	calculated earlier (4.19)
k_{α}	preliminary centre-distance coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$a_w \approx k_{\alpha} \sqrt[3]{M_{вих}}, \; M_{вих} = M_{emax} u_1 \tag{10.8}$$

$$a_w \approx 9.10 \sqrt[3]{275.0} = 59.2 \text{ mm}$$

- $M_{вих}$ = 207.8 · 1.323 = 275.0 N·m
- k_{α} = 8.9…9.3 for passenger cars.

- Equation (53) is printed only for three-shaft gearboxes.
- This value is approximate; the book explicitly describes it as such.

Source: [10]

Gearbox axial dimension

Sum gear, bearing and coupling widths. Read their counts from the actual kinematic layout; do not assign synchronisers to a non-shifting electric reducer. Opanasiuk's formula gives an internal layout length excluding walls and flanges.

Table Д.10.9 — Inputs for (10.9): Gearbox axial dimension

Symbol	Quantity	Unit	Provenance
m,n,c	counts of gears, bearings and couplings on the shafts	—	adopted by the designer
is,B,H	relative widths of gearbox components	—	adopted by the designer
a_w	geometric centre distance	mm	calculated earlier (10.7)

Example — conventional internal-combustion vehicle

$$l = m\text{is} + nB + cH \tag{10.9}$$

$$l = 2 \cdot 24.0 + 4 \cdot 26.2 + 1 \cdot 67.1 = 220 \text{ mm}$$

- Here m is the NUMBER of gears, not module; the book uses the same letter differently in adjacent equations.
- Widths use midpoint proportions: в ≈ 24.0 mm, B ≈ 26.2 mm, H ≈ 67.1 mm.

- Housing internal length excluding walls and flanges; the actual gearbox is longer.
- Only Opanasiuk prints equation (54), so this step is independent of the selected gearbox method: the course guide gives no separate axial-length expression.

- Counts follow the kinematic layout (Figure 10.1) and gear count; enter them manually if the design differs.

Source: [10]

Table M.10.6 — Approximate gearbox component widths

Component	Approximate width as a centre-distance fraction
Gear	0,19…0,25
Bearing	0,20…0,28
Synchroniser toothed coupling	0,45…0,78

Axial proportions follow Opanasiuk, p. 33, equation (54). The letter denoting gear count in the length formula is not tooth module. Refine walls, supports, lubrication passages and service clearances after preliminary layout; those dimensions do not follow from these proportions.

Chapter 11. Propeller shaft

Maximum and critical speeds, standard shaft section, design torque, torsional stress and twist angle.

The propeller shaft connects gearbox and final drive and must withstand two different hazards. The first is **high-speed instability**: at a certain speed a long, slender shaft bends progressively, independently of transmitted torque. The second is **torque overload**: the shaft twists, and excessive twist impairs rotational uniformity and loads the joints.

Calculate this chapter in two stages. First use rotational speeds to select a section from the GOST series. Once the section is known, calculate torque and check strength and stiffness.

Stage one: speeds and section

Maximum propeller-shaft speed.

Maximum shaft speed occurs with maximum source speed and minimum gearbox reduction, in top gear. A safety factor raises this value to cover brief overspeed as well as the design operating condition.

Table Д.11.1 — Inputs for (11.1): Maximum propeller-shaft speed.

Symbol	Quantity	Unit	Provenance
ω_{src} $\omega_{дж,max}$	maximum angular speed of the traction source Chapter 4 selects the machine driving the wheels and defines its quantities independently of whether it is an engine or electric motor.	s^{-1}	quantity defined in another chapter
u_{min} $u_{кmin}$	lowest gearbox ratio Top gear provides the least reduction, giving the highest shaft speed.	—	quantity defined in another chapter
k_{ω} k_{ω}	rotational-speed safety factor	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$\omega_{max} = \frac{k_{\omega} \cdot \omega_{src}}{u_{min}} \tag{11.1}$$

$$\omega_{max} = \frac{1.15 \cdot 691.15}{1} = 794{,}8 \ s^{-1}$$

- $\omega_{дmax}$ denotes the traction source, and ω_{max} denotes the propeller shaft. The source's textual definitions reverse these symbols; their printed equation meanings are retained here.
- Factor 1.1…1.2 allows for traction-source overspeed above the rating.

Source: [6]

Critical shaft speed.

Critical speed is where shaft instability occurs. Keep it above operating speed by a margin to avoid operation too close to the dangerous region.

Table Д.11.2 — Inputs for (11.2): Critical shaft speed.

Symbol	Quantity	Unit	Provenance
$K_{r}es$ $K_{зап}$	critical-speed margin above operating speed	—	adopted by the designer
$\omega_{m}ax$	maximum shaft speed	s^{-1}	calculated earlier (11.1)

Example — conventional internal-combustion vehicle

$$\omega_{kr}=K_{res}\cdot\omega_{max} \tag{11.2}$$

$$\omega_{kr}=1.6\cdot794.823=1271,7\ s^{-1}$$

- Margin $K_{зап}$ = 1.2…2 above operating speed: the shaft begins to whirl at critical speed, so nearby operation is unsuitable.

Source: [6]

Required outer diameter from critical speed.

Solve for required outer diameter: specified critical speed and length determine the section needed to achieve it.

A negative square-root argument is not necessarily an error: it can mean any wall thickness satisfies the critical-speed requirement. Diameter then imposes no lower constraint, and other conditions determine the section.

Table Д.11.3.2 — Inputs for (11.3): Required outer diameter from critical speed.

Symbol	Quantity	Unit	Provenance
L_{v} $L_{в}$	propeller-shaft length	m	adopted by the designer
$d_{i}n$ d	shaft inner diameter	mm	adopted by the designer
C_{w}	constant coefficient in the critical-speed formula	mm/s	input data
$A_{z}ero$	zero area used to clip a negative square-root argument	mm^2	input data
$\omega_{k}r$	critical shaft speed	s^{-1}	calculated earlier (11.2)

Example — conventional internal-combustion vehicle

$$D_{потр}=\sqrt{\max\left(A_{zero},\left(\frac{\omega_{kr}\cdot L_{v}^{2}}{C_{w}}\right)^{2}-d_{in}^{2}\right)} \tag{11.3}$$

$$D_{потр}=\sqrt{\max\left(0,\left(\frac{1271.72\cdot0.9^{2}}{1.13\cdot10^{+7}}\right)^{2}-71^{2}\right)}=57,174\ mm$$

- This is a REQUIREMENT rather than an adopted dimension: rearrange the printed expression for D. Select the dimension next from GOST 13758-89. If no complete section meets all three checks, explicitly identify an out-of-table preliminary tube. Searching candidates is not an equation.
- Zero means every tabulated wall thickness exceeds the required critical speed: the negative square-root argument represents a low-speed drive, not an error.
- The source prints the whirling coefficient as $1,13\cdot10^{7}$ for millimetres. In SI the same coefficient is $1,13\cdot10^{4}$ m/s; the worksheet includes its unit to ensure dimensional consistency.

Source: [6]

Propeller-shaft outer diameter

The required diameter rarely matches a manufactured tube. Select the nearest suitable standard row: no smaller than required, but not unnecessarily larger, which adds mass.

This chapter's only step that **is not a formula**selects a table value rather than calculating one. Application code therefore handles it instead of an engdoc expression: an expression evaluator does not select the suitable row of a series.

Selection rule. Select the smallest row whose outer diameter meets the requirement; its inner diameter defines the wall.

GOST — 45/49…51, 55/59…62, 71/74,6…76, 85/90…92, 110/114…116 mm

Table Д.11.3.1 — Inputs for (11.3): Propeller-shaft outer diameter

Symbol	Quantity	Unit	Provenance
$D_{потр}$	required shaft outer diameter	mm	calculated earlier (11.3)
$d_i n\ d$	shaft inner diameter	mm	adopted by the designer
$M_р$	design shaft torque	N·m	calculated earlier (11.4)

Example — conventional internal-combustion vehicle

$$(d,D)\in \text{ГОСТ 13758-89}:\ D\geq D_{потр},\ \tau\leq[\tau],\ \varphi/L\leq[\varphi/L] \qquad (11.3)$$

$$d=71\ \text{мм},\ L_в=900\ \text{мм}:\ D\geq 57.2\Rightarrow 74.6\ \text{мм}$$

- Adopted complete section d=71 mm, D=74.6 mm satisfies three conditions: critical speed, τ=27.3 MPa ≤ 300 MPa and φ/L=0.52°/m ≤ 6°/m. The same engdoc worksheet evaluates all candidates.
- Selection is limited to the listed tube series. Checks are preliminary; welds, fatigue, balancing and supports are not separately checked.

Source: [6]

Stage two: torque and checks

Design propeller-shaft torque.

Design torque is the greatest torque the shaft will carry, occurring in the lowest gear with the largest ratio.

The **source** of that torque depends on the design, so the method branches here.

(11.4) **Applicability:** The engine drives the wheels through a clutch.

A friction clutch limits transmitted torque when its friction capacity is exceeded. Reserve factor β indicates how much more than rated engine torque the clutch can transmit; that increased torque reaches the shaft.

Table Д.11.4.1 — Inputs for (11.4): Design propeller-shaft torque.

Symbol	Quantity	Unit	Provenance
$M_{\text{s}}rc$ $M_{\text{дж,max}}$	maximum torque of the traction source	N·m	quantity defined in another chapter
u_1 $u_{\text{к1}}$	highest gearbox ratio	—	quantity defined in another chapter
β	clutch torque reserve factor The factor by which clutch torque capacity exceeds rated engine torque.	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$M_{\text{p}}=\beta\cdot M_{\text{src}}\cdot u_1 \tag{11.4}$$

$$M_{\text{p}}=1.475\cdot 207.841\cdot 1.30155=399\ \text{N·m}$$

- Use FIRST-gear torque: this is the greatest shaft load.

Source: [6]

(11.4) **Applicability:** A traction motor drives the wheels without a clutch.

There is no clutch and therefore no frictional clutch-torque limit. The motor supplies maximum torque directly, with no clutch reserve multiplier.

Multiplying by unity would be less clear than omitting the factor because it would show a quantity absent from this vehicle.

Table Д.11.4.2 — Inputs for (11.4): Design propeller-shaft torque.

Symbol	Quantity	Unit	Provenance
$M_{\text{s}}rc$ $M_{\text{дж,max}}$	maximum torque of the traction source	N·m	quantity defined in another chapter
u_1 $u_{\text{к1}}$	highest gearbox ratio	—	quantity defined in another chapter

Example — electric vehicle

$$M_{\text{p}}=M_{\text{src}}\cdot u_1 \tag{11.4}$$

$$M_{\text{p}}=298.416\cdot 9.7=2894,6\ \text{N·m}$$

- Use FIRST-gear torque: this is the greatest shaft load.

Source: [6]

Propeller-shaft torsional stress.

The first check is strength. Torsional stress in the tube increases towards the outer surface, where it is evaluated.

Table Д.11.5 — Inputs for (11.5): Propeller-shaft torsional stress.

Symbol	Quantity	Unit	Provenance
M_p	design shaft torque	N·m	calculated earlier (11.4)
d_in d	shaft inner diameter	mm	adopted by the designer
$D_o$ut D	adopted shaft outer diameter	mm	calculated earlier (11.3)

Example — conventional internal-combustion vehicle

$$\tau = \frac{16 \cdot M_p \cdot D_{out}}{(\pi \cdot (D_{out}^4 - d_{in}^4))} \tag{11.5}$$

$$\tau = \frac{16 \cdot 399.008 \cdot 74.6}{(\pi \cdot (74.6^4 - 71^4))} = 27{,}269 \text{ MPa}$$

- The source specifies 100…300 MPa.
- Stress below 100 MPa does not exceed the allowable limit. The section must also satisfy critical speed and stiffness; comparing all conditions is necessary to explain its reserve.

Source: [6]

Shaft twist angle.

The second check is stiffness. Even a strong shaft is unsuitable if too flexible: twist disturbs rotation and adds joint loading.

Table Д.11.6.2 — Inputs for (11.6): Shaft twist angle.

Symbol	Quantity	Unit	Provenance
M_p	design shaft torque	N·m	calculated earlier (11.4)
L_v $L_в$	propeller-shaft length	m	adopted by the designer
d_in d	shaft inner diameter	mm	adopted by the designer
$D_o$ut D	adopted shaft outer diameter	mm	calculated earlier (11.3)
G_s	shear modulus of the shaft material	Pa	input data
$k_d$eg	conversion factor from radians to degrees	deg	input data

Example — conventional internal-combustion vehicle

$$\varphi = \frac{k_{deg} \cdot M_p \cdot L_v}{\left(\frac{G_s \cdot \pi \cdot (D_{out}^4 - d_{in}^4)}{32} \right)} \tag{11.6}$$

$$\varphi = \frac{57.2958 \cdot 399.008 \cdot 0.9}{\left(\frac{8 \cdot 10^{+10} \cdot \pi \cdot (74.6^4 - 71^4)}{32} \right)} = 0{,}4712 \text{ deg}$$

- The limit is 6° per metre, so check angle per metre rather than total shaft twist.
- The adopted shear modulus is 80 GPa, a common structural-steel value. The source prints 85 GPa. Since twist is inversely proportional to G, this difference is material rather than rounding; it is a project decision, not source data.

Source: [6]

Twist angle per unit shaft length.

Twist per unit length makes shafts of different lengths comparable. Compare this value with the allowable limit.

Table Д.11.6.1 — Inputs for (11.6): Twist angle per unit shaft length.

Symbol	Quantity	Unit	Provenance
L_{v} $L_{\text{в}}$	propeller-shaft length	m	adopted by the designer
φ	shaft twist angle	deg	calculated earlier (11.6)

Example — conventional internal-combustion vehicle

$$\varphi_{\text{L}} = \frac{\varphi}{L_{\text{v}}} \tag{11.6}$$

$$\varphi_{\text{L}} = \frac{0.471232}{0.9} = 0{,}5236 \ \text{deg/m}$$

- The limit is 6° per metre; compare that quantity rather than total shaft twist.

Source: [6]

Chapter 12. Final drive

Kinematic reduction split, tooth bending, contact stresses and compatible bevel or hypoid geometry.

This chapter concerns a separate final drive in the selected mechanical path. An integrated electric drive may contain all reduction in Chapter 10's gearbox. Do not add another final drive and multiply torque twice. For a parallel hybrid, first identify which traction path contains this assembly.

Drive type, torque and tooth counts

Design torque at the driving pinion

Take final-drive input torque from the traction source through the lowest gearbox ratio, including dry-clutch reserve only if the clutch belongs to that path. Use it to check the pair's peak strength.

Table Д.12.1 — Inputs for (12.1): Design torque at the driving pinion

Symbol	Quantity	Unit	Provenance
M_{e} $M_{\text{к,max}}$	maximum internal-combustion engine torque	N·m	calculated earlier
$u_{\text{кj}}$	adopted gear-ratio series	—	calculated earlier (4.19)
$u_{\text{рк,min}}$	top-range transfer-case ratio	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$M_{\text{розр}}=\beta M_{\text{кmax}}\,u_{\text{к1}}u_{\text{рк}} \tag{12.1}$$

$$M_{\text{розр}}=1.475\cdot 207.8\cdot 1.302\cdot 1.000=399\ \text{N·m}$$

Source: [6]

Final-drive tooth counts

In a single-stage final drive, the bevel pair provides the entire ratio. In a double-stage drive, the method adopts 1,7 for the bevel stage and assigns the remainder to the cylindrical stage. Round tooth counts to integers and recalculate each stage's actual reduction.

(12.2) **Applicability:** Single-stage final drive.

One bevel or hypoid pair supplies the complete reduction. Check both integer tooth counts after rounding.

Table Д.12.2.1 — Inputs for (12.2): Final-drive tooth counts

Symbol	Quantity	Unit	Provenance
—	adopted vehicle type for range selection	—	adopted by the designer
—	adopted final-drive type	—	adopted by the designer
Z_1	adopted final-drive pinion tooth count	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$u_0 = \frac{Z_2}{Z_1}, \qquad \delta = \operatorname{arctg}\frac{Z_1}{Z_2} \tag{12.2}$$

$$u_0 = \frac{55}{9} = 6.11111$$

$$\delta = \operatorname{arctg}\frac{9}{55} = 9.293^{\circ}$$

- Single-stage drive: the bevel pair supplies the whole ratio.
- The source gives Z_1 = 5...11, with smaller values for trucks.

Source: [6]

(12.2) **Applicability:** Double-stage final drive with bevel and cylindrical stages.

The substitution lists integer tooth counts for both stages. Their actual ratio product defines the adopted final drive; the required ratio remains a separate target. Determine separate modules and check bending and contact for both pairs at their own torques.

Table Д.12.2.2 — Inputs for (12.2): Final-drive tooth counts

Symbol	Quantity	Unit	Provenance
—	adopted vehicle type for range selection	—	adopted by the designer
—	adopted final-drive type	—	adopted by the designer
Z_1	adopted final-drive pinion tooth count	—	adopted by the designer

Example — double-stage final drive

$$u_0 = \frac{Z_2}{Z_1}\,\frac{Z_4}{Z_3}, \qquad \delta = \operatorname{arctg}\frac{Z_1}{Z_2} \tag{12.2}$$

$$u_0 = \frac{15}{9}\cdot\frac{55}{15} = 6.11111$$

$$\delta = \operatorname{arctg}\frac{9}{15} = 30.964^{\circ}$$

- Double-stage drive: bevel target 1.7, actual integer-tooth ratio 1.6667; cylindrical ratio 3.6667. Overall target u_0 = 6.1420.
- The source gives Z_1 = 5...11, with smaller values for trucks.

Source: [6]

Table M.12.1 — Reference final-drive design parameters

Assumption	Model range or rule
Driving bevel-pinion tooth count	5...11; smaller values for trucks
Approximate overall ratio for a passenger car	3,5...4,5, adopted teaching-project guideline
Approximate truck ratio	5...7, adopted teaching-project guideline
Pitch-cone distance	90...150 mm
Gear face width	Usually one-third of cone distance
Allowable bending stress	420...550 MPa; lower for an overhung pinion

Ratios 3,5…4,5 and 5…7 provide a plausibility check. An unusual result calls for checking target maximum speed, traction-source speed and top gear. It does not justify arbitrarily replacing the reduction required by traction calculation.

Module and geometric verification

Gear module at the outer radius

Determine the outer-cone module from bending. Take tooth-form factor from virtual tooth count using the same Table 12 as Chapter 10. Adopted cone distance sets face width, so changing it affects the result and requires rechecking.

Table Д.12.3 — Inputs for (12.3): Gear module at the outer radius

Symbol	Quantity	Unit	Provenance
σ	allowable final-drive tooth bending stress	MPa	adopted by the designer
$M_{\text{розр}}$	design torque at the final-drive input	N·m	calculated earlier (12.1)
Z	tooth counts of the final-drive stages	—	calculated earlier (12.2)
β_1	adopted pinion spiral angle	deg	adopted by the designer
L	adopted cone distance	mm	adopted by the designer

Example — conventional internal-combustion vehicle

$$m_{\text{req,i}}=\sqrt{\frac{6\cdot10^{3}M_{\text{i}}\cos\beta_{1}}{\pi y_{\text{i}}Z_{\text{i}}L(1-\lambda^{3})\sigma}},\qquad \lambda=1-\frac{b}{L} \tag{12.3}$$

$$\lambda=1-\frac{37.1543}{111.4630}=0.666667$$

$$m_{\text{req,1}}=\sqrt{\frac{6\cdot399.008\cdot10^{3}\cdot\cos(50.000^{\circ})}{\pi\cdot0.1280\cdot9\cdot111.4630\cdot(1-0.666667^{3})\cdot485.00}}=3.3432\ \text{mm}$$

$$m_{\text{req,2}}=\sqrt{\frac{6\cdot2438.383\cdot10^{3}\cdot\cos(20.000^{\circ})}{\pi\cdot0.1500\cdot55\cdot111.4630\cdot(1-0.666667^{3})\cdot485.00}}=3.7341\ \text{mm}$$

$$m_{\text{req}}=\max(3.3432,\ 3.7341)=3.7341\ \text{mm}$$

$$m_{\text{n}}=4.00\ \text{mm}\ \text{(adopted після перевірок)}$$

- In this formula M_i is in N·m, converted to N·mm by 1000; L and b are in mm and σ in MPa. Subscripts 1 and 2 denote driving and driven gears; M_1 is design torque (12.1).
- Bending requires m ≥ 3.73 mm; adopt 4.00 mm with preliminary 0,25 mm increments after the contact check. Recalculate L and b from adopted module. This is teaching selection of equivalent cones, not full ISO/AGMA hypoid verification.
- Tooth-form factor comes from Table 12 at virtual tooth count $Z_{\text{пр}}$ = 34.3, using the same table as Chapter 10.

Source: [6]

The teaching project retains spiral angles 30…40° for bevel gearing and 40…50° driving / 20…30° driven for hypoid gears. The cosine ratio enlarges the driving-pinion diameter; Buriennikov, Kashkanov and Rebedailo give this relationship on p. 81. The multiplier range 1,2…1,5 remains a model parameter; the text's exact ranges differ and are not the source of these adopted numbers.

Compare adopted cone distance with the distance implied by module and tooth counts. If they differ, preliminary geometry is inconsistent; refine dimensions and repeat the calculation.

Satisfying stress alone does not resolve this.

Final-drive tooth contact stress

Check contact at peak design torque. The range 1000…1200 MPa remains a teaching-project assumption. Allowable stress needs material, heat-treatment and life justification; the new reference text gives different limits.

The adopted denominator uses face width, as in Buriennikov, Kashkanov and Rebedailo, p. 80, (6.9), rather than the course guide's contact-line length. The hypoid model includes unequal spiral angles and enlarged pinion diameter. Both departures must be explicit beside the result. Unlike the printed Chapter 10 expression, this source uses the sum of curvatures.

Table Д.12.4 — Inputs for (12.4): Final-drive tooth contact stress

Symbol	Quantity	Unit	Provenance
m_{n}	final-drive gear module	mm	calculated earlier (12.3)
$M_{\text{розр}}$	design torque at the final-drive input	N·m	calculated earlier (12.1)

Example — conventional internal-combustion vehicle

$$\sigma_{\text{ст}}=0{,}418\sqrt{\frac{PE}{b\sin\alpha\cos\alpha}\left(\frac{1}{r_{\text{e1}}}+\frac{1}{r_{\text{e2}}}\right)} \tag{12.4}$$

$$\sigma_{\text{ст}}=0{,}418\sqrt{\frac{17114.353\cdot 210000}{37.1543\cdot\sin(20^{\circ})\cdot\cos(20^{\circ})}\left(\frac{1}{57.1774}+\frac{1}{642.8328}\right)}=1000.77\ \text{MPa}$$

- Use DESIGN torque, a peak rather than continuous operating torque. A preliminary pair may therefore exceed the range, indicating a need for more pinion teeth, a larger module or longer cone distance rather than an arithmetic error.
- The book prints PLUS here but minus for the cylindrical pair in Chapter 10. Both are external pairs with the same contact physics. The document calculates each printed expression and identifies the discrepancy.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

Source: [6]; [22]

If the limit is exceeded, increase pinion tooth count, module or geometrically compatible face width and repeat both checks. The final conclusion must include strength, actual ratio and dimensional consistency, not contact stress alone.

Bending verification of both final-drive gears

Check both gears using adopted module by rearranging the same formula used for module sizing. Increase driven-gear torque by actual tooth ratio without reducing it for losses. Show both stresses, the governing gear, allowable stress and margin. This preliminary equivalent-cone check does not replace full durability analysis.

Table Д.12.5 — Inputs for (12.5): Bending verification of both final-drive gears

Symbol	Quantity	Unit	Provenance
m_n	final-drive gear module	mm	calculated earlier (12.3)
$M_{розр}$	design torque at the final-drive input	N·m	calculated earlier (12.1)
Z	tooth counts of the final-drive stages	—	calculated earlier (12.2)
β_1	adopted pinion spiral angle	deg	adopted by the designer
L	adopted cone distance	mm	adopted by the designer
σ	allowable final-drive tooth bending stress	MPa	adopted by the designer

Example — conventional internal-combustion vehicle

$$\sigma_{F,i}=\frac{6\cdot10^3 M_i\cos\beta_i}{\pi y_i Z_i L(1-\lambda^3)m_n^2},\ \sigma_{F,max}=\max(\sigma_{F,1},\sigma_{F,2})\le[\sigma_F] \tag{12.5}$$

$$\sigma_{F,1}=\frac{6\cdot399.008\cdot10^3\cdot\cos(50.000^\circ)}{\pi\cdot0.1280\cdot9\cdot111.4630\cdot(1-0.666667^3)\cdot4.0000^2}=338.811\ \text{MPa}$$

$$\sigma_{F,2}=\frac{6\cdot2438.383\cdot10^3\cdot\cos(20.000^\circ)}{\pi\cdot0.1500\cdot55\cdot111.4630\cdot(1-0.666667^3)\cdot4.0000^2}=422.663\ \text{MPa}$$

$$\sigma_{F,max}=\max(338.811,422.663)=422.663\ \text{MPa}$$

$$\sigma_{F,max}\le[\sigma_F]=485\ \text{MPa}$$

- Form factors: $y_1 = 0.128$, $y_2 = 0.150$; bending passes with margin 1.15.
- This checks adopted module using rearranged formula (12.3); multiply torque by 1000 to convert N·m to N·mm.
- This preliminary check is not a complete ISO 10300 bevel or hypoid calculation.

Source: [6]; [7]

Cylindrical stage of a double-stage final drive

Normal module being checked for the cylindrical stage.

Select the second pair's module from the adopted series after contact and bending checks. Its torque already includes clutch reserve and bevel-stage multiplication; do not apply these again. Face width, helix angle and allowable stresses are explicit preliminary assumptions to refine for the eventual material and duty.

Table Д.12.6 — Inputs for (12.6): Normal module being checked for the cylindrical stage.

Symbol	Quantity	Unit	Provenance
$m_candidate$	normal module of a candidate cylindrical pair	mm	input data
$M_{розр}$	design torque at the final-drive input	N·m	calculated earlier (12.1)
Z	tooth counts of the final-drive stages	—	calculated earlier (12.2)

Example — double-stage final drive

$$m_n=m_{cand} \tag{12.6}$$

$$m_n=(5\ \text{mm})=5\ \text{mm}$$

- M_3 = 665 N·m; b = 50.0 mm; β = 25.0°; y_3 = 0.106, y_4 = 0.150.
- Bending requires m ≥ 4.63 mm; adopt 5.00 mm from the first source series after checks (12.7), (12.8).
- This is preliminary static bending/contact selection, not full ISO 6336 verification. Specify material, heat treatment, life, accuracy, profile and face-load distribution separately.
- Virtual tooth count exceeds the table; the nearest endpoint form factor is explicitly used as a preliminary approximation.

Source: [7]

Cylindrical-stage contact stress.

Use summed curvatures and face width in the external cylindrical-pair contact formula. This differs from Chapter 10's historical expression; the adopted engineering model and assumptions are explicit. Compare the result with this pair's allowable contact stress.

Table Д.12.7 — Inputs for (12.7): Cylindrical-stage contact stress.

Symbol	Quantity	Unit	Provenance
M_3	design torque at the cylindrical-stage input	N·m	input data
Z_3	cylindrical-stage pinion tooth count	—	input data
Z_4	cylindrical-stage wheel tooth count	—	input data
β	helix angle of the cylindrical pair	deg	input data
k_b	face-width to normal-module ratio	—	input data
α	pressure angle of the cylindrical pair	deg	input data
E	elastic modulus of the gear material	MPa	input data
k_H	adopted Hertz contact-formula coefficient	—	input data
m_n	adopted normal module of the cylindrical stage	mm	calculated earlier (12.6)
$M_{розр}$	design torque at the final-drive input	N·m	calculated earlier (12.1)
Z	tooth counts of the final-drive stages	—	calculated earlier (12.2)

Example — double-stage final drive

$$\sigma_H = k_H \cdot \cos(\beta) \cdot \sqrt{\frac{\left(\frac{M_3}{\left(\frac{m_n \cdot Z_3}{(2 \cdot \cos(\beta))}\right)}\right) \cdot E}{(k_b \cdot m_n \cdot \sin(\alpha) \cdot \cos(\alpha))} \cdot \left(\left(\frac{m_n \cdot Z_3}{(2 \cdot \cos(\beta))}\right) + \left(\frac{m_n \cdot Z_4}{(2 \cdot \cos(\beta))}\right)\right)} \qquad (12.7)$$

$$\sigma_H = 0.418 \cdot \cos(25^\circ) \cdot \sqrt{\frac{\left(\frac{665.014\ \text{N·m}}{\left(\frac{5\ \text{mm} \cdot 15}{(2 \cdot \cos(25^\circ))}\right)}\right) \cdot (210000\ \text{MPa})}{(10 \cdot (5\ \text{mm}) \cdot \sin(20^\circ) \cdot \cos(20^\circ))} \cdot \left(\left(\frac{1}{\left(\frac{5\ \text{mm} \cdot 15}{(2 \cdot \cos(25^\circ))}\right)}\right) + \left(\frac{1}{\left(\frac{5\ \text{mm} \cdot 55}{(2 \cdot \cos(25^\circ))}\right)}\right)\right)} = 962{,}9\ \text{MPa}$$

- M_3 = 665 N·m; b = 50.0 mm; β = 25.0°; y_3 = 0.106, y_4 = 0.150.
- F_t = 16072 N; r_3 = 41.38 mm, r_4 = 151.71 mm; limit 1000 MPa; margin 1.04.
- This is preliminary static bending/contact selection, not full ISO 6336 verification. Specify material, heat treatment, life, accuracy, profile and face-load distribution separately.

- Virtual tooth count exceeds the table; the nearest endpoint form factor is explicitly used as a preliminary approximation.

Source: [6]; [22]; [7]

Maximum cylindrical-stage bending stress.

Check both second-stage gears using that pair's torque, module, width and form factors. A double-stage final drive passes only after both stages are verified. Selection remains educational: separately confirm material, heat treatment, accuracy, load spectrum, supports and durability.

Table Д.12.8 — Inputs for (12.8): Maximum cylindrical-stage bending stress.

Symbol	Quantity	Unit	Provenance
M_3	design torque at the cylindrical-stage input	N·m	input data
Z_3	cylindrical-stage pinion tooth count	—	input data
Z_4	cylindrical-stage wheel tooth count	—	input data
β	helix angle of the cylindrical pair	deg	input data
k_b	face-width to normal-module ratio	—	input data
y_3	tooth-form factor of the driving cylindrical pinion	—	input data
y_4	tooth-form factor of the driven cylindrical wheel	—	input data
m_n	adopted normal module of the cylindrical stage	mm	calculated earlier (12.6)
$M_{розр}$	design torque at the final-drive input	N·m	calculated earlier (12.1)
Z	tooth counts of the final-drive stages	—	calculated earlier (12.2)

Example — double-stage final drive

$$\sigma_F=\max\left(\frac{2\cdot M_3\cdot\cos(\beta)}{(\pi\cdot Z_3\cdot y_3\cdot k_b\cdot m_n\cdot m_n^2)},\frac{2\cdot\left(\frac{M_3\cdot Z_4}{Z_3}\right)\cdot\cos(\beta)}{(\pi\cdot Z_4\cdot y_4\cdot k_b\cdot m_n\cdot m_n^2)}\right) \tag{12.8}$$

$$\sigma_F=\max\left(\frac{2\cdot(665.014\ \text{N·m})\cdot\cos(25^\circ)}{(\pi\cdot 15\cdot 0.106\cdot 10\cdot(5\ \text{mm})\cdot(5\ \text{mm})^2)},\frac{2\cdot\left(\frac{(665.014\ \text{N·m})\cdot 55}{15}\right)\cdot\cos(25^\circ)}{(\pi\cdot 55\cdot 0.15\cdot 10\cdot(5\ \text{mm})\cdot(5\ \text{mm})^2)}\right)=193{,}1\ \text{MPa}$$

- M_3 = 665 N·m; b = 50.0 mm; β = 25.0°; y_3 = 0.106, y_4 = 0.150.
- σ_3 = 193.1 MPa, σ_4 = 136.4 MPa; limit 225 MPa; margin 1.17.
- This is preliminary static bending/contact selection, not full ISO 6336 verification. Specify material, heat treatment, life, accuracy, profile and face-load distribution separately.
- Virtual tooth count exceeds the table; the nearest endpoint form factor is explicitly used as a preliminary approximation.

Source: [7]

Chapter 13. Differential

Adopting differential type and locking factor, and determining maximum axle-torque share at one wheel.

A mechanical differential permits different wheel speeds on one axle. This chapter covers a symmetric interwheel differential whether supplied by an engine or electric machine. Independent wheel motors have no mechanical differential; electronic torque distribution requires a separate control model.

Table M.13.1 — Mechanical-differential locking factors

Differential type	Course-guide locking factor
Low-friction bevel differential	0…0,2
Limited-slip	0,2…0,6
Self-locking	0,6

A teaching example may adopt the relevant range midpoint. The table and formula come from the course guide, subsection 4.1.5, printed pp. 52–53. Prefer a manufacturer's value when its determination conditions are known. This locking factor is not tyre adhesion.

Torque split between driven wheels

A symmetric differential has internal ratio unity. The result gives the larger axle-torque share used for one half-shaft. With zero locking, shares are equal and the poorer-adhesion wheel limits the entire axle's tractive capability.

Table Д.13.1 — Inputs for (13.1): Torque split between driven wheels

Symbol	Quantity	Unit	Provenance
—	adopted interwheel differential type	—	adopted by the designer
$K_б$	differential locking factor	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$K_M = \frac{U + K_б}{1 + U} \tag{13.1}$$

$$K_M = \frac{1 + 0.40}{1 + 1} = 0.700$$

- The differential is symmetric, so U = 1.
- Type: limited slip, $K_б$ = 0.2…0.6.
- At $K_б$ = 0 the wheels receive equal shares, so the slippery-side wheel limits whole-axle traction to its available adhesion.

Source: [6]

For multiple driven axles, determine each axle's torque first. An interwheel differential share does not distribute total torque between axles.

Chapter 14. Drive to the driven wheels

Load-bearing and fully floating half-shafts: diameter selection for characteristic loads and torsional-stiffness verification.

The hub-support layout determines half-shaft type. A load-bearing half-shaft transmits torque and carries bending from road reactions. In a fully floating layout, hub bearings and axle housing carry bending while the half-shaft carries torque. Show the components that isolate it from bending on the drawing.

Take the wheel-plane-to-critical-section arm and flange-to-spline-midpoint length from the layout. Wheel load follows axle-weight distribution. For multi-axle electric or hybrid drive, check the relevant axle and its torque source.

Table M.14.1 — Reference driven-wheel-drive parameters

Assumption	Range
Allowable load-bearing half-shaft bending stress	100…140 MPa
Adhesion coefficient for the limiting condition	0,8…0,9
Passenger-car dynamic-load factor	1,7…2,0
Truck dynamic-load factor	2,0…2,5
Allowable fully floating half-shaft torsional stress	500…700 MPa
Allowable twist per unit length	9 deg/m

Load cases and ranges follow the course guide, subsection 4.1.6, printed pp. 53–55.

Load-bearing half-shaft diameter

Determine diameters for three load cases: combined traction bending/torsion, bending under limiting lateral force and bending under dynamic vertical load. Adopt the largest and name its governing case. Traction alone does not fully check a load-bearing half-shaft.

Table Д.14.1 — Inputs for (14.1): Load-bearing half-shaft diameter

Symbol	Quantity	Unit	Provenance
σ	allowable half-shaft bending stress	MPa	adopted by the designer
b	distance from the wheel plane to the critical section	m	adopted by the designer
—	adopted half-shaft support layout	—	adopted by the designer
$\gamma_д$	half-shaft dynamic-load factor	—	adopted by the designer
G_i	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
φ_{max}	adhesion coefficient for the limiting half-shaft load	—	adopted by the designer
$r_д$	dynamic wheel radius	m	calculated earlier

Example — conventional internal-combustion vehicle

$$d_{\text{і}}=\sqrt[3]{\frac{32M_{\text{і}}}{\pi[\sigma]}},\qquad d=\max(d_{\text{тяга}},d_{\text{бічна}},d_{\text{перешкода}}) \tag{14.1}$$

$$d_{\text{тяга}}=\sqrt[3]{\frac{32\cdot10^{3}\cdot\sqrt{(4713.5\cdot0.080)^{2}(1+0.850^{2})+(4713.5\cdot0.850\cdot0.296/1.000)^{2}}}{\pi\cdot120.0}}=47.79\ \text{mm}$$

$$d_{\text{бічна}}=\sqrt[3]{\frac{32\cdot10^{3}\cdot4713.5(0.080+0.850\cdot0.296)}{\pi\cdot120.0}}=51.01\ \text{mm}$$

$$d_{\text{перешкода}}=\sqrt[3]{\frac{32\cdot10^{3}\cdot4713.5\cdot1.850\cdot0.080}{\pi\cdot120.0}}=38.98\ \text{mm}$$

$$d=\max(47.79,51.01,38.98)=51.01\ \text{mm}$$

- Three cases give three diameters; the source requires the LARGEST. Here lateral force with bending only governs.
- Adopted bending stress is 120 MPa from the range 100…140 MPa.

Source: [6]

Fully floating half-shaft diameter

First size the fully floating half-shaft for torsional strength. Compare with the diameter required by allowable twist and adopt the larger. This example is stiffness-governed, so final diameter exceeds the strength minimum.

Table Д.14.4 — Inputs for (14.4): Fully floating half-shaft diameter

Symbol	Quantity	Unit	Provenance
$[\tau]$	allowable torsional stress of a fully floating half-shaft	MPa	adopted by the designer
$G_{\text{і}}$	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
$\varphi_{\max}$	adhesion coefficient for the limiting half-shaft load	—	adopted by the designer
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier
$M_{\text{розр}}$	design torque in one half-shaft	N·m	calculated earlier (14.5)

Example — conventional vehicle with a plateau characteristic

$$M_{\text{кр}}=\max\left(\frac{Z_{\text{к}}\varphi_{\max}r_{\text{д}}}{u_{\text{кол}}},M_{\text{прив}}\right),\ d=\max\left(\sqrt[3]{\frac{16M_{\text{кр}}}{\pi[\tau]}},d_{\theta}\right) \tag{14.4}$$

$$M_{\text{кр}}=\max(4713.5\cdot0.850\cdot0.296/1.000,1706.87)=1706.87\ \text{N·m}$$

$$d_{\tau}=\sqrt[3]{\frac{16\cdot1706.87\cdot10^{3}}{\pi\cdot600.0}}=24.38\ \text{mm}$$

$$d_{\theta}=10^{34}\sqrt{\frac{32\cdot1706.87\cdot180}{\pi^{2}\cdot85\cdot10^{9}\cdot9}}=33.78\ \text{mm}$$

$$d=\max(24.38,33.78)=33.78\ \text{mm}$$

- A fully floating half-shaft carries no bending; hub bearings carry it, leaving torsion.
- Allowable torsional stress is 600 MPa from the range 500…700 MPa.
- STIFFNESS governs. Stress alone would allow 24.4 mm, but twist would exceed the source's 9°/m limit. Stiffness requires 33.8 mm, which is adopted.
- The source requires identification of the design components that isolate the half-shaft from bending.

Source: [6]

Design half-shaft torque

Torque reaches one wheel through the selected path reduction and differential share. Use consistent design torque for strength and stiffness; when comparing source torque with adhesion-limited torque, identify the governing condition explicitly.

(14.5) **Applicability:** Mechanical engine path with dry clutch.

Use maximum engine torque, first gear, active transfer range, final drive and differential torque share. Clutch reserve belongs to this path and uses the same basis as Chapter 9.

Table Д.14.5.1 — Inputs for (14.5): Design half-shaft torque

Symbol	Quantity	Unit	Provenance
M_{e} $M_{\text{к,max}}$	maximum internal-combustion engine torque	N·m	calculated earlier
$u_{\text{кj}}$	adopted gear-ratio series	—	calculated earlier (4.19)
K_{M}	maximum fraction of axle torque on one wheel	—	calculated earlier (13.1)
u_0	required final-drive ratio	—	calculated earlier (4.12)

Example — conventional internal-combustion vehicle

$$M_{\text{розр}}=\beta M_{\text{кmax}} u_{\text{к1}} u_{\text{рк}} u_0 K_{\text{M}} \tag{14.5}$$

$$M_{\text{розр}}=1.475\cdot 208\cdot 1.30\cdot 1.000\cdot 6.11\cdot 0.700=1707 \text{ N·m}$$

- K_{M} from Chapter 13: this determines how much axle torque reaches one wheel.

Source: [6]

(14.5) **Applicability:** Electric wheel path without a dry clutch, including a series hybrid.

Take torque from the electric-machine characteristic. No clutch reserve factor exists in this layout. If overall reduction already includes final drive, count it once; an additional unity factor is not another stage.

Table Д.14.5.2 — Inputs for (14.5): Design half-shaft torque

Symbol	Quantity	Unit	Provenance
$P_{\text{м}}(\omega)$	traction electric-machine characteristic	—	calculated earlier (4.23)
$u_{\text{кj}}$	adopted gear-ratio series	—	calculated earlier
K_{M}	maximum fraction of axle torque on one wheel	—	calculated earlier (13.1)

Example — electric vehicle

$$M_{\text{розр}}=\beta M_{\text{кmax}} u_{\text{к1}} u_{\text{рк}} u_0 K_{\text{M}} \tag{14.5}$$

$$M_{\text{розр}}=1.000\cdot 298\cdot 9.70\cdot 1.000\cdot 1.00\cdot 0.700=2026 \text{ N·m}$$

- K_{M} from Chapter 13: this determines how much axle torque reaches one wheel.

- There is no clutch reserve factor β because the electric drive has no clutch. The source does not cover this case.

Source: [6]

Half-shaft twist angle

Calculate twist from adopted diameter, length and shear modulus. Polar second moment depends on diameter to the fourth power, so even a small diameter increase changes twist significantly. The 9 deg/m limit applies to both designs.

(14.6) **Applicability:** Load-bearing half-shaft sized for three loading cases.

Use the load-bearing half-shaft diameter. Allowable total twist equals the per-length limit multiplied by actual length.

Table Д.14.6.1 — Inputs for (14.6): Half-shaft twist angle

Symbol	Quantity	Unit	Provenance
ℓ	design half-shaft length	m	adopted by the designer
$M_{\text{розр}}$	design torque in one half-shaft	N·m	calculated earlier (14.5)
d	diameter of a load-bearing half-shaft	mm	calculated earlier (14.1)

Example — conventional internal-combustion vehicle

$$\theta = \frac{M_{\text{розр}}\ell}{GJ_{\text{p}}} \cdot \frac{180}{\pi}, \qquad J_{\text{p}} = \frac{\pi d^4}{32} \tag{14.6}$$

$$J_{\text{p}} = \frac{\pi \cdot 51.012^4}{32} = 664805.75 \ \text{mm}^4$$

$$\theta = \frac{1706.87 \cdot 10^3 \cdot 0.600 \cdot 10^3}{85 \cdot 10^3 \cdot 664805.75} \cdot \frac{180}{\pi} = 1.04^\circ$$

- The limit is 9° per metre. The propeller shaft in Chapter 11 uses 6°/m: different components have different requirements.
- Twist per length θ/ℓ = 1.038° / 0.600 m = 1.731°/m. Total degrees and degrees per metre are different quantities.

Source: [6]

(14.6) **Applicability:** Fully floating half-shaft checked for torsion and stiffness.

Use its own fully floating half-shaft diameter; transferring the load-bearing diameter would check a different component.

Table Д.14.6.2 — Inputs for (14.6): Half-shaft twist angle

Symbol	Quantity	Unit	Provenance
ℓ	design half-shaft length	m	adopted by the designer
$M_{\text{розр}}$	design torque in one half-shaft	N·m	calculated earlier (14.5)
d	adopted fully floating half-shaft diameter	mm	calculated earlier (14.4)

Example — conventional vehicle with a plateau characteristic

$$\theta = \frac{M_{\text{розр}}\ell}{GJ_{\text{p}}} \cdot \frac{180}{\pi}, \qquad J_{\text{p}} = \frac{\pi d^4}{32} \tag{14.6}$$

$$J_{\text{p}} = \frac{\pi \cdot 33.780^4}{32} = 127838.35 \ \text{mm}^4$$

$$\theta = \frac{1706.87 \cdot 10^3 \cdot 0.600 \cdot 10^3}{85 \cdot 10^3 \cdot 127838.35} \cdot \frac{180}{\pi} = 5.40^\circ$$

- The limit is 9° per metre. The propeller shaft in Chapter 11 uses 6°/m: different components have different requirements.
- Twist per length θ/ℓ = 5.400° / 0.600 m = 9.000°/m. Total degrees and degrees per metre are different quantities.

Source: [6]

Steel shear modulus is assumed as 85 GPa {{cite:course-methodic}}.

Chapter 15. Handling and cornering

Steered-wheel lateral-sliding limit, tyre slip effects and oversteer critical speed.

These estimates depend on wheelbase, axle weights, adhesion and lateral tyre stiffness. They use the same adopted wheel model for engine, electric and hybrid drives. Battery and component mass distribution must already be included in the layout. The equations do not model active torque control or stability control.

Critical speed from the handling condition

At steering angles 5, 10, 15, 20, 25, 30, 35 and 40 degrees, calculate the speed above which steered wheels slide laterally. The result is a curve, not a universal cornering limit. Adhesion must exceed rolling resistance; a negative square-root argument means no solution at that angle rather than zero speed.

Table Д.15.1 — Inputs for (15.1): Critical speed from steering controllability

Symbol	Quantity	Unit	Provenance
B	vehicle wheelbase	m	input data
f_0	rolling-resistance coefficient at low speed	—	input data
φ	tyre–road adhesion coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$V_{\text{кер}}=\sqrt{\left(\frac{\sqrt{\varphi^2-f^2}}{\operatorname{tg}\theta}-f\right)gL\cos\theta} \tag{15.1}$$

$$V_{\text{кер}}(5.00^\circ)=\sqrt{\left(\frac{\sqrt{0.700^2-0.018^2}}{\operatorname{tg}(5.00^\circ)}-0.018\right)\cdot 9.81\cdot 2.700\cdot\cos(5.00^\circ)}=14.511\ \text{m/s}$$

$$=3{,}6\cdot 14.511=52.24\ \text{km/h}$$

$$V_{\text{кер}}(40.00^\circ)=\sqrt{\left(\frac{\sqrt{0.700^2-0.018^2}}{\operatorname{tg}(40.00^\circ)}-0.018\right)\cdot 9.81\cdot 2.700\cdot\cos(40.00^\circ)}=4.069\ \text{m/s}$$

$$=3{,}6\cdot 4.069=14.65\ \text{km/h}$$

- Above this speed steered wheels slide laterally; further steering no longer changes direction. This is a controllability limit rather than another arbitrary speed.

Source: [6]

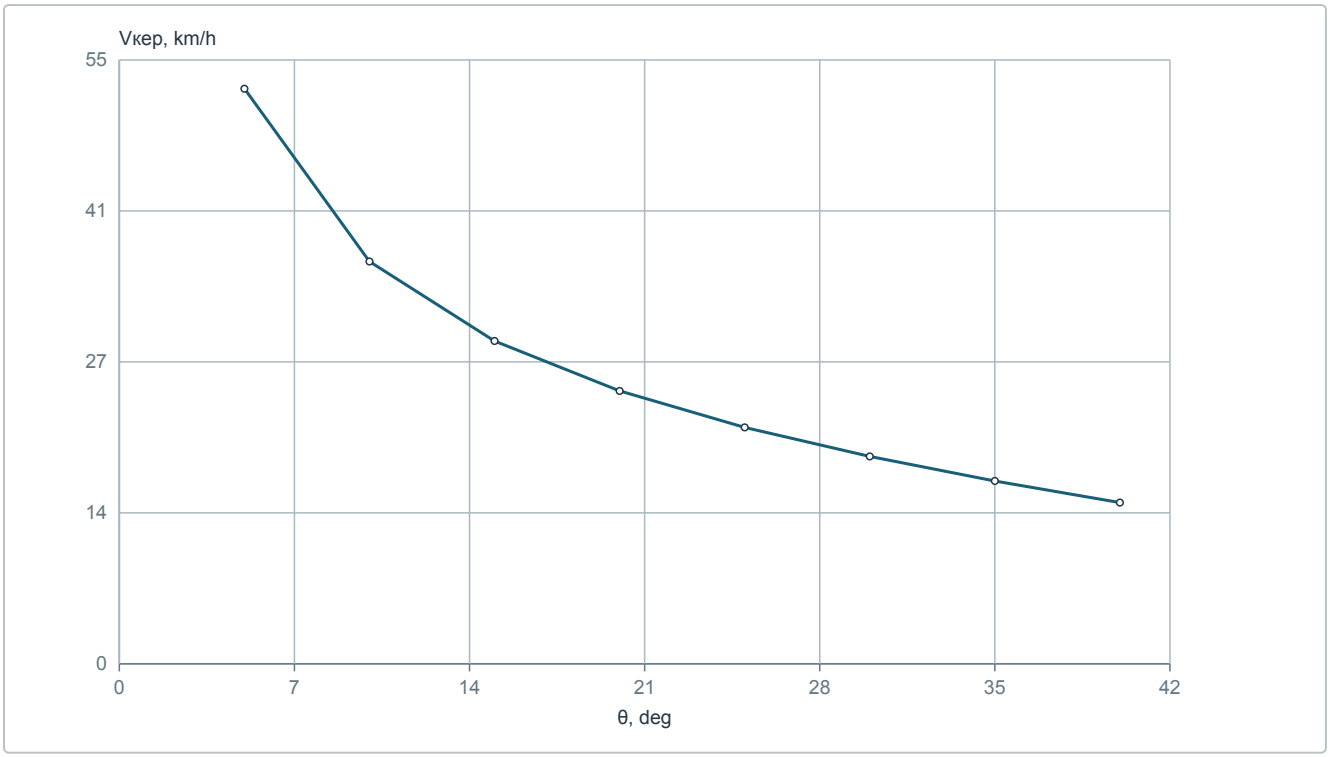


Figure 15.1 — Critical controllability speed at eight wheel-steering angles. Every point uses adopted adhesion, rolling resistance and wheelbase.
Source: [6].

Tyre slip and cornering radius

Stiffness ranges and checking sequence follow the course guide, subsection 3.4, printed pp. 26–28.

Table M.15.1 — Lateral stiffness per single tyre by vehicle group

Vehicle type	Lateral stiffness per single tyre, N/deg
Passenger car	500...1000
Cargo	800...1500

Sum tyre stiffnesses on each axle. The teaching model takes limiting lateral force as 0,4 times adhesion coefficient times axle weight. Dividing force by total stiffness gives slip angle. N/deg requires explicit angle conversion in trigonometric functions. The cornering-radius example uses 20 degrees mean steer angle.

Cornering radius and steering tendency

Compare the compliant-tyre radius with the geometric no-slip radius. A larger radius indicates understeer, a smaller one oversteer and equality neutral steer. The source gives no neutral-steer tolerance, so report the radius ratio rather than hiding a small difference through arbitrary rounding.

Table Д.15.2 — Inputs for (15.2): Cornering radius and steering tendency

Symbol	Quantity	Unit	Provenance
—	adopted vehicle type for range selection	—	adopted by the designer
$K_{\text{від}}$	adopted lateral stiffness of one tyre	N/deg	adopted by the designer
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)

Example — conventional internal-combustion vehicle

$$R_{\text{e}}=\frac{L}{\text{tg}(\theta-\delta_1)+\text{tg}\delta_2}, \qquad R=\frac{L}{\text{tg}\theta} \tag{15.2}$$

$$R_{\text{e}}=\frac{2.700}{\text{tg}((20.00-1.440)^{\circ})+\text{tg}(1.760^{\circ})}=7.367\ \text{m}$$

$$R=\frac{2.700}{\text{tg}(20.00^{\circ})}=7.418\ \text{m}$$

- Oversteer: R_{e} is less than R: the vehicle turns more sharply than commanded.
- Radius ratio R_e/R = 0.9931.
- The source specifies no neutral-steer tolerance, so comparison is exact; R_e/R shows the distance from neutral behaviour.

Source: [6]

Critical speed from lateral tyre slip

Total lateral stiffness is entered in N/deg, so multiply the weight-to-stiffness ratio difference by π/180 to convert degrees to radians. Multiply m/s by 3,6 for km/h. A finite critical speed exists only with a positive denominator, corresponding to oversteer in this model. Understeer or neutral steer has no finite root here. Its absence does not imply safety at every speed: adhesion and other motion limits remain.

Table Д.15.3 — Inputs for (15.3): Critical speed from lateral tyre slip

Symbol	Quantity	Unit	Provenance
B	vehicle wheelbase	m	input data
R_{e}	cornering radius including tyre slip	m	calculated earlier (15.2)
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)

Example — conventional internal-combustion vehicle

$$V_{\text{відв}}=\sqrt{\frac{gL}{\left(\frac{G_2}{\Sigma K_{\text{від2}}}-\frac{G_1}{\Sigma K_{\text{від1}}}\right)\frac{\pi}{180}}} \tag{15.3}$$

$$V_{\text{відв}}=\sqrt{\frac{9.81\cdot 2.700}{\left(\frac{9427.0}{1500.0}-\frac{7713.0}{1500.0}\right)\frac{\pi}{180}}}=36.443\ \text{m/s}$$

$$=3,6\cdot 36.443=131.20\ \text{km/h}$$

- Total cornering stiffness is in N/deg. Convert $G_2/\Sigma K_від2 - G_1/\Sigma K_від1$ from degrees to radians using п/180. Speed is initially in m/s; multiply by 3,6 for km/h.
- Calculate this speed only for oversteer. With understeer the denominator is nonpositive and this model has no speed-induced instability.

Source: [6]

The course guide requires critical speed 20…30 % above target maximum speed. If margin is insufficient, refine weight distribution, tyre properties and suspension design, then repeat both handling estimates rather than only the final inequality.

Chapter 16. Ride comfort

Consistent suspension deflection, natural frequencies and speeds over road undulations.

Body, occupants, cargo and body-mounted components form sprung mass. Wheels, tyres and parts of axle/suspension assemblies form unsprung mass. These groups have different natural frequencies. First adopt static deflection for the vehicle group, then check frequencies and find speeds where road undulations excite resonance.

The sequence is the same for engine vehicles, BEVs and plug-in hybrids; masses and layout capture the differences. A body-mounted traction battery is sprung mass; wheel motors require separately justified unsprung mass. The BEV label alone does not specify that mass.

16.1. Static-deflection selection

The table comes from the course vehicle-calculation guidelines, subsection 3.5, printed pp. 28–30. These are recommended design ranges. The midpoint is an explicit starting assumption to refine for the suspension design.

Table M.16.1 — Recommended sprung-mass deflections and frequencies

Vehicle group	Static deflection, m	Desired sprung-mass frequency, Hz
Passenger car	0,15–0,25	0,80–1,30
Truck and urban bus	0,08–0,13	1,30–1,80
Intercity bus	0,12–0,18	0,70–1,35

The source recommends smaller deflection for rear suspension and larger for front. Calculate the selected suspension; one averaged number does not prove consistency of both axles. This example transfers adopted deflection to Chapter 18 without an independent reselection.

Sprung-mass natural frequency.

Specify static deflection in metres. Compare calculated frequency with the same vehicle group's table row. A result outside the range requires reviewing stiffness or adopted deflection, not suppressing the result. This model omits damping and coupling of vertical body motion with pitch.

Table Д.16.1 — Inputs for (16.1): Sprung-mass natural frequency.

Symbol	Quantity	Unit	Provenance
$f_s t\ f_{ст}$	adopted static suspension deflection	m	adopted by the designer
g	gravitational acceleration	m/s²	input data

Example — conventional internal-combustion vehicle

$$\nu_n = \frac{\sqrt{\frac{g}{f_{st}}}}{(2 \cdot \pi)} \tag{16.1}$$

$$\nu_n = \frac{\sqrt{\frac{9.81}{0.2}}}{(2 \cdot \pi)} = 1{,}115 \text{ Hz}$$

- Frequency 1.11 Hz lies within the 0.8…1.3 Hz range associated with satisfactory ride.

Source: [6]

16.2. Unsprung-mass frequency

Use the formula when total axle-tyre radial stiffness and unsprung axle mass are known, in N/m and kg respectively. Otherwise, pp. 29–30 permit adopting 6,7–8,5 Hz; this is not a tyre-stiffness measurement.

Unsprung-mass natural frequency

The branches distinguish calculation from known properties and adoption of a reference frequency. Both feed the subsequent resonance-speed calculation.

(16.2) **Applicability:** Tyre stiffness and unsprung mass are unspecified; adopt the midpoint of the source's 6,7–8,5 Hz range.

The 7,6 Hz value is adopted. This branch's formula and substitution show the range midpoint; do not infer tyre stiffness or mass backwards from a desired result.

Table Д.16.2.1 — Inputs for (16.2): Unsprung-mass natural frequency

Symbol	Quantity	Unit	Provenance
$\nu_{в,min}$	lower source limit for unsprung-mass natural frequency	Hz	input data
$\nu_{в,max}$	upper source limit for unsprung-mass natural frequency	Hz	input data

Example — conventional internal-combustion vehicle

$$\nu_{в} = \frac{\nu_{в,min} + \nu_{в,max}}{2} \tag{16.2}$$

$$\nu_{в} = \frac{6.7 + 8.5}{2} = 7.60 \text{ Hz}$$

- Total tyre stiffness and unsprung axle mass are unspecified. Adopt the midpoint of 6,7…8,5 Hz as a preliminary reference frequency, not a stiffness-based result.
- This value is selected from 6.7…8.5 Hz rather than calculated from tyre stiffness.

Source: [6]

(16.2) **Applicability:** Total axle-tyre stiffness and unsprung mass are known.

For demonstration, the author assumes 180 000 N/m and 75 kg. These are illustrative inputs, not the selected prototype's specifications. Replace them with test data or a cited source in your project.

Table Д.16.2.2 — Inputs for (16.2): Unsprung-mass natural frequency

Symbol	Quantity	Unit	Provenance
$\Sigma C_{ш}$	total radial tyre stiffness on the axle	N/m	input data
$m_{м}$	unsprung axle mass	kg	input data

Example — separately specified tyre stiffness and unsprung mass

$$\nu_{в} = \frac{1}{2\pi}\sqrt{\frac{\Sigma C_{ш}}{m_{м}}} \qquad (16.2)$$

$$\nu_{в} = \frac{1}{2\cdot\pi}\sqrt{\frac{180000}{75.00}} = 7.80 \text{ Hz}$$

- The author assumes stiffness 180 000 N/m and mass 75 kg only for the teaching example; no manufacturer measurements are claimed.

Source: [6]; [2]

Resonant vehicle speeds

Determine resonance speed separately for sprung and unsprung masses. Each table row gives wavelength and two associated speeds. The source considers 0,5–5 m waves. Frequency in Hz times wavelength in metres gives m/s; multiply by 3,6 for km/h. Each of the four substitutions identifies mass group, wavelength, frequency and speed. These are possible resonance conditions, not real-road amplitude predictions.

Table Д.16.3 — Inputs for (16.3): Resonant vehicle speeds

Symbol	Quantity	Unit	Provenance
$\nu_{п}$	sprung-mass natural frequency	Hz	calculated earlier (16.1)
$\nu_{в}$	unsprung-mass natural frequency	Hz	calculated earlier (16.2)

Example — conventional internal-combustion vehicle

$$V_{р,н} = 3{,}6\nu_{н}S, \qquad V_{р,в} = 3{,}6\nu_{в}S \text{ [km/h]} \qquad (16.3)$$

$$V_{р,н}(S=0.5 \text{ m}) = 3{,}6\cdot 1.11465\cdot 0.5 = 2.01 \text{ km/h}$$
$$V_{р,в}(S=0.5 \text{ m}) = 3{,}6\cdot 7.60000\cdot 0.5 = 13.68 \text{ km/h}$$
$$V_{р,н}(S=5.0 \text{ m}) = 3{,}6\cdot 1.11465\cdot 5.0 = 20.06 \text{ km/h}$$
$$V_{р,в}(S=5.0 \text{ m}) = 3{,}6\cdot 7.60000\cdot 5.0 = 136.80 \text{ km/h}$$

- Resonance occurs when road excitation frequency equals natural frequency. The source uses undulation wavelengths 0.5…5 m. Subscript н denotes sprung mass and в unsprung mass; factor 3,6 converts m/s to km/h.

Source: [6]

16.3. Vibration-intensity indices

On p. 30, the source expresses these indices using frequency in Hz without the factor 2π. The following retains those teaching indices at bump height 0,05 m. Do not equate them with exact derivative amplitudes of harmonic displacement; harmonic motion requires angular frequency $\omega = 2\pi\nu$. This difference affects the numbers and must remain explicit in the conclusion.

Source-based teaching vibration-velocity index.

The first index is proportional to frequency. Within the source expression, it compares alternatives using the same assumed bump height.

Table Д.16.4.3 — Inputs for (16.4): Source-based teaching vibration-velocity index.

Symbol	Quantity	Unit	Provenance
Z_0	bump height for the teaching comparison	m	input data
ν_n	sprung-mass natural frequency	Hz	calculated earlier (16.1)

Example — conventional internal-combustion vehicle

$$I_v = Z_0 \cdot \nu_n \tag{16.4}$$

$$I_v = 0.05 \cdot 1.11465 = 0{,}0557 \text{ m/s}$$

- Source index $Z_{0\nu}$ uses frequency in Hz. Harmonic velocity amplitude requires 2π; the displayed index is not that amplitude.

Source: [6]

Source-based teaching vibration-acceleration index.

The second index increases with frequency squared. Compare only results sharing frequency and amplitude definitions; this teaching index does not independently verify allowable human vibration exposure.

Table Д.16.4.2 — Inputs for (16.4): Source-based teaching vibration-acceleration index.

Symbol	Quantity	Unit	Provenance
Z_0	bump height for the teaching comparison	m	input data
ν_n	sprung-mass natural frequency	Hz	calculated earlier (16.1)

Example — conventional internal-combustion vehicle

$$I_a = Z_0 \cdot \nu_n^2 \tag{16.4}$$

$$I_a = 0.05 \cdot 1.11465^2 = 0{,}0621 \text{ m/s}^2$$

- Bump height is 0.05 m as specified by the source. Z_0ν² using Hz is not harmonic acceleration amplitude; that requires an additional factor (2π)².

Source: [6]

Source-based teaching vibration-jerk index.

The third index represents sensitivity to frequent undulations and grows with frequency cubed. State the adopted bump height and the frequency qualification preceding the formulas.

Table Д.16.4.1 — Inputs for (16.4): Source-based teaching vibration-jerk index.

Symbol	Quantity	Unit	Provenance
Z_0	bump height for the teaching comparison	m	input data
ν_n	sprung-mass natural frequency	Hz	calculated earlier (16.1)

Example — conventional internal-combustion vehicle

$$I_j = Z_0 \cdot \nu_n^3 \tag{16.4}$$

$$I_j = 0.05 \cdot 1.11465^3 = 0{,}0692 \text{ m/s}^3$$

- Source index $Z_0\nu^3$ increases with frequency cubed at fixed bump height. The third-derivative amplitude of harmonic displacement requires an additional $(2\pi)^3$; this index does not replace human vibration-exposure assessment.

Source: [6]

Chapter 17. Off-road capability

Gradient, side-slope, adhesive-weight and wheel ground-pressure checks.

Geometric off-road capability depends on ground clearance, overhangs, approach/departure angles and component positions. Traction capability depends on adhesion, driven-wheel loading and ground contact. The parametric outline explains layout but does not turn unknown overhangs or angles into manufacturer data.

Use weight forces and centre-of-mass position from Chapter 3, dynamic wheel radius from tyre selection, and road coefficients for one consistent scenario. The source gives these expressions in subsection 3.6 after ride calculation.

Maximum climbable gradient

Calculate wheelspin-onset and longitudinal-rollover angles separately. The lower angle governs this model. Here a is the front-axle to centre-of-mass distance, b is the centre-of-mass to rear-support distance, L is wheelbase, and h_g is centre-of-mass height. The slip expression containing the fraction a/L applies to rear drive. Other layouts require reactions of their actual driven axles; changing the powertrain label is insufficient. This result excludes insufficient traction torque in the selected gear, which needs a separate traction-characteristic check.

Table Д.17.1 — Inputs for (17.1): Maximum climbable gradient

Symbol	Quantity	Unit	Provenance
B	vehicle wheelbase	m	input data
a	distance from the front axle to the centre of mass	m	calculated earlier (3.6)
h_g	adopted centre-of-mass height above the road	m	input data
f_0	rolling-resistance coefficient at low speed	—	input data
φ	tyre–road adhesion coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$\alpha_{\text{ковз}}=\operatorname{arctg}\frac{\varphi a/L-f}{1-\varphi h_g/L}\cdot\frac{180}{\pi}$$
$$\alpha_{\text{пер}}=\operatorname{arctg}\frac{b}{h_g}\cdot\frac{180}{\pi} \qquad (17.1)$$
$$\alpha_{\max}=\min(\alpha_{\text{ковз}},\alpha_{\text{пер}})$$
$$\alpha_{\text{ковз}}=\operatorname{arctg}\left(\frac{0.700\cdot1.485/2.700-0.0180}{1-0.700\cdot0.550/2.700}\right)\cdot\frac{180}{\pi}=23.17^{\circ}$$
$$\alpha_{\text{пер}}=\operatorname{arctg}\frac{1.215}{0.550}\cdot\frac{180}{\pi}=65.64^{\circ}$$
$$\alpha_{\max}=\min(23.17,65.64)=23.17^{\circ}$$

- ADHESION limits climbing: driven wheels spin before the vehicle overturns rearwards.

Source: [6]

Maximum side-slope angle

Compare lateral sliding and transverse rollover. The symbol B in this source formula denotes track, not layout wheelbase. Use centre-of-mass height for the same load state. This static estimate excludes wheel impacts, cornering acceleration and soil settlement; it is not a safe operating slope.

Table Д.17.2 — Inputs for (17.2): Maximum side-slope angle

Symbol	Quantity	Unit	Provenance
$K_{\text{ср}}$	mean front and rear track width	m	input data
h_{g}	adopted centre-of-mass height above the road	m	input data
φ	tyre–road adhesion coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$\beta_{\text{ковз}} \approx \operatorname{arctg} \varphi \cdot \frac{180}{\pi}$$
$$\beta_{\text{пер}} \approx \operatorname{arctg} \frac{B}{2h_{\text{g}}} \cdot \frac{180}{\pi} \tag{17.2}$$
$$\beta_{\max} = \min(\beta_{\text{ковз}}, \beta_{\text{пер}})$$

$$\beta_{\text{ковз}} = \operatorname{arctg}(0.700) \cdot \frac{180}{\pi} = 34.99^{\circ}$$
$$\beta_{\text{пер}} = \operatorname{arctg} \frac{1.500}{2 \cdot 0.550} \cdot \frac{180}{\pi} = 53.75^{\circ}$$
$$\beta_{\max} = \min(34.99, 53.75) = 34.99^{\circ}$$

- Lateral sliding occurs first in the static model. This estimate does not assess its consequences during motion or contact with an obstacle.

Source: [6]

Adhesive-weight coefficient

Adhesive weight sums all driven-axle loads: both axles of a driven rear bogie, or all driven supports for all-wheel drive. The source's soft-ground scenario uses rolling resistance 0,04, gradient 0,06 and adhesion 0,22. Do not silently combine these with a dry-surface coefficient. This condition checks adhesive-weight reserve for launch, not a complete power balance.

Table Д.17.3 — Inputs for (17.3): Adhesive-weight coefficient

Symbol	Quantity	Unit	Provenance
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)

Example — conventional internal-combustion vehicle

$$K_{\varphi} = \frac{G_{\text{зч}}}{G_{\text{a}}}, \qquad K_{\varphi} \geq \frac{f+i}{\varphi} \tag{17.3}$$

$$K_{\varphi} = \frac{9427.0}{17140.0} = 0.550 \geq \frac{0.04+0.06}{0.22} = 0.455$$

- Soft-ground check: f = 0.04, i = 0.06, φ = 0.22, exactly as specified by the source.
- The adhesive-weight condition passes. Check available launch torque separately.

Source: [6]

Ground contact pressure

Calculate pressure for one axle. The denominator uses that axle's wheel count and contact area per wheel; do not substitute whole-vehicle load. Contact-patch calculation and tyre-pressure approximation have different evidential bases.

(17.4) **Applicability:** Contact area is unspecified; approximate ground pressure by the specified tyre inflation pressure.

The source approximation gives 210 kPa here. It does not determine contact-pressure distribution or replace footprint measurement on the actual soil.

Table Д.17.4.1 — Inputs for (17.4): Ground contact pressure

Symbol	Quantity	Unit	Provenance
$p_{ш}$	adopted tyre pressure for the preliminary ground-pressure estimate	kPa	input data

Example — conventional internal-combustion vehicle

$$p_0 \approx p_{ш} \tag{17.4}$$

$$p_0 \approx p_{ш} = 210.0 \text{ kPa}$$

- Contact-area data are unavailable, so ground pressure is equated to tyre pressure as prescribed by the source.
- Tyre pressure is a separate project assumption; refine it using tyre and axle-load data.
- Pressure is equated to tyre pressure: a source approximation rather than a calculated value.

Source: [6]

(17.4) **Applicability:** Axle load, wheel count and contact area per wheel are specified.

The example assumes two wheels with 0,025 m² contact area each; load comes from the rear-axle calculation. For twin tyres, use actual contact count and an area definition consistent with it.

Table Д.17.4.2 — Inputs for (17.4): Ground contact pressure

Symbol	Quantity	Unit	Provenance
G_i	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
$F_к$	contact area of one wheel on the supporting surface	m²	input data
$n_к$	number of wheels on the design axle	—	input data

Example — specified wheel contact area

$$p_0 = \frac{Z}{1000 n_к F_к} \text{ [kPa]} \tag{17.4}$$

$$p_0 = \frac{9427.0}{1000 \cdot 2 \cdot 0.02500} = 188.54 \text{ kPa}$$

- The author assumes 0,025 m² per wheel; load comes from this vehicle's rear-axle calculation.

Source: [6]; [2]

Chapter 18. Running gear

From wheel load and required deflection to spring characteristics and leaf section.

Justify running gear through the load-bearing structure, axles, suspension and wheels. Describe load paths and supports for the frame or monocoque. Select dependent or independent suspension and its elastic element. Apply the leaf-spring calculation below only where a leaf spring is actually specified; Chapter 22 covers coil springs and torsion bars.

18.1. Loads and displacements

Ranges come from the guidelines, subsection 4.2.3, printed pp. 58–60. Derive static load per spring element from the relevant axle force, support count and adopted kinematics. It must concern the same element whose deflection was selected in Chapter 16.

Table M.18.1 — Reference suspension-spring load multipliers

Condition or parameter	Recommended multiplier
Dynamic-to-static load ratio on hard surfaces	2–3
Dynamic-to-static load ratio on earth roads	3–4
Auxiliary-spring engagement load relative to static load	0,6–0,7
Passenger-car dynamic-to-static deflection ratio	0,5–0,7
Bus dynamic-to-static deflection ratio	0,7–0,9
Truck dynamic-to-static deflection ratio	0,8–1,0

Specify the road group explicitly. Do not silently classify gravel, sand or earth as asphalt. If a surface cannot be classified, refine the inputs first.

Suspension elastic characteristic

Use static load to determine dynamic load and auxiliary-spring engagement. Mark all three states on the load–deflection characteristic. Range midpoints are initial example assumptions, not suspension-test results.

Table Д.18.1.1 — Inputs for (18.1): Suspension elastic characteristic

Symbol	Quantity	Unit	Provenance
—	adopted road conditions	—	adopted by the designer
—	adopted vehicle type for range selection	—	adopted by the designer
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
$f_{\text{s}}t\ f_{\text{ст}}$	adopted static suspension deflection	m	adopted by the designer

Example — conventional internal-combustion vehicle

$$Z_{дин}=(2…4)Z_{ст},\qquad Z_0=(0{,}6…0{,}7)Z_{ст} \tag{18.1}$$

$$Z_{дин}=\frac{3+4}{2}\cdot 4713.51=16497.28\ \text{N}$$

$$Z_0=\frac{0.6+0.7}{2}\cdot 4713.51=3063.78\ \text{N}$$

- Dynamic deflection $f_{дин}$ = 120 mm, a fraction 0.5…0.7 of the static deflection adopted in Chapter 16.
- The source uses a greater fraction for earth roads and off-road use.
- Static deflection is the same value adopted for ride in Chapter 16; changing it here also changes ride performance.

Source: [6]

Total suspension deflection

Add vehicle-group dynamic deflection to the existing static deflection. Total travel is needed for installation space, bump stops and coil count. Changing static deflection here requires repeating the Chapter 16 frequency check.

Table Д.18.1.2 — Inputs for (18.1): Total suspension deflection

Symbol	Quantity	Unit	Provenance
f_st $f_{ст}$	adopted static suspension deflection	m	adopted by the designer
—	adopted vehicle type for range selection	—	adopted by the designer
—	adopted road conditions	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$f=f_{ст}+f_{дин},\qquad f_{дин}=k_f f_{ст} \tag{18.1}$$

$$f_{дин}=\frac{0.5+0.7}{2}\cdot 0.20000=0.12000\ \text{m}$$

$$f=0.20000+0.12000=0.32000\ \text{m}=320.00\ \text{mm}$$

- Dynamic fraction k_f = 0.5…0.7 of static deflection by vehicle type; the midpoint is adopted.
- This is the spring's travel to its stop and determines coil count in Chapter 22.
- Static deflection is the same as adopted for ride in Chapter 16; changing it also changes ride performance.

Source: [6]

18.2. Leaf-spring length and section

For preliminary length, source p. 60 specifies a wheelbase fraction. There is no separate front passenger-car spring row; do not substitute the rear-spring row without explaining the design.

Table M.18.2 — Reference leaf-spring lengths relative to wheelbase

Spring position	Spring length as a wheelbase fraction
Rear, passenger car	0,40–0,55
Front, truck	0,26–0,35
Rear, truck	0,35–0,45

Adopt 6–14 leaves and width-to-thickness ratio 6–10. The leaf-end shape factor is 1,25–1,35 for trucks and 1,35–1,40 for passenger cars. The example uses steel elastic modulus 210 000 MPa. An asymmetric spring requires separately specified short and long arm lengths.

For an asymmetric leaf spring, use $f_{ст}=\delta Z_{ст}l_1^2l_2^2/(3EJ_\Sigma L_р)$, where $L_р=l_1+l_2$. This is deflection under a static concentrated load between two pinned supports. Substituting $x=a$, $L=a+b$ into the UW ME354 solution, p. 3.20, gives $|y(a)|=Pa^2b^2/(3EIL)$; the leaf-shape factor δ remains the teaching method's correction. Missing squared lever arms have been restored to ensure dimensions of J_Σ in mm⁴ and recovery of the symmetric expression when $l_1=l_2=L_р/2$. The beam analogy assumes small angles, constant EJ_Σ and negligible shear; mounting compliance, friction and nonlinear leaf contact are not checked. University of Washington, ME354.

Leaf-spring section

First derive total leaf second moment from required deflection. Adopt leaf count and width-to-thickness ratio to determine thickness, then width. Model the set as separate load-sharing leaves whose second moments add. The guide's printed cube of total thickness would represent a solid beam. The working relationship substitutes summed second moments into the beam-deflection formula. State calculated thickness separately from the adopted standard section.

Table Д.18.2 — Inputs for (18.2): Leaf-spring section

Symbol	Quantity	Unit	Provenance
$h\times b$	adopted standard leaf-spring section	—	adopted by the designer
$L_р$	adopted leaf-spring length	m	adopted by the designer
$Z_{дин}$	design dynamic suspension load	N	calculated earlier (18.1)

Example — conventional internal-combustion vehicle

$$f_{ст}=\delta\frac{Z_{ст}L_р^{\,3}}{48EJ_\Sigma},\qquad J_\Sigma=\frac{n_л b h^3}{12}$$
$$h=\sqrt[4]{\frac{12J_\Sigma}{k_b n_л}},\qquad b=k_b h \tag{18.2}$$

$$J_\Sigma=\frac{1.375\cdot4713.51\cdot1282.50^3}{48\cdot210000\cdot200.000}=6781.54\ \text{mm}^4$$
$$h=\sqrt[4]{\frac{12\cdot6781.54}{8.00\cdot10}}=5.647\ \text{mm}$$
$$b=8.00\cdot5.647=45.180\ \text{mm}$$

- A leaf-spring set consists of leaves that SLIDE over one another, equivalent to n parallel beams: $J_\Sigma = n_л\cdot b\cdot h^3/12$. The guide prints $J_\Sigma = b(h\cdot n_л)^3/12$, a solid beam as thick as the entire stack. A solid beam is much stiffer than the same material divided into leaves.
- The printed formula gives a 1.8 mm leaf, three times thinner than the thinnest entry in Table 16 used by the same guide. A separate-leaf model gives 5.6 mm, within the table.
- Adopt 10 leaves and b/h = 8; the source gives 6…14 leaves and b/h = 6…10.
- Table 16 gives a 6×55 mm leaf for passenger cars and buses/trucks based on passenger cars.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

Source: [6]; [22]

The following series reproduce guide Table 16, p. 61, presented as an extract from OH 8027-86. This source series supports teaching selection; citing it does not establish the standard's current validity or availability of every section.

Table M.18.3 — Reference leaf-spring section series

Leaf-spring group	Leaf widths, mm	Leaf thicknesses, mm
Passenger cars and buses/trucks based on them	45; 55	4,5; 5; 6; 6,5; 7; 7,5
Buses with nonstandardised leaf springs	65; 75; 90	7; 8; 9; 10; 12
Trucks and buses	55; 65; 75; 90; 100	6; 6,5; 7; 8; 9; 10; 11; 12; 14; 16

After section selection, recheck stiffness, deflection and installation. Matching a table size alone proves neither strength, durability nor adequate suspension travel.

Chapter 19. Steering system

Ratios, steering resistance torque and driver effort.

Calculate steering of a stationary vehicle, a demanding actuator condition. Inputs are steered-axle load, wheel radius, scrub radius and steering-gear dimensions. The method is unchanged for BEVs and hybrids, though axle loads, geometry and assistance can differ.

Ranges follow the guidelines, subsections 4.3.1–4.3.2, printed pp. 65–66.

Table M.19.1 — Reference steering-parameter ranges

Parameter	Source range
Steering-wheel angle	540–1080°
Road-wheel steering angle	35–40°
Steering-wheel radius	0,20–0,25 m
Passenger-car scrub radius	0,03–0,06 m
Truck scrub radius	0,05–0,12 m
Force ratio	100–300
Steering-gear efficiency	0,70–0,90
Steering-linkage efficiency	0,65–0,95

Steering ratios

Kinematic ratio is steering-wheel angle divided by road-wheel angle. Force ratio also includes steering-wheel radius and scrub radius. A large ratio reduces effort but increases required steering-wheel movement; check both.

Table Д.19.1 — Inputs for (19.1): Steering ratios

Symbol	Quantity	Unit	Provenance
$R_{\text{р.к}}$	adopted steering-wheel radius	m	adopted by the designer
—	adopted vehicle type for range selection	—	adopted by the designer
C	adopted steered-wheel scrub radius	m	adopted by the designer

Example — conventional internal-combustion vehicle

$$U_{\omega}=\frac{\alpha_{\text{р.к}}}{\alpha_{\text{к.к}}},\qquad U_{\text{р}}=U_{\omega}\,\frac{R_{\text{р.к}}}{C} \tag{19.1}$$

$$U_{\omega}=\frac{810}{37.50}=21.600$$

$$U_{\text{р}}=21.600\cdot\frac{0.2250}{0.0450}=108.00$$

- The source requires $U_{\text{р}}$ within 100…300.
- The expression excludes friction losses; they enter the subsequent steering effort calculation.

Source: [6]

Stationary-wheel turning resistance uses a separate coefficient $f_{\text{пов}}$: 0,03–0,06 on hard surfaces and 0,10–0,30 off-road. It differs from Chapter 2's straight-line rolling coefficient. Adopt

adhesion 0,6–0,8 and slip radius 0,12–0,15 times dynamic wheel radius for this calculation.

Steered-wheel resistance torque

Calculate total resistance torque for the steered wheels. Enter steered-axle weight force in newtons, converting kilonewtons first. The factor 1,25 is the method's approximation. Retain slip radius and scrub radius among the inputs to keep the result reproducible.

Table Д.19.2 — Inputs for (19.2): Steered-wheel resistance torque

Symbol	Quantity	Unit	Provenance
G_{i}	axle weight forces This is a list rather than a single number: each axle has its own force.	kN	calculated earlier (3.5)
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier

Example — conventional internal-combustion vehicle

$$\Sigma M_{\text{к.к}} \approx 1{,}25 G (f_{\text{пов}} c + \varphi r_{\text{ковз}}) \tag{19.2}$$

$$r_{\text{ковз}} = 0.1350 \cdot 0.29622 = 0.03999 \ \text{m}$$

$$\Sigma M_{\text{к.к}} \approx 1{,}25 \cdot 7713.01 (0.0450 \cdot 0.0450 + 0.700 \cdot 0.03999) = 289.41 \ \text{N·m}$$

- $f_{\text{пов}} = 0.045$ is resistance when steering STATIONARY wheels. It is NOT Chapter 2's driving coefficient f: the wheel turns on its footprint, scrubbing the tread rather than rolling forwards. The source gives separate ranges with differences up to fivefold.
- Slip radius $r_{\text{ковз}} \approx (0.12 \ldots 0.15) r_{\text{к}}$.

Source: [6]

Steering-wheel effort

Account for losses through separate gear and linkage efficiencies. Calculated effort excludes additional assistance. The source's 400 N threshold indicates a need for assistance, not compliance with modern certification requirements. If assistance is specified, justify its characteristic, power supply and failure behaviour separately.

Table Д.19.3 — Inputs for (19.3): Steering-wheel effort

Symbol	Quantity	Unit	Provenance
$R_{\text{р.к}}$	adopted steering-wheel radius	m	adopted by the designer
$\eta_{\text{р.м}}$	adopted steering-gear efficiency	—	adopted by the designer
$\eta_{\text{р.п}}$	adopted steering-linkage efficiency	—	adopted by the designer
$\Sigma M_{\text{к.к}}$	total steering resistance moment at the steered wheels	N·m	calculated earlier (19.2)
$U_{\text{р}}$	steering-system force ratio	—	calculated earlier (19.1)

Example — conventional internal-combustion vehicle

$$P_{\text{р.к}} = \frac{\Sigma M_{\text{к.к}}}{U_{\omega}\eta_{\text{р.м}}\eta_{\text{р.п}}R_{\text{р.к}}} \tag{19.3}$$

$$P_{\text{р.к}} = \frac{289.41}{21.600 \cdot 0.800 \cdot 0.800 \cdot 0.2250} = 93.05 \text{ N}$$

- Effort is below 400 N, the source criterion's assistance threshold. Final assistance selection requires requirements for all steering modes.
- Effort is calculated with the vehicle STATIONARY, the most demanding case.

Source: [6]

Chapter 20. Braking system

Brake-force distribution, mechanism selection and preliminary actuation assessment.

Start with kinematic layout, circuit split, brake mechanism and actuation type. The following calculation links force distribution with support load transfer, determines one wheel's brake torque and mechanism actuation force. Account for regeneration separately in the BEV/hybrid energy balance. Regeneration does not automatically justify reducing friction-brake design capacity: define permitted blending and a condition where regeneration is unavailable first.

20.1. Road reactions and force distribution

The source is the guidelines, subsections 4.4.1–4.4.2, printed pp. 70–74. The coefficient K_τ is front-to-rear braking-force ratio. Adopt 1,5–1,6 for passenger cars and trucks with similar centre-of-mass lever arms; for the source's specified truck case, $a>b$ use 0,6–1,0. Check the adopted coefficient against actual geometry: the table row alone does not guarantee a feasible distribution.

Use normal wheelbase for two axles. For three axles with equally loaded rear bogie axles, place the equivalent rear support at the bogie centre; equivalent wheelbase is $B+B_1/2$. Subtract front-axle to centre-of-mass distance to obtain the rear lever arm. This reduction specifically requires equal adopted rear-axle loads.

Design adhesion coefficient and road reactions

Derive design adhesion from the ratio of dynamic support reactions. The guide's printed numerator multiplies the coefficient by the lever-arm difference; the derivation multiplies only the front lever arm. Apply the correction consistently even when the printed expression is positive. Also determine front and rear wheel reactions. Adhesion 0,8 is this dry-surface scenario's adopted limit, not a universal upper bound for every tyre and road.

Table Д.20.1 — Inputs for (20.1): Design adhesion coefficient and road reactions

Symbol	Quantity	Unit	Provenance
—	adopted vehicle type for range selection	—	adopted by the designer
a	distance from the front axle to the centre of mass	m	calculated earlier (3.6)
K_τ	adopted braking-force distribution coefficient	—	adopted by the designer
G_a	total vehicle weight force	kN	calculated earlier (3.5)
h_g	adopted centre-of-mass height above the road	m	input data

Example — conventional internal-combustion vehicle

$$\varphi_{\text{розр}} = \frac{K_\tau a - b}{(K_\tau + 1) h_g},$$
$$Z_{\tau 1} = \frac{G_a}{2} \cdot \frac{b + \varphi_{\text{розр}} h_g}{L}, \qquad (20.1)$$
$$Z_{\tau 2} = \frac{G_a}{2} \cdot \frac{a - \varphi_{\text{розр}} h_g}{L}$$

$$\varphi_{\text{розр}} = \frac{1.550 \cdot 1.485 - 1.215}{(1.550+1) \cdot 0.550} = 0.774866,$$
$$Z_{\tau 1} = \frac{17140.032}{2} \cdot \frac{1.215 + 0.774866 \cdot 0.550}{2.700} = 5209.225 \text{ N},$$
$$Z_{\tau 2} = \frac{17140.032}{2} \cdot \frac{1.485 - 0.774866 \cdot 0.550}{2.700} = 3360.791 \text{ N}$$

- The source prints $\varphi_{\text{розр}} = K_\tau(a - b)/((K_\tau+1)h_g)$. For this vehicle, that gives 0.298 instead of 0.775: its K_τ has lost its multiplier on a.
- The error is clearer when a < b, as in a front-heavy vehicle: numerator $K_\tau(a - b)$ becomes negative. It is also wrong when a > b, merely returning a different positive result.
- Derivation from the reactions printed later in the same source: K_τ = Z_τ1/Z_τ2 = (b + φh_g)/(a − φh_g). Solving for φ gives $(K_\tau \cdot a - b)/((K_\tau+1)h_g)$, which is used here. The reader can verify the two-line rearrangement.
- Here a is front-axle to centre-of-mass distance (1.485 m) and b rear-support distance (1.215 m), the same as Chapter 3.
- Departure from the source: the printed expression $\varphi_{\text{розр}} = K_\tau(a - b)/((K_\tau+1)h_g)$ gives 0.298; the reaction-derived expression $(K_\tau \cdot a - b)/((K_\tau+1)h_g)$ = 0.775.

Source: [6]

20.2. Brake mechanism

Required torque comes from one wheel's reaction, design adhesion and dynamic radius. Equate this with the selected mechanism's friction torque to find actuation force. Adopt lining friction 0,30–0,35 for all layouts below.

Table M.20.1 — Reference brake-mechanism friction-torque relationships

Mechanism	Friction-torque expression
Disc	$M_{\text{тр}} = P\mu R_{\text{ср}} n$
Drum brake with equal forces and different wrap angles	$M_{\text{тр}} = \mu h R_б \frac{2Pa}{a^2 - \mu^2 b^2}$
Drum brake with two actuating forces	$M_{\text{тр}} = \mu h R_б \frac{2P}{a - \mu b}$
Drum brake with one actuating force	$M_{\text{тр}} = \mu h^2 R_б \frac{P}{(a - \mu b)^2}$

The table reproduces source subsection 4.4.1. Disc-brake inputs are mean friction radius and working-surface count. In drum layouts the symbols a, b, h denote local mechanism lever arms, not the vehicle centre-of-mass arms. The preliminary example assumes $a = b = 0{,}8R_б$, $h = R_б$; source wrap angles are 60–140°. Denominators must remain positive; approaching self-locking requires reviewing geometry.

Brake torque and mechanism actuation force

The examples retain vehicle loading while changing the mechanism. Required wheel torque therefore remains equal while actuation force changes, allowing comparison without changing the load case.

(20.2) **Applicability:** Disc brake with specified mean friction radius and surface count.

The example uses friction radius 0,11 m and two surfaces. These are teaching design assumptions, not prototype manufacturer data.

Table Д.20.2.1 — Inputs for (20.2): Brake torque and mechanism actuation force

Symbol	Quantity	Unit	Provenance
$\varphi_{\text{розр}}$	design braking adhesion coefficient	—	calculated earlier (20.1)
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier
—	adopted brake-mechanism design	—	adopted by the designer
μ	adopted brake friction coefficient	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$M_{\text{т}}=\varphi_{\text{розр}}Z_{\text{т}}r_{\text{d}},\quad M_{\text{тр}}=M_{\text{т}},\quad P=\frac{M_{\text{тр}}}{\mu R_{\text{ср}}n} \tag{20.2}$$

$$M_{\text{т}}=0.774866\cdot 5209.225\cdot 0.296219=1195.673\ \text{N·m},\quad P=\frac{1195.673143}{0.325000\cdot 0.110000\cdot 2}=16722.701\ \text{N}$$

- Equate mechanism friction torque with required brake torque: $M_{\text{тр}} = M_{\text{т}}$. Solve for mechanism actuation force P. A preliminary pedal-force estimate from whole-vehicle brake force does not replace matching this force to cylinders, levers and assistance.
- Lining friction μ = 0.3...0.35; wrap angle β = 60...140°.
- For drum brakes, use lever arms a ≈ b ≈ 0.8R_б.

Source: [6]

(20.2) **Applicability:** Drum brake with equal actuation forces and different wrap angles.

The example's drum working radius is 0,16 m. Find force by rearranging the relevant tabulated friction-torque expression.

Table Д.20.2.2 — Inputs for (20.2): Brake torque and mechanism actuation force

Symbol	Quantity	Unit	Provenance
$\varphi_{\text{розр}}$	design braking adhesion coefficient	—	calculated earlier (20.1)
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier
—	adopted brake-mechanism design	—	adopted by the designer
μ	adopted brake friction coefficient	—	adopted by the designer

Example — drum brakes with equal actuation forces

$$M_{\tau}=\varphi_{\text{розр}}Z_{\tau}r_{\text{d}},$$
$$M_{\text{тр}}=M_{\tau},$$
$$P=\frac{M_{\text{тр}}(a^2-\mu^2b^2)}{2\mu hR_{\text{б}}a} \tag{20.2}$$

$$M_{\tau}=0.774866\cdot5209.225\cdot0.296219=1195.673\ \text{N·m},$$
$$P=\frac{1195.673143\cdot(0.128000^2-0.325000^2\cdot0.128000^2)}{2\cdot0.325000\cdot0.160000\cdot0.160000\cdot0.128000}=8226.001\ \text{N}$$

- Equate mechanism friction torque with required brake torque: $M_{\text{тр}} = M_{\tau}$. Solve for mechanism actuation force P. A preliminary pedal-force estimate from whole-vehicle brake force does not replace matching this force to cylinders, levers and assistance.
- Lining friction μ = 0.3…0.35; wrap angle β = 60…140°.
- For drum brakes, use lever arms a ≈ b ≈ 0.8R_б.

Source: [6]

(20.2) **Applicability:** Drum brake with two actuation forces.

Drum radius remains 0,16 m with the same local lever arms. The mechanism's kinematic layout causes the force difference.

Table Д.20.2.4 — Inputs for (20.2): Brake torque and mechanism actuation force

Symbol	Quantity	Unit	Provenance
$\varphi_{\text{розр}}$	design braking adhesion coefficient	—	calculated earlier (20.1)
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier
—	adopted brake-mechanism design	—	adopted by the designer
μ	adopted brake friction coefficient	—	adopted by the designer

Example — drum brakes with two leading shoes

$$M_{\tau}=\varphi_{\text{розр}}Z_{\tau}r_{\text{d}},$$
$$M_{\text{тр}}=M_{\tau},$$
$$P=\frac{M_{\text{тр}}(a-\mu b)}{2\mu hR_{\text{б}}} \tag{20.2}$$

$$M_{\tau}=0.774866\cdot5209.225\cdot0.296219=1195.673\ \text{N·m},$$
$$P=\frac{1195.673143\cdot(0.128000-0.325000\cdot0.128000)}{2\cdot0.325000\cdot0.160000\cdot0.160000}=6208.303\ \text{N}$$

- Equate mechanism friction torque with required brake torque: $M_{\text{тр}} = M_{\tau}$. Solve for mechanism actuation force P. A preliminary pedal-force estimate from whole-vehicle brake force does not replace matching this force to cylinders, levers and assistance.
- Lining friction μ = 0.3…0.35; wrap angle β = 60…140°.
- For drum brakes, use lever arms a ≈ b ≈ 0.8R_б.

Source: [6]

(20.2) **Applicability:** Drum brake with one actuation force.

Assess self-energising action using this layout's formula. A lower calculated force does not remove friction-stability, temperature or reliability checks.

Table Д.20.2.3 — Inputs for (20.2): Brake torque and mechanism actuation force

Symbol	Quantity	Unit	Provenance
$\varphi_{\text{розр}}$	design braking adhesion coefficient	—	calculated earlier (20.1)
$r_{\text{д}}$	dynamic wheel radius	m	calculated earlier
—	adopted brake-mechanism design	—	adopted by the designer
μ	adopted brake friction coefficient	—	adopted by the designer

Example — drum brakes with one shoe actuator

$$M_{\text{т}}=\varphi_{\text{розр}}Z_{\text{т}}r_{\text{d}},$$
$$M_{\text{тр}}=M_{\text{т}},$$
$$P=\frac{M_{\text{тр}}(a-\mu b)^2}{\mu h^2 R_{\text{б}}} \tag{20.2}$$

$$M_{\text{т}}=0.774866\cdot 5209.225\cdot 0.296219=1195.673\ \text{N·m},$$
$$P=\frac{1195.673143\cdot(0.128000-0.325000\cdot 0.128000)^2}{0.325000\cdot 0.160000^2\cdot 0.160000}=6704.967\ \text{N}$$

- Equate mechanism friction torque with required brake torque: $M_{\text{тр}} = M_{\text{т}}$. Solve for mechanism actuation force P. A preliminary pedal-force estimate from whole-vehicle brake force does not replace matching this force to cylinders, levers and assistance.
- Lining friction μ = 0.3…0.35; wrap angle β = 60…140°.
- For drum brakes, use lever arms a ≈ b ≈ 0.8R_б.

Source: [6]

20.3. Brake actuation

The table ranges follow source subsection 4.4.2, pp. 73–74. Specify pedal and shoe travel in the same units. A force ratio may initially be selected from the range; final cylinder diameters and lever arms must realise that ratio.

Table M.20.2 — Initial brake-actuation parameters

Parameter	Passenger car	Truck
Service pedal travel, mm	70–75	80–100
Shoe displacement, mm	5–6	5–6
Hydraulic actuation force ratio	30–40	50–60
Hydraulic actuation efficiency	0,95–0,97	0,95–0,97
Pneumatic actuation force ratio	100–600	100–600
Pneumatic actuation efficiency	0,92–0,95	0,92–0,95

Brake actuation and pedal force

Kinematic ratio relates movements; force ratio relates forces. Both actuator types follow. This is a preliminary estimate using adopted ratio and total vehicle brake force. Complete matching to actual mechanism forces requires a separate cylinder, lever and assistance layout; this number does not replace that calculation.

(20.3) **Applicability:** Hydraulic actuation is adopted; its geometric force ratio depends on cylinder-area and pedal-arm ratios.

The example uses the passenger-car force-ratio midpoint. If effort exceeds the working range, state that the actuator must change or assistance must be justified.

Table Д.20.3.2 — Inputs for (20.3): Brake actuation and pedal force

Symbol	Quantity	Unit	Provenance
—	adopted vehicle type for range selection	—	adopted by the designer
$\varphi_{\text{розр}}$	design braking adhesion coefficient	—	calculated earlier (20.1)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)
$S_{\text{пед.роб}}$	adopted service brake-pedal travel	mm	adopted by the designer
$U_{\text{р}}$	adopted brake-actuation force ratio	—	adopted by the designer
$\eta_{\text{пр}}$	adopted brake-actuation efficiency	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$U_{\text{к}} = \frac{S_{\text{пед.роб}}}{n_{\text{к}} S_{\text{к}}},$$
$$U_{\text{р}} = \frac{U_{\text{р,min}} + U_{\text{р,max}}}{2}, \tag{20.3}$$
$$P_{\text{пед}} = \frac{\Sigma P_{\tau}}{U_{\text{р}} \eta_{\text{пр}}}$$

$$U_{\text{к}} = \frac{72.500}{2 \cdot 5.500} = 6.590909,$$
$$U_{\text{р}} = \frac{30.000 + 40.000}{2} = 35.000000,$$
$$P_{\text{пед}} = \frac{13281.233}{35.000000 \cdot 0.960000} = 395.275 \text{ N}$$

- $U_{\text{р}}$ is adopted at the source-range midpoint. This selects a ratio rather than calculating actuator geometry.
- $n_{\text{к}} = 2$ is the shoe count in the displacement sum; all travel is entered in millimetres.
- For service braking, the source allows $P_{\text{пед}} \le 100…150$ N, and up to 500 N for emergency braking.
- Above 400 N the source requires assistance.
- Hydraulic actuation: $\eta_{\text{пр}} = 0.95…0.97$; $U_{\text{р}} = 30…40$ for passenger cars and 50…60 for trucks.
- Unassisted effort of 395 N exceeds the service-braking guideline.

Source: [6]

(20.3) **Applicability:** Pneumatic truck actuation with its corresponding force-ratio and efficiency ranges.

This example's reactions use the equivalent rear support of a three-axle vehicle. Do not transfer pneumatic force-ratio ranges to a hydraulic layout.

Table Д.20.3.1 — Inputs for (20.3): Brake actuation and pedal force

Symbol	Quantity	Unit	Provenance
—	adopted vehicle type for range selection	—	adopted by the designer
$\varphi_{\text{розр}}$	design braking adhesion coefficient	—	calculated earlier (20.1)
G_{a}	total vehicle weight force	kN	calculated earlier (3.5)
$S_{\text{пед.роб}}$	adopted service brake-pedal travel	mm	adopted by the designer
$U_{\text{р}}$	adopted brake-actuation force ratio	—	adopted by the designer
$\eta_{\text{пр}}$	adopted brake-actuation efficiency	—	adopted by the designer

Example — three-axle truck (KamAZ-53215, 6×4)

$$U_{\text{к}} = \frac{S_{\text{пед.роб}}}{n_{\text{к}} S_{\text{к}}},$$
$$U_{\text{р}} = \frac{U_{\text{р,min}} + U_{\text{р,max}}}{2}, \tag{20.3}$$
$$P_{\text{пед}} = \frac{\Sigma P_{\text{т}}}{U_{\text{р}} \eta_{\text{пр}}}$$

$$U_{\text{к}} = \frac{90.000}{2 \cdot 5.500} = 8.181818,$$
$$U_{\text{р}} = \frac{100.000 + 600.000}{2} = 350.000000,$$
$$P_{\text{пед}} = \frac{261966.582}{350.000000 \cdot 0.935000} = 800.509 \text{ N}$$

- $U_{\text{р}}$ is adopted at the source-range midpoint. This selects a ratio rather than calculating actuator geometry.
- $n_{\text{к}} = 2$ is the shoe count in the displacement sum; all travel is entered in millimetres.
- For service braking, the source allows $P_{\text{пед}} \leq 100…150$ N, and up to 500 N for emergency braking.
- Without assistance, 801 N > 400 N. Preliminary rounded assistance ×6.5 gives approximately 123 N; check the final booster against manufacturer data.
- Pneumatic actuation: $U_{\text{р}} = 100…600$, $\eta_{\text{пр}} = 0.92…0.95$.
- Assistance ratio is a preliminary design recommendation; check the final assembly using its specifications and redundancy requirements.

Source: [6]

The source gives 100–150 N for service braking, 500 N for emergencies and 400 N as the assistance threshold. This preliminary model treats 100–150 N as target service effort and values above 400 N as requiring assistance. Do not call 150–400 N satisfactory merely because it is below the latter threshold. Temperature, fluid/air reserve, parking brake and circuit failure require separate design checks.

Chapter 21. Steering linkage

Steering-trapezoid dimensions, inner/outer wheel angles and rack geometry.

The inner wheel must steer further in a turn. Select the trapezoid so actual angle relationships approximate rolling around a common centre. Animation or a neat drawing is not verification; conclusions require an angle table, mechanism closure and numerical deviation.

The source is the guidelines, subsection 4.3.3 and the end of 4.3.2, printed pp. 66–69. Symbols are distinguished: $m_{тр}$ is steering-arm length, $m_{р}$ is rack module; M is kingpin spacing, and $M_{рол}$ is rack measurement over rollers. These are different quantities.

21.1. Initial geometry

Table M.21.1 — Initial steering-linkage geometry

Adopted parameter	Value or range
Rear-trapezoid position coefficient	0,7
Front-trapezoid position coefficient	1,0
Steering-arm to tie-rod length ratio	0,12–0,16
Inner-wheel angles to check	5; 10; 15; 20; 25; 30; 35; 40°

Derive kingpin spacing from track and stub-axle lengths. For toroidal tyres, the source permits preliminary stub length equal to tyre section width. Replace this construction approximation with the actual assembly dimension when known.

Principal steering-trapezoid dimensions

Use wheelbase, track, stub length and adopted position to find arm angle, tie-rod length and steering-arm length. Then check closure: tie-rod length plus both horizontal arm projections must reproduce kingpin spacing.

(21.1) **Applicability:** The tie rod lies behind the front axle; position coefficient 0,7 is adopted.

This is the worked example's initial layout. Its numerical dimensions feed the angle check without an independent redrawing.

Table Д.21.1.1 — Inputs for (21.1): Principal steering-trapezoid dimensions

Symbol	Quantity	Unit	Provenance
$K_{ср}$	mean front and rear track width	m	input data
B	vehicle wheelbase	m	input data
x	steering-trapezoid position relative to the front axle	—	adopted by the designer
$m_{тр}/n$	adopted steering-arm to tie-rod length ratio	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$M=B-2l_{ц},\qquad \operatorname{ctg}\theta=\frac{M}{2xL},\qquad n=\frac{M}{1+2\dfrac{m_{тр}}{n}\cos\theta} \tag{21.1}$$

$$M=1.500-2\cdot0.195=1.110\ \text{m}$$
$$\theta=\operatorname{arctg}\frac{2\cdot0.70\cdot2.700}{1.110}\cdot\frac{180}{\pi}=73.63508^{\circ}$$
$$n=\frac{1.110}{1+2\cdot0.1400\cdot\cos(73.63508^{\circ})}=1.02883\ \text{m}$$
$$m_{тр}=0.1400\cdot1.02883=0.14404\ \text{m}$$

- Arm-to-rod ratio m_тр/n = 0.12…0.16; position factor x = 0.7 for a rear trapezoid and 1 for a front trapezoid.
- The n expression only appears implicit: multiplying by its denominator gives n + 2m_тр·cos θ = M, the trapezoid geometry in which each arm subtracts $m_{тр}$·cos θ. Direct check: M − 2m_тр·cos θ = 1.029 m.
- The steering stub length $l_{ц}$ may be approximated by tyre section width for toroidal tyres according to the source $B_{ш}$.

Source: [6]

(21.1) **Applicability:** The tie rod lies ahead of the front axle; position coefficient 1,0 is adopted.

Moving the linkage changes geometry and kinematic verification. Mirroring the rear layout alone does not replace a new calculation.

Table Д.21.1.2 — Inputs for (21.1): Principal steering-trapezoid dimensions

Symbol	Quantity	Unit	Provenance
$K_{ср}$	mean front and rear track width	m	input data
B	vehicle wheelbase	m	input data
x	steering-trapezoid position relative to the front axle	—	adopted by the designer
$m_{тр}/n$	adopted steering-arm to tie-rod length ratio	—	adopted by the designer

Example — front steering-trapezoid arrangement

$$M=B-2l_{ц},\qquad \operatorname{ctg}\theta=\frac{M}{2xL},\qquad n=\frac{M}{1+2\dfrac{m_{тр}}{n}\cos\theta} \tag{21.1}$$

$$M=1.500-2\cdot0.195=1.110\ \text{m}$$
$$\theta=\operatorname{arctg}\frac{2\cdot1.00\cdot2.700}{1.110}\cdot\frac{180}{\pi}=78.38433^{\circ}$$
$$n=\frac{1.110}{1+2\cdot0.1400\cdot\cos(78.38433^{\circ})}=1.05076\ \text{m}$$
$$m_{тр}=0.1400\cdot1.05076=0.14711\ \text{m}$$

- Arm-to-rod ratio m_тр/n = 0.12…0.16; position factor x = 0.7 for a rear trapezoid and 1 for a front trapezoid.
- The n expression only appears implicit: multiplying by its denominator gives n + 2m_тр·cos θ = M, the trapezoid geometry in which each arm subtracts $m_{тр}$·cos θ. Direct check: M − 2m_тр·cos θ = 1.051 m.

- The steering stub length $l_{ц}$ may be approximated by tyre section width for toroidal tyres according to the source $B_{ш}$.

Source: [6]

Steering-trapezoid angle verification

Find theoretical outer-wheel angle from a common turning centre. Determine actual angle by closing the four-bar linkage with fixed tie-rod length. This numerically implements the source's drawing construction at 5° increments and scale at least 1:10. The source requires agreement through 20° and no more than 2–3° difference thereafter. The application's 0,5° agreement tolerance is a project assumption, not a source tolerance.

(21.2) **Applicability:** Check the calculated rear trapezoid over the full 5–40° series.

Report the largest deviation and its angle. Check the region through 20° separately.

Table Д.21.2.1 — Inputs for (21.2): Steering-trapezoid angle verification

Symbol	Quantity	Unit	Provenance
n	transverse tie-rod length	m	calculated earlier (21.1)

Example — conventional internal-combustion vehicle

$$\begin{gathered} \operatorname{ctg}\alpha_{\text{зовн}}-\operatorname{ctg}\alpha_{\text{вн}}=\frac{M}{L} \\ \Delta\alpha_{\max}=\max\left|\alpha_{\text{факт}}-\alpha_{\text{теор}}\right| \\ \frac{M}{L}=\frac{1.110}{2.70}=0.4111 \\ \Delta\alpha_{\max}=\left|31.2662-31.9593\right|=0.6931^{\circ}\ (\alpha_{\text{вн}}=40^{\circ}) \end{gathered} \tag{21.2}$$

- The source checks the trapezoid by DRAWING: at scale at least 1:10, turn the inner wheel in 5° steps and measure the outer angle. The document solves the same four-bar closure exactly: arm ends remain n apart. This implements the same measurement without ruler error, rather than a different method.
- Source criterion: agreement through 20°, then no more than 2…3° difference.
- Here deviation is at most 0.33° through 20° and 0.69° thereafter.

Source: [6]

(21.2) **Applicability:** Check the calculated front trapezoid using its own lengths and initial angle.

If the criterion fails, retain the result in the table, change geometry and recheck. An unsuccessful initial layout is also a useful worked example.

Table Д.21.2.2 — Inputs for (21.2): Steering-trapezoid angle verification

Symbol	Quantity	Unit	Provenance
n	transverse tie-rod length	m	calculated earlier (21.1)

Example — front steering-trapezoid arrangement

$$\operatorname{ctg}\alpha_{\text{зовн}}-\operatorname{ctg}\alpha_{\text{вн}}=\frac{M}{L}$$
$$\Delta\alpha_{\max}=\max|\alpha_{\text{факт}}-\alpha_{\text{теор}}| \tag{21.2}$$
$$\frac{M}{L}=\frac{1.110}{2.70}=0.4111$$
$$\Delta\alpha_{\max}=|30.2368-28.5328|=1.7040^{\circ}\ \ (\alpha_{\text{вн}}=35^{\circ})$$

- The source checks the trapezoid by DRAWING: at scale at least 1:10, turn the inner wheel in 5° steps and measure the outer angle. The document solves the same four-bar closure exactly: arm ends remain n apart. This implements the same measurement without ruler error, rather than a different method.
- Source criterion: agreement through 20°, then no more than 2…3° difference.
- Here deviation is at most 0.95° through 20° and 1.70° thereafter.
- The curves do not agree through 20°: deviation is 0.95°.

Source: [6]

21.2. Rack-and-pinion steering

Adopt rack module 3,0–3,5 mm, pinion count 6–10, rack height 20–80 mm and pressure angle 20°. Addendum coefficient is 1,0 and clearance coefficient 0,25, giving tooth height 2,25 modules and pitch-line thickness half the circular pitch. Do not confuse pinion tooth count with the rack's working tooth count.

Rack-and-pinion steering

Find pitch from module and required travel from steering-arm length and maximum wheel angle. After rounding rack tooth count, recalculate actual travel and compare it with demand. The double-angle sine expression applies only through 45°; beyond that it no longer increases with required travel. Measurement over rollers checks the rack profile, not kingpin spacing.

Table Д.21.3 — Inputs for (21.3): Rack-and-pinion steering gear

Symbol	Quantity	Unit	Provenance
Z_1	adopted rack-and-pinion pinion tooth count	—	adopted by the designer
H	adopted rack height	mm	adopted by the designer
n	transverse tie-rod length	m	calculated earlier (21.1)
m_{p}	adopted rack tooth module	mm	adopted by the designer

Example — conventional internal-combustion vehicle

$$P_{\text{п}}=\pi m_{\text{р}},\qquad L_{\text{поп}}=1000OA_3\sin(2\alpha_{\max})\ \ [\text{mm}]$$
$$Z_{\text{поп}}=\frac{L_{\text{поп}}}{P_{\text{п}}}+0{,}5,\qquad L=(Z_{\text{пр}}-0{,}5)P_{\text{п}} \tag{21.3}$$
$$P_{\text{п}}=\pi\cdot3.25=10.21018\ \text{mm}$$
$$L_{\text{поп}}=1000\cdot0.14404\cdot\sin(2\cdot37.50^{\circ})=139.1288\ \text{mm}$$
$$Z_{\text{поп}}=\frac{139.1288}{10.21018}+0{,}5=14.1265\rightarrow Z_{\text{пр}}=14$$
$$L=(14-0{,}5)\cdot10.21018=137.837\ \text{mm}$$

- The printed symbol m denotes TWO quantities here: rack module (3,0…3,5 mm) and steering-arm length OA₃ (hundreds of millimetres). L = OA₃·sin2α_max uses the arm, while P_n = πm uses module. They are distinguished here: m_p and $m_{тр}$.
- Tooth height h = (2h*_a + c*)m_p = 7.31 mm at h*_a = 1 and c* = 0.25; thickness S̄_y = 0,5πm_p = 5.11 mm.
- Measurement over rollers $M_{рол}$ = 50.58 mm at rack height H = 50 mm and reference-profile angle α = 20°. The printed source also calls it M, as it does kingpin spacing.
- Pinion tooth count Z₁ = 8; the source gives 6…10.

Source: [6]

For the rack inspection drawing, the source gives $M_{рол}=H+0,85m_p(1/\sin\alpha+1)-[\pi/(4\tan\alpha)+1]m_p$. Here α is tooth pressure angle. Enter all lengths in millimetres. The trapezoid drawing shows kingpin axes, arms, tie rod and both angles; a static position with the verification table is sufficient in print.

Chapter 22. Coil spring and torsion members

Elastic-element dimensions from suspension load and deflection.

A coil spring, round torsion bar and laminated torsion member are alternatives. Compare them at the same design load and required displacement, then retain the chosen design in the final layout. Loads and total deflection come from Chapter 18; do not reselect them to obtain a desired diameter.

The underlying relationships appear in the guidelines, subsection 4.2.3, printed pp. 63–64. The cube root follows from torque equilibrium and the circular-section modulus. Allowable stresses below are adopted teaching-project parameters. Schematic ends are acceptable in drawings, but working length, arm and diameter must match calculation.

Table M.22.1 — Spring and torsion-member calculation inputs

Quantity	Adopted basis
Allowable spring stress under static load	500 MPa, project assumption
Allowable spring stress under dynamic load	800 MPa, project assumption
Allowable round torsion-bar stress	800–1000 MPa, project assumption
Allowable laminated torsion-member stress	900 MPa, project assumption
Inactive spring coils	1,5–2,0, guidelines p. 63
Splined-end diameter relative to torsion-bar working diameter	1,2–1,3, guidelines p. 64

The example adopts shear modulus 80 GPa; the original guide prints 85 GPa. Using 80 GPa is an explicit project assumption, not a correction established by checking a specific material. Allowable stresses require the corresponding material, heat treatment and load type.

22.1. Coil spring

Suspension coil spring

Find wire diameter from dynamic suspension load, specified mean spring diameter and allowable stress. Mean diameter follows the wire centreline, not the spring's outside envelope. The printed root lacks an exponent: the argument has dimensions of length cubed, requiring a cube root. Torque and circular-section modulus give the same result. Dynamic allowable stress is assumed as 800 MPa; the guide's 100 MPa is retained for comparison. A real spring's allowable stress needs material, heat-treatment and fatigue justification.

Table Д.22.1.2 — Inputs for (22.1): Suspension coil spring

Symbol	Quantity	Unit	Provenance
$Z_{\text{дин}}$	design dynamic suspension load	N	calculated earlier (18.1)
$D_{\text{ср}}$	mean coil-spring diameter	m	input data
$[\tau]_{\text{пр}}$	adopted allowable spring shear stress	MPa	adopted by the designer

Example — conventional internal-combustion vehicle

$$d=\sqrt[3]{\frac{8P_{\text{пр}}D_{\text{ср}}}{\pi[\tau]}} \tag{22.1}$$

$$d=\sqrt[3]{\frac{8\cdot16497\cdot0.120}{\pi\cdot800\cdot10^{6}}}=0.018471\ \text{m}=18.47\ \text{mm}$$

- The source prints a SQUARE root. Its argument has units m³: N·m divided by N/m². A square root would give m^1,5 rather than length. The printed form yields 2.5 mm, a dimensionally meaningless result. The correct root is cubic.
- The guide prints 100 MPa. Lower allowable stress increases required wire diameter and changes coil count at fixed deflection. A different stress basis is selected here; verify spring feasibility using geometry, material and duty.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

- Departure from the source: [τ] = 800 MPa is adopted instead of the guide's 100 MPa. Its provenance is recorded as a separate decision.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

Source: [6]; [22]

Spring coil count

Determine active coils from required total deflection, calculated wire diameter, load, mean diameter and shear modulus. Add inactive coils for total count. A noninteger result is preliminary geometry; after adoption, check actual stiffness and coil-bind margin. The project guideline of 4–14 active coils prompts layout review rather than universally forbidding other designs.

Table Д.22.1.1 — Inputs for (22.1): Spring coil count

Symbol	Quantity	Unit	Provenance
d	calculated spring-wire diameter	m	calculated earlier (22.1)
f	total static and dynamic suspension deflection	m	calculated earlier (18.1)
$Z_{\text{дин}}$	design dynamic suspension load	N	calculated earlier (18.1)
$D_{\text{ср}}$	mean coil-spring diameter	m	input data
$n_{\text{п}}$-$n_{\text{р}}$	adopted number of inactive spring coils	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$n_{\text{р}}=\frac{fGd^{4}}{8P_{\text{пр}}D_{\text{ср}}{}^{3}},\qquad n_{\text{п}}=n_{\text{р}}+n_{\text{непр}} \tag{22.1}$$

$$n_{\text{р}}=\frac{0.32000\cdot80.0\cdot10^{9}\cdot0.018471^{4}}{8\cdot16497.28\cdot0.1200^{3}}=13.065$$

$$n_{\text{п}}=13.065+1.75=14.815\ \text{coils}$$

- Deflection determines active coils; add inactive end coils separately. They support the spring in its seats and do not contribute active deflection.
- Adopted inactive coil count: 1.8.
- f = $f_{дин}$ + $f_{ст}$; inactive coils 1.5…2.

Source: [6]; [22]

22.2. Round torsion bar

Round torsion bar

Torsion torque equals dynamic load times lever arm. Use it to find working diameter. The cube root follows circular-section modulus and dimensional consistency. The 800–1000 MPa range remains a teaching-project assumption requiring material, heat-treatment and fatigue justification.

Table Д.22.2.1 — Inputs for (22.2): Round torsion bar

Symbol	Quantity	Unit	Provenance
$Z_{дин}$	design dynamic suspension load	N	calculated earlier (18.1)
h_θ	torsion-bar lever arm	m	input data
$[\tau]_т$	adopted allowable shear stress of the round torsion bar	MPa	adopted by the designer

Example — conventional internal-combustion vehicle

$$d=\sqrt[3]{\frac{16M_{кр}}{\pi[\tau]}},\qquad M_{кр}=Z_{дин}h_\theta \tag{22.2}$$

$$M_{кр}=16497.28\cdot0.3000=4949.18\ \text{N·m}$$

$$d=\sqrt[3]{\frac{16\cdot4949.18}{\pi\cdot900.0\cdot10^{6}}}=0.030368\ \text{m}=30.37\ \text{mm}$$

- Splined ends $d_ш$ = (1.2…1.3)d = 36.4…39.5 mm; the source requires DIFFERENT sizes: $d_{ш1} \neq d_{ш2}$to prevent installing the torsion bar with its ends reversed.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

Source: [6]; [22]

Torsion-bar twist angle

After selecting diameter, calculate twist using working length and circular polar second moment. In the formula $J_p=\pi d^4/32$, use metres for lengths and pascals for shear modulus. Report total twist and twist per metre. The source's 15°/m limit is not automatically applied after changing the allowable-stress basis; this means that limit remains unchecked, not that every resulting angle is acceptable. Establish working deformation and reserve for the adopted material and design.

Table Д.22.2.2 — Inputs for (22.2): Torsion-bar twist angle

Symbol	Quantity	Unit	Provenance
d	calculated round torsion-bar diameter	m	calculated earlier (22.2)
$Z_{дин}$	design dynamic suspension load	N	calculated earlier (18.1)
h_{θ}	torsion-bar lever arm	m	input data
l	adopted torsion-bar working length	m	adopted by the designer

Example — conventional internal-combustion vehicle

$$\varphi = \frac{M_{кр}\ell}{GJ_{р}} \cdot \frac{180}{\pi} \tag{22.2}$$

$$\varphi = \frac{4949.18 \cdot 1.2000}{80.0 \cdot 10^{9} \cdot (\pi \cdot 0.030368^{4}/32)} \cdot \frac{180}{\pi} = 50.941^{\circ}$$

$$\varphi/\ell = \frac{50.941}{1.2000} = 42.451 \ ^{\circ}/\text{m}$$

- Twist increases directly with working length: a longer bar twists further at the same torque, which is how its compliance is selected.
- The guide gives a 15°/m limit. Changing the allowable-stress basis gives 42.5°/m here. This variant does not apply the printed limit. It does not establish acceptable twist: working deformation, material and durability require verification for the actual bar.
- The source's 15°/m check is not applied here. Acceptability of the calculated angle is not separately confirmed.

Source: [6]; [22]

Splined ends are thicker than the working section; unequal end sizes can ensure unique assembly. Show working length between fixings, not total length including splines. Angular animation may explain deformation, but verification uses the stated torque, stiffness and angle.

22.3. Laminated torsion member

Laminated torsion bar

Find stress in the specified leaf stack and compare it with the adopted 900 MPa design limit. In the formula, h is the smaller side, the strip thickness, while b is width. The original text reverses these descriptions; the rectangular geometry used here is consistent with the expression. Strip count must be a positive integer. Retain the same load, arm and torque definition as the round-bar calculation.

Table Д.22.3 — Inputs for (22.3): Laminated torsion member

Symbol	Quantity	Unit	Provenance
$Z_{дин}$	design dynamic suspension load	N	calculated earlier (18.1)
h_{θ}	torsion-bar lever arm	m	input data
h	torsion-strip thickness	m	input data
b	torsion-strip width	m	input data
$n_{п}$	number of torsion strips	—	input data

Example — conventional internal-combustion vehicle

$$\tau = 3M_{кр}\frac{1+0{,}6h/b}{n_{п}h^{2}b} \tag{22.3}$$

$$\tau = \frac{3\cdot 4949\cdot 10^{3}(1+0{,}6\cdot \frac{10.0}{60.0})}{6\cdot 10.0^{2}\cdot 60.0} = 454 \text{ MPa}$$

- The source labels h and b as the lengths of the LARGER and SMALLER sides, respectively. Reading the labels in reverse, with h the THIN dimension and b the wide dimension, makes the printed expression identical to the classical rectangular-section torsion solution τ = 3M(1 + 0,6t/a)/(n·a·t²). The label, rather than the formula, is reversed; the expression is used as written here.
- Taking the label literally would give 316 MPa instead of 454 MPa, a 30 % underestimate. This would make the torsion member appear stronger than it is, an unsafe direction of error.
- Calculated stresses must not exceed 900 MPa.
- The adopted limits are 800…1000 MPa for a round torsion bar and 900 MPa for a laminated torsion member. Comparing these values does not replace material selection or deflection checks for both designs.
- Component-calculation reference: Buriennikov, Kashkanov and Rebedailo (2013). Numerical allowances in this example are retained as design assumptions; refine them for the material and loading conditions.

Source: [6]; [22]

Finally compare diameters, lengths, required installation space and the adopted strength basis. Acceptable stress does not replace durability checks, spring stability, end-connection checks or verification of the required suspension travel.

Chapter 23. Wheel reduction gears

Preliminary selection of the wheel-reduction ratio, tooth counts and dimensions.

A wheel reduction gear moves part of the overall reduction to the wheel. This can reduce torque in upstream components or provide the required axle layout. An electric motor alone does not establish the need for wheel reduction; justify it using the kinematic layout and ratio distribution.

The guidelines, subsection 4.1.6, printed pp. 55–57, provide an initial selection of tooth counts, widths and standard sizes. The calculation below is preliminary. A final planetary design must also reconcile actual ratio after rounding tooth counts, mesh geometry, equal planet-spacing assembly and component strength.

23.1. Tooth counts and transmission ratio

Wheel-reduction tooth counts

The source requires at least 12 sun-gear teeth. The adopted ring-gear count can differ from the initial product, so calculate actual reduction from the adopted counts. Divisibility of ring-gear teeth by the planet count is insufficient: with a common module and no profile shift, $Z_c=(Z_k-Z_{по})/2$ must be a positive integer, and $(Z_k+Z_{по})/n_c$ must be an integer. These checks establish coaxial compatibility and equally spaced assembly. Check planet clearance, interference, undercutting and strength separately.

Selection rule. Calculate the initial ring-gear tooth count, select an integer candidate and check the minimum sun-gear count, a positive integer planet tooth count and the equal-spacing assembly condition. Reject an incompatible candidate.

Table Д.23.1 — Inputs for (23.1): Wheel-reduction tooth counts

Symbol	Quantity	Unit	Provenance
U_p	adopted wheel-reduction ratio	—	adopted by the designer
n_c	adopted number of wheel-reduction planet gears	—	adopted by the designer
$Z_{по}$	adopted sun-gear tooth count	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$Z_{по}\geq 12, \qquad Z_k=Z_{по}(U_p-1) \tag{23.1}$$

$$Z_k=18(3.00-1)=36.0\rightarrow 36$$

- The sun gear has at least 12 teeth.
- After rounding according to the guide, the planet count is verified as an integer: Zc = (Zк − Zпо)/2 = 9. Equal spacing is also checked: (Zпо + Zк)/nc = 18. Divisibility of Zк by planet count alone is insufficient (КНК, §17.1).
- The adopted tooth counts give U_p = 3 instead of the specified 3: the effect of rounding.
- Only integer tooth counts and the assembly phasing condition are checked. Adjacent-planet clearance, undercutting, interference, load sharing and strength require separate verification.

Source: [6]; [23]

23.2. Centre distance

The printed expression in subsection 4.1.6 contains inconsistent units and a size estimate not confirmed for the example. This project explicitly adopts an analogy: Opanasiuk's preliminary centre-distance relationship on p. 33 is applied to torque per planet. Adopt a coefficient of 8,6–9,6, enter torque in N·m and obtain distance in millimetres. Extending the relationship to wheel reduction is a project assumption, not a substitute for gear-contact analysis.

Preliminary minimum wheel-reduction centre distance

The series appears on p. 56 of the guidelines. If the nearest value is below the calculated lower bound, choose the next larger value. If the result exceeds the whole series, its last entry does not establish compliance. In comparison with the printed expression, 1000 MPa replaces the printed 1000 GPa. This separate unit correction does not establish the correctness of the rest of the printed expression.

Selection rule. Calculate preliminary distance using the adopted relationship and choose the first source-series value no smaller than the result.

40; 50; 63; 71; 80; 90; 100; 112; 125; 140; 160; 180; 200; 225; 250; 280; 315 mm

Table Д.23.2 — Inputs for (23.2): Preliminary minimum wheel-reduction centre distance

Symbol	Quantity	Unit	Provenance
$M_{\text{розр}}$	design torque in one half-shaft	N·m	calculated earlier (14.5)
U_{p}	adopted wheel-reduction ratio	—	adopted by the designer
n_{c}	adopted number of wheel-reduction planet gears	—	adopted by the designer
$[\sigma]$	adopted allowable tooth contact stress	MPa	adopted by the designer

Example — conventional internal-combustion vehicle

$$a_{\text{w}} \approx k_{\alpha} \sqrt[3]{\frac{M_{\text{по}}}{n_{\text{c}}}} \tag{23.2}$$

$$a_{\text{w}} \approx 9.1 \sqrt[3]{\frac{1707}{3}} = 75.4 \rightarrow 80 \text{ mm}$$

- Calculate centre distance using Opanasiuk's rule (p. 33): $a_{\text{w}} \approx k_{\alpha} \cdot \sqrt[3]{\text{M}}$, k_{α} = 8.6…9.6. Extending the gearbox rule to wheel reduction is a project analogy for initial sizing, not a completed contact check of this planetary gear.
- Use torque PER PLANET: 1 707 / 3 = 569 N·m. This preliminary estimate assumes equal sharing; load unevenness needs separate verification.
- The expression printed in the guide, $a_{\text{w}} \geq K_{\text{a}}(U_{\text{p}}-1)\sqrt{(\Omega \text{K_HM_по}/(\text{n_c}\psi\text{_a}[\sigma]^2))}$, would give 370 mm, 4,9 times greater. The reason for the difference, whether a constant, root exponent or another assumption, is not established here. Size alone does not prove every possible design unsuitable; both numbers are given to examine the adopted method.
- The source prints [σ] = 1000 GPa; the comparison uses 1000 MPa. This explicit unit correction does not validate the rest of the printed expression.
- Centre-distance series: 40, 50, 63, 71, 80, 90, 100, 112, 125, 140, 160, 180, 200, 225, 250, 280, 315 mm. The nearest value is 71 mm; 80 mm is chosen because a_{w} is a lower bound.

- Departure from the source: centre distance uses Opanasiuk's rule $a_w \approx k_\alpha \cdot \sqrt[3]{}(M_по/n_c)$, rather than the expression in subsection 4.1.6. The printed expression gives 370 mm versus 75 mm. Applicability of the adopted analogy requires a separate check of the actual reduction gear.
- Departure from the source: the comparison reads [σ] as 1000 MPa rather than 1000 GPa.
- Departure from the source: the nearest series value, 71 mm, is below the calculated 75.4 mm, so the next value, 80 mm, is selected. The source asks for nearest-value rounding but also states $a_w \geq \ldots$; choosing the nearest value below the result violates that inequality.

Source: [6]; [10]

23.3. Gear module

Reduction-gear module

The series comes from Table 15 of the guidelines and is also used in Chapter 10. With a fixed ring gear, sun input and carrier output, $U_{р,факт}=1+Z_к/Z_{по}$. Coaxial geometry gives $a_w=m(Z_{по}+Z_к)/4$, hence $m_{потр}=4a_{w,min}/(Z_{по}+Z_к)$, and $D_{по}=4a_{w,факт}/U_{р,факт}$. The expression $2a_w/(U_р+1)$ applies to a separate external gear pair, not to this overall planetary reduction. The adopted module determines every pitch diameter as mZ, the actual centre distance and gear face widths. Actual centre distance must meet or exceed the preliminary target. This preliminary geometric selection does not verify contact or bending strength, interference, clearances or durability.

Selection rule. Calculate the required module from planetary geometry and minimum centre distance, then select the first source-series module no smaller than required. Recalculate actual diameters and centre distance using that module.

2; 2,25; 2,5; 2,75; 3; 3,5; 4; 4,5; 5; 5,5; 6; 7; 8; 9; 10; 11; 12; 14; 16; 18 mm

Table Д.23.3 — Inputs for (23.3): Reduction-gear module

Symbol	Quantity	Unit	Provenance
a_w	adopted wheel-reduction centre distance	mm	calculated earlier (23.2)
$Z_{по}$	adopted sun-gear tooth count	—	adopted by the designer
$Z_к$	Adopted wheel-reduction ring-gear tooth count	—	calculated earlier (23.1)

Example — conventional internal-combustion vehicle

$$m_{потр}=\frac{4a_{w,min}}{Z_{по}+Z_к}, \qquad m \geq m_{потр} \tag{23.3}$$

$$m_{потр}=\frac{4\cdot 80.00}{18+36}=5.93\rightarrow 6 \text{ mm}$$

- The source prints $D_{по}$ = 2Z_по/($U_р$ + 1), with TOOTH COUNT in the numerator. Dimensionally, this is not a diameter. Replacing $Z_{по}$ with distance repairs the dimensions, but the formula 2a_w/($U_р$ + 1) belongs to an external pair, not the adopted planetary layout.
- First, the dimensions: 2Z_по/($U_р$+1) is dimensionless, whereas $D_{по}$ is a diameter. Second, the consequence: the module would be m = D_по/Z_по = 2/($U_р$+1) = 0,5 mm, identical for every gear with this ratio, from a passenger car to a dump truck. The centre distance calculated immediately above would be unused.

- For the planetary layout, a_w = m(Zпо + Zc)/2 = m(Zпо + Zк)/4. The adopted module gives афакт = 81 mm, Dпо = 108 mm, Dc = 54 mm and Dк = 216 mm. The tooth ratio is unchanged: Uфакт = 3.
- The first series module no smaller than required is adopted; actual diameters, centre distance and widths are then recalculated.

- Departure from the source: dimensional geometry of the adopted planetary mesh replaces tooth count in the diameter formula. An external gear pair is not combined with the planetary ratio.
- The module is selected from a preliminary geometric requirement. Wheel-reduction contact, bending, undercutting, interference and durability checks have not been performed.

Source: [6]; [23]; [7]

23.4. Gear face widths

Table M.23.1 — Wheel-reduction face width by transmission ratio

Adopted transmission ratio	Face-width to centre-distance coefficient
No greater than 6,8	0,5
Above 6,8	0,4

The threshold 6,8 and multipliers appear on source p. 56. Increase each successive gear face width by a factor of 1,04–1,05. The example adopts the midpoint; calculated widths remain preliminary until a justified manufacturing series is specified.

Wheel-reduction gear face widths

Starting from actual centre distance determined by the adopted module and integer tooth counts, calculate ring-gear, planet and sun-gear widths in sequence. The previous calculation gives a lower target; the actual distance is $m_{пр}(Z_{по}+Z_к)/4$. All three widths belong to the same geometry. Recalculate them together when module, tooth counts or the width multiplier change.

(23.4) **Applicability:** The adopted ratio is no greater than 6,8; the face-width coefficient is 0,5.

The worked example adopts a ratio of 3. At this stage the sequentially calculated widths are shown without claiming final standardisation.

Table Д.23.4.1 — Inputs for (23.4): Wheel-reduction gear face widths

Symbol	Quantity	Unit	Provenance
m	Adopted wheel-reduction gear module	mm	calculated earlier (23.3)
$Z_к$	Adopted wheel-reduction ring-gear tooth count	—	calculated earlier (23.1)
$Z_{по}$	adopted sun-gear tooth count	—	adopted by the designer
U_p	adopted wheel-reduction ratio	—	adopted by the designer
b_c/b_2	adopted progressive face-width multiplier	—	adopted by the designer

Example — conventional internal-combustion vehicle

$$a_{\text{факт}} = \frac{m(Z_{\text{по}} + Z_{\text{k}})}{4}, \qquad b_2 = \psi_{\text{a}} a_{\text{факт}},\ b_{\text{c}} = (1{,}04 \ldots 1{,}05) b_2,\ b_{\text{по}} = (1{,}04 \ldots 1{,}05) b_{\text{c}} \tag{23.4}$$

$$a_{\text{факт}} = \frac{6.00000 \cdot (18 + 36)}{4} = 81.00000 \text{ mm}$$

$$b_2 = 0.50000 \cdot 81.00000 = 40.50000 \text{ mm}$$

$$b_{\text{c}} = 1.04500 \cdot 40.50000 = 42.32250 \text{ mm}$$

$$b_{\text{по}} = 1.04500 \cdot 42.32250 = 44.22701 \text{ mm}$$

- Widths use the actual centre distance of 81 mm after adopting the module and integer tooth counts. The preliminary lower bound from (23.2) is not reused as finished geometry.
- Widths are calculated sequentially using the adopted multiplier. No rounding to a separate manufacturing series has been performed; justify that series and final dimensions during detailed design.
- Face-width coefficient $\psi_{\text{a}} = 0.5$: the source gives 0.5 at $U_{\text{p}} \le 6.8$ and 0.4 above it.

Source: [6]

(23.4) **Applicability:** The adopted ratio exceeds 6,8; the face-width coefficient is 0,4.

This separate example adopts a ratio of 7. Using the same torque and calculation engine isolates the effect of changing the table row.

Table Д.23.4.2 — Inputs for (23.4): Wheel-reduction gear face widths

Symbol	Quantity	Unit	Provenance
m	Adopted wheel-reduction gear module	mm	calculated earlier (23.3)
Z_{k}	Adopted wheel-reduction ring-gear tooth count	—	calculated earlier (23.1)
$Z_{\text{по}}$	adopted sun-gear tooth count	—	adopted by the designer
U_{p}	adopted wheel-reduction ratio	—	adopted by the designer
b_c/b_2	adopted progressive face-width multiplier	—	adopted by the designer

Example — high-ratio wheel reduction

$$a_{\text{факт}} = \frac{m(Z_{\text{по}} + Z_{\text{k}})}{4}, \qquad b_2 = \psi_{\text{a}} a_{\text{факт}},\ b_{\text{c}} = (1{,}04 \ldots 1{,}05) b_2,\ b_{\text{по}} = (1{,}04 \ldots 1{,}05) b_{\text{c}} \tag{23.4}$$

$$a_{\text{факт}} = \frac{2.75000 \cdot (18 + 108)}{4} = 86.62500 \text{ mm}$$

$$b_2 = 0.40000 \cdot 86.62500 = 34.65000 \text{ mm}$$

$$b_{\text{c}} = 1.04500 \cdot 34.65000 = 36.20925 \text{ mm}$$

$$b_{\text{по}} = 1.04500 \cdot 36.20925 = 37.83867 \text{ mm}$$

- Widths use the actual centre distance of 86,63 mm after adopting the module and integer tooth counts. The preliminary lower bound from (23.2) is not reused as finished geometry.
- Widths are calculated sequentially using the adopted multiplier. No rounding to a separate manufacturing series has been performed; justify that series and final dimensions during detailed design.
- Face-width coefficient $\psi_{\text{a}} = 0.4$: the source gives 0.5 at $U_{\text{p}} \le 6.8$ and 0.4 above it.

Source: [6]

The final reduction-gear drawing must show the sun, planets, ring gear, input and output members, and adopted dimensions. If it contains only preliminary schematic outlines, label it accordingly; drawn teeth do not establish a verified working tooth profile.

Final running-gear checks and presentation of results

How to combine individual calculations into a consistent course-project design.

After Chapters 16–23, prepare a short summary of adopted designs and checks. For each assembly state the input load, adopted dimensions, check result and required refinement. Retain a failing alternative in the comparison with its rejection reason; do not include it in the final design merely because a numerical result exists.

Table П.2 — Consistency of final verification results

Check	Quantities that must agree
Masses and supports	Load state, axle forces, driven axles, centre of mass and equivalent bogie support
Ride and elastic components	The same static deflection in Chapters 16 and 18; the same loads and total travel in Chapters 18 and 22
Steering	Steered-axle load, scrub radius, trapezoid geometry, angle series and required rack travel
Brakes	The same load state, wheel reactions, selected mechanism, actuation and explicitly defined assistance
Wheel reduction gear	Adopted tooth counts, actual ratio, module, diameters, assembly and transmitted torque
Figures	Calculated dimensions, prototype reference dimensions and schematic proportions have different, explicitly explained bases

For every formula in the report, include symbols, units, source or derivation, substitution and conclusion. If the substitution already contains the result, do not duplicate it in a separate Result row. For an adopted parameter give the table or range, selected row and reason. Explain any source correction and its basis; do not present an original assumption as source data.

The interactive version may show mechanism motion, parameter changes or a sequence of check points. The printed version must remain self-contained: a static vector figure, caption, symbols and numerical table must explain the same state. Unknown standard dimensions, unperformed checks and adopted simplifications must remain visible in both versions.

Sources

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[2] Input data specified for this project. Interactive project input.

[3] Перевірений набір даних прототипів автомобілів. Локальна база даних automethods.com «Автомобілі». Курований спільний набір даних: прототипи, вимірювання та шини. Значення, умови вимірювань і повне походження записів збережено в локальній базі.

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[6] Методичні вказівки до виконання курсового проекту з дисципліни «Автомобілі». Київ: УТУ, 1995. Упоряд.: Сахно В. П., Основенко М. Ю., Вельбівець А. Ф., Філіпова Г. А. Методичні вказівки до виконання курсового проекту з дисципліни «Автомобілі». Для спеціальності 7.090215 «Автомобілі та автомобільне господарство», денна форма навчання. Київ : УТУ, 1995. Розділи 1—2; 4.1.1 (зчеплення), с. 37—40.

[7] Побудова цього проєкту — співвідношення, якого джерела не друкують.

[8] Ukrainian Road Traffic Rules, пункт 22.5 — граничні mass. Ukrainian Traffic Rules, paragraph 22.5.

[9] Reference table for the selected road surface. Reference table.

[10] Опанасюк Є. Г. Методичні вказівки до курсового проєктування з дисципліни «Автомобілі». с. 28—32, рівняння (46)—(52); підрозділи 2.1 і 2.4.

[11] Правила ООН № 54, додаток 4: індекси навантаження tyres; робочі таблиці проєкту. United Nations Economic Commission for Europe UN Regulation No. 54. Tyres for commercial vehicles and their trailers. Annex 4: list of load-capacity indices and corresponding maximum mass. Позначення категорій швидкості та індексів навантаження. URL: https://unece.org/transport/vehicle-regulations-wp29/standards/addenda-1958-agreement-regulations-41-60

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